

# Set-Sized Quotients of the Omnific Integers

The universal constant-term quotient, exact cardinal thresholds,  
support thresholds, and what survives in large quotients

Merged research report built from thirteen manuscripts (03, 04, 06–11 of one batch,  
12–16 of the next)

22–23 September 2026; merged 23 September 2026

## Abstract

Write an omnific integer uniquely as  $n + s$  with  $n \in \mathbb{Z}$  and  $s$  purely infinite, and let  $\text{ct}(n + s) = n$ . Every additive and multiplicative map from  $\mathbf{Oz}$  to a set-sized ring, even a noncommutative, nonreduced or nonunital one, has the form  $x \mapsto \text{ct}(x)e$  with  $e$  idempotent; every set-sized  $\mathbf{Oz}$ -module is an abelian group with scalar action through  $\text{ct}$ . The same holds for  $D + \Pi_K$  over any set field  $K$  and unital  $D \subseteq K$ , for the Gaussian ring  $\mathbf{Oz}[i]$  (with  $\mathbb{Z}[i]$ ), and, on the finite side, for standard part on the finite surreal and surcomplex rings. Countable supports already force this collapse while finite supports do not.

Over set-sized exponent groups the theorem becomes a detection threshold. For each of five explicit set-sized rings (four integer parts inside  $\mathbf{Oz}$ , of sizes  $\kappa^{\aleph_0}$ ,  $\kappa^{<\kappa}$ ,  $\kappa$  and  $\lambda$ , and a lexicographic Hahn ring of size  $\kappa^{<\kappa}$ ), the least size of a ring or module detecting any nonzero purely infinite element equals the size of the ring; two of these values, and the exact thresholds of two examples in fixed exponent groups, are new to this merge. The report also keeps the large-quotient side (a copy of  $\mathbf{No}$  and a Cantor algebra inside  $\mathbf{Oz}/(1 + \omega^a)$ ), the integral and complete integral closures of  $\mathbf{Oz}$ , the arithmetic of integer-valued polynomials on  $\mathbf{Oz}$ , and a negative answer to a posted question on natural ordinal arithmetic. Five later manuscripts add the ordinary homological and tensor algebra of subrings of  $\mathbf{Oz}$ : for set-sized finite-support cores the arithmetic quotient has flat dimension  $d$  when the exponents form a coordinatewise cone of rank  $d$  and 1 when they form a totally ordered cone, and coefficient-lattice ideals have explicit tensor, Rees and duality algebra; and they add a fresh-scale image-gap theorem for polynomial and rational maps. It is an AI-assisted merge of unrefereed manuscripts; nothing in it is verified in Lean.

**Status.** Proof-bearing research report assembled from thirteen AI-assisted manuscripts written on 22 and 23 September 2026. The sources call their main results candidate original; priority is not certified, no named conjecture is claimed solved, the proofs have not been refereed, and no statement is formalized. Results added in the merge are marked [merge]; the source of every other statement is tagged [NN].

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# 1 Introduction

## 1.1 The question and the answer

The omnific integers  $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$  form a discretely ordered subring of  $\mathbf{No}$  whose fraction field is all of  $\mathbf{No}$ . Their purely infinite ideal  $\Pi$  is a proper class; each of its elements has a set-sized normal form, but below any positive exponent there is a proper class of smaller exponent scales. The eight manuscripts of the first merge (03, 04, 06–11) ask the same question: *what can an ordinary, set-sized ring or module see of an omnific integer?* All eight give the same answer, by different proofs, and two later manuscripts (13, 15) reprove it independently: exactly its ordinary integer constant term. This is much stronger than the observation that a proper-class field cannot embed in a set: the source is a ring, its homomorphisms have large kernels, and the targets may have zero divisors, nilpotents, noncommutative multiplication and any characteristic (06, 11). Nor is it a consequence of divisibility of  $\Pi$  by every integer, which only disposes of targets of positive characteristic (04, 07).

The sources then go in different directions: exact cardinal thresholds for set-sized integer parts (03, 06, 08, 09, 11), the finite side and the support threshold (04, 11), the large quotients in which a purely infinite element becomes a unit (07), integral closure (03), polynomial maps (10), and natural ordinal arithmetic (11). Three later manuscripts add the ordinary homological and tensor algebra of subrings of  $\mathbf{Oz}$  with finite supports or with coefficient lattices (12, 14; [Section 13](#)), and fresh-scale arguments for polynomial and rational maps (16; [Sections 16.6](#) and [16.7](#)). This report prints the common spine once and keeps every result of every source.

## 1.2 Main theorems

**Theorem A** (Universal set-sized quotient). *Let  $K$  be a set field and  $D \subseteq K$  a unital subring. Every additive multiplicative map from  $D + \Pi_K$  to a set-sized ring kills  $\Pi_K$  and factors uniquely through  $\text{ct}$ ; unital maps correspond bijectively to unital maps from  $D$ ; every set-sized module is annihilated by  $\Pi_K$ . For  $\mathbf{Oz}$  the unique unital map to a set-sized ring  $S$  is  $x \mapsto \text{ct}(x)1_S$ , and unital maps  $\mathbf{Oz}[i] \rightarrow S$  correspond to  $j \in S$  with  $j^2 = -1$ . The kernels admitting set-sized quotients are exactly the  $\text{ct}^{-1}(\mathbf{a})$ ; for  $\mathbf{Oz}$  they are  $\Pi$  and  $n\mathbf{Oz}$  ( $n \geq 1$ ). ([Theorems 5.1](#) and [6.2](#) and [Corollaries 5.2](#) and [5.3](#); the eight sources of the first merge, manuscripts 13 and 15, and manuscript 05 of the sibling report.)*

**Theorem B** (The finite side and the support threshold). *Standard part is the universal set-sized quotient of the finite surreal and surcomplex rings, and more generally of  $D + \mathfrak{m}_k$  (04). For every infinite  $\lambda$ , the rings of normal forms with supports of size at most  $\lambda$  (full surreal exponents) already have universal set-sized quotient  $D$  on both sides, while the finite-support rings admit the augmentation  $\sum c_g \omega^g \mapsto \sum c_g$ ; a map on the finite-support ring extends to the countable-support ring if and only if it factors through  $\text{ct}$  (the failure for the augmentation is 04's and 11's, the criterion is 11's). ([Theorems 8.2](#) and [9.1](#).)*

**Theorem C** (Detection threshold equals size). *For each of five explicit set-sized rings  $A$ , namely four integer parts inside  $\mathbf{Oz}$ , of cardinality  $\kappa^{\aleph_0}$  (cf  $\kappa > \aleph_0$ ; 03),  $\kappa^{<\kappa}$  ( $\kappa$  regular uncountable; 06),  $\kappa$  (every infinite  $\kappa$ ; 08) and  $\lambda$  ( $\lambda^{\aleph_0} = \lambda$ ; 09), and 11's lexicographic Hahn ring of cardinality  $\kappa^{<\kappa}$  (coefficient field with at most  $\max(\kappa, 2^{\aleph_0})$  elements), the least cardinality of a ring target or module detecting any nonzero purely infinite element is exactly  $|A|$ , in ZFC. ([Theorem 10.10](#).) The values for 06's and 11's rings are new here: 06 and 11 prove exactness only under  $\kappa^{<\kappa} = \kappa$ , and the unconditional value  $\kappa^{<\kappa}$  follows from 06's own collision lemma ([Theorem 11.11](#) and [Proposition 11.23](#)). The same thresholds hold for the first and second Ext groups of the homological package of 06, which extends to all five rings ([Theorems 12.2](#) and [12.3](#)).*

**Theorem D** (The field bound in a fixed exponent group). *In a set-sized Hahn integer part, a ring or module detecting an element divisible by  $X^{2^c}$  has at least as many elements as the field  $L_c(F)$  of elements of the ambient field  $F$  supported in the scale subgroup  $H_c$  (03, generalized through 06's lemma). This bound is exact in 04's example  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ , where it equals  $2^{\aleph_0}$ , and in an example with rational coefficients it exceeds by far all the monomial and weighted counts of 04, 07 and 10, showing that 10's bound need not be attained. (Proposition 10.3 and Examples 10.4 and 10.5.)*

**Theorem E** (Large internal fields). *If a nonzero purely infinite element becomes a unit in a nonzero ring, the ring contains an explicitly embedded copy of  $\mathbf{No}$ . The nonzero quotient  $\mathbf{Oz}/(1 + \omega^a)$ , which has no nonzero set-sized image, contains an algebraically closed field containing  $\mathbf{No}[i]$ , exact cyclotomic algebras over it, and the Cantor algebra of locally constant functions on  $\mathbb{Z}_2$  (07; one proposition of 06). (Theorems 14.7, 14.15 and 14.19.)*

**Theorem F** (Closures and invisible equations). *The complete integral closure of  $\mathbf{Oz}$  is  $\mathbf{No}$ , while its integral closure is a proper, fine-dense subring with zero conductor that no set generates; its constant slice is the real algebraic integers, it has no finite quotients, and no nontrivial set-valued valuation of  $\mathbf{No}$  contains  $\mathbf{Oz}$  (03). For  $\gamma > 0$  and  $m \geq 2$  the equation  $X^m = \omega^\gamma + 1$  has no root in  $\mathbf{Oz}[i]$ , yet its image in every set-sized ring has the root 1 (08). (Theorems 15.5, 15.12, 15.22 and 15.25.)*

**Theorem G** (Polynomial maps). *The ring of finite-degree surreal polynomials preserving  $\mathbf{Oz}^r$  is  $\Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$  with universal set-sized image  $\text{Int}(\mathbb{Z}^r)$ ; a Newton least-common-multiple condition decides congruence preservation modulo every omnific integer; the  $p$ -adic completion is  $C(\mathbb{Z}_p^r, \mathbb{Z}_p)$  and most of its characters are not evaluations; rational self-maps of  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  are polynomial and the bijections are affine (10). For every unital  $D \subseteq k \in \{\mathbb{R}, \mathbb{C}\}$  and every polynomial  $f$  of degree at least two with coefficients in a set-sized Hahn field  $K_G$ , a fresh monomial  $t$  has  $f(D + \Pi_k) \cap (f(t) + K_G) = \{f(t)\}$ , every root of  $f(Y) = f(t) + c$  ( $c \neq 0$ ) having an explicit forbidden block of negative exponents; rational self-maps of  $D + \Pi_k$  are polynomials and the surjective ones are the affine maps  $uX + a$ ,  $u \in D^\times$ ;  $d + 1$  points test degree  $d$  and no fewer do, while no set-sized sample tests all rational self-maps (16). (Theorems 16.7, 16.14, 16.36, 16.43, 16.46 and 16.51 and Corollaries 16.20 and 16.29.)*

**Theorem H** (Natural ordinal arithmetic). *The Grothendieck ring of the ordinals under natural operations is not a quotient of  $\mathbf{Oz}$ ; every unital ring map from  $\mathbf{Oz}$  to it has image  $\mathbb{Z}$ . This answers negatively the quotient question in MathOverflow question 188430 of Jesse Elliott (11; independently 13). (Theorem 17.1.)*

**Theorem I** (Homological and tensor algebra of omnific cores). *Let  $D \subseteq K$  be  $\mathbb{Z} \subset \mathbb{R}$  or  $\mathbb{Z}[i] \subset \mathbb{C}$ . The set-sized subring  $A_d = D + \bigoplus_{0 \neq \alpha \in \mathbb{Q}_{\geq 0}^d} KX^\alpha$  of  $\mathbf{Oz}$  (or  $\mathbf{Oz}[i]$ ), whose exponents form a coordinatewise cone of rank  $d$ , has  $\text{Tor}_{>0}^{A_d}(D, D) = 0$  but  $\text{fd}_{A_d} D = d$ , with a complete multigraded computation of  $\text{Tor}^{A_d}(D, A_d/(X_i : i \in F))$ , and its finite quotients are those of  $D$  (12). Enlarging the cone to all positive exponents of the same ordered group is a nonflat extension over which  $\text{fd } D = 1$  and  $\text{pd } D = 2$ , as for the totally ordered cones of Section 12 (12, with the merge). For a coefficient lattice  $M \subset K$  of rank  $r \geq 2$ , the module  $L(M) = M + \Pi$  over  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  (or over the set-sized model  $A_1$ ) has first syzygy  $\Pi^{r-1}$ , tensor square  $\Pi \oplus (M \otimes M)$  whose torsion is  $\ker(M \otimes M \rightarrow M^2)$ , nonlinear Rees equations exactly the ordinary relations among the generators (the first one in degree  $d$  for  $(\omega, 2^{1/d}\omega)$ ), isomorphism type determined by  $M$  up to a scalar, ideal classes realizing  $\text{Pic}$  of an order, and dual  $\Pi$  (14). (Theorems 13.7, 13.9, 13.11, 13.14, 13.18, 13.21, 13.24, 13.25 and 13.27 and Corollary 13.8.)*

### 1.3 Sources, provenance and where the merge chose

The report is built from thirteen manuscripts. Eight (03, 04, 06–11), dated 22 September 2026 and placed in commit `be06fc8`, were merged first; their numbers are arrival numbers in their batch. Five (12–16), dated 23 September 2026 and placed in commit `cf350b1`, are items 02–06 of the following batch; they are numbered here by the file prefixes under which their programs and data are stored in this directory. None has a reviewer, and none is shipped with this report.

|    | Manuscript (pages)                                                            | Pin     | Contribution and placement                                                                                                                                                                                                                                        |
|----|-------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03 | <i>Cardinal Visibility and Normalization of Omnific Integers</i> (25)         | cd80e5e | field bound ( <a href="#">Proposition 10.3</a> ); model $A_\kappa^{03}$ with threshold $\kappa^{\aleph_0}$ ( <a href="#">Section 11.1</a> ); flatness, Tor; integral and complete integral closure, valuations ( <a href="#">Section 15</a> )                     |
| 04 | <i>What Set-Sized Algebra Can See of Surreal Arithmetic</i> (24)              | cd80e5e | $D + \Pi_k$ theorem via separation and telescope; finite side ( <a href="#">Section 8</a> ); support threshold with negative side ( <a href="#">Section 9</a> ); fixed-group bounds; localizations; modules, derivations, algebras                                |
| 06 | <i>The Universal Set-Sized Quotient of the Omnific Integers</i> (25)          | a5c2a97 | <b>base</b> : coefficient-ring theorem for any set field, scaled-field collision lemma, direct module argument; model $A_\kappa^{06}$ ; homological package ( <a href="#">Section 12</a> ); matrix representations; polynomial tests; prime-reservoir proposition |
| 07 | <i>Small Quotients and Large Internal Fields of the Omnific Integers</i> (27) | 37eefca | theorem for any set field via weighted certificates; completions; compression $\mathbf{No} \hookrightarrow \mathcal{F}_a$ , transfer, binomial kernel, Cantor algebra ( <a href="#">Section 14</a> )                                                              |
| 08 | <i>What Small Rings Can See of Omnific Integers</i> (26)                      | 37eefca | Hartogs formulation, nonunital maps; closure formula; invisible equation; class-indexed derivations; criterion; model $A_\kappa^{08}$ for every infinite $\kappa$ ; cyclic-module lemma                                                                           |
| 09 | <i>Set-Sized Shadows of Omnific Integers</i> (20)                             | 37eefca | direct division of $D + \Pi_k$ elements; model $A_\lambda^{09}$ , residue fields, non-arithmetic maximal ideals, $\aleph_1$ generators; Gaussian ramification                                                                                                     |
| 10 | <i>Set-Sized Shadows of Omnific Arithmetic</i> (26)                           | a5c2a97 | theorem for any set field incl. maps on $\Pi_K$ ; quantitative bound; endomorphisms; numerical polynomials, $p$ -adic completion and characters, lattice rigidity ( <a href="#">Section 16</a> )                                                                  |
| 11 | <i>What Set-Sized Algebra Can See of the Omnific Integers</i> (20)            | 37eefca | pairwise-difference families; localization and presentation calculus; extension criterion; coinitality theorem; residual targets; ordinal ring ( <a href="#">Section 17</a> )                                                                                     |

|    | Later manuscript (pages)                                                               | Pin     | Contribution and placement                                                                                                                                                                                     |
|----|----------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | <i>Arithmetic Invisibility and Homological Dimension in Omnific Integer Cores</i> (21) | none    | coordinate-cone cores $A_d$ : Boolean flat resolution, multigraded Tor, $\text{fd } D = d$ ; noncoherence certificate; ordered-cone collapse to $\text{fd } D = 1$ ; projective bounds (Section 13)            |
| 13 | <i>Small-Target Rigidity and Arithmetic Specialization</i> (21)                        | 71e9606 | independent proofs of the universal theorem and of the ordinal answer (credited in Section 5.1 and Section 17); its fiber and norm theorems belong to the Diophantine report [30]                              |
| 14 | <i>Arithmetic Tensors in the Omnific Integers</i> (19)                                 | 71e9606 | coefficient-lattice modules: syzygy, tensor normal form, symmetric and exterior powers, Rees equations, Hom, ideal classes, duals, matrix kernels, $\text{fd}$ over $A_0$ (Section 13)                         |
| 15 | <i>Omnific Integers Have Only Ordinary Set-Sized Representations</i> (21)              | 71e9606 | an independent proof of the universal theorem and its consequences (credited throughout); two one-line additions (Corollary 5.5 and Proposition 7.3)                                                           |
| 16 | <i>Fresh-Scale Image Gaps and Rational Rigidity for Omnific Integers</i> (23)          | 71e9606 | image gap and forbidden layers, fresh fibers, rational collapse and surjective maps over any constant ring, optimal and Gaussian finite tests, rational probes, sampling obstructions (Sections 16.6 and 16.7) |

The pins are the repository commits the manuscripts inspected (cd80e5e0adc2, a5c2a97df4b5, 37eefca1a309, all of 22 September 2026, Pacific time; and 71e9606d297c, 23 September 2026, for 13–16). The commit 71e9606 precedes the placement of the first eight manuscripts, so 13–16 saw neither this report nor its sibling; 12 records no commit, only access to the repository through a GitHub connector on 23 September 2026. Manuscript 05 of the same batch, *Omnific Integers and Diophantine Rigidity*, is merged into the sibling report [30]. It also proves the unital universal theorem for **Oz** (its “set-target rigidity”), the clearing of set supports and common monomial divisors, and it asks the “exact size boundary” question that 03, 06, 08, 09 and 11 answer; 03 credits the first three to 05 and 04 credits its finite-quotient background to manuscript 02 of the same sibling report. Those credits are kept.

### Where the merge chose.

- *Base.* 06 is the base: its coefficient-ring theorem over any set field (also proved by 07 and 10) is the most general statement, and its scaled-field collision lemma (with 03’s field bound, a special case) is the mechanism that yields the sharp cardinal bounds where counting monomials does not suffice (03, 06, 11 and the fixed-group examples; 08 and 09 reach their thresholds by counting monomials). The proof is written in 08’s choice-free Hartogs form, with 08’s and 10’s nonunital forms and 06’s direct module argument. 04’s explicit telescope, 09’s and 10’s direct division, 11’s pairwise-difference families and 07’s weighted certificates are kept as second routes (Section 5.1).
- *Support threshold.* Printed once, in 11’s formulation (every infinite support bound, with the extension criterion) together with 04’s negative side (Theorem 9.1).
- *Thresholds.* Organized by one theorem (Theorem 10.10) with the five constructions of Section 11; generator counts are listed per construction and no general law is stated.
- *Homological algebra.* 06’s package is proved under three hypotheses that all five constructions satisfy (Theorem 12.2).



- *Foundations.* NBG with choice for sets; global choice, stated by 04, 06, 09 (and 05), is not used anywhere (Section 2.1).
- *Later manuscripts.* 13 and 15 reprove the universal theorem and its consequences; they are credited where each result is printed, and only two one-line additions of 15 are printed (Corollary 5.5 and Proposition 7.3); 13’s answer to Elliott’s question is credited in Section 17. 12 and 14 form one new section (Section 13), placed after the homological package it is to be compared with; the ring  $A_0 = A_1$  and the two-generator syzygy are printed once, the syzygy in 14’s rank generality over 12’s rings. 16’s results sit with 10’s polynomial maps; where they overlap (rational collapse, affine surjections, transfer, grids) the general- $D$  statement is printed once with 16’s fresh-scale proof and 10’s statements keep their finite-difference and lattice proofs as a second route. 14’s criterion for unimodular projective directions and 13’s fiber and norm theorems are printed in the Diophantine report [30], not here.

**Results added in the merge.** Each carries the tag [merge] and a complete proof: the field bound for an arbitrary subfield of a fixed-group Hahn field (Proposition 10.3); the exact thresholds  $2^{\aleph_0}$  in 04’s example and in the rational-coefficient example (Examples 10.4 and 10.5), which answer 04’s question for its own example and show that 10’s bound need not be attained; the criterion (b’) in Theorem 10.7; the exact value  $\kappa^{<\kappa}$  of the sizes and thresholds of  $A_\kappa^{06}$  and of 11’s example, in ZFC (Lemma 11.10, Theorem 11.11, and Proposition 11.23), which answers the first half of 06’s question on cardinal arithmetic (08 answers the second half); the organizing theorem itself (Theorem 10.10) and the reconciliation of the realized cardinals (Remark 10.11); the homological package and the Ext thresholds for all five constructions (Theorems 12.2 and 12.3); the negative-side extension criterion (Theorem 9.1(iv)); the embedding  $\mathbf{No} \hookrightarrow \text{Frac}(\mathbf{Oz}/P)$  for primes not containing  $\Pi$  (Remark 14.13); and the explicit elements of the normalization provided by 08’s equation (Remark 15.27). From the later manuscripts: the syzygy, noncoherence and nonflatness theorems in the common generality of 12’s and 14’s rings (Theorem 13.9 and Proposition 13.10); the flat dimensions of lattice modules and cyclic quotients over every core, which removes a superfluous hypothesis of 12 (Theorem 13.11); exact projective dimensions where 12 gives bounds, and  $\text{pd } D = 2$  for every ring of Section 12 with countably generated tail (Proposition 13.12); and the comparison of the two kinds of flat-dimension statements (Remark 13.16). Each follows from lemmas of the sources; none is a new method.

## 1.4 Relation to the repository

The foundations report defines  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  (found:eq:omnific in [26]), and omnific integers appear as kernels and integer parts in the trigonometry, gamma-zeta, analysis and euclidean-three-space reports; the sibling report [30] treats their Diophantine geometry. The euclidean-three-space report proves the group analogue of the finite-side theorem for  $\text{SO}(3, \mathbf{No})$  (Section 8). The first-kappa report proves closedness of bounded-support Hahn fields, which contains the closure step of 06’s construction (06 proves that step itself; Section 11.2). The  $\kappa$ -bounded coefficient rings  $\mathbb{Z} + K((G^{<0}))_\kappa$  of L’Innocente–Mantova [16] are the published precedent for the bounded-support rings of 06 and 11. The Hahn-evaluation report [34], cited by 12, separates finite polynomial evaluation at  $\omega$  from evaluation of unrestricted series; 12’s embeddings use finite normal forms only. No statement of this report has a Lean counterpart; the repository has no omnific module.

**Stale statements corrected.** Three sources report a repository code search for “omnific” with no hits (06, 08, 11; 09 reports an explicitly incomplete result). At their pins the repository already



defined **Oz** at `found:eq:omnific` and used it in the reports listed above; what it lacked was a report devoted to omnific arithmetic, and the collection now has two (this one and [30]). 03’s companion draft and 04’s “prior manuscript” were not in the repository at their pins; they are now manuscripts 05 and 02 of [30]. 07’s description of the staging directory (“Gamma–zeta ZIP archives”) was true at its pin and is a snapshot statement. Of the later manuscripts, 13 and 15 (and 15’s audit file) report that a repository search for “omnific” returned no matches; at their pin 71e9606 the foundations report already defined **Oz** at `found:eq:omnific`. 13 and 15 call the universal theorem a proposed, respectively candidate-original, contribution, and 13 calls its answer to Elliott’s question one; the same theorem and the same answer are in the eight manuscripts of the first merge, dated a day earlier (Theorems 5.1 and 17.1), which 13 and 15 could not see, so both are independent re-derivations. 16 found no integer-valued-polynomial treatment in the repository; at its pin that was accurate, and the collection now has 10’s (Section 16). 14’s “companion manuscripts” are manuscripts 02 and 05 of the Diophantine report [30] and 03 of this one. The shipped audit files keep these statements verbatim; the README records the corrections.

**Corrections to the sources.** 03’s source lost a relation symbol and prints “ $C \text{ e } B[i]$ ” for  $C \neq B[i]$  (Proposition 15.18). 09 attributes the separated-scale division to Proposition 8.2.1 of [16]; the displayed separated-scale factorization is Fact 8.0.1 (Gonshor’s Theorem 8.6) with its example on p. 46, as 11 cites, while Proposition 8.2.1 is the truncation criterion for divisibility cited by 08 (Section 4.2). 09’s hypothesis  $\lambda \geq 2^{\aleph_0}$  is implied by  $\lambda^{\aleph_0} = \lambda$  (Section 11.4). 04’s remark on its  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$  example is true but misleading (Remark 10.6). 06’s remark that  $|A_\kappa^{06}/xA_\kappa^{06}| \geq \kappa$  “under the cardinal-arithmetic hypothesis” needs no hypothesis and in fact gives  $\kappa^{<\kappa}$  (Section 12). 06, 04 and 09 state global choice, which is never used (Section 2.1). 12 assumes  $c^2 \in D$  for its flat dimensions  $d+1$  and 2 of cyclic modules, but needs it only for its  $\text{Tor}_2$  formula (Theorem 13.11), and gives only bounds for projective dimensions that are exact in rank one (Proposition 13.12). 14 credits the Diophantine report’s example  $[1 : \sqrt{2}]$  but not its rationality theorem for primitive directions, of which 14’s criterion is the Gaussian and principal-ideal extension (Section 13.8). 16’s rational collapse has the D-ring precedent that 16 itself credits (Theorem 16.42). No source contains a false theorem.

## 2 Conventions, notation, and the constant-term retraction

### 2.1 Foundations

We work in von Neumann–Bernays–Gödel class theory (NBG) with the axiom of choice for sets. Every surreal number, every normal-form support and every coefficient field or target ring called *set-sized* is a set; the carriers **No**, **Oz**, **No**[ $i$ ] and most rings of Sections 3 and 5 are proper classes. A homomorphism with a class source is a class function obeying the usual identities, and its restriction to a set is a set.

The sources state different foundations. Manuscript 06 works in Gödel–Bernays set theory with global choice, 09 in GBC, and 04 in “GBC, with ordinary choice for sets” (GBC already contains global choice); 07, 08 and 11 work in NBG with choice for sets, 10 in NBG with choice, and 03 in NBG for the classes and ZFC for its set-sized constructions; 07 remarks that global choice is unnecessary. *No argument of any source uses global choice.* The collision step of the universal theorem (Theorem 5.1) is choice-free in the formulation of 08, which we adopt: for a set  $S$ , its Hartogs ordinal is an ordinal that does not inject into  $S$ ; it exists in ZF. The same device is used in the rotation-group theorem `e3:cut:thm:smallquotient` of [27]. Choice for sets enters only

where it is visibly needed: maximal ideals of set-sized rings (Section 11), cardinal arithmetic such as  $\kappa^{\aleph_0} = \max(2^{\aleph_0}, |\kappa|^{\leq \aleph_0})$ , and the existence of residue fields.

**Definition 2.1** (Set-sized quotients and detection). A class ideal  $J$  of a class ring  $R$  *admits a set-sized quotient* if there is a surjective class ring homomorphism  $R \rightarrow S$  onto a set-sized ring with kernel  $J$ . A ring homomorphism  $\varphi$  *detects*  $x$  if  $\varphi(x) \neq 0$ ; a module  $M$  *detects*  $x$  if  $xM \neq 0$ .

This is the convention of 04 (“set-realizable quotient”), 08, 09 (“set-sized realization”), 10 (“set-sized quotient presentation”) and 11. Manuscripts 06 and 07 alternatively represent a class quotient by Scott codes (the members of a coset of least set-theoretic rank form a nonempty set); every quotient statement below may be read in either way, and no set or class whose members are arbitrary proper classes is formed. Notation such as  $\text{Hom}(\mathbf{Oz}, S)$  abbreviates a statement about individual class maps and their factorizations; after the universal theorem these maps are coded by their restrictions to a set-sized coefficient ring.

**The universe-relative reading.** Manuscripts 04, 10 and 11 add an equivalent reading: fix a lower universe containing the coefficient fields, targets and supports, and place the surreal carrier in a higher universe; “set-sized” then means small in the lower universe. The size restriction must be carried in the statement. A set-sized fragment of surreal arithmetic viewed in a larger universe has its own identity map as a set-sized target, so it does not satisfy the full-class theorem (11 insists that its main theorem is relative to one fixed universe, not to fragments viewed in a larger one).

## 2.2 Normal forms and the orientation of exponents

A nonzero surreal has a unique Conway normal form  $x = \sum_{\gamma \in S} r_\gamma \omega^\gamma$  with nonzero real coefficients and a *reverse well-ordered set*  $S$  of exponents (every nonempty subset has a greatest element) [9, 13, 4]. We use *large monomials*:  $\omega^\gamma$  is infinitely large when  $\gamma > 0$ , and  $\omega^{\gamma+\delta} = \omega^\gamma \omega^\delta$  is Conway’s monomial law, not an exponential. For an abstract ordered abelian group  $G$  we write formal monomials  $X^g$  with the same orientation, and  $k((X^G))$  for the Hahn series over a set field  $k$  with reverse well-ordered set supports in  $G$  (for  $G = \mathbf{No}$  a proper class of series). The usual small-monomial notation is recovered by  $t^g = X^{-g}$ ; sources that write  $t$  with well-ordered supports reverse all signs. A family is strongly summable if the union of its supports is reverse well ordered and each exponent receives finitely many contributions; strong sums are coefficientwise and are *not* limits of partial sums in the fine order topology.

*Purely infinite* refers to support, not to size or sign:  $-\omega$  is purely infinite, while  $\omega - 1$  is infinite but not purely infinite. This is the convention of the foundations report, which defines  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  with  $\Pi$  the purely infinite normal forms (`found:eq:omnific` in [26]); the gamma-zeta report uses the same symbol [33]. The trigonometry report writes the same class  $\Pi$  as the normal forms supported at *negative  $t$ -exponents* (`trigonometry:eq:split` in [28]), because it uses  $t = \omega^{-1}$ ; the two descriptions agree after the sign change  $t^g = \omega^{-g}$ .

## 2.3 The rings

Let  $k$  be a set field,  $G$  an ordered abelian group (a set, or the class  $\mathbf{No}$ ), and  $D \subseteq k$  a unital subring. Put

$$\Pi_k(G) = \{f \in k((X^G)) : \text{supp } f \subseteq G_{>0}\}, \quad \mathbf{m}_k(G) = \{f \in k((X^G)) : \text{supp } f \subseteq G_{<0}\}, \quad (2.1)$$

$$\mathcal{A}_{D,k}(G) = D + \Pi_k(G), \quad \mathcal{B}_{D,k}(G) = D + \mathbf{m}_k(G). \quad (2.2)$$

Both sums are direct as additive groups, and the zero series belongs to both ideals. We drop  $(G)$  when  $G = \mathbf{No}$  and write  $\Pi = \Pi_{\mathbb{R}}$ . For  $k = \mathbb{R}$  and  $k = \mathbb{C}$  the identification  $X^g \mapsto \omega^g$  gives  $\mathbb{R}((X^{\mathbf{No}})) = \mathbf{No}$  and  $\mathbb{C}((X^{\mathbf{No}})) = \mathbf{No}[i]$  (a union of two reverse well-ordered supports is reverse well ordered, so complexification is coefficientwise), and

$$\mathbf{Oz} = \mathcal{A}_{\mathbb{Z}, \mathbb{R}} = \mathbb{Z} \oplus \Pi, \quad \mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz} = \mathcal{A}_{\mathbb{Z}[i], \mathbb{C}} = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}, \quad \Pi_{\mathbb{C}} = \Pi + i\Pi. \quad (2.3)$$

We call  $\mathbf{Oz}[i]$  the *Gaussian omnific ring*; it is a subring of  $\mathbf{No}[i]$ , not the surcomplex field, and not the integral closure of  $\mathbb{Z}$  in  $\mathbf{No}[i]$  (08, 09). The finite surreal and surcomplex rings are  $\mathcal{O}_{\mathbb{R}} = \mathcal{B}_{\mathbb{R}, \mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}} = \mathcal{B}_{\mathbb{C}, \mathbb{C}}$ , and  $\mathcal{A}_{\mathbb{R}, \mathbb{R}} = \mathbb{R} + \Pi$  is the nonnegative-support ring (03's  $P_{\mathbb{R}}$ ). For a general set field  $k$  (06, 07, 10) the class field  $k((X^{\mathbf{No}}))$  need not be realized inside the surreals; nothing below requires such a realization.

**Lemma 2.2** (Constant term [03, 04, 06, 07, 08, 09, 10, 11, 13, 14, 15, 16]). *The map  $\text{ct} : \mathcal{A}_{D,k}(G) \rightarrow D$  extracting the coefficient of  $X^0$  is a surjective unital ring homomorphism with kernel  $\Pi_k(G)$ , split by the inclusion of constants. On  $\mathcal{B}_{D,k}(G)$  the same coefficient is a split surjective ring homomorphism with kernel  $\mathbf{m}_k(G)$ ; on  $\mathcal{O}_{\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}}$  it is the standard part, written  $\text{st}$ .*

*Proof.* All exponents occurring in  $\mathcal{A}_{D,k}(G)$  are nonnegative, and a sum of two nonnegative exponents is zero only when both are zero; so the constant coefficient of a product is the product of the constant coefficients. Additivity, the kernel and the section are immediate from uniqueness of the decomposition (2.2). The negative side is identical with nonpositive exponents.  $\square$

*Remark 2.3* (What  $\text{ct}$  is not). The map  $\text{ct}$  is defined on a ring of generally infinite elements; it is not the standard part, and coefficient extraction at 0 is not multiplicative on  $\mathbf{No}$ :  $\omega \cdot \omega^{-1} = 1$  although both factors have zero constant coefficient. This is the warning recorded in [31] and `diff:warn:ct` of [35]. Nor is  $\text{ct}$  order preserving:  $\omega - N > 0$  but  $\text{ct}(\omega - N) = -N$  for every ordinary  $N \geq 1$  (04, 08, 09, 11). On  $\mathcal{A}_{D,k}$  the two maps  $\text{ct}$  and  $\text{st}$  receive different names because they have different domains (04), and (8.1) below shows that they cannot be combined.

**Lemma 2.4** (Degree and units [03, 04, 07, 08, 09, 10, 11, 13, 14, 16]). *For  $x \neq 0$  in  $k((X^G))$  put  $\deg_{\omega}(x) = \max \text{supp } x$ . Then  $\deg_{\omega}(xy) = \deg_{\omega}(x) + \deg_{\omega}(y)$  for  $x, y \neq 0$ , so  $\mathcal{A}_{D,k}(G)$  and  $\mathcal{B}_{D,k}(G)$  are domains, and*

$$\mathcal{A}_{D,k}(G)^{\times} = D^{\times}, \quad \mathbf{Oz}^{\times} = \{\pm 1\}, \quad \mathbf{Oz}[i]^{\times} = \{\pm 1, \pm i\}.$$

*In particular every nonconstant element of  $\mathcal{A}_{D,k}(G)$ , such as  $1 + \omega^a$  or  $\omega^{\sqrt{2}} + \omega + 1$ , is a nonunit.*

*Proof.* The leading coefficient of a product is the product of the leading coefficients, which is nonzero because  $k$  is a field; the degree identity follows. A nonzero element of  $\mathcal{A}_{D,k}(G)$  has nonnegative degree, so  $xy = 1$  forces both degrees to be 0, i.e. both factors are constants in  $D$  with  $xy = 1$  in  $D$ . The two unit lists are the units of  $\mathbb{Z}$  and  $\mathbb{Z}[i]$ .  $\square$

Here  $\deg_{\omega}$  is the largest growth exponent (04); it is not the ordinal-valued degree of the order type of a support used in parts of the factorization literature [16].

## 2.4 Notation and renamings

The manuscripts use many symbols for the purely infinite ideal and several for the constant term. We use the repository's  $\Pi$  and the common  $\text{ct}$ , and resolve every collision by renaming in the new material. The renamings, and the tags distinguishing the different set-sized constructions, are listed below; Section 13 and Section 16.6 list the further renamings of 12, 14 and 16.

| Object                                          | In the sources                                                                                                                                                                                                 | Here                                                                                                    |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| purely infinite ideal                           | $\mathcal{I}$ (03, 06, 09, 14), $\mathcal{J}_k$ (04), $\mathfrak{J}$ (07, 08, 11),<br>$\mathfrak{P}$ (10), $J_k$ (13), $\mathcal{I}_K$ (15), $\mathcal{P}_k$ (16)                                              | $\Pi$ , $\Pi_k$ , $\Pi_{\mathbb{C}}$                                                                    |
| constant term                                   | $c_0$ (06), $\pi$ , $\pi_D$ (10), $\varepsilon_D$ (09), $\varepsilon$ (12), ct<br>(others)                                                                                                                     | ct                                                                                                      |
| coefficientwise constant term on<br>polynomials | $\Pi$ (10), $c_0$ (06)                                                                                                                                                                                         | $\text{ct}_*$                                                                                           |
| $D + \Pi_k$                                     | $\mathcal{A}_{D,k}$ (04), $A_{D,k}$ (06), $A(D, K)$ (07), $R_D$<br>(08), $\mathcal{A}_D$ (09), $R(D, K)$ (10), $\mathcal{Z}_{K,D}$ (11),<br>$A(D, k, G)$ (03), $A_D$ (13), $R_A$ (15), $\mathcal{A}_D$<br>(16) | $\mathcal{A}_{D,k}$ , $\mathcal{A}_{D,k}(G)$                                                            |
| Gaussian omnific ring                           | $\text{GOz}$ (06), $\mathbf{Og}$ (09)                                                                                                                                                                          | $\mathbf{Oz}[i]$                                                                                        |
| surcomplex field                                | $\mathbf{No}(i)$ (04, 07, 08, 09, 12, 15, 16)                                                                                                                                                                  | $\mathbf{No}[i] = \mathbf{No}[i]$ [31]                                                                  |
| Hahn field over $H_a$                           | $L_a$ (06), $\mathcal{F}_a$ (07), $L_h$ (03, set version)                                                                                                                                                      | $\mathcal{F}_a(k)$ ; $L_h$ for 03's set<br>field                                                        |
| localization $\mathbf{Oz}[\omega^{-a}]$         | $L_a$ (07)                                                                                                                                                                                                     | $\mathbf{Oz}[\omega^{-a}]$ (never $L_a$ )                                                               |
| binomial quotient $\mathbf{Oz}/(1 + \omega^a)$  | $B_a$ (07)                                                                                                                                                                                                     | $\mathcal{Q}_a$                                                                                         |
| 07's compression map $a\Psi$                    | $E_a$                                                                                                                                                                                                          | $\Psi_a$                                                                                                |
| integral closure                                | $B, C, \text{Int}_F(A)$ (03)                                                                                                                                                                                   | $\mathcal{N}, \mathcal{N}_{\mathbb{C}}, \text{IC}_F(A)$                                                 |
| integer-valued polynomials                      | $\text{Int}(\mathbb{Z}^r)$ , $I_r$ (10)                                                                                                                                                                        | $\text{Int}(\mathbb{Z}^r)$                                                                              |
| set-sized integer parts                         | $A_{\kappa}$ (03, 08), $R_{\kappa}$ (06), $A_{\lambda,D}$ (09), $R_{\Gamma, < \kappa}$ (11)                                                                                                                    | $A_{\kappa}^{03}, A_{\kappa}^{06}, A_{\kappa}^{08}, A_{\lambda,D}^{09},$<br>$R_{\Gamma, < \kappa}^{11}$ |
| their ideals                                    | $I_{\kappa}, \mathfrak{J}_{\kappa}, I_{\lambda,k}$                                                                                                                                                             | $\Pi_{\kappa}^{03}, \Pi_{\kappa}^{06}, \Pi_{\kappa}^{08}, \Pi_{\lambda}^{09}$                           |
| small exponents below a support                 | $\text{Tiny}(S)$ (09), $U(S)$ (10), $G_{\ll h}$ (03)                                                                                                                                                           | $U_G(S), H_h(G)$                                                                                        |
| binomial series                                 | $B_m(t)$ (08)                                                                                                                                                                                                  | $\mathbf{b}_m(t)$                                                                                       |

Throughout, for  $a > 0$  in an ordered group  $G$ ,

$$H_a(G) = \{h \in G : n|h| < a \text{ for every ordinary } n \geq 1\}, \quad H_a = H_a(\mathbf{No}), \quad (2.4)$$

and  $h \ll a$  means  $h > 0$  and  $h \in H_a(G)$ . This is a comparison of *exponents*:  $h \ll a$  does not say that  $\omega^h$  is small (it is infinite whenever  $h > 0$ ). For a set  $S \subseteq G_{>0}$  we write  $U_G(S) = \{a > 0 : na < s \text{ for all } s \in S, n \geq 1\}$  (09's  $\text{Tiny}(S)$ , 10's  $U(S)$ ). Letters  $\kappa, \lambda, \mu, \theta$  denote cardinals and are restated locally;  $\mu = \text{cf } \kappa$  and  $\theta = \kappa^{\aleph_0}$  in the construction of 03.

### 3 Support gaps and the purely infinite ideal

This section collects the facts that use the full surreal exponent class. All eight manuscripts of the first merge prove them, as do 13 and 15 and manuscript 05 of the sibling report [30], and 14 and 16 use them; they are background for the new results, not claimed as new. Throughout,  $k$  is a set field and  $D \subseteq k$  a unital subring.

**Lemma 3.1** (Gaps below and above a set [03, 04, 06, 07, 08, 09, 10, 11, 13, 15, 16]).

- (i) Every set of surreal numbers has a strict upper and a strict lower bound in  $\mathbf{No}$ .
- (ii) For every set  $S \subseteq \mathbf{No}_{>0}$  there is  $h > 0$  with  $nh < s$  for all  $s \in S$  and all ordinary  $n \geq 1$ ; in particular  $0 < h < S$ .
- (iii) For such  $S$  the class  $U_{\mathbf{No}}(S)$  contains the interval  $(0, h)$ , which is not a set, and for every ordinal  $\Lambda$  the exponents

$$a_{\alpha} = \frac{h}{\omega^{\alpha+1}} \quad (\alpha < \Lambda) \quad (3.1)$$

are distinct elements of  $U_{\mathbf{No}}(S)$ , each  $\ll h$ .

*Proof.* For a set  $T$ , the cuts  $\{T \mid \emptyset\}$  and  $\{\emptyset \mid T\}$  are set cuts and give (i). For (ii), apply the set cut  $\{0 \mid T\}$  to the set  $T = \{s/n : s \in S, n \geq 1\}$  of positive surreals; for  $S = \emptyset$  take  $h = 1$ . For (iii),  $(0, h)$  is not a set: a positive lower bound of it would have a positive half still in the interval. The denominators  $\omega^{\alpha+1}$  are the Conway monomials of the ordinals  $\alpha + 1$  in the *exponent* field; they strictly increase with  $\alpha$  and exceed every ordinary integer, so the  $a_\alpha$  are distinct, positive and satisfy  $na_\alpha < h$ .  $\square$

Only the existence of a separator is used, not its simplicity or birthday (08). The sources use several equivalent families in place of (3.1):  $a/\omega^{\alpha+1}$  (04, 05, 06, 08, 09, 03),  $a/(\omega + \alpha + 1)$  with the field sum of embedded ordinals (07),  $\delta\omega^{-(\alpha+1)}$  (11),  $\epsilon\xi/\kappa$  (10),  $\epsilon/(\bar{\alpha} + 2)$  with  $\bar{\alpha}$  the surreal ordinal (13), and  $g\omega^{-(\xi+1)}$  (15). Each is a set whenever  $\alpha$  ranges over a set.

**Example 3.2** (Surreal gaps where real gaps fail [03, 06, 08, 10, 11]). The series  $f = \sum_{n \geq 1} \omega^{1/n}$  lies in  $\Pi$ ; its support has no positive *real* lower bound but has positive surreal lower bounds, for instance  $1/\omega$ . With  $a = 2/\omega$  and  $b = 1/\omega$  one has  $na < 1/m$  for all ordinary  $m, n$  (10). In the fixed exponent group  $\mathbb{R}$  the same series is purely infinite but is not divisible by any  $X^h$ ,  $h > 0$ , inside the nonnegative-support ring (03); this is where the passage from monomials to the whole ideal fails in a fixed group (Section 10).

**Proposition 3.3** (Common monomial divisors [03, 04, 06, 07, 08, 09, 10, 11, 13, 14, 15]). *Let  $A = A_{D,k}$  and  $\Pi_k$  its purely infinite ideal.*

- (i) *Every set-sized family  $\mathcal{F} \subseteq \Pi_k$  satisfies  $\mathcal{F} \subseteq X^a \Pi_k$  for some  $a > 0$ .*
- (ii) *Each  $f \in \Pi_k$  is a single product of two elements of  $\Pi_k$ ; hence  $\Pi_k^2 = \Pi_k$ , and  $\Pi_k = \bigcup_{a > 0} X^a A$  is a directed union of principal ideals.*
- (iii)  *$\Pi_k$  is not generated, as an ideal of  $A$ , by any set of its elements.*
- (iv)  *$d\Pi_k = \Pi_k$  for every  $0 \neq d \in D$ ; in particular  $n\Pi_k = \Pi_k$  for every nonzero ordinary integer  $n$ .*

*Proof.* (i) The union of the supports of the members of  $\mathcal{F}$  is a set of positive surreals; choose  $0 < a <$  that union by Lemma 3.1. Translating each support by  $-a$  leaves it positive and reverse well ordered, so  $X^{-a}f \in \Pi_k$ . (ii) Apply (i) to  $\{f\}$ :  $f = X^a(X^{-a}f)$  with both factors in  $\Pi_k$ . If  $0 < b < a$  then  $X^a A \subseteq X^b A$ , so the union is directed. (iii) If a set generated  $\Pi_k$ , (i) would place the generated ideal inside  $X^a A$ ; but  $X^{a/2} \in \Pi_k$  is not in  $X^a A$ , since its unique quotient by  $X^a$  in the ambient field is  $X^{-a/2}$ , which has a negative exponent. (iv) Divide every coefficient by  $d$ , which is invertible in  $k$ .  $\square$

Idempotence and divisibility do not by themselves prove the universal theorem: an idempotent, rationally divisible ideal can act nontrivially on infinite set-sized modules, and a homomorphism to an infinite characteristic-zero ring need not kill an arbitrarily divisible additive subgroup (04, 06, 07, 08). In positive characteristic the last item already suffices (Remark 5.10).

$$n\mathbf{Oz} = n\mathbb{Z} + \Pi = \text{ct}^{-1}(n\mathbb{Z}) \quad (n \geq 1), \quad d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}} \quad (0 \neq d \in \mathbb{Z}[i]). \quad (3.2)$$

Indeed  $n\Pi = \Pi$  and  $d\Pi_{\mathbb{C}} = \Pi_{\mathbb{C}}$  by Proposition 3.3(iv). Consequently

$$\Pi = \bigcap_{n \geq 1} n\mathbf{Oz} = \bigcap_{p \text{ prime}} p\mathbf{Oz}, \quad \Pi_{\mathbb{C}} = \bigcap_{n \geq 1} n\mathbf{Oz}[i], \quad (3.3)$$

since a nonzero ordinary integer (or Gaussian integer) is not divisible by every  $n$  or every prime (08, 10, 11). For a general  $D$  the intersection formula can fail: for  $D = \mathbb{Q}$  every element of  $\mathcal{A}_{\mathbb{Q}, \mathbb{R}}$  is divisible by every ordinary positive integer (11).

**Proposition 3.4** (Clearing and fraction fields [03, 04, 06, 07, 08, 09, 11, 14, 15, 16]).

- (i) For every set  $E \subseteq k((X^{\mathbf{No}}))$  there is  $h > 0$  with  $X^h E \subseteq \Pi_k$ . For  $E \subseteq \mathbf{No}$  or  $E \subseteq \mathbf{No}[i]$  this reads  $\omega^h E \subseteq \Pi$ , respectively  $\omega^h E \subseteq \Pi_{\mathbb{C}}$ .
- (ii)  $\text{Frac}(\mathcal{A}_{D,k}) = k((X^{\mathbf{No}}))$ ; in particular  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  and  $\text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}[i]$ .
- (iii) Inverting the positive monomials gives the whole field: if  $\mathcal{U} = \{X^a : a > 0\} \cup \{1\}$  then  $\mathcal{U}^{-1} \mathcal{A}_{D,k} = k((X^{\mathbf{No}}))$ .
- (iv) Neither  $\mathbf{No}$  nor  $\mathbf{No}[i]$ , nor any localization of  $\mathcal{A}_{D,k}$  in which some nonzero element of  $\Pi_k$  becomes a unit, admits a unital homomorphism to a nonzero set-sized ring or a nonzero set-sized unital module.

*Proof.* (i) The union of the supports of the members of  $E$  is a set; by Lemma 3.1(i) choose  $h > 0$  larger than  $-g$  for each exponent  $g$  in it. Then every exponent of  $X^h x$ ,  $x \in E$ , is positive. (ii) and (iii) For  $x \in k((X^{\mathbf{No}}))$  apply (i) to  $\{x\}$ :  $x = (X^h x)/X^h$  with numerator and denominator in  $\mathcal{A}_{D,k}$ ; the reverse inclusions hold because  $\mathcal{A}_{D,k}$  is a subring of the field. (iv) is proved after the universal theorem (Corollary 5.9).  $\square$

Part (i) is manuscript 05's "Clearing supports" theorem, now in the sibling report [30]; 03 credits it to 05 and adds the algebra-wide formulation of Section 15. Clearing one element is weaker than clearing all powers of it with one denominator, which Section 15 also supplies (03).

## 4 The collision mechanism

Every proof of the universal theorem combines two finite multiplicative identities with a cardinal comparison. This section states the identities once, in the forms used by the different sources. The first (Lemma 4.1, from 06, and in a special form from 03) gives a uniform proof of the sharp cardinal bounds of Sections 10 and 11, including those that counting monomials cannot reach; the others are genuinely different routes and are kept as such.

### 4.1 Scaled-field collision

**Lemma 4.1** (Scaled-field collision [06; 03]). *Let  $A$  be a commutative subring of a field  $F$ , let  $L \subseteq F$  be a subfield and let  $0 \neq m \in F$  satisfy  $mL \subseteq A$ .*

- (i) *If  $\varphi : A \rightarrow S$  is a ring homomorphism,  $S$  any ring (not necessarily commutative, unital or reduced), and  $\varphi(m^2) \neq 0$ , then  $x \mapsto \varphi(mx)$  is injective on  $L$ .*
- (ii) *If  $M$  is a unital  $A$ -module,  $v \in M$  and  $m^2 v \neq 0$ , then  $x \mapsto (mx)v$  is injective on  $L$ .*

*Consequently, if  $|L| > |S|$  then  $\varphi(m^2) = 0$ , and if  $|L| > |M|$  then  $m^2 M = 0$ .*

*Proof.* Since  $1 \in L$ ,  $m \in A$ . For distinct  $x, y \in L$  the element  $m/(x - y)$  lies in  $mL \subseteq A$ , and in  $A$

$$(mx - my) \cdot \frac{m}{x - y} = m^2. \quad (4.1)$$



If  $\varphi(mx) = \varphi(my)$ , applying  $\varphi$  gives  $\varphi(m^2) = 0$ ; no cancellation in  $S$  is used. For the module, apply  $m/(x - y)$  to  $(mx)v = (my)v$ ; commutativity of  $A$  turns the result into  $m^2v = 0$ .  $\square$

The proof of (i) uses only that  $A$  is closed under addition and multiplication and contains  $mL$ ; so it applies verbatim when  $A$  is a nonunital subring such as  $\Pi_K$ , which is how it is used in [Theorem 5.1\(iii\)](#). The module version bounds  $|M|$  itself. Deriving it through the ring  $\text{End}_{\mathbb{Z}}(M)$  would only bound  $|\text{End}_{\mathbb{Z}}(M)|$ , which can be much larger than  $|M|$  (06, 03, 09); this matters for every set-sized threshold below. In 03 the lemma appears as the identity  $(X^{h/2}(f - f'))(X^{h/2}/(f - f')) = X^h$  for  $f \neq f'$  in a smaller-scale field (03's “smaller-scale field obstruction”), i.e. the case  $m = X^{h/2}$  of (4.1).

**Lemma 4.2** (The scale subgroup and its Hahn field [03, 06, 07]). *Let  $G$  be an ordered abelian group (a set or **No**) and  $a \in G_{>0}$ .*

- (i)  $H_a(G)$  is a convex subgroup of  $G$ , divisible if  $G$  is, and  $H_c(G) = H_{2c}(G)$  for every  $c \in G_{>0}$ .
- (ii)  $k((X^{H_a(G)}))$  is a subfield of  $k((X^G))$ , and  $X^c k((X^{H_a(G)})) \subseteq \Pi_k(G)$  for every  $c \in G$  with  $2c \geq a$ ; for instance  $c = a$ , or  $c = a/2$  when  $a/2 \in G$ .
- (iii) For  $G = \mathbf{No}$  the field  $\mathcal{F}_a(k) = k((X^{H_a}))$  is a proper class: it contains the distinct monomials  $X^{a_\alpha}$  of (3.1) for every ordinal  $\alpha$ .

*Proof.* (i) If  $h, h' \in H_a(G)$  then  $2n|h| < a$  and  $2n|h'| < a$  give  $n|h + h'| < a$ ; negation and convexity are immediate, and if  $G$  is divisible then  $h/m$  satisfies the same inequalities. For every  $c > 0$ ,  $n|h| < c$  for all  $n$  is equivalent to  $n|h| < 2c$  for all  $n$ . (ii) Supports of sums and products of series supported in a subgroup stay in it, and the Hahn inverse of  $f = cX^g(1 + u)$ ,  $\text{supp } u < 0$ , is  $c^{-1}X^{-g}\sum_{n \geq 0}(-u)^n$ , whose support lies in the subgroup generated by  $\text{supp } f$ . If  $2c \geq a$  and  $h \in H_a(G)$  then  $2|h| < a \leq 2c$ , so  $c + h > 0$ ; hence  $X^c f$  has positive support for  $f \in k((X^{H_a(G)}))$ . (iii) Each  $a_\alpha \ll a$  lies in  $H_a$ .  $\square$

The field  $\mathcal{F}_a = \mathcal{F}_a(\mathbb{R})$  is 07's  $\mathcal{F}_a$  and 06's  $L_a$  with  $k = \mathbb{R}$ ; it is real closed and contains a copy of all of **No** ([Proposition 14.1](#) and [Theorem 14.4](#)).

## 4.2 The geometric telescope

**Lemma 4.3** (Bounded-exponent telescope [03, 04, 05, 06, 07, 08, 09, 10, 11, 15]). *Let  $G$  be an ordered abelian group, and let  $0 < b < a$  and  $a \ll c$  in  $G$ . Put  $d = a - b$ . Then*

$$q_{c;a,b} = \sum_{n \geq 0} X^{c-a-nd} \in \Pi_k(G), \quad (X^a - X^b) q_{c;a,b} = X^c. \quad (4.2)$$

*The support of  $q_{c;a,b}$  is countable and strictly decreasing. For every ordinary  $N \geq 0$ ,*

$$(X^a - X^b) \sum_{n=0}^N X^{c-a-nd} = X^c - X^{c-(N+1)d}, \quad (4.3)$$

*and the remainder monomial  $X^{c-(N+1)d}$  is still infinitely large.*

*Proof.* For each ordinary  $n \geq 0$ ,  $a + nd \leq (n+1)a < c$ , so every exponent is positive; the exponents strictly decrease with  $n$ , so every nonempty set of them has a largest element. Multiplication by  $X^a$  gives the exponents  $c - nd$  and multiplication by  $X^b$  gives  $c - (n+1)d$ ; at  $c$  the coefficient is 1 and

at every  $c - md$ ,  $m \geq 1$ , it is  $1 - 1 = 0$ . Each coefficient receives at most two contributions, so this is a Hahn identity. The finite identity is the same computation truncated; its remainder exponent  $c - (N + 1)d \geq c - (N + 1)a > 0$ .  $\square$

*Remark 4.4* (No limit is taken and no sum is transported). The remainders in (4.3) are positive monomials; they do not tend to 0 in the order, valuation or fine topology, not even at the ordinary scale 1. The infinite identity is a coefficient identity, not a limit of (4.3) (04, 06, 07, 08, 10, 03). A target homomorphism is never applied term by term to  $q_{c;a,b}$ : the series is first a single element of the source ring, and only the finite product identity is mapped (06, 08, 09, 11). This is why no continuity, valuation or strong-additivity hypothesis on homomorphisms appears anywhere. It is also why a ring of finite-support series behaves differently (Section 9): there  $q_{c;a,b}$  is absent and the nonzero remainder of (4.3) cannot be removed.

**Example 4.5** (A witness for  $\omega$  [04, 10, 15]). With  $c = 1$ ,  $b = 1/\omega$  and  $a = 2/\omega$ ,  $(\omega^{2/\omega} - \omega^{1/\omega}) \sum_{n \geq 0} \omega^{1-(n+2)/\omega} = \omega$ , while the finite partial products leave the nonzero remainders  $\omega^{1-(N+1)/\omega}$  (04). In the lexicographic group  $\mathbb{Z} \times \mathbb{Z}$  with  $c = (1, 0)$ ,  $a = (0, 2)$ ,  $b = (0, 1)$  the quotient is  $\sum_{n \geq 0} X^{(1, -n-2)}$  and the  $N$ -th remainder is  $X^{(1, -N-1)}$  (10); under a compatible monomial realization each remainder is still infinite. 15 uses the same series  $q = \sum_{n \geq 0} \omega^{1-(n+2)/\omega}$  to block the extension of the finite-support augmentation (Section 9).

**Precedents.** The countable geometric-series product with a monomial result is already in L’Innocente–Mantova [16, Remark 3.4.4], used there to show that  $\deg \circ \text{ot}$  is not multiplicative when the value group does not embed in  $\mathbb{R}$ ; 04 names it a direct precursor. The factorization of a series whose support meets two distinct nonzero Archimedean classes is [16, Fact 8.0.1] (Gonshor’s [13, Thm. 8.6]), with the geometric factorization displayed on p. 46 of the arXiv version; 11 cites it as the precedent of the division below. Manuscript 09 credited “Proposition 8.2.1 on separated-scale division”; that proposition is the truncation criterion ( $b$  divides  $c$  if and only if  $b$  divides a truncation  $T_\Delta(c)$ ), whose proof divides the higher-scale part of  $c$  by  $b$ , and 08 cites it as the precedent for higher-scale divisibility. We credit all three places; none of the sources claims the geometric mechanism as new.

**Lemma 4.6** (Weighted certificate [07]). *Let  $K$  be a set field,  $r, s \in K^\times$ ,  $a > 0$  and  $\delta > \eta > 0$  with  $\delta \ll a$ . Then*

$$\frac{X^a}{rX^\delta - sX^\eta} = \frac{1}{r} \sum_{k \geq 0} \left(\frac{s}{r}\right)^k X^{a-\delta-k(\delta-\eta)} \in \Pi_K, \quad (4.4)$$

and if  $\delta = \eta$  and  $r \neq s$  the quotient is  $(r - s)^{-1} X^{a-\delta}$ .

*Proof.* The exponents are those of Lemma 4.3; multiplying by  $rX^\delta - sX^\eta$  gives, at exponent  $a - k(\delta - \eta)$  with  $k \geq 1$ , the two contributions  $r \cdot r^{-1}(s/r)^k$  and  $-s \cdot r^{-1}(s/r)^{k-1}$ , which cancel.  $\square$

### 4.3 Division of an arbitrary purely infinite element

The telescope divides a monomial. Manuscripts 09, 10 and 11 divide an arbitrary element directly, which removes the separate reduction to monomials.

**Lemma 4.7** (Subordinate division [09, 10, 11, 13, 15]). *Let  $G$  be an ordered abelian group,  $0 \neq s = \sum_{g \in S} c_g X^g \in \Pi_k(G)$  and  $a > b > 0$  in  $U_G(S)$ . Put  $d = a - b$ . Then*

$$\frac{s}{X^a - X^b} = \sum_{g \in S} \sum_{n \geq 0} c_g X^{g-a-nd} \in \Pi_k(G), \quad (4.5)$$

so  $s = (X^a - X^b)q$  with  $q \in \Pi_k(G)$ . The support of  $q$  lies in  $\{g - a - nd : g \in S, n \in \mathbb{N}\}$ , of cardinality at most  $\max(|S|, \aleph_0)$ ; it is countable when  $S$  is.

*Proof.* In the ambient Hahn field  $(X^a - X^b)^{-1} = X^{-a}(1 - X^{-d})^{-1} = X^{-a} \sum_{n \geq 0} X^{-nd}$ . The product with  $s$  is a Hahn product: the sum of two reverse well-ordered sets is reverse well ordered and each exponent has finitely many representations (11 proves this two-support lemma in full; it is the standard Neumann support lemma). Every exponent that can occur satisfies  $g - a - nd \geq g - (n+1)a > 0$  because  $(n+1)a < g$ ; cancellation cannot create new exponents. Hence the quotient lies in  $\Pi_k(G)$ .  $\square$

The inverse of  $X^a - X^b$  is not in the ring; only its product with the larger-scale  $s$  is (11).

#### 4.4 Separation principles

**Lemma 4.8** (Separation [04, 09, 11; 03]). *Let  $R$  be a ring (possibly a proper class),  $e \in R$ , and  $(x_t)_{t \in T}$  a set-indexed family with  $e \in (x_s - x_t)R$  for all  $s \neq t$ .*

- (i) *If  $\varphi : R \rightarrow S$  is a ring homomorphism with  $\varphi(e) \neq 0$ , then  $t \mapsto \varphi(x_t)$  is injective; hence  $|T| \leq |S|$  if  $S$  is a set.*
- (ii) *If  $R$  is commutative,  $M$  an  $R$ -module,  $v \in M$  and  $ev \neq 0$ , then  $t \mapsto x_tv$  is injective.*
- (iii) *If for every cardinal  $\lambda$  such a family with  $|T| = \lambda$  exists, then every homomorphism from  $R$  to a set-sized ring kills  $e$ , and  $e$  annihilates every set-sized  $R$ -module.*

*Proof.* If  $\varphi(x_s) = \varphi(x_t)$  with  $e = (x_s - x_t)q$ , then  $\varphi(e) = (\varphi(x_s) - \varphi(x_t))\varphi(q) = 0$ ; neither commutativity of  $S$  nor preservation of 1 is used. For (ii),  $ev = q(x_s - x_t)v = 0$ . For (iii), take  $\lambda$  larger than the target, or than the Hartogs ordinal of the target, and apply the pigeonhole principle to a set.  $\square$

Item (i) is 04's separation lemma, (ii) is 09's orbit separation and 03's exponent-family obstruction, and (iii) is 11's collision-divisibility principle.

**Theorem 4.9** (Divisibility-separated families [11, 15]). *Let  $\mathcal{S} \subseteq \Pi_k$  be a set and  $\Lambda$  an ordinal. There are distinct monomials  $(u_\alpha)_{\alpha < \Lambda}$  in  $\mathcal{A}_{D,k}$  with*

$$s \in (u_\alpha - u_\beta)\Pi_k \quad (s \in \mathcal{S}, \alpha \neq \beta).$$

*Proof.* Choose  $\delta > 0$  strictly below the union of the supports of the members of  $\mathcal{S}$  (Lemma 3.1), put  $a_\alpha = \delta\omega^{-(\alpha+1)}$  and  $u_\alpha = X^{a_\alpha}$ . For  $\alpha < \beta$ ,  $0 < a_\beta < a_\alpha \leq \delta/\omega$ , so  $na_\alpha < \delta$  for every  $n$ , and Lemma 4.7 applies to each  $s$ ; reversing the indices only changes the sign of the quotient.  $\square$

The coefficients of the  $u_\alpha$  are 1; no independence of the coefficients of  $s$  is used, and cancellations inside  $s$  or the quotient are harmless (11). 15 proves the same with the family  $u_\xi = \omega^{g\omega^{-(\xi+1)}}$ ,  $\xi < \theta$ , for every ordinal  $\theta$ , and simultaneously for a set of elements  $s$ .

## 5 The universal set-sized quotient

### 5.1 The theorem

**Theorem 5.1** (Universal set-sized quotient [04, 06, 07, 08, 09, 10, 11, 13, 15; 03, 05]). *Let  $K$  be a set field,  $D \subseteq K$  a unital subring, and  $A = \mathcal{A}_{D,K} = D + \Pi_K$  with the full surreal exponent class.*

(i) *Let  $S$  be a set-sized associative ring, not necessarily unital or commutative, and let  $\varphi : A \rightarrow S$  be an additive and multiplicative class map, not necessarily preserving 1. Then  $\varphi(\Pi_K) = 0$ , and  $\varphi = \psi \circ \text{ct}$  for a unique ring homomorphism  $\psi : D \rightarrow S$ , namely  $\psi = \varphi|_D$ .*

(ii) *For every set-sized unital ring  $S$ , restriction to  $D$  and composition with  $\text{ct}$  are mutually inverse bijections, natural in  $S$ ,*

$$\text{Hom}_1(A, S) \longleftrightarrow \text{Hom}_1(D, S).$$

(iii) *Every additive and multiplicative map from the nonunital ring  $\Pi_K$  to a set-sized ring is zero.*

(iv) *Every set-sized unital  $A$ -module  $M$  satisfies  $\Pi_K M = 0$ , and its action is  $f \cdot v = \text{ct}(f)v$ .*

*Proof.* We follow 06's scaled-field argument in 08's choice-free Hartogs form.

*Positive monomials are killed.* Let  $\varphi : A \rightarrow S$  be as in (i), or an additive multiplicative map defined only on  $\Pi_K$  as in (iii). Fix  $g > 0$ , put  $m = X^{g/2}$  and  $L = \mathcal{F}_g(K) = K((X^{H_g}))$ . By Lemma 4.2(ii) with  $c = g/2$ ,  $mL \subseteq \Pi_K$ ; also  $m/(x - y) \in mL$  for distinct  $x, y \in L$ , so the identity (4.1) takes place inside  $\Pi_K$ . Let  $\lambda$  be the Hartogs ordinal of the set  $S$ , and put  $b_\alpha = g/\omega^{\alpha+1}$  for  $\alpha < \lambda$ ; these are distinct elements of  $H_g$  (Lemma 3.1(iii)), so  $\alpha \mapsto X^{b_\alpha}$  is an injection  $\lambda \rightarrow L$ . If  $\varphi(X^g) = \varphi(m^2) \neq 0$ , Lemma 4.1(i) makes  $x \mapsto \varphi(mx)$  injective on  $L$ , and composing gives an injection  $\lambda \rightarrow S$ , which is impossible. Hence  $\varphi(X^g) = 0$  for every  $g > 0$ .

*The whole ideal is killed.* For  $f \in \Pi_K$ , Proposition 3.3 gives  $f = X^a h$  with  $a > 0$  and  $h \in \Pi_K$ , so  $\varphi(f) = \varphi(X^a)\varphi(h) = 0$ . This proves (iii), and for (i) it gives  $\varphi(d + f) = \varphi(d)$  for  $d \in D$ ,  $f \in \Pi_K$ . Thus  $\varphi = \varphi|_D \circ \text{ct}$ , and  $\psi = \varphi|_D$  is forced because  $\text{ct}$  is surjective.

*The correspondence.* If  $\varphi$  is unital, so is  $\psi$ . Conversely, for a unital  $\psi : D \rightarrow S$  the composite  $\psi \circ \text{ct}$  is a unital homomorphism by Lemma 2.2, and its restriction to  $D$  is  $\psi$ . Post-composition with a map of targets preserves both constructions, which proves (ii).

*Modules.* Let  $M$  be a set-sized unital  $A$ -module and  $v \in M$ . If  $X^g v \neq 0$ , Lemma 4.1(ii) injects  $L$ , hence the Hartogs ordinal of  $M$ , into  $M$ ; so  $X^g M = 0$  for every  $g > 0$ , and the factorization  $f = X^a h$  gives  $\Pi_K M = 0$ . Writing  $f = \text{ct}(f) + (f - \text{ct}(f))$  gives  $f \cdot v = \text{ct}(f)v$ .  $\square$

**Sources and routes.** The statement is proved in the following generality by the sources: any set field  $K$  and unital  $D \subseteq K$  (06, 07, 10);  $k \in \{\mathbb{R}, \mathbb{C}\}$  and any unital  $D \subseteq k$  (04, 09, 11); **Oz**, **Oz**[ $i$ ] and complex coefficients with any unital  $D \subseteq \mathbb{C}$  (08). Nonunital maps are treated by 08 (as a theorem), 06 (as a remark) and 10 (including maps defined on  $\Pi_K$  alone). Noncommutative targets are allowed in all eight sources. Manuscript 05 of the sibling report [30] proves the unital case for **Oz** with the telescope and a cardinal  $\kappa > |A|$ , and 03 reproves that case (credited to 05) through its exponent-family obstruction. Two later manuscripts prove the theorem again, independently and without seeing the others: 13 for any set field  $k$  and unital  $D \subseteq k$  (unital maps, noncommutative targets), by direct division of an arbitrary purely infinite element with the Hartogs family  $\epsilon/(\bar{\alpha} + 2)$  (the route of 09, 10 and 11 below), and 15 for  $k \in \{\mathbb{R}, \mathbb{C}\}$  and unital  $D \subseteq k$ , with nonunital maps in a remark, by the separating families of Theorem 4.9 and a Hartogs ordinal, which is 08's formulation; 15 also states the universe-relative reading of Section 2.1. Neither adds a

hypothesis-free generalization. The proofs differ, and the following are genuinely different routes, kept here in brief.

- *Explicit telescope* (04, 05, 06 appendix, 08). If  $\varphi(X^c) \neq 0$ , then for distinct  $a_\alpha, a_\beta \ll c$  the telescope (4.2) puts  $X^c$  in  $(X^{a_\alpha} - X^{a_\beta})A$ , so Lemma 4.8(i) makes  $\alpha \mapsto \varphi(X^{a_\alpha})$  injective, which is impossible for  $\alpha$  ranging over a set larger than  $S$ ; Proposition 3.3 then kills all of  $\Pi_K$ . This is manuscript 05's proof, which the sibling report [30] quotes and refers here for; 05 chooses a cardinal  $\kappa > |S|$  with global choice, where the Hartogs ordinal suffices. The route uses only countable supports, which is why it survives in the support-bounded rings of Section 9.
- *Direct division of an arbitrary element* (09, 10, 11). If  $\varphi(s) \neq 0$  for  $s \in \Pi_K$ , Lemma 4.7 gives  $s \in (X^a - X^b)\Pi_K$  for all  $a > b$  in  $U_{\mathbf{No}}(\text{supp } s)$ , and Theorem 4.9 supplies such families of every set size, so Lemma 4.8(iii) applies to  $s$  itself; no separate reduction to monomials is needed. 10 phrases this as an injection  $U(\text{supp } s) \rightarrow S$ ,  $a \mapsto \varphi(X^a)$ , valid also for nonunital maps on  $\Pi_K$ ; 11 phrases it as divisibility-separated families.
- *Weighted certificates* (07). With  $\delta_\alpha = a/(\omega + \alpha + 1)$ ,  $\alpha < \kappa^+$  for a cardinal  $\kappa \geq |S|$ , pairwise differences of the monomials  $X^{\delta_\alpha}$  divide  $X^a$ , and (4.4) even separates all  $rX^\delta$ ,  $r \in K^\times$ .
- *Modules through the endomorphism ring* (04, 07, 08, 10, 11). A set-sized module gives a unital ring map  $A \rightarrow \text{End}_{\mathbb{Z}}(M)$  into a set-sized, generally noncommutative ring, and (ii) applies. This is valid for the full class but bounds only  $|\text{End}_{\mathbb{Z}}(M)|$ ; the direct argument above (06, 09, 03) and 08's cyclic-module argument (Lemma 10.9) are the ones that give set-sized thresholds.

## 5.2 Consequences for rings and representations

**Corollary 5.2** (Omnific integers [03, 04, 05, 06, 07, 08, 09, 10, 11, 13, 15]). *For every set-sized unital ring  $S$  the only unital homomorphism  $\mathbf{Oz} \rightarrow S$  is  $x \mapsto \text{ct}(x)1_S$ . Additive multiplicative maps  $\mathbf{Oz} \rightarrow S$  into a set-sized ring correspond bijectively to idempotents  $e \in S$ , via  $x \mapsto \text{ct}(x)e$ .*

*Proof.* There is one unital map  $\mathbb{Z} \rightarrow S$ . A ring map  $\psi : \mathbb{Z} \rightarrow S$  is determined by  $e = \psi(1)$ , which satisfies  $e^2 = e$ , and every idempotent defines one. Apply Theorem 5.1(i),(ii).  $\square$

**Corollary 5.3** (Gaussian omnific integers [04, 06, 07, 08, 09, 10, 11, 13, 15]). *For a set-sized unital ring  $S$ , unital maps  $\mathbf{Oz}[i] \rightarrow S$  correspond bijectively to elements  $j \in S$  with  $j^2 = -1_S$ , via  $x + iy \mapsto \text{ct}(x)1_S + \text{ct}(y)j$  ( $x, y \in \mathbf{Oz}$ ). Every such map discards all positive-exponent complex coefficients, not only their real parts.*

*Proof.* Apply Theorem 5.1 with  $(D, K) = (\mathbb{Z}[i], \mathbb{C})$ . A unital map from  $\mathbb{Z}[i] = \mathbb{Z}[T]/(T^2 + 1)$  is a choice of a root of  $T^2 + 1$ ; in a noncommutative  $S$  the subring generated by  $1_S$  and  $j$  is commutative, so evaluation is valid.  $\square$

The image of  $i$  is part of the data: these are ring maps, not maps of complex algebras. For  $(D, K) = (k, k)$  with  $k \in \{\mathbb{R}, \mathbb{C}\}$ , every unital map from  $\mathcal{A}_{\mathbb{R}, \mathbb{R}}$  or  $\mathcal{A}_{\mathbb{C}, \mathbb{C}}$  to a set-sized ring factors through the constant coefficient in  $k$ , and if the maps are required to fix  $k$ , the constant term is the unique such map to  $k$  (04). For any unital  $D \subseteq \mathbb{C}$ , the ring of complex-coefficient normal forms with positive support apart from a constant in  $D$  has universal set-sized quotient  $D$  (08); this is the case  $K = \mathbb{C}$  of the theorem.

**Corollary 5.4** (What all small observations identify [04]). *For  $x, y \in \mathcal{A}_{D,k}$ ,  $\text{ct}(x) = \text{ct}(y)$  if and only if  $\varphi(x) = \varphi(y)$  for every unital homomorphism  $\varphi$  from  $\mathcal{A}_{D,k}$  to a set-sized ring. Thus  $\Pi_k$  is the common kernel of all set-sized ring observations.*

*Proof.* The forward direction is the theorem; for the converse use the target  $D$  and the map  $\text{ct}$ .  $\square$

This does not say that the elements of  $\Pi_k$  are zero:  $\omega \neq 0$  in  $\mathbf{Oz}$ , yet every set-target ring homomorphism sends it to 0 (04).

**Corollary 5.5** (A common monomial in a set of kernels [15]). *Let  $(\varphi_j : \mathcal{A}_{D,k} \rightarrow B_j)_{j \in J}$  be a set-indexed family of ring homomorphisms, the targets possibly proper classes, together with specified nonzero  $b_j \in \ker \varphi_j$ . Then some nonzero monomial  $X^\delta$  lies in every  $\ker \varphi_j$ ; in particular the family is not jointly injective.*

*Proof.* Apply Proposition 3.4(i) to the set  $\{b_j^{-1}\}$ : some  $X^\delta$  has  $X^\delta/b_j \in \Pi_k$  for all  $j$ , so  $X^\delta \in b_j \mathcal{A}_{D,k} \subseteq \ker \varphi_j$ .  $\square$

The witnesses are specified so that no choice over a proper class of families is needed; nothing is claimed about an intersection indexed by the class of all nonzero elements (15).

**Corollary 5.6** (No faithful set-sized representation [04, 06, 07, 08, 11]). *No nonzero element of  $\Pi_k$  acts nontrivially on a set-sized unital  $\mathcal{A}_{D,k}$ -module; in particular  $\mathcal{A}_{D,k}$  has no faithful set-sized module. For every set-sized field  $F$  and set-sized  $F$ -vector space  $V$  (of any dimension), every unital ring map  $\mathbf{Oz} \rightarrow \text{End}_F(V)$  is  $x \mapsto \text{ct}(x) \text{id}_V$ , and every unital ring map  $\mathbf{Oz}[i] \rightarrow \text{End}_F(V)$  is  $x + iy \mapsto \text{ct}(x) \text{id}_V + \text{ct}(y)J$  with  $J^2 = -\text{id}_V$ .*

**Proposition 5.7** (Finite matrix representations [06, 11]). *Let  $n \geq 1$ .*

- (i) *The only unital map  $\mathbf{Oz} \rightarrow M_n(F)$ ,  $F$  a set-sized field, is  $x \mapsto \text{ct}(x)I_n$ .*
- (ii) *Up to conjugacy in  $\text{GL}_n(\mathbb{C})$ , the unital maps  $\mathbf{Oz}[i] \rightarrow M_n(\mathbb{C})$  are indexed by  $p \in \{0, \dots, n\}$ , with  $i \mapsto \text{diag}(iI_p, -iI_{n-p})$ .*
- (iii) *A unital map  $\mathbf{Oz}[i] \rightarrow M_n(\mathbb{R})$  exists exactly when  $n$  is even, and then all such maps are conjugate to the one sending  $i$  to  $n/2$  diagonal blocks  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .*

*Proof.* (i) is Corollary 5.2. By Corollary 5.3 a map in (ii) or (iii) is a matrix  $J$  with  $J^2 = -I_n$ . Over  $\mathbb{C}$  its minimal polynomial divides  $(T - i)(T + i)$ , which has distinct roots, so  $J$  is diagonalizable with eigenvalues  $\pm i$ , and the multiplicity  $p$  of  $i$  determines the conjugacy class. Over  $\mathbb{R}$ ,  $J$  makes  $\mathbb{R}^n$  a complex vector space by  $(u + iv)w = uw + vJw$ , so  $n$  is even, and a complex basis gives the block form; conversely the blocks square to  $-I_n$ .  $\square$

No norm bound, analytic hypothesis or finite-dimensionality is used to kill the purely infinite part; finite dimension enters only the classification of the possible images of  $i$  (06). For an infinite-dimensional set-sized space the reduction to one operator  $J$  with  $J^2 = -1$  remains valid.

**Algebras.** A set-sized associative  $\mathcal{A}_{D,k}$ -algebra is a set-sized ring with a unital map from  $\mathcal{A}_{D,k}$  to its center; by the theorem this map factors through  $D$ , so set-sized  $\mathcal{A}_{D,k}$ -algebras and  $D$ -algebras coincide, with the same underlying rings and morphisms; noncentral structure maps are covered by the noncommutative form (04). Enlarging the set-sized algebra in which a calculation takes place therefore never yields a faithful algebraic model of the full ring, although such calculations



can be faithful on a chosen set-sized subring (04). Nor does the theorem forbid set-sized rings from embedding *into*  $\mathbf{Oz}$ : the inclusion of  $\mathbb{Z}[\omega]$  detects  $\omega$ ; it concerns maps defined on all of  $\mathbf{Oz}$ , and it makes no claim that  $\mathbf{Oz}$  and  $\mathbb{Z}$  have the same first-order theory (15).

**Corollary 5.8** (No ordered specialization, no small retract, rigid endomorphisms [08, 10, 11; 05]).

- (i) *There is no order-preserving unital map from  $\mathbf{Oz}$  to a nonzero ordered set-sized ring (11; 09 records the relevant value  $\text{ct}(\omega - 1) = -1$ ).*
- (ii) *A set-sized unital subring  $A \subseteq \mathbf{Oz}$  is the image of a ring retraction  $\mathbf{Oz} \rightarrow A$  if and only if  $A = \mathbb{Z}$  (08).*
- (iii) *Every unital class ring endomorphism  $\sigma$  of  $\mathbf{Oz}$  satisfies  $\text{ct} \circ \sigma = \text{ct}$ ; hence every automorphism preserves  $\Pi$ . Every automorphism of  $\mathbf{Oz}[i]$  induces the identity or complex conjugation on  $\mathbb{Z}[i]$  (10; the first assertion is also 05's, see [30]).*

*Proof.* (i) The unique map sends the positive element  $\omega - 1$  to  $-1 < 0$ . (ii) A retraction is  $f \mapsto \text{ct}(f)1_A$  by Corollary 5.2, whose image is  $\mathbb{Z}$ ; and  $\text{ct}$  is a retraction onto  $\mathbb{Z}$ . (iii)  $\text{ct} \circ \sigma$  is a unital map to the set  $\mathbb{Z}$ , hence equals  $\text{ct}$ . For an automorphism apply this to  $\sigma$  and  $\sigma^{-1}$ . In the Gaussian case  $\sigma$  and  $\sigma^{-1}$  induce mutually inverse ring automorphisms of  $\mathbb{Z}[i]$  fixing  $\mathbb{Z}$ , and  $i$  goes to a root of  $T^2 + 1$  in  $\mathbb{Z}[i]$ , i.e. to  $\pm i$ .  $\square$

Passing to a small workspace is therefore not a homomorphic truncation of the full ring; the workspace may embed without being a retract (08). The endomorphism statement does not say that every endomorphism is the identity:  $\mathbb{Z} \hookrightarrow \mathbf{Oz}$  composed with  $\text{ct}$  is a noninjective endomorphism.

**Corollary 5.9** (Fields and unit-forcing localizations [04, 06, 09, 11]). *Let  $0 \neq u \in \Pi_k$ . The class rings  $k((X^{\mathbf{No}}))$  (in particular  $\mathbf{No}$  and  $\mathbf{No}[i]$ ) and  $\mathcal{A}_{D,k}[u^{-1}]$  admit no unital homomorphism to a nonzero set-sized ring and no nonzero set-sized unital module.*

*Proof.* A unital map from either ring to a set-sized  $S$  restricts to  $\mathcal{A}_{D,k}$  and so kills  $u$ ; but  $u$  is invertible in the source, so  $1_S = \varphi(u)\varphi(u^{-1}) = 0$ . A nonzero set-sized module would give such a map to the nonzero set-sized ring  $\text{End}_{\mathbb{Z}}(M)$ .  $\square$

This is not merely the observation that a proper-class field cannot inject into a set: the localizations need not be fields (04). For example  $\mathbf{Oz}[\omega^{-1}]$  is a nonzero class domain, not a field (Proposition 6.11), with no nonzero set-sized module. This completes the proof of Proposition 3.4(iv).

*Remark 5.10* (Positive characteristic and additive observations). If the target has positive characteristic  $n$ , the conclusion needs no cardinal argument:  $f = n(f/n)$  with  $f/n \in \Pi_k$  gives  $\varphi(f) = n\varphi(f/n) = 0$  (04, 07); more generally an ideal of  $\mathcal{A}_{D,k}$  containing a nonzero element of  $D$  contains  $\Pi_k$  (07, 15). The cardinal argument is needed exactly for characteristic-zero targets such as  $\mathbb{Q}$ ,  $\mathbb{R}$  or  $\text{End}_{\mathbb{Z}}(M)$  for infinite torsion-free  $M$  (11). The theorem concerns multiplicative maps: for a fixed  $\gamma > 0$  the coefficient extraction  $x \mapsto [\omega^\gamma]x$  is a nonzero additive map  $\mathbf{Oz} \rightarrow \mathbb{R}$  detecting  $\omega^\gamma$ , but it is not multiplicative, since  $\omega^{\gamma/2}$  has coefficient 0 at  $\gamma$  while its square has coefficient 1 (04, 07, 09, 10, 11, 15). Valuations also detect nonconstant parts, but not by unital ring maps into set-sized rings (04).

*Remark 5.11* (What the theorem does not say). The theorem is a universal property relative to set-sized targets, not an isomorphism  $\mathbf{Oz} \cong \mathbb{Z}$ . The ideal  $\Pi$  is nonzero, idempotent, not nilpotent and not generated by any set, and  $\mathbf{Oz}$  is a domain (09, 08). The regular module  $\mathbf{Oz}$  and the ideal  $\Pi$  are proper-class modules on which  $\Pi$  acts nontrivially, so the module correspondence is not a

Morita equivalence of unrestricted module theories (06). The identity of  $\mathbf{Oz}$  and the inclusion  $\mathbf{Oz} \hookrightarrow \mathbf{No}$  are faithful class-valued maps outside the theorem (09). Individual monomials can be represented faithfully in small rings, for instance in polynomial rings or Hahn subfields, and every finite calculation lies in a set-sized subring with a faithful regular representation; the obstruction concerns one global homomorphism defined on *all* of the class ring, including the collision witnesses at all lower scales (04, 07, 08). In the language of 11, the set-sized reflection is  $\text{Ref}_{\text{set}}(\mathbf{Oz}) = \mathbb{Z}$  and  $\text{Ref}_{\text{set}}(\mathbf{Oz}[i]) = \mathbb{Z}[i]$ : a surjection  $q : R \rightarrow D$  onto a set-sized ring is a universal set-sized quotient if every map from  $R$  to a set-sized ring factors uniquely through  $q$ ; this refers to the explicit universal property, not to a category of all class rings with set-sized hom-sets.

**Theorem 5.12** (Residually set-sized targets [11, 13]). *Call a class ring  $B$  residually set-sized if for each nonzero  $b \in B$  there are a set-sized ring  $A_b$  and a unital map  $\pi_b : B \rightarrow A_b$  with  $\pi_b(b) \neq 0$  (the quantifier is elementwise; no set of detecting maps is formed). Every unital map  $h : \mathcal{A}_{D,k} \rightarrow B$  into a residually set-sized ring kills  $\Pi_k$  and factors uniquely through  $D$ . For a set-sized commutative ring  $E$  and any class  $X$  of indeterminates, the ring  $E[X]$  of finite polynomials is residually set-sized.*

*Proof.* If  $h(s) \neq 0$  for some  $s \in \Pi_k$ , then  $\pi \circ h$  detects  $s$  for a detecting map  $\pi$  with set-sized target, contrary to [Theorem 5.1](#). A nonzero polynomial  $f \in E[X]$  uses a finite set  $Y \subseteq X$  of variables; sending the other variables to 0 gives a map  $E[X] \rightarrow E[Y]$  fixing  $f$ , with set-sized target.  $\square$

## 6 Quotients, closures, completions and localizations

Throughout,  $A = \mathcal{A}_{D,k}$  with  $k$  a set field and  $D \subseteq k$  a unital subring, and  $\Pi_k = \ker \text{ct}$ .

### 6.1 Every quotient and its small image

**Theorem 6.1** (Small reflection of an arbitrary quotient [06, 07, 09, 11, 15]). *Let  $J$  be a class ideal of  $A$ . Then  $\text{ct}(J)$  is an ideal of  $D$ , and for every set-sized unital ring  $S$*

$$\text{Hom}_1(A/J, S) \cong \text{Hom}_1(D/\text{ct}(J), S),$$

*the universal map being  $[x] \mapsto \text{ct}(x) \bmod \text{ct}(J)$ . Its kernel is  $(\Pi_k + J)/J$ , and  $\text{ct}^{-1}(\text{ct}(J)) = \Pi_k + J$ .*

*Proof.*  $\text{ct}$  is surjective, so the image of an ideal is an ideal. A map from  $A/J$  is a map  $\varphi : A \rightarrow S$  vanishing on  $J$ ; by [Theorem 5.1](#),  $\varphi = \psi \circ \text{ct}$  for a unique  $\psi$ , and  $\varphi(J) = 0$  if and only if  $\psi(\text{ct}(J)) = 0$ . If  $\text{ct}(a) = \text{ct}(b)$  with  $b \in J$ , then  $a - b \in \Pi_k$ , which gives the kernel and the preimage.  $\square$

Thus a quotient can be a nonzero ring while all of its set-sized images are zero; a quotient presentation of a proper-class ring is not automatically a set-sized ring (09). “Reflection” is only a name for the displayed mapping property; no category of all proper classes is used (07).

**Theorem 6.2** (Ideals with set-sized quotient [04, 06, 07, 08, 09, 10, 11, 13, 15]). *For a class ideal  $J$  of  $A$  the following are equivalent:*

- (i)  $J$  admits a set-sized quotient ([Definition 2.1](#));
- (ii)  $\Pi_k \subseteq J$ ;
- (iii)  $J = \text{ct}^{-1}(\mathfrak{a}) = \Pi_k + \mathfrak{a}$  for an ideal  $\mathfrak{a}$  of  $D$ .

Then  $\mathfrak{a} = J \cap D = \text{ct}(J)$  is unique and  $A/J \cong D/\mathfrak{a}$ . For  $\mathbf{Oz}$  the kernels are exactly

$$\Pi \text{ (quotient } \mathbb{Z}) \quad \text{and} \quad n\mathbf{Oz} = n\mathbb{Z} + \Pi \text{ (quotient } \mathbb{Z}/n\mathbb{Z}, n \geq 1),$$

the case  $n = 1$  giving the zero ring. For  $\mathbf{Oz}[i]$  they are  $\Pi_{\mathbb{C}}$  (quotient  $\mathbb{Z}[i]$ ) and  $d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}}$  for  $0 \neq d \in \mathbb{Z}[i]$ , with finite quotient  $\mathbb{Z}[i]/(d)$  of  $|d|^2$  elements; associates give the same ideal.

*Proof.* (i) $\Rightarrow$ (ii) by [Theorem 5.1](#) applied to the quotient map. (ii) $\Rightarrow$ (iii): if  $\Pi_k \subseteq J$  then  $d + f \in J$  ( $f \in \Pi_k$ ) if and only if  $d \in J \cap D$ . (iii) $\Rightarrow$ (i):  $A \rightarrow D \rightarrow D/\mathfrak{a}$  realizes the quotient. For  $\mathbf{Oz}$  the ideals of  $\mathbb{Z}$  are  $n\mathbb{Z}$ ,  $n \geq 0$ ;  $n = 0$  gives  $\Pi$  and  $n \geq 1$  gives  $n\mathbf{Oz}$  by (3.2). For  $\mathbf{Oz}[i]$ ,  $\mathbb{Z}[i]$  is a principal ideal domain (division with remainder by rounding the real and imaginary parts of a complex quotient), and (3.2) gives  $d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}}$ . Multiplication by  $d = b + ci$  on  $\mathbb{Z}[i] \cong \mathbb{Z}^2$  has matrix  $\begin{pmatrix} b & -c \\ c & b \end{pmatrix}$  of determinant  $b^2 + c^2 = |d|^2$ , the index of its image; more generally a nonzero ideal containing  $d$  contains the positive integer  $d\bar{d}$ , so it has finite index without any appeal to unique factorization (07, 06, 08).  $\square$

The distinction between  $n = 0$  and  $n \geq 1$  matters: the characteristic-zero kernel is  $\Pi$ , not  $0\mathbf{Oz}$ ; one must not set  $n = 0$  in  $n\mathbf{Oz} = n\mathbb{Z} + \Pi$  (04, 06, 07, 08, 09, 10, 11). For  $D = k$ , the only proper kernel with a set-sized quotient is  $\Pi_k$ , with quotient  $k$  (04).

**Corollary 6.3** (Small primes, maximal ideals and fields [06, 07, 08, 10, 11, 13, 15]). *The prime class ideals of  $\mathbf{Oz}$  with set-sized quotient are  $\Pi$  and  $p\mathbf{Oz}$  ( $p$  an ordinary prime), with quotient domains  $\mathbb{Z}$  and  $\mathbb{F}_p$  and fraction fields  $\mathbb{Q}$  and  $\mathbb{F}_p$ ; the maximal ones are the  $p\mathbf{Oz}$ . A unital map from  $\mathbf{Oz}$  to a set-sized field has kernel  $\Pi$  in characteristic 0 (its image is  $\mathbb{Z}$ , not a field) and  $p\mathbf{Oz}$  in characteristic  $p$ ; the only nonzero set-sized field quotients are the  $\mathbb{F}_p$ . For  $\mathbf{Oz}[i]$  replace these by  $\Pi_{\mathbb{C}}$  and the ideals generated by Gaussian primes, with quotient domains  $\mathbb{Z}[i]$  and the corresponding finite fields.*

In particular  $\Pi$  is prime but not maximal (it lies in  $p\mathbf{Oz}$ ). These results classify only ideals with set-sized quotients: (0) is prime in  $\mathbf{Oz}$  with a proper-class quotient, and no Zorn argument for a proper class of ideals is used to assert that proper class ideals lie in maximal class ideals; a maximal ideal outside this list, if one is given, has a proper-class residue field (07, 08, 11).

**Proposition 6.4** (Principal quotients [04]). *Let  $f \in A$  be nonconstant and  $\beta = \deg_{\omega}(f) > 0$ . For distinct  $\gamma, \delta \in (0, \beta)$ ,  $X^{\gamma} - X^{\delta} \notin fA$ ; hence  $A/(f)$  has sets of pairwise distinct residues of every set cardinality and admits no set-sized quotient. Consequently  $(f)$  admits a set-sized quotient exactly when  $f \in D \setminus \{0\}$ , and then  $fA = fD + \Pi_k$  and  $A/(f) \cong D/fD$ . Among the ideals with set-sized quotient,  $\Pi_k$  is the only one not generated by a set; every other one is generated by a subset of  $D$ , and for  $\mathbf{Oz}$  these are the principal ideals  $n\mathbf{Oz}$ ,  $n \geq 1$ .*

*Proof.* The difference has degree  $\max(\gamma, \delta) < \beta$ , while a nonzero multiple  $fq$ ,  $q \in A$ , has degree  $\beta + \deg_{\omega}(q) \geq \beta$  ([Lemma 2.4](#)); the interval  $(0, \beta)$  contains sets of every cardinality ([Lemma 3.1](#)). For  $0 \neq f \in D$ ,  $f\Pi_k = \Pi_k$  ([Proposition 3.3](#)); the zero ideal would inject the class  $A$  into a set. For the last assertion write the ideal as  $\Pi_k + \mathfrak{a}$ ; if  $\mathfrak{a} = 0$  use [Proposition 3.3\(iii\)](#); otherwise some  $0 \neq d \in \mathfrak{a}$  gives  $d\Pi_k = \Pi_k$ , so the ideal generated by the set  $\mathfrak{a}$  already contains  $\Pi_k$ .  $\square$

**Theorem 6.5** (Quotient-size alternative [08, 13]). *Let  $J$  be a class ideal of  $\mathbf{Oz}$  with  $\Pi \not\subseteq J$ . For every set cardinal  $\mu$  there is a set of at least  $\mu$  elements of  $\mathbf{Oz}$  pairwise incongruent modulo  $J$ .*

*Proof.* Choose  $f \in \Pi \setminus J$  and write  $f = \omega^c h$  with  $c > 0$ ,  $h \in \Pi$ ; then  $\omega^c \notin J$ . For an ordinal  $\lambda$  of cardinality at least  $\mu$  use the exponents  $c/\omega^{\beta+1}$ ,  $\beta < \lambda$ : if two of the monomials were congruent modulo  $J$ , the telescope (4.2) would put  $\omega^c$  in  $J$ .  $\square$

A statement that a quotient “is not small” always has this concrete meaning: for every set cardinal it contains a larger set of distinct elements; no “cardinality of  $\mathbf{No}$ ” is used (07).

## 6.2 Finite congruences, closures and invisible primes

For a class ideal  $J$  of  $\mathbf{Oz}$  define its finite-congruence closure  $\bar{J}^{\text{fin}} = \bigcap_{n \geq 1} (J + n\mathbf{Oz})$ ; the witness in  $J$  may depend on  $n$  (08).

**Theorem 6.6** (Closure of every ideal [08, 09]). *If  $\text{ct}(J) = d\mathbb{Z}$  with  $d \geq 0$ , then  $\bar{J}^{\text{fin}} = J + \Pi = \text{ct}^{-1}(d\mathbb{Z})$ . In particular  $J$  is dense for finite congruences exactly when  $\text{ct}(J) = \mathbb{Z}$ , and the closure of 0 is  $\Pi$ .*

*Proof.*  $\text{ct}(J)$  is an ideal of  $\mathbb{Z}$ , and  $J + n\mathbf{Oz} = \text{ct}^{-1}(d\mathbb{Z} + n\mathbb{Z}) = \Pi + \gcd(d, n)\mathbb{Z}$  by (3.2). If  $d > 0$  the term  $n = d$  is  $d\mathbb{Z}$  and all terms contain it; if  $d = 0$  the intersection is  $\bigcap_n n\mathbb{Z} = 0$ .  $\square$

**Corollary 6.7** (Dense proper principal ideals [08, 09]). *If  $u \in \mathbf{Oz}$  is nonconstant with  $\text{ct}(u) = \pm 1$ , then  $(u)$  is proper and dense for finite congruences, and every unital map from  $\mathbf{Oz}$  to a nonzero set-sized ring sends  $u$  to  $\pm 1$ . Every nonconstant irreducible  $p \in \mathbf{Oz}$  has  $\text{ct}(p) = \pm 1$ ; hence  $(p)$  is proper, profinitely dense, and  $\mathbf{Oz}/(p)$  has no nonzero set-sized ring image.*

*Proof.*  $u$  is a nonunit by Lemma 2.4; apply Theorem 6.6 and Corollary 5.2. For an irreducible  $p = c + f$ ,  $0 \neq f \in \Pi$ : if  $c = 0$  then  $p = 2(p/2)$ , and if  $|c| > 1$  then  $p = c(p/c)$  with  $p/c = 1 + f/c$  nonconstant, both factorizations into nonunits; so  $c = \pm 1$ . A map killing  $p$  would kill  $\pm 1$ .  $\square$

The converse of the second statement is false:  $\omega^2 - 1 = (\omega - 1)(\omega + 1)$  has constant term  $-1$  and is reducible. The constant term is a complete invariant for set-sized ring observations, not for factorization inside  $\mathbf{Oz}$  (09).

**Corollary 6.8** (A prime quotient invisible to all set-sized rings [06, 08, 09, 11, 15]). *Let  $q = \omega^{\sqrt{2}} + \omega + 1$ . Then  $\mathbf{Oz}/(q)$  is a nonzero proper-class integral domain of characteristic 0; it admits no unital map to a nonzero set-sized ring and no nonzero set-sized unital module.*

*Proof.* L’Innocente and Mantova proved that  $q$  is prime in  $\mathbf{Oz}$  [16, Theorem B], answering Gonshor’s question [13, p. 117]; this primality is imported, not reproved. So  $(q)$  is a proper prime ideal and  $\mathbf{Oz}/(q)$  is a nonzero domain. Since  $\text{ct}(q) = 1$ , Theorem 6.1 gives the zero ring as its universal set-sized image. On a set-sized module, inflated to  $\mathbf{Oz}$ ,  $q$  acts both as  $\text{ct}(q) = 1$  and as 0, so the module is zero. If  $\mathbf{Oz}/(q)$  were a set, its identity map would contradict the first assertion. No nonzero integer  $n$  lies in  $(q)$ :  $n = qa$  would force  $a = n/q$ , a nonzero infinitesimal, which is not omnific; so  $\mathbb{Z} \hookrightarrow \mathbf{Oz}/(q)$ .  $\square$

This is no contradiction with the maximal-ideal theorem for set-sized rings: the source quotient is not a set, and the corollary does not say that a particular maximal class ideal fails to exist (06, 09). The same obstruction needs no primality.

**Proposition 6.9** (Unit-constant ideals [06, 07]). *If a class ideal  $J$  of  $\mathbf{Oz}$  contains an element with constant term 1, then  $\mathbf{Oz}/J$  has no nonzero set-sized unital module and no unital map to a nonzero set-sized ring; if  $J$  is proper,  $\mathbf{Oz}/J$  is a nonzero proper-class ring. More generally, if  $\text{ct}(f) = c$ , the small reflection of  $\mathbf{Oz}/(f)$  is  $\mathbb{Z}/c\mathbb{Z}$ .*

*Proof.* The element acts as 1 by [Theorem 5.1](#) and as 0 by the quotient relation. For a principal ideal  $\text{ct}((f)) = c\mathbb{Z}$ ; apply [Theorem 6.1](#).  $\square$

For example  $\omega + 1$  and  $1 + \omega^a$  ( $a > 0$ ) are nonunits, so  $\mathbf{Oz}/(1 + \omega^a)$  is a nonzero class ring with no nonzero set-sized image; [Section 14](#) shows how much it contains. The benefit of  $q$  in [Corollary 6.8](#) is that the quotient is a domain (06). Under any map to a set-sized ring,  $\omega^a$  becomes 0 and  $1 + \omega^a$  becomes 1, which is the arithmetic reason (07).

**Theorem 6.10** (Arithmetic completions [04, 06, 07, 08, 09, 10, 11, 13, 15]). *All finite quotient rings of  $\mathbf{Oz}$  are the  $\mathbb{Z}/n\mathbb{Z}$ , and*

$$\widehat{\mathbf{Oz}} := \varprojlim_{n \geq 1} \mathbf{Oz}/n\mathbf{Oz} \cong \widehat{\mathbb{Z}}, \quad \varprojlim_{r \geq 1} \mathbf{Oz}/p^r \mathbf{Oz} \cong \mathbb{Z}_p \quad (p \text{ prime}).$$

*The canonical maps are  $\text{ct}$  followed by the usual injections; their kernel is  $\Pi$  and their image is the diagonal copy of  $\mathbb{Z}$ . For the Gaussian ring,  $\widehat{\mathbf{Oz}[i]} \cong \widehat{\mathbb{Z}[i]} = \widehat{\mathbb{Z}}[T]/(T^2 + 1)$  and  $\varprojlim_n \mathbf{Oz}[i]/\ell^n \mathbf{Oz}[i] \cong \mathbb{Z}_\ell[i] = \mathbb{Z}_\ell[T]/(T^2 + 1)$  for an ordinary prime  $\ell$  (not necessarily a domain), with kernel  $\Pi_{\mathbb{C}}$ .*

*Proof.* By [Theorem 6.2](#) the finite quotients and their transition maps are those of  $\mathbb{Z}$  (respectively  $\mathbb{Z}[i]$ ), identified by  $\text{ct}$ ; a nonzero integer is not divisible by every  $n$  or every  $p^r$ . For  $\mathbf{Oz}[i]$ , every finite quotient has some positive characteristic  $n$ , so its kernel contains  $n\mathbf{Oz}[i] = n\mathbb{Z}[i] + \Pi_{\mathbb{C}}$ ; the ideals  $n\mathbb{Z}[i]$  are cofinal among finite-index ideals, and the limit is free of rank two over  $\widehat{\mathbb{Z}}$  (respectively  $\mathbb{Z}_\ell$ ) with  $i^2 = -1$ .  $\square$

The finite quotient system is an ordinary set-indexed system; no inverse limit over a proper class of targets is formed, and  $\mathbf{Oz}$  is not claimed to be an object of the category of small topological rings (08, 09). The finite-congruence topology on  $\mathbf{Oz}$  is not Hausdorff:  $\Pi$  is the closure of 0. It is unrelated to the fine order topology and to Hahn valuation topologies (07, 10, 11). Every  $g \in \Pi$  is divisible in  $\mathbf{Oz}$  by every power of  $p$ , but this does not represent  $g$  as a nonzero  $p$ -adic integer; for example  $\omega^\omega + \pi\omega + 17$  maps to 17 in every finite quotient and in every  $\mathbb{Z}_p$  (07). The finite-quotient and completion statements were already in the Diophantine manuscripts of the sibling report [30] (04 cites them as prior); the universal theorem explains why passing from finite targets to arbitrary set-sized targets recovers nothing more.

### 6.3 Localizations and presentations

**Proposition 6.11** (Adjoining one monomial inverse [04, 07]). *For  $a > 0$ ,*

$$\mathcal{A}_{D,k}[X^{-a}] = \{x \in k((X^{\mathbf{No}})) : \text{supp } x \subseteq [-Na, \infty) \text{ for some ordinary } N\},$$

*independently of  $D$ . This is a proper subring of  $k((X^{\mathbf{No}}))$  and not a field; it has no nonzero set-sized ring target ([Corollary 5.9](#)). In particular  $\mathbf{Oz}[\omega^{-a}]$  is a nonzero class domain, not a field, with no nonzero set-sized unital module.*

*Proof.* An element of the localization is  $fX^{-Na}$  with  $f \in \mathcal{A}_{D,k}$ , whose support is bounded below by  $-Na$ . Conversely, if  $\text{supp } x \subseteq [-Na, \infty)$ , then  $X^{(N+1)a}x$  has strictly positive support, so  $x = X^{(N+1)a}x/X^{(N+1)a}$  lies in the localization; the extra factor  $X^a$  removes the restriction on the coefficient at the lower endpoint, which explains the independence from  $D$ . The exponent  $a\omega$  exceeds every  $Na$ , so  $X^{-a\omega}$  is not in the ring although  $X^{a\omega}$  is; the ring is proper and not a field.  $\square$

The same argument shows  $\omega^{-\omega} \notin \mathbf{Oz}[1/\omega]$ :  $\omega^{-\omega} = a/\omega^n$  would force  $a = \omega^{n-\omega} \notin \mathbf{Oz}$  (11).

**Theorem 6.12** (Quotient-localization calculus [08, 09, 11, 15]). *Let  $R = \mathcal{A}_{D,k}$ ,  $J$  a class ideal and  $S \subseteq R$  a multiplicative class containing 1. Put  $D_0 = D/\text{ct}(J)$  and  $T = \{\text{ct}(s) + \text{ct}(J) : s \in S\} \subseteq D_0$ . For every set-sized commutative ring  $B$ ,*

$$\text{Hom}((R/J)[S^{-1}], B) \cong \text{Hom}(D_0[T^{-1}], B);$$

*if  $0 \in T$  the right side is empty for  $B \neq 0$ . For  $R = \mathbf{Oz}$ ,  $J = 0$  and  $S$  a set-sized multiplicative subset, the same holds for noncommutative  $B$ , with universal formula  $f/s \mapsto \text{ct}(f)1_B(\text{ct}(s)1_B)^{-1}$ .*

*Proof.* A map from the left ring is a map  $h : R \rightarrow B$  killing  $J$  and sending each  $s \in S$  to a unit. By Theorem 5.1,  $h = g \circ \text{ct}$  uniquely, and the conditions become  $g(\text{ct}(J)) = 0$  and  $g(\text{ct}(s)) \in B^\times$ , which are exactly the conditions for a map  $D_0[T^{-1}] \rightarrow B$ ; the sets  $\text{ct}(J)$  and  $T$  are sets even when  $J$  and  $S$  are classes. The natural map to the reflection is surjective on fractions of constants. For the noncommutative statement, the denominators  $\text{ct}(s)1_B$  are central, so the ordinary universal property of localization applies (08).  $\square$

**Example 6.13** (Three effects [08, 11, 15]).  $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/(\omega + 2)]) = \mathbb{Z}[1/2]$ ;  $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/\omega]) = 0$ ;  $\text{Ref}_{\text{set}}(\mathbf{Oz}/(\omega + 1)) = 0$ ; and  $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/n]) = \mathbb{Z}[1/n]$  for a nonzero ordinary integer  $n$ . The second ring is a nonzero domain inside  $\mathbf{No}$  and the third is nonzero because  $\omega + 1$  is not a unit; both have no nonzero set-sized representation. A nonzero class ring can therefore have no nonzero set-sized unital module, so set-sized existence arguments for modules cannot be applied to a proper class of class ideals (11). If  $\text{ct}(s) = 0$  for some  $s \in S$ , the displayed localization of  $\mathbb{Z}$  is the zero ring (08). In the language of 09: no unital map to a nonzero set-sized ring can make a nonzero  $f \in \Pi_k$  invertible or make an element with unit constant term zero.

**Theorem 6.14** (Algebraic points forget infinite coefficients [06, 09, 11, 15]). *Let  $X$  be a set of commuting variables,  $(f_j)_{j \in E}$  a set-indexed family in  $\mathcal{A}_{D,k}[X]$ ,  $B = \mathcal{A}_{D,k}[X]/(f_j)$  and  $B_0 = D[X]/(\text{ct}_* f_j)$ , where  $\text{ct}_*$  applies  $\text{ct}$  to coefficients. For every set-sized commutative ring  $S$  the natural map  $B \rightarrow B_0$  induces  $\text{Hom}(B_0, S) \cong \text{Hom}(B, S)$ ; thus  $\text{Ref}_{\text{set}}(B) = B_0$ .*

*Proof.* A map  $\mathcal{A}_{D,k}[X] \rightarrow S$  is a map on  $\mathcal{A}_{D,k}$ , which factors through  $\text{ct}$ , together with images of the variables; for such data  $f_j$  vanishes exactly when  $\text{ct}_* f_j$  does.  $B_0$  is a set because  $D$ ,  $X$  and the relations are sets, and  $B \rightarrow B_0$  is surjective.  $\square$

06 states this for finitely many variables and 11 for a set of variables; 09 states the corresponding reduction of localization and relation problems. The relation  $\omega T = 1$  reduces to  $0 = 1$ , so it has no solution in a nonzero set-sized ring under any representation of its coefficients, although  $T = \omega^{-1}$  solves it in  $\mathbf{No}$  (06). For  $B = \mathbf{Oz}[T]/(T^2 - \omega)$  one gets  $\text{Ref}_{\text{set}}(B) = \mathbb{Z}[T]/(T^2)$ : a set-sized field sends  $T$  to 0 and a set-sized ring may send it to any square-zero element, while the actual root  $\omega^{1/2} \in \mathbf{Oz}$  gives a map to the class ring  $\mathbf{Oz}$  (11). These are universal-property statements, not quantifier elimination, not a reconstruction of class-valued solutions, and not an assertion that a scheme with a proper-class coordinate ring is an ordinary scheme (06, 11).

## 6.4 Diophantine observation

If  $P \in \mathbb{Z}[X_1, \dots, X_m]$ ,  $x \in \mathbf{Oz}^m$  and  $\varphi : \mathbf{Oz} \rightarrow S$  is a unital set-target map, then  $\varphi(P(x)) = P(\text{ct}(x_1)1_S, \dots, \text{ct}(x_m)1_S)$ , even for noncommutative  $S$ , since the coordinates are integer scalars (04). This does *not* say that  $P(x) = 0$  whenever  $P(\text{ct}(x)) = 0$  (take  $P(X) = X$  and  $x = \omega$ ), and it



does not classify infinite points of curves; inequalities, supports, valuations and nonmultiplicative coefficient observations remain available (04).

**Proposition 6.15** (Conservativity for integer coefficients [08, 13]). *A finite system of polynomial equations with coefficients in  $\mathbb{Z}$  has a solution in  $\mathbf{Oz}$  if and only if it has one in  $\mathbb{Z}$ ; more generally, positive existential ring formulas with parameters in  $\mathbb{Z}$  have the same truth value in  $\mathbf{Oz}$  and  $\mathbb{Z}$ . With an inequation this fails:  $\exists x \exists y (x^2 = 2y^2 \wedge y \neq 0)$  holds in  $\mathbf{Oz}$  (take  $x = \sqrt{2}\omega$ ,  $y = \omega$ ) but not in  $\mathbb{Z}$ .*

*Proof.* Apply the homomorphisms  $\text{ct}$  and  $\mathbb{Z} \hookrightarrow \mathbf{Oz}$ , which preserve positive existential formulas (induction on their construction). A nonzero integer solution of  $x^2 = 2y^2$  would give a rational square root of 2;  $\text{ct}$  maps the omnific witnesses to  $x = y = 0$ , preserving the equation but not the inequation.  $\square$

The parameter restriction is essential (the coefficient  $\omega^\gamma + 1$  of Theorem 15.25 is not an integer), and so is positivity (08). The sibling report [30] proves the same constant-term transfer for equations in its Diophantine setting.

## 7 Modules and derivations over the full class ring

**Theorem 7.1** (Set-sized modules [04, 06, 07, 08, 09, 10, 11, 13, 15]). *Let  $A = \mathcal{A}_{D,k}$ . Restriction along  $D \hookrightarrow A$  and inflation along  $\text{ct}$  are mutually inverse equivalences between set-sized unital  $A$ -modules and set-sized unital  $D$ -modules; the action of  $A$  on a set-sized module is  $(d + f)m = dm$  ( $d \in D$ ,  $f \in \Pi_k$ ). The equivalence is the identity on underlying abelian groups and on morphisms, and it is exact. In particular set-sized  $\mathbf{Oz}$ -modules are exactly abelian groups with  $xm = \text{ct}(x)m$ , and set-sized  $\mathbf{Oz}[i]$ -modules are exactly  $\mathbb{Z}[i]$ -modules inflated along  $\text{ct}$ .*

*Proof.* Theorem 5.1(iv) gives  $\Pi_k M = 0$  and the action formula. Conversely, a  $D$ -module inflates along  $\text{ct}$ , and restriction recovers the  $D$ -action because  $\text{ct}|_D = \text{id}$ . An additive map between two such modules is  $A$ -linear exactly when it is  $D$ -linear. Kernels, cokernels and exact sequences are computed on underlying groups with the induced actions.  $\square$

A  $\mathbf{Oz}[i]$ -module structure on an abelian group is the choice of an endomorphism  $J$  with  $J^2 = -\text{id}$ , automatically invertible (08). Category language abbreviates the correspondence of individual objects and morphisms: each action is coded by its restriction to the set  $D \times M$ , and no class of arbitrary class actions is formed (04, 08, 11). The regular module  $A$  is a proper class on which  $\Pi_k$  acts nontrivially, so there is no conflict with recovering a ring from its regular module (04). The equivalence can transport extension problems between the two small-module categories, but it should not be rephrased as a statement about homological dimensions computed with arbitrary class-sized modules: the rank-one free module  $A$  is not an object of the small-module category (07). Section 12 treats the homological algebra over set-sized integer parts.

**Proposition 7.2** (An additive splitting that is not a module splitting [08, 15]). *The exact sequence of class  $\mathbf{Oz}$ -modules  $0 \rightarrow \Pi \rightarrow \mathbf{Oz} \xrightarrow{\text{ct}} \mathbb{Z} \rightarrow 0$ , with  $\mathbb{Z}$  carrying the action  $x \cdot n = \text{ct}(x)n$ , does not split as a sequence of  $\mathbf{Oz}$ -modules, although it splits as a sequence of additive groups and of rings.*

*Proof.* A module section  $s$  gives  $e = s(1)$  with  $\text{ct}(e) = 1$  and  $ue = s(u \cdot 1) = s(0) = 0$  for every  $u \in \Pi$ . With  $u = \omega$  and  $\mathbf{Oz}$  a domain,  $e = 0$ , a contradiction.  $\square$

**Proposition 7.3** (Projectivity depends on the size convention [15]). *With the constant-term action,  $\mathbb{Z}$  is projective in the category of set-sized  $\mathbf{Oz}$ -modules, although  $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$  has no  $\mathbf{Oz}$ -linear section (Proposition 7.2). For every set-sized  $\mathbf{Oz}$ -module  $M$ ,  $\text{Hom}_{\mathbf{Oz}}(M, \mathbf{Oz}) = 0$ .*

*Proof.* By Theorem 7.1 the set-sized modules are abelian groups, and  $\mathbb{Z}$  is free on one generator. If  $f : M \rightarrow \mathbf{Oz}$  is linear and  $m \in M$ , then  $\omega f(m) = f(\omega m) = f(0) = 0$ , so  $f(m) = 0$  in the domain  $\mathbf{Oz}$ .  $\square$

No nonzero free  $\mathbf{Oz}$ -module is set-sized, yet the set-sized module category has many projectives; this is a warning against importing the size convention of ordinary module theory, not a Morita statement (15).

**Theorem 7.4** (Derivations into set-sized modules [04, 06, 08, 09, 10, 11, 13]). *Let  $A = \mathcal{A}_{D,k}$  and  $M$  a set-sized  $A$ -module. Every derivation  $\partial : A \rightarrow M$  (an additive class map with  $\partial(xy) = x\partial(y) + y\partial(x)$ ) vanishes on  $\Pi_k$ , and restriction gives a natural bijection  $\text{Der}_{\mathbb{Z}}(A, M) \cong \text{Der}_{\mathbb{Z}}(D, M)$  with inverse  $\delta \mapsto \delta \circ \text{ct}$ . Consequently:*

- (i) every derivation  $\mathbf{Oz} \rightarrow M$  is zero;
- (ii)  $\text{Der}_{\mathbb{Z}}(\mathbf{Oz}[i], M) \cong M[2] = \{m \in M : 2m = 0\}$  via  $\partial \mapsto \partial(i)$ , and  $\text{Der}_{\mathbb{Z}[i]}(\mathbf{Oz}[i], M) = 0$ ;
- (iii) every  $k$ -linear derivation of  $\mathcal{A}_{k,k}$  or of  $\mathcal{O}_k$  ( $k = \mathbb{R}, \mathbb{C}$ ) into a set-sized module is zero, and  $\text{Der}_D(A, M) = 0$ .

*Proof.*  $\Pi_k M = 0$  by Theorem 7.1, and each  $x \in \Pi_k$  is a product  $uv$  with  $u, v \in \Pi_k$  (Proposition 3.3), so  $\partial(x) = u\partial(v) + v\partial(u) = 0$ . Thus  $\partial(d + x) = \partial(d)$ . Conversely, for a derivation  $\delta$  of  $D$ ,  $\delta \circ \text{ct}$  satisfies the Leibniz rule because  $A$  acts on  $M$  through  $\text{ct}$ . A derivation of  $\mathbb{Z}$  vanishes ( $\partial(1) = 2\partial(1)$ ). A derivation of  $\mathbb{Z}[i]$  is determined by  $m = \partial(i)$ ; differentiating  $i^2 + 1 = 0$  gives  $2im = 0$ , equivalently  $2m = 0$  since  $i$  acts invertibly; conversely  $\partial(a + bi) = bm$  satisfies the Leibniz rule, the discrepancy for a product being  $2b\dim = 0$ . It kills  $\mathbb{Z}[i]$  exactly when  $m = 0$ . A  $k$ -linear derivation of  $k$  vanishes; the finite side uses Theorem 8.2 and  $\mathfrak{m}_k^2 = \mathfrak{m}_k$  (Lemma 8.4).  $\square$

**A second route (09).** Since  $M$  is a  $D$ -module, the square-zero extension  $D \ltimes M$  with  $(d, m)(e, n) = (de, dn + em)$  is a set-sized ring, and  $a \mapsto (\text{ct}(a), \partial(a))$  is a unital ring map  $A \rightarrow D \ltimes M$ ; the universal theorem forces it to factor through  $D$ .

**Example 7.5** (The ramification at 2 survives [09, 11]). Give  $\mathbb{F}_2$  the  $\mathbb{Z}[i]$ -module structure  $i \mapsto 1$ . Then  $\partial(a + bi + f) = b \bmod 2$  is a nonzero  $\mathbb{Z}$ -derivation  $\mathbf{Oz}[i] \rightarrow \mathbb{F}_2$ . The statement “all set-valued derivations of the Gaussian omnific ring vanish” is false; only the positive-support directions vanish, and the surviving derivation reflects ordinary ramification of  $\mathbb{Z}[i]$  at 2.

The coefficient-linearity qualification in (iii) cannot be dropped: the theorem reduces arbitrary derivations of  $\mathcal{A}_{\mathbb{R}, \mathbb{R}}$  or  $\mathcal{O}_{\mathbb{R}}$  to derivations of  $\mathbb{R}$  over  $\mathbb{Q}$ , which are not zero (04). For a set-sized commutative  $D$ -algebra  $B$ , every  $D$ -algebra map  $A \rightarrow B[\varepsilon]/(\varepsilon^2)$  kills  $\Pi_k$ , so relative first-order deformations with set-sized coefficients cannot detect  $\Pi_k$ ; no assertion about an absolute proper-class module of Kähler differentials is made (11).

**Proposition 7.6** (Class-valued derivations [06, 08, 10]). *For  $\eta \in \mathbf{No}$  let  $\ell_\eta : \mathbf{No} \rightarrow \mathbb{R}$  extract the coefficient of  $\omega^\eta$  from the normal form of its argument; it is additive. The formula*

$$D_\eta \left( \sum_x r_x \omega^x \right) = \sum_x \ell_\eta(x) r_x \omega^x$$

defines a derivation  $\mathbf{No} \rightarrow \mathbf{No}$  which restricts to a derivation  $\mathbf{Oz} \rightarrow \Pi$  into the class module  $\Pi$ , preserves strongly summable families, and satisfies  $D_\eta(\omega^{\omega^\theta}) = \omega^{\omega^\theta}$  if  $\eta = \theta$  and 0 otherwise. Every finite family of distinct  $D_\eta$  is linearly independent over  $\mathbb{R}$ , and  $D_0(\omega) = \omega$ . Coefficientwise complexification gives the analogous derivations of  $\mathbf{No}[i]$  and  $\mathbf{Oz}[i]$ .

*Proof.* The output support is a subset of the input support, so the formula defines a surreal; the exponent 0 is killed because  $\ell_\eta(0) = 0$ , and no negative exponent appears for an omnific input. Additivity and preservation of strong sums are coefficientwise. At a fixed exponent  $z$  the coefficient of a product is a finite sum over pairs  $x + y = z$ , and  $\ell_\eta(z) = \ell_\eta(x) + \ell_\eta(y)$ ; multiplying and summing gives the Leibniz rule. The exponent  $\omega^\theta$  has coefficient 1 at  $\theta$  and 0 elsewhere, which gives the displayed values and, by evaluation at the monomials  $\omega^{\omega^\theta}$ , linear independence. Finally  $\ell_0(1) = 1$ .  $\square$

The proposition combines three sources, none of which states all of it: 08 introduces the family  $D_\eta$  for every  $\eta$ , as derivations  $\mathbf{Oz} \rightarrow \Pi$ , with the diagonal values and linear independence; the case  $\eta = 0$  is 06's Euler-type derivation  $\mathcal{D}$  (06 defines it on  $\mathbf{No}$ , with the complex extension) and 10's exponent-weighted derivation  $D_h$  ( $h = \ell_0$ ; 10 records preservation of strongly summable families). The general- $\eta$  statements on  $\mathbf{No}$  and on strong sums follow by the same coefficientwise proof. There is no conflict with [Theorem 7.4](#): the target  $\Pi$  is a proper-class module with its internal multiplication. Berarducci and Mantova's surreal derivation with  $\partial\omega = 1$  [5], restricted to  $\mathbf{Oz}$  with values in  $\mathbf{No}$ , is another nonzero class-valued boundary example (09). None of the  $D_\eta$  is asserted to be the Berarducci–Mantova derivation, a fine derivative, or compatible with surreal exponentiation (06, 08, 10), and no uniqueness is claimed for the established derivation (09). The word “small” in [Theorem 7.4](#) cannot be dropped.

## 8 The finite side: standard part is universal

This section is manuscript 04's; no other source treats the negative-support side. Its proof is shorter because the ambient valuation produces the needed quotients directly, and it is a different route rather than a sign reversal of the positive side (04). Let  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $D \subseteq k$  a unital subring, and for  $x \neq 0$  let  $v(x) = -\deg_\omega(x)$ , so that  $v(x/y) = v(x) - v(y)$ ; a nonzero normal form lies in  $\mathfrak{m}_k$  exactly when  $v(x) > 0$ .

**Lemma 8.1** (Division of an infinitesimal by small differences [04]). *Let  $0 \neq \varepsilon \in \mathfrak{m}_k$  and  $a = v(\varepsilon) > 0$ . For  $0 < \gamma < \delta < a$ ,  $\varepsilon/(\omega^{-\gamma} - \omega^{-\delta}) \in \mathfrak{m}_k$ .*

*Proof.* The denominator has largest exponent  $-\gamma$ , so valuation  $\gamma$ ; the quotient has valuation  $a - \gamma > 0$ , i.e. all of its exponents are negative.  $\square$

**Theorem 8.2** (Universal set-sized residue on the finite side [04]). *For every set-sized unital ring  $S$ , restriction gives a bijection  $\text{Hom}_1(\mathcal{B}_{D,k}, S) \rightarrow \text{Hom}_1(D, S)$  with inverse  $\psi \mapsto \psi \circ \text{st}$ . Every such homomorphism annihilates  $\mathfrak{m}_k$ . In particular every unital map from  $\mathcal{O}_{\mathbb{R}}$  or  $\mathcal{O}_{\mathbb{C}}$  to a set-sized ring factors through  $\text{st} : \mathcal{O}_{\mathbb{R}} \rightarrow \mathbb{R}$ , respectively  $\text{st} : \mathcal{O}_{\mathbb{C}} \rightarrow \mathbb{C}$ .*

*Proof.* Suppose  $0 \neq \varepsilon \in \mathfrak{m}_k$  and  $\varphi(\varepsilon) \neq 0$ , put  $a = v(\varepsilon)$  and choose a set of exponents in  $(0, a)$  of cardinality larger than  $|S|$  ([Lemma 3.1](#)). By [Lemma 8.1](#),  $\varepsilon$  is a multiple, by an element of  $\mathfrak{m}_k \subseteq \mathcal{B}_{D,k}$ , of the difference of any two of the monomials  $\omega^{-\gamma}$ , so [Lemma 4.8\(i\)](#) injects the chosen set into  $S$ , a contradiction. Hence  $\varphi(\mathfrak{m}_k) = 0$  and  $\varphi(d + \varepsilon) = \varphi(d)$ ; conversely  $\psi \circ \text{st}$  extends every  $\psi$ .  $\square$

For a target without a specified coefficient-field structure, the induced map on  $\mathbb{R}$  or  $\mathbb{C}$  is not presumed to be the identity: the assertion is factorization through the residue field, not uniqueness of maps from  $\mathcal{O}_{\mathbb{C}}$  to an arbitrary ring (04). The statement involves no topology on  $S$  and so is stronger than a statement about continuous maps into topological rings; it must also be distinguished from the existence of standard part, which is classical and part of the repository's formal infrastructure [32].

**Corollary 8.3** (Quotients, modules and derivations on the finite side [04]). *The kernels of  $\mathcal{B}_{D,k}$  admitting a set-sized quotient are exactly  $\mathfrak{m}_k + \mathfrak{a}$  for ideals  $\mathfrak{a}$  of  $D$ ; the only nonzero set-sized quotients of  $\mathcal{O}_{\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}}$  are  $\mathbb{R}$  and  $\mathbb{C}$ , up to the residue identification. Set-sized  $\mathcal{B}_{D,k}$ -modules are exactly  $D$ -modules inflated along  $\text{st}$  (set-sized  $\mathcal{O}_k$ -modules are  $k$ -vector spaces), no faithful set-sized module exists, and  $\text{Der}_{\mathbb{Z}}(\mathcal{B}_{D,k}, M) \cong \text{Der}_{\mathbb{Z}}(D, M)$  for set-sized  $M$ . For  $0 \neq \varepsilon \in \mathfrak{m}_k$ , the localization  $\mathcal{B}_{D,k}[\varepsilon^{-1}]$  has no nonzero set-sized ring target and no nonzero set-sized module.*

*Proof.* Repeat the proofs of Theorems 6.2, 7.1 and 7.4 and Corollary 5.9 with Theorem 8.2 and  $\text{st}$  in place of the positive-side theorem and  $\text{ct}$ ; for  $D = k$  the only ideals are 0 and  $k$ . Derivations use Lemma 8.4.  $\square$

**Lemma 8.4** (Idempotence on the finite side [04]).  $\mathfrak{m}_k^2 = \mathfrak{m}_k$  as an ideal of  $\mathcal{B}_{D,k}$ .

*Proof.* For  $0 \neq \varepsilon \in \mathfrak{m}_k$  with  $a = v(\varepsilon)$ ,  $\varepsilon = \omega^{-a/2}(\omega^{a/2}\varepsilon)$  with both factors of valuation  $a/2 > 0$ .  $\square$

**Two residues that cannot be combined (04).** Every element of  $k((X^{\text{No}}))$  has a unique decomposition along  $\Pi_k \oplus k \oplus \mathfrak{m}_k$ , obtained by splitting its support at 0; so  $\mathcal{A}_{k,k} \cap \mathcal{O}_k = k$  and  $\mathcal{A}_{k,k} + \mathcal{O}_k$  is the whole field additively. The two constant-coefficient maps agree on the intersection, yet no unital ring map on the field extends both:

$$\text{ct}(\omega) = 0, \quad \text{st}(\omega^{-1}) = 0, \quad \omega \cdot \omega^{-1} = 1. \quad (8.1)$$

The obstruction is multiplicative, not an ambiguity of the additive decomposition, and Corollary 5.9 sharpens it: adjoining one inverse to the positive-support ring already destroys every nonzero set-sized target.

**Relation to the rotation groups.** The euclidean-three-space report proves that every class homomorphism from  $\text{SO}(3, \text{No})$  to a set-sized group factors through standard part (`e3:cut:thm:smallquotient`), that every nontrivial set-sized image is  $\text{SO}(3, \mathbb{R})$  (`e3:cut:cor:onlyimage`), and the analogues for  $\text{SO}(n, \text{No})$ ,  $n \geq 3$ , and  $\text{SU}(n, \text{No}[i])$ ,  $n \geq 2$  [27]. Theorem 8.2 is the ring analogue on the finite side; neither result implies the other. The group proof uses exact commutators and normal generation before a choice-free Hartogs contradiction; the ring proof uses divisibility by differences of monomials. Manuscripts 06, 07, 08, 09, 10 and 11 acknowledge the rotation theorem as the repository's conceptual predecessor for universal set-sized quotients, 04 refers to related size obstructions in other algebraic settings, and none of them uses the rotation theorem as a lemma.

## 9 The support threshold: finite supports fail, countable supports suffice

The universal theorem does not need normal forms of every set cardinality. Keep the *full* exponent class **No** and bound only the size of each individual support. For  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $D \subseteq k$  unital, and an

infinite cardinal  $\lambda$ , put

$$\mathcal{A}_{D,k}^{<\lambda} = D + \{f \in \Pi_k : |\text{supp } f| < \lambda\}, \quad \mathcal{B}_{D,k}^{<\lambda} = D + \{f \in \mathfrak{m}_k : |\text{supp } f| < \lambda\},$$

and, in 11's notation,  $R_{\leq \lambda} = \mathcal{A}_{D,k}^{<\lambda+}$  and  $R_{\text{fin}} = \mathcal{A}_{D,k}^{<\aleph_0}$ . These are rings: the support of a sum lies in a finite union and that of a product in the image of the product of the two supports under addition, and for an infinite  $\lambda$  finite unions and products of sets of size  $< \lambda$  (or  $\leq \lambda$ ) keep that bound; regularity of  $\lambda$  is not needed. They remain proper classes, since a one-term series may have any positive exponent (04, 11).

**Theorem 9.1** (Support threshold [04, 11; 03, 07, 09]).

(i) For every uncountable cardinal  $\lambda$  and every set-sized unital ring  $S$ , restriction induces bijections

$$\text{Hom}_1(\mathcal{A}_{D,k}^{<\lambda}, S) \cong \text{Hom}_1(D, S), \quad \text{Hom}_1(\mathcal{B}_{D,k}^{<\lambda}, S) \cong \text{Hom}_1(D, S),$$

with inverses given by *ct* and *st*. In particular countably supported normal forms ( $\lambda = \aleph_1$ , i.e.  $R_{\leq \aleph_0}$ ) already have universal set-sized quotient  $D$  on both sides, and so does  $R_{\leq \mu}$  for every infinite  $\mu$ .

(ii) For finite supports both assertions fail: the augmentation

$$E\left(d + \sum_{g \in F} c_g \omega^g\right) = d + \sum_{g \in F} c_g \quad (F \text{ finite})$$

is a unital ring homomorphism from each finite-support ring to  $k$ , sending every monomial, including  $\omega$  and  $\omega^{-1}$ , to 1; it does not factor through *ct* or *st*.

(iii) (Extension criterion.) For a set-sized unital ring  $S$  and a unital map  $h : R_{\text{fin}} \rightarrow S$  the following are equivalent: (a)  $h$  extends to  $R_{\leq \aleph_0}$ ; (b)  $h$  extends to  $R_{\leq \mu}$  for some infinite  $\mu$ ; (c)  $h = g \circ \text{ct}$  for a map  $g : D \rightarrow S$ ; (d)  $h$  extends to  $\mathcal{A}_{D,k}$ . Whenever an extension exists it is unique at each of these support levels. In particular  $E$  does not extend to  $R_{\leq \aleph_0}$ .

(iv) [merge] The same equivalence holds on the finite side, with  $\mathcal{B}_{D,k}^{<\aleph_0}$ ,  $\mathcal{B}_{D,k}^{<\aleph_1}$ ,  $\mathcal{B}_{D,k}^{<\lambda}$ ,  $\mathcal{B}_{D,k}$  and *st* in place of  $R_{\text{fin}}$ ,  $R_{\leq \aleph_0}$ ,  $R_{\leq \mu}$ ,  $\mathcal{A}_{D,k}$  and *ct*.

*Proof.* (i), positive side. Fix  $s \in \Pi_k$  with  $|\text{supp } s| < \lambda$ . For  $a > b$  in  $U_{\mathbf{No}}(\text{supp } s)$ , the quotient of Lemma 4.7 has support of cardinality at most  $\max(|\text{supp } s|, \aleph_0) < \lambda$ , and the monomials  $\omega^a$  are in the ring; so the families of Theorem 4.9 and all their quotients lie in  $\mathcal{A}_{D,k}^{<\lambda}$ , and Lemma 4.8(iii) gives  $\varphi(s) = 0$  (11). Alternatively (04), every monomial and every telescope  $q_{c;a,b}$  has at most countable support, so the explicit-telescope route (Lemma 4.3 with Lemma 4.8(i); see the second routes after Theorem 5.1) kills all positive monomials inside the bounded ring (the field-collision proof of Theorem 5.1 does not apply here, since the Hahn field over  $H_g$  is not contained in the bounded ring), and the factorization  $s = \omega^a(\omega^{-a}s)$  of Proposition 3.3 only translates the support, keeping its cardinality.

(i), finite side (04). For  $0 \neq \varepsilon$  with  $|\text{supp } \varepsilon| < \lambda$ ,  $a = v(\varepsilon)$ ,  $0 < \gamma < \delta < a$  and  $d = \delta - \gamma$ ,

$$\frac{\varepsilon}{\omega^{-\gamma} - \omega^{-\delta}} = \varepsilon \omega^\gamma \sum_{n \geq 0} \omega^{-nd},$$

the geometric series being the inverse of  $1 - \omega^{-d}$ . Its support lies in  $\{\beta + \gamma - nd : \beta \in \text{supp } \varepsilon, n \in \mathbb{N}\}$ , of cardinality at most  $|\text{supp } \varepsilon| \cdot \aleph_0 < \lambda$ , and its largest exponent is  $-a + \gamma < 0$ . So the quotient of

**Lemma 8.1** stays in the bounded ring, and the proof of **Theorem 8.2** applies. A valuation inequality alone would not control the support size; this check is needed.

(ii) Additivity and  $E(1) = 1$  are clear; for finite supports  $E(xy) = \sum_{\alpha, \beta} c_\alpha d_\beta = E(x)E(y)$  by expanding the finite double sum.  $E(\omega) = 1 \neq 0 = \text{ct}(\omega)$  and  $E(\omega^{-1}) = 1 \neq 0 = \text{st}(\omega^{-1})$ .

(iii) An extension to some  $R_{\leq \mu}$  factors through  $\text{ct}$  by (i), giving (a) $\Rightarrow$ (c) and (b) $\Rightarrow$ (c); an extension to  $\mathcal{A}_{D,k}$  factors by **Theorem 5.1**, giving (d) $\Rightarrow$ (c). Conversely  $g \circ \text{ct}$  is defined on every ring in the list and extends  $h$ , and two extensions to one ring both factor through  $\text{ct}$  and agree on  $D$ . Since  $E(\omega) = 1 \neq \text{ct}(\omega)$ ,  $E$  violates (c).

(iv) The same argument with (i) on the finite side, **Theorem 8.2** and  $\text{st}$ ; the extension  $g \circ \text{st}$  exists on every ring of the list, and uniqueness follows as before.  $\square$

*Remark 9.2* (Why the threshold is about closure, not continuity). No axiom that target maps preserve countable sums is added: certain countable sums lie *inside* the source ring, where they satisfy finite identities. An extension of  $E$  to the countable-support ring would send both monomials in **Example 4.5** to 1 and the product to 1, forcing  $(1 - 1)E(q) = 1$  (04, 11). A map that assigns different values to some monomials is handled by the full theorem: a large family forces a collision somewhere (11). The separating families may have cardinality far above  $\lambda$ ; their members are distinct elements, not terms of one series, and each witness separately has countable support (04, 07, 11).

The finite-support augmentation appears independently in 03 (“two necessary warnings”), 04, 09, 11 and 15 (which blocks its extension with the single series of **Example 4.5**), and 12’s cores carry it too (**Proposition 13.2**); 07 describes the same boundary in words (on a finite-support fragment the certificate (4.2) need not exist, and a faithful interpretation in a set-sized target can exist), and manuscript 05 of the sibling report [30] observes that  $\mathbb{Z} + \omega\mathbb{R}[\omega]$  has many real evaluation maps that do not extend to **Oz**. Proper-class size alone therefore does not prove the universal theorem: the finite-support rings have a proper class of monomials and arbitrarily small positive exponents, yet  $E$  sees every monomial as 1 (04). Neither size nor additive divisibility alone suffices; the proof needs many subordinate scales together with enough support closure for geometric division (11). For  $D = \mathbb{Z}$ , finite targets still kill the positive-support ideal of  $R_{\text{fin}}$  ( $f = n(f/n)$  in characteristic  $n$ ), so its finite quotient theory is ordinary congruence theory; having profinite completion  $\hat{\mathbb{Z}}$  is much weaker than having  $\mathbb{Z}$  universal for all set-sized targets (09).

The quotient and module classifications extend to the bounded rings of (i), and so does the derivation statement, because the monomial factorizations proving idempotence preserve support cardinality (04). 04 does *not* infer that these smaller rings have fraction field  $k((X^{\mathbf{No}}))$ : arbitrary normal forms need not have the required support size. The result is also different from completion or omitted-type questions for bounded-support series over a *fixed* set-sized exponent group (compare [29]): here the exponent class is still large enough to supply a new separating family for every proposed target (04). The fixed-group question is the subject of the next section.

## 10 Fixed exponent groups: cardinal lower bounds and exact thresholds

Over a fixed set-sized exponent group the full-class theorem cannot hold verbatim: the ring is itself a set, and its identity map is a set-sized target that retains every element (03, 04, 07, 08, 10, 11). What survives is a lower bound on the size of a target that detects a given element. This section compares the bounds proved by the sources, proves a field bound that dominates them, and states



the organizing theorem of the report: for each of the set-sized constructions of [Section 11](#), the exact detection threshold is the size of the ring.

Throughout,  $\Gamma$  is an ordered abelian set group,  $k$  a set field,  $D \subseteq k$  a unital subring,  $F \subseteq k((X^\Gamma))$  a subfield containing  $k$  and every monomial  $X^g$ ,  $\Pi(F) = \{f \in F : \text{supp } f \subseteq \Gamma_{>0}\}$  and  $A = D + \Pi(F)$ . The case  $F = k((X^\Gamma))$  is the ring  $\mathcal{A}_{D,k}(\Gamma)$  of [\(2.2\)](#); the constructions of [Section 11](#) take  $F$  to be a countable-support, a  $< \kappa$ -support, or a controlled subfield.

## 10.1 The bounds of the sources

**Proposition 10.1** (Monomial-count bounds [03, 04, 07, 10, 13, 15]). *Let  $A = \mathcal{A}_{D,k}(\Gamma)$  and  $\varphi : A \rightarrow S$  a ring homomorphism.*

- (i) *If  $\varphi(X^a) \neq 0$  for some  $a > 0$ , then  $\gamma \mapsto \varphi(X^\gamma)$  is injective on  $H_a(\Gamma)_{>0}$  (04); more generally  $rX^\delta \mapsto \varphi(rX^\delta)$  is injective on  $\{rX^\delta : r \in k^\times, 0 < \delta \in H_a(\Gamma)\}$ , and the same holds if  $X^a$  divides some  $g$  with  $\varphi(g) \neq 0$  (07).*
- (ii) *If  $0 \neq x \in \Pi_k(\Gamma)$  and  $\varphi(x) \neq 0$ , then  $a \mapsto \varphi(X^a)$  is injective on  $U_\Gamma(\text{supp } x)$ ; this also holds for a nonunital map defined on  $\Pi_k(\Gamma)$  alone (10; 15's local criterion is this bound, and 13's criterion is the same injection on any set  $T \subseteq \Gamma_{>0}$  with  $s > a + n(a - b)$  for  $a > b$  in  $T$ ,  $s \in \text{supp } x$ ).*
- (iii) *On the negative side, if  $0 \neq \varepsilon \in \mathfrak{m}_k(\Gamma)$ ,  $a = v(\varepsilon)$  and  $\varphi(\varepsilon) \neq 0$  for a map  $\varphi$  on  $\mathcal{B}_{D,k}(\Gamma)$ , then  $\gamma \mapsto \varphi(X^{-\gamma})$  is injective on  $(0, a)_\Gamma$  (04).*
- (iv) *For a module  $M$  and  $v \in M$  with  $X^h v \neq 0$ ,  $a \mapsto X^a v$  is injective on  $\{a \in \Gamma : a \ll h\}$  (03).*

*No divisibility of  $\Gamma$  is needed.*

*Proof.* The telescope, the weighted certificate, the division lemma and the finite-side division ([Lemmas 4.3, 4.6, 4.7](#) and [8.1](#)) use only  $a, \gamma, \delta$ , subtraction and ordinary integer multiples, so their witnesses lie in  $\mathcal{A}_{D,k}(\Gamma)$  or  $\mathcal{B}_{D,k}(\Gamma)$ ; apply [Lemma 4.8](#).  $\square$

**Example 10.2** (Where these bounds are vacuous [04, 08, 10, 13]). If  $\Gamma$  is Archimedean,  $H_a(\Gamma) = \{0\}$  and the telescopes do not exist. For  $\Gamma = \mathbb{Z}$  a reverse well-ordered positive support is finite,  $\mathcal{A}_{\mathbb{Z},\mathbb{R}}(\mathbb{Z}) = \mathbb{Z} + X\mathbb{R}[X]$ , and evaluation at  $X = 1$  is a map to the set field  $\mathbb{R}$  detecting  $X$  (10). An arbitrary fixed Hahn workspace cannot inherit the full theorem merely because its notation resembles **No** (08).

## 10.2 The field bound

The monomial counts are far from optimal. The following bound is 03's “smaller-scale field obstruction”, obtained there for countable-support fields over divisible groups; the general form below is a direct application of 06's scaled-field collision lemma and is recorded as a merge statement.

**Proposition 10.3** (Field bound [03, 06] [merge]). *Let  $c \in \Gamma_{>0}$  and  $L_c(F) = F \cap k((X^{H_c(\Gamma)}))$ .*

- (i)  *$L_c(F)$  is a subfield of  $F$  and  $X^c L_c(F) \subseteq \Pi(F) \subseteq A$ .*
- (ii) *If  $\varphi : A \rightarrow S$  is a ring homomorphism with  $\varphi(X^{2c}) \neq 0$ , then  $f \mapsto \varphi(X^c f)$  is injective on  $L_c(F)$ ; if  $M$  is an  $A$ -module and  $X^{2c} v \neq 0$ , then  $f \mapsto (X^c f)v$  is injective on  $L_c(F)$ .*

(iii) Consequently, if  $x \in X^{2c}A$  (for instance  $x = X^{2c}$ , or  $0 \neq x \in \Pi(F)$  with  $2c$  a strict lower bound of  $\text{supp } x$ ), every ring homomorphism and every module detecting  $x$  has cardinality at least  $|L_c(F)|$ .

*Proof.* (i)  $k((X^{H_c(\Gamma)}))$  is a subfield by Lemma 4.2(ii), so  $L_c(F)$  is an intersection of two subfields. For  $h \in H_c(\Gamma)$ ,  $|h| < c$ , so  $c + h > 0$  and  $X^c f \in F$  has positive support. (ii) is Lemma 4.1 with  $m = X^c$  and  $L = L_c(F)$ , inside the field  $k((X^\Gamma))$ . (iii) If  $x = X^{2c}y$  with  $y \in A$  and  $\varphi(x) \neq 0$ , then  $\varphi(X^{2c}) \neq 0$ ; if  $xv \neq 0$  then  $X^{2c}(yv) \neq 0$ . If  $2c < \text{supp } x$  then  $X^{-2c}x \in F$  has positive support, so it lies in  $\Pi(F) \subseteq A$ .  $\square$

For  $F = H_\omega(k, \Gamma)$  and  $c = h/2$  this is exactly 03's theorem ( $H_{h/2} = H_h$ ). The bound contains 04's and 10's monomial counts whenever those monomials lie in  $H_c(\Gamma)$ , and 07's weighted count, because  $L_c(F)$  contains every  $rX^\delta$  with  $r \in k$  and  $\delta \in H_c(\Gamma)$ ; it also contains  $k$  itself, so with real coefficients every detecting target has at least  $2^{\aleph_0}$  elements (03). The following examples show that it can be strictly and decisively larger.

**Example 10.4** (The exact threshold in 04's example [merge]). Let  $\Gamma = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$  (first coordinate dominant),  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $A = \mathcal{A}_{D,k}(\Gamma)$  with full Hahn supports, and  $a = (1, 0)$ . 04 computes the positive part  $H_a(\Gamma)_{>0} = \{(0, q) : q \in \mathbb{Q}_{>0}\}$ , so  $H_a(\Gamma) = \{0\} \oplus \mathbb{Q}$ , and notes that retaining  $X^a$  requires only a countable family of distinct images. With  $c = (1/2, 0)$ ,  $H_c(\Gamma) = \{0\} \oplus \mathbb{Q}$  and  $L_c = k((X^{\{0\} \oplus \mathbb{Q}})) \cong k((t^\mathbb{Q}))$  has cardinality  $2^{\aleph_0}$ ; and  $|A| \leq |k((X^\Gamma))| \leq 2^{\aleph_0} \cdot (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ , since  $\Gamma$  is countable. Hence the least cardinality of a ring target, and of a module, detecting any nonzero  $x \in X^{(1,0)}A$  is exactly  $2^{\aleph_0}$ , attained by the identity map and the regular module. The same holds on the negative side: if  $\varepsilon \in \mathfrak{m}_k(\Gamma)$  has  $v(\varepsilon) = (1, 0)$ , then with  $c = (1/4, 0)$  the element  $X^{-2c}$  divides  $\varepsilon$  in  $\mathcal{B}_{D,k}(\Gamma)$  (the cofactor  $X^{2c}\varepsilon$  has largest exponent  $(-1/2, 0) < 0$ ),  $X^{-c}L_c \subseteq \mathfrak{m}_k(\Gamma)$ , and Lemma 4.1 gives the bound  $2^{\aleph_0}$ , which is again attained.

In this example 07's weighted bound ( $|k^\times| \cdot \aleph_0 = 2^{\aleph_0}$ ) is also attained, while 04's and 10's monomial counts give only  $\aleph_0$ . With a countable coefficient field all three source bounds fail:

**Example 10.5** (None of the monomial or weighted counts is attained [merge]). Let  $k = D = \mathbb{Q}$ ,  $\Gamma = \bigoplus_{n < \omega} \mathbb{Q}e_n$  with the least nonzero index dominant, and  $A = \mathcal{A}_{\mathbb{Q},\mathbb{Q}}(\Gamma)$  with full Hahn supports. For  $x = X^{e_0}$  and  $c = e_0/2$ ,  $H_c(\Gamma) = \bigoplus_{n \geq 1} \mathbb{Q}e_n$ , and  $L_c$  contains the  $2^{\aleph_0}$  distinct series  $\sum_{n \in B} X^{e_n}$ ,  $B \subseteq \omega \setminus \{0\}$  (the exponents  $e_n$  decrease with  $n$ , so each support is reverse well ordered). Since  $\Gamma$  is countable,  $|A| \leq 2^{\aleph_0} \cdot \aleph_0^{\aleph_0} = 2^{\aleph_0}$ . So the exact detection threshold of  $X^{e_0}$  is  $2^{\aleph_0}$ , while 04's bound  $|H_{e_0}(\Gamma)_{>0}|$ , 07's bound  $|\mathbb{Q}^\times| \cdot |H_{e_0}(\Gamma)_{>0}|$  and 10's bound  $|U_\Gamma(\{e_0\})|$  are all  $\aleph_0$ .

These examples settle the two example-level questions of the sources and leave the general ones open: 04's question on optimal small targets in a fixed exponent group is answered for its own example  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$  (the answer is  $2^{\aleph_0}$ ), and for 10's question (when is the bound of Proposition 10.1(ii) attained?) Example 10.5 shows, in ZFC, that the bound need not be attained. The same phenomenon appears in 03's ring  $A_{\aleph_1}^{03}$  of Section 11.1: there  $|U_\Gamma(S)| \leq \aleph_1$ , while the exact threshold is  $2^{\aleph_0}$ , so if the continuum hypothesis fails, 10's bound is not attained there either. Remark 10.6 records the corresponding correction to a sentence of 04.

*Remark 10.6* (A correction to 04's commentary). 04 remarks that in  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$  “this is only a countable family” which “cannot contradict the existence of a sufficiently large infinite set target, such as the ring itself”. The statement is true, but the ring itself has  $2^{\aleph_0}$  elements and, by Example 10.4, no smaller target detects  $X^{(1,0)}$ ; the countable family is not the obstruction that decides the answer.

### 10.3 Criteria that kill the whole ideal below a cardinal

To pass from monomials to the whole ideal one also needs a gap: every element must be divisible by a positive monomial inside the ring. 03's example  $\sum_{n \geq 1} X^{1/n}$  over  $\Gamma = \mathbb{R}$  shows that this can fail (Example 3.2), and 09's uncountable-support series  $f_*$  (Remark 11.21) shows the same for full supports over a long group.

**Theorem 10.7** (Quantitative criterion [08] [merge]). *Let  $\Gamma$  be divisible,  $\kappa$  an infinite cardinal, and suppose*

- (a) *every nonzero  $f \in \Pi(F)$  has a positive exponent  $\epsilon \in \Gamma$  strictly below its support;*
- (b) *for every  $c > 0$  in  $\Gamma$ , the set  $\{a \in \Gamma : a \ll c\}$  has at least  $\kappa$  elements.*

*Then every ring homomorphism from  $A$  to a ring of cardinality  $< \kappa$  factors uniquely through  $\text{ct} : A \rightarrow D$ . [merge] The same conclusion holds, with the same proof, if (b) is replaced by (b'):*  $|L_c(F)| \geq \kappa$  *for every  $c > 0$ .*

*Proof.* Fix  $c > 0$ . Under (b), for two distinct  $a > b \ll c$  the series  $q_{c;a,b}$  equals the rational expression  $X^c/(X^a - X^b)$  in  $k((X^\Gamma))$ , whose numerator and denominator lie in the field  $F$ ; so  $q_{c;a,b} \in \Pi(F) \subseteq A$ , and a nonzero image of  $X^c$  would make the images of  $\kappa$  monomials distinct. Under (b') use Proposition 10.3(ii) with  $c/2$  in place of  $c$ . So every positive monomial is killed. For  $f \in \Pi(F)$ , (a) gives  $f = X^\epsilon g$  with  $g = X^{-\epsilon} f \in F$  of positive support, so  $f$  is killed.  $\square$

Condition (a) is a gap property of the chosen field and (b) a supply of lower Archimedean scales; neither says that homomorphisms preserve strongly summable families (08). 08 states the theorem for a subfield  $K \subseteq \mathbb{R}$  and constants  $\mathbb{Z}$ ; the proof uses neither restriction, and it is stated here in the setting of this section.

**Theorem 10.8** (Coinitiality criterion [11]). *Let  $\Gamma$  be divisible,  $\kappa$  uncountable, and  $\text{coin}(\Gamma_{>0}) \geq \kappa$  (every set of fewer than  $\kappa$  positive elements has a positive strict lower bound). Put  $R_{\Gamma, < \kappa} = D + \{s \in \Pi_k(\Gamma) : |\text{supp } s| < \kappa\}$ , a ring for every infinite  $\kappa$ . Then every unital map  $R_{\Gamma, < \kappa} \rightarrow S$  with  $|S| < \kappa$  factors uniquely through  $\text{ct}$ .*

*Proof.* Let  $s \neq 0$  have  $|\text{supp } s| < \kappa$ . Choose  $\delta > 0$  strictly below  $\text{supp } s$ , and then a positive strict lower bound  $\eta$  of the countable set  $\{\delta/n : n \geq 1\}$  (possible since  $\kappa > \aleph_0$ ); every  $a \in E = (0, \eta)$  satisfies  $na < \delta$ . The set  $E$  has at least  $\kappa$  elements: otherwise it would have a positive strict lower bound  $b$ , but then  $b \in E$  (as  $b < \eta/2 \in E$ ), which is absurd. For distinct  $a, b \in E$ , Lemma 4.7 gives  $s \in (X^a - X^b)R_{\Gamma, < \kappa}$  with a quotient of support size at most  $\max(|\text{supp } s|, \aleph_0) < \kappa$ . Since  $|S| < |E|$ , two of the monomials collide and  $s$  is killed.  $\square$

This theorem concerns an ordinary set-sized source and cannot exclude its identity representation; the target bound is indispensable. Regularity of  $\kappa$  is not needed for the theorem itself, only for the examples of Section 11.5 (11).

**Lemma 10.9** (Cyclic modules [08]). *Let  $A$  be a commutative ring,  $I$  an ideal and  $\kappa$  a cardinal such that every ring homomorphism from  $A$  onto a ring of cardinality  $< \kappa$  kills  $I$ . Then  $IM = 0$  for every  $A$ -module  $M$  with  $|M| < \kappa$ .*

*Proof.* For  $m \in M$ ,  $A/\text{Ann}(m) \rightarrow Am$ ,  $a \mapsto am$ , is a bijection, so the quotient ring  $A/\text{Ann}(m)$  has fewer than  $\kappa$  elements, and  $I \subseteq \text{Ann}(m)$ .  $\square$

Applying the ring statement to  $\text{End}_{\mathbb{Z}}(M)$  would need an estimate of  $|\text{End}_{\mathbb{Z}}(M)|$ , which may exceed  $\kappa$  even when  $|M| < \kappa$ ; the lemma avoids that hidden cardinal-arithmetic assumption (08). [Proposition 10.3\(ii\)](#) gives the module bound directly (06, 03, 09).

## 10.4 The organizing theorem: the threshold is the size of the ring

The five constructions of [Section 11](#) are summarized here; their notation is tagged by source because they are different rings.

| ring                        | exponent group                                                | supports              | coefficients                            | hypotheses                     | size                | gen. of $\Pi_A$ |
|-----------------------------|---------------------------------------------------------------|-----------------------|-----------------------------------------|--------------------------------|---------------------|-----------------|
| $A_{\kappa}^{03}$           | $\bigoplus_{\kappa^{\text{op}} + \kappa} \mathbb{Q}$ , 2 arms | countable             | $\mathbb{R}$                            | cf $\kappa > \aleph_0$         | $\kappa^{\aleph_0}$ | cf $\kappa$     |
| $A_{\kappa}^{06}$           | $\text{span}_{\mathbb{Q}}\{\omega^{\pm(\alpha+1)}\}$ , 2 arms | $< \kappa$            | $\mathbb{R}$                            | $\kappa$ reg. unctbl.          | $\kappa^{<\kappa}$  | $\kappa$        |
| $A_{\kappa}^{08}$           | $\bigoplus_{\xi < \kappa} \mathbb{Q}$ , 1 arm                 | countable, controlled | $\mathbb{R} \cap \overline{\mathbb{Q}}$ | $\kappa$ infinite              | $\kappa$            | cf $\kappa$     |
| $A_{\lambda, D}^{09}$       | $\bigoplus_{\omega_1 \times \lambda} \mathbb{Q}$ , 1 arm      | countable             | $k \in \{\mathbb{R}, \mathbb{C}\}$      | $\lambda^{\aleph_0} = \lambda$ | $\lambda$           | $\aleph_1$      |
| $R_{\Gamma, < \kappa}^{11}$ | $\bigoplus_{\alpha < \kappa} \mathbb{Q}$ , 1 arm              | $< \kappa$            | $ K  \leq \max(\kappa, 2^{\aleph_0})$   | $\kappa$ reg. unctbl.          | $\kappa^{<\kappa}$  | $\kappa$        |

All are ZFC constructions. The sizes are proved in [Section 11](#); for  $A_{\kappa}^{06}$  and  $R_{\Gamma, < \kappa}^{11}$  the size  $\kappa^{<\kappa}$  is a merge result (06 and 11 compute the size only under  $\kappa^{<\kappa} = \kappa$ ).

**Theorem 10.10** (Detection threshold equals size [03, 08, 09] [merge]). *Let  $A$  be any of the five rings of the table, with purely infinite ideal  $\Pi_A$ , and let  $0 \neq x \in \Pi_A$ . Then*

$$\min\{|S| : \varphi : A \rightarrow S, \varphi(x) \neq 0\} = \min\{|M| : M \text{ an } A\text{-module, } xM \neq 0\} = |A|.$$

*Every homomorphism from  $A$  to a ring of cardinality  $< |A|$  factors uniquely through ct, every  $A$ -module of cardinality  $< |A|$  is annihilated by  $\Pi_A$ , and every quotient  $A/J$  either is  $D/\text{ct}(J)$  with  $\Pi_A \subseteq J$  or has cardinality exactly  $|A|$ . The same holds for the Gaussian versions of  $A_{\kappa}^{03}$ ,  $A_{\kappa}^{06}$  and  $A_{\lambda, \mathbb{Z}[i]}^{09}$ .*

*Proof.* The identity map and the regular module detect  $x$  and have cardinality  $|A|$ . For the lower bound, each construction has a divisible exponent group and the gap property:  $x$  has a positive strict lower bound  $g$  of its support ([Lemmas 11.3, 11.7, 11.15 and 11.17](#) and, for  $R_{\Gamma, < \kappa}^{11}$ ,  $\text{coin}(\Gamma_{>0}) = \kappa$  with [Proposition 11.22](#)). With  $c = g/2$ , [Proposition 10.3\(iii\)](#) shows that every ring or module detecting  $x$  has at least  $|L_c(F)|$  elements, and in each construction  $|L_c(F)| = |A|$  for every  $c > 0$  ([Lemmas 11.4, 11.10 and 11.17](#) and [Proposition 11.23](#); for  $A_{\kappa}^{08}$ ,  $L_c(F) \subseteq F$  has at most  $\kappa$  elements and contains the  $\kappa$  monomials  $X^{e_{\xi}}$  with  $\xi$  beyond the least index of  $c$ , by [Lemma 11.13](#)). A homomorphism into a smaller ring therefore kills  $\Pi_A$  and factors through ct, and a smaller module is annihilated. If  $\Pi_A \not\subseteq J$  the quotient map detects an element of  $\Pi_A$ , so  $|A/J| \geq |A|$ ; the reverse inequality is trivial.  $\square$

For  $A_{\kappa}^{03}$ ,  $A_{\kappa}^{08}$  and  $A_{\lambda, D}^{09}$  the theorem is proved in the sources (03 with the field bound; 08 and 09 by counting monomials, which suffices because there the number of small monomials already equals  $|A|$ ). For  $A_{\kappa}^{06}$  06 proves the lower bound  $\kappa$  and exactness under  $\kappa^{<\kappa} = \kappa$ , and for  $R_{\Gamma, < \kappa}^{11}$  11 proves sharpness under  $\kappa^{<\kappa} = \kappa$  and  $|K| \leq \kappa$ ; the unconditional exact value  $\kappa^{<\kappa}$  is new here ([Theorem 11.11](#) and [Proposition 11.23](#)). The quotient alternative is 08's and 09's for their rings and a one-line consequence for the others.

*Remark 10.11* (Reconciling the thresholds [merge]).

- *Realized cardinals.* Since  $(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$ , and conversely a cardinal  $\theta$  with  $\theta^{\aleph_0} = \theta$  has  $\text{cf } \theta > \aleph_0$  by König’s theorem (so  $\kappa = \theta$  is allowed in 03), the thresholds realized by 03 are exactly the cardinals  $\theta$  with  $\theta^{\aleph_0} = \theta$ ; these are 09’s cardinals  $\lambda$ . 06 and 11 realize  $\kappa^{<\kappa}$  for regular uncountable  $\kappa$ . 08 realizes every infinite cardinal, and is the only construction reaching cardinals below  $2^{\aleph_0}$  or with  $\kappa^{\aleph_0} > \kappa$ : if  $rX^c$  lies in the ring for every real  $r$ , every target detecting  $X^{2^c}$  has at least  $2^{\aleph_0}$  elements by Lemma 4.1 with  $L = \mathbb{R}$ , and 08 escapes this with real algebraic coefficients and a countable-stage closure.
- *The same ring, different bounds.*  $A_{\aleph_1}^{06}$  and  $A_{\aleph_1}^{03}$  are isomorphic: both are the countable-support real Hahn integer parts over the lexicographic sum  $\bigoplus_J \mathbb{Q}$  with  $J$  of order type  $\omega_1^{\text{op}} + \omega_1$  (06’s generators satisfy  $\epsilon_\beta \ll \epsilon_\alpha \ll h_\alpha \ll h_\beta$  for  $\alpha < \beta$ ). 06 proves the threshold is at least  $\aleph_1$  and exactly  $\aleph_1$  under CH; 03 proves it is exactly  $2^{\aleph_0}$  in ZFC; 06’s own example already notes that without CH its ring need not have size  $\aleph_1$ . These agree.
- *Generators.* The generator counts are  $\text{cf } \kappa$  (03, 08),  $\kappa$  (06, and for 11’s ring by the argument of (H3) in Section 12) and  $\aleph_1$  (09). In each case the proof exhibits a coinital chain of monomials in the positive cone of the exponent group of that length; the counts differ because the constructions differ, and no general formula is asserted for other exponent groups or support restrictions.
- *Fraction fields.*  $\text{Frac } A = F$  is proved for 03 and 06, which use a second (reversed) arm to clear negative exponents; 08 explicitly does not claim it for its one-armed controlled field, and 09 and 11 do not address it.
- *Hypotheses.* Regularity (06) and  $\text{cf } \kappa > \aleph_0$  (03) are used for simultaneous gaps below fewer than  $\kappa$  (respectively countably many) full supports; 08 avoids both through its finite-coordinate invariant, which confines each element to finitely many coordinates.

## 11 Set-sized integer parts with exact thresholds

Five constructions replace the proper-class supply of scales by a cardinal. They are different rings and are tagged by source. Each is an explicit subring of  $\mathbf{Oz}$  (or of  $\mathbf{Oz}[i]$ ), and each is a ZFC construction of sets. For every one of them Theorem 10.10 gives the exact detection threshold; this section supplies the constructions and the size computations it uses. Manuscript 05 (sibling report [30]) asked which set-sized exponent groups admit analogues of the common-divisor and set-target theorems (its “exact size boundary” question); 03, 06, 08, 09 and 11 each answer it with a sharp family, and none claims a classification of all exponent groups or support restrictions.

### 11.1 Countable supports over a two-armed group [03]

For an ordered abelian group  $G$  and  $k \in \{\mathbb{R}, \mathbb{C}\}$  let  $H_\omega(k, G)$  be the Hahn series over  $k$  with *at most countable* reverse well-ordered support (any countable reverse order type is allowed; the subscript means countable support, not order type  $\omega$ ). It is the  $\aleph_1$ -bounded Hahn field in the terminology of [16].

**Lemma 11.1** (Countable-support Hahn fields [03]).  *$H_\omega(k, G)$  is a field. If  $G$  is divisible,  $H_\omega(\mathbb{R}, G)$  is real closed,  $H_\omega(\mathbb{C}, G)$  is algebraically closed, and  $H_\omega(\mathbb{C}, G) = H_\omega(\mathbb{R}, G)[i]$ .*

*Proof.* Finitely many series lie in the full Hahn field over the countable subgroup  $H$  generated by their supports, all of whose supports are countable; so sums, products and inverses stay in  $H_\omega(k, G)$ .

For closedness, place the coefficient supports of a polynomial in their divisible hull, still countable; the full Hahn field over a divisible group with real closed (algebraically closed) coefficients is real closed (algebraically closed) [16, Fact 2.1.1], and it lies inside  $H_\omega(k, G)$ . Real and imaginary parts split a countable support into two subsets.  $\square$

With  $A(D, k, G) = D + \{f \in H_\omega(k, G) : \text{supp } f \subseteq G_{>0}\}$ , the constant term is a split surjection, the units are  $D^\times$  (Lemma 2.4), and for  $(D, k) = (\mathbb{Z}, \mathbb{R})$  the ring is a discretely ordered integer part of  $H_\omega(\mathbb{R}, G)$ : write  $x = P + r + \epsilon$  by positive, zero and negative exponents and take  $P + \lfloor r \rfloor$ , except  $P + r - 1$  when  $r \in \mathbb{Z}$  and  $\epsilon < 0$  (discarding all negative exponents would send  $-X^{-g}$  to 0, which is not its floor). The same floor rule gives the omnific integer part of **No** (03, 06, 08; compare [28]).

**The construction.** Let  $\kappa$  be an infinite cardinal with  $\mu = \text{cf } \kappa > \aleph_0$ ,  $\theta = \kappa^{\aleph_0}$ ,  $J_\kappa = \kappa^{\text{op}} + \kappa$  with indices  $p_\alpha$  (reversed arm) and  $q_\alpha$  (forward arm), and

$$G_\kappa = \bigoplus_{j \in J_\kappa} \mathbb{Q}e_j,$$

ordered lexicographically with the *least* index of a nonzero vector deciding its sign; earlier indices are larger scales ( $ne_{j'} < e_j$  for  $j < j'$ ).  $\iota(e_{p_\alpha}) = \omega^\alpha$ ,  $\iota(e_{q_\alpha}) = \omega^{-(\alpha+1)}$  extends to an ordered additive embedding  $G_\kappa \hookrightarrow \mathbf{No}$ , and  $\sum a_g X^g \mapsto \sum a_g \omega^{\iota(g)}$  is an ordered field embedding. Put

$$F_\kappa^{03} = H_\omega(\mathbb{R}, G_\kappa), \quad A_\kappa^{03} = A(\mathbb{Z}, \mathbb{R}, G_\kappa) = F_\kappa^{03} \cap \mathbf{Oz}, \quad \Pi_\kappa^{03} = \ker(\text{ct} \upharpoonright_{A_\kappa^{03}}),$$

and use the superscript  $\mathbb{C}$  for the parallel construction over  $\mathbb{C}$  and  $\mathbb{Z}[i]$ :  $F_\kappa^{03, \mathbb{C}} = F_\kappa^{03}[i]$ ,  $A_\kappa^{03, \mathbb{C}} = A_\kappa^{03}[i]$ ,  $\Pi_\kappa^{03, \mathbb{C}} = \Pi_\kappa^{03} + i\Pi_\kappa^{03}$ .

**Lemma 11.2** (Room on both sides [03]).

- (i) Every family of fewer than  $\mu$  elements of  $G_\kappa$  is bounded above and below in  $G_\kappa$ .
- (ii) Every family of fewer than  $\mu$  positive elements has a positive strict lower bound.
- (iii) For every  $h > 0$ , a final segment of the forward arm, of cardinality  $\kappa$ , consists of basis vectors  $e_{q_\alpha} \ll h$ .

*Proof.* Fewer than  $\mu$  vectors have fewer than  $\mu$  support indices, bounded below  $\kappa$  in each arm. If  $\beta$  exceeds all reversed-arm ordinals occurring,  $p_\beta$  precedes all occurring indices, so  $e_{p_\beta}$  exceeds the absolute value of each vector; this gives (i). If  $\gamma$  exceeds all forward-arm ordinals occurring,  $q_\gamma$  follows every occurring index, and each positive vector, having a positive leading coefficient at an earlier index, exceeds  $e_{q_\gamma} > 0$ ; this gives (ii), and (iii) is the case of one vector.  $\square$

The reversed arm supplies a denominator large enough to shift a countable support upward; the forward arm supplies a positive divisor below a countable positive support and many indistinguishable smaller scales (03).

**Lemma 11.3** (Small families have common divisors [03]). *Every family of fewer than  $\mu$  elements of  $\Pi_\kappa^{03}$  (or  $\Pi_\kappa^{03, \mathbb{C}}$ ) has a common divisor  $X^g$ ,  $g > 0$ , with all quotients in the ideal.*

*Proof.* The union of fewer than  $\mu$  countable supports has fewer than  $\mu$  elements,  $\mu$  being regular and uncountable; apply Lemma 11.2(ii) and translate by  $-g$ .  $\square$



**Lemma 11.4** (Every smaller-scale field has size  $\theta$  [03]). *For either coefficient pair, the field, the ring, its purely infinite ideal and every  $L_h = H_\omega(k, H_h(G_\kappa))$  ( $h > 0$ ) have cardinality  $\theta = \kappa^{\aleph_0}$ .*

*Proof.* There are at most  $\kappa^{\aleph_0}$  countable supports and at most  $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0} \leq \theta$  coefficient choices for each. Conversely, for a forward tail  $T \subseteq \kappa$  with  $e_{q_\alpha} \ll h$  ( $\alpha \in T$ ), each countable  $S \subseteq T$  gives  $f_S = \sum_{\alpha \in S} X^{e_{q_\alpha}} \in L_h$  (increasing ordinals give decreasing exponents), and  $L_h \supseteq k$ ; since  $\kappa^{\aleph_0} = \max(2^{\aleph_0}, |\kappa|^{\leq \aleph_0})$  (Lemma 11.6),  $|L_h| \geq \theta$ . Multiplication by  $X^{h/2}$  embeds  $L_h$  in the ideal.  $\square$

**Theorem 11.5** (Cardinal realization [03]). *For  $A = A_\kappa^{03}$  (or  $A_\kappa^{03, \mathbb{C}}$ ) with ideal  $\Pi_A$ :*

- (i)  $|F_\kappa^{03}| = |A| = \theta$  and  $\text{Frac}(A) = F_\kappa^{03}$ ;
- (ii) every unital map  $A \rightarrow R$  with  $|R| < \theta$  factors uniquely through  $\text{ct}$  (for the Gaussian ring the factorization through the Gaussian constant term is unique for a given map; different square roots of  $-1$  give different maps);
- (iii) every  $A$ -module of cardinality  $< \theta$  is annihilated by  $\Pi_A$ , and for every  $0 \neq f \in \Pi_A$  the least cardinality of a ring target detecting  $f$ , and of a module on which  $f$  acts nontrivially, is exactly  $\theta$ ;
- (iv)  $\Pi_A$  needs exactly  $\mu$  generators; it is flat with  $\Pi_A^2 = \Pi_A$ , the module  $D = A/\Pi_A$  is not flat, and  $\text{Tor}_n^A(D, N) = 0$  for every  $D$ -module  $N$  and  $n > 0$ .

*No inaccessible cardinal, CH or other additional axiom is used.*

*Proof.* (i) Sizes by Lemma 11.4. For  $x \in F_\kappa^{03}$ , Lemma 11.2(i) gives  $h \in G_\kappa$  larger than  $-g$  for every exponent  $g$  of  $x$ , so  $x = X^h x / X^h$  with numerator and denominator in  $A$ . (ii) and (iii) are Theorem 10.10: by Lemma 11.3 each  $f \in \Pi_A$  is  $X^g b$  with  $b \in \Pi_A$ , and Proposition 10.3 with Lemma 11.4 gives the lower bound  $\theta$  (03's own proof is exactly this). (iv) Generators: choose a cofinal sequence  $(\alpha_\xi)_{\xi < \mu}$  in  $\kappa$ ; the countably many index supports of the exponents of  $f \in \Pi_A$  are bounded on the forward arm, so  $X^{e_{q_{\alpha_\xi}}}$  divides  $f$  in  $A$  for large  $\xi$ , and these  $\mu$  monomials generate; a family of fewer than  $\mu$  elements lies in some  $X^h A$  by Lemma 11.3, which misses  $X^{h/2}$ . Flatness, idempotence and Tor are Theorem 12.2.  $\square$

For  $\kappa = \aleph_1$ :  $2^{\aleph_0} \leq \aleph_1^{\aleph_0} \leq (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ , so  $A_{\aleph_1}^{03}$  has the size of the continuum, every target of size below the continuum sees only the constant term, whether or not CH holds, and the ideal needs  $\aleph_1$  generators. For every infinite  $\rho$ ,  $\kappa = 2^\rho$  gives  $\theta = 2^\rho$  with  $\text{cf}(2^\rho) > \rho$ ; so arbitrarily large thresholds are realized in ZFC. This is a realization theorem for this family, not a classification of ordered exponent groups. The case  $\text{cf } \kappa = \aleph_0$  is excluded because countable supports can then run cofinally through an arm (03).

**Lemma 11.6** (Cardinal facts [03]). *For every infinite  $\kappa$ ,  $\kappa^{\aleph_0} = \max(2^{\aleph_0}, |\kappa|^{\leq \aleph_0})$ ; for every infinite  $\rho$ ,  $(2^\rho)^{\aleph_0} = 2^\rho$  and  $\text{cf}(2^\rho) > \rho$ .*

*Proof.* A countable sequence is determined by its countable range and a sequence in that range; conversely binary sequences embed in  $\kappa^{\aleph}$  and each countable subset is the range of an enumeration (choice).  $(2^\rho)^{\aleph_0} = 2^{\rho \aleph_0} = 2^\rho$ . If  $2^\rho$  were a union of  $\tau \leq \rho$  families  $E_\xi$  of binary functions on  $\rho$ , each of size  $< 2^\rho$ , split  $\rho$  into  $\tau$  blocks of size  $\rho$  and on block  $\xi$  choose a pattern that is not the restriction of any member of  $E_\xi$ ; the assembled function lies in no  $E_\xi$  (König).  $\square$

## 11.2 Supports of size $< \kappa$ over a two-sided group [06]

Fix an uncountable regular cardinal  $\kappa$ . In the additive group of **No** put

$$h_\alpha = \omega^{\alpha+1}, \quad \epsilon_\alpha = \omega^{-(\alpha+1)} \quad (\alpha < \kappa), \quad \Gamma_\kappa^{06} = \text{span}_{\mathbb{Q}}\{h_\alpha, \epsilon_\alpha : \alpha < \kappa\}$$

(finite rational combinations of distinct Conway monomials, hence linearly independent). This is a divisible ordered group of cardinality  $\kappa$ , and  $\epsilon_\beta \ll \epsilon_\alpha \ll h_\alpha \ll h_\beta$  for  $\alpha < \beta$ ; here  $\ll$  refers to the outer exponents.

**Lemma 11.7** (Simultaneous upper and lower scales [06]). *For every  $T \subseteq \Gamma_\kappa^{06}$  with  $|T| < \kappa$  there is  $\beta < \kappa$  with  $\epsilon_\beta \ll |t| \ll h_\beta$  for all  $t \in T \setminus \{0\}$ .*

*Proof.* The generators occurring in the finite expressions of the members of  $T$  have fewer than  $\kappa$  indices, bounded below some  $\beta$  by regularity. The leading generator of a nonzero expression determines its Archimedean class, which lies strictly between those of  $\epsilon_\beta$  and  $h_\beta$ .  $\square$

For  $k \in \{\mathbb{R}, \mathbb{C}\}$  put  $K_{\kappa, k}^{06} = \{f \in k((X^{\Gamma_\kappa^{06}})) : |\text{supp } f| < \kappa\}$ ,  $K_\kappa^{06} = K_{\kappa, \mathbb{R}}^{06}$ ,  $\Pi_\kappa^{06} = \{f \in K_\kappa^{06} : \text{supp } f > 0\}$  and  $A_\kappa^{06} = \mathbb{Z} + \Pi_\kappa^{06}$  (06's  $R_\kappa$ );  $X^g \mapsto \omega^g$  makes them subsets of **No** and **No**[ $i$ ].

**Proposition 11.8** (Field closure and integer part [06]).  *$K_{\kappa, \mathbb{R}}^{06}$  is real closed,  $K_{\kappa, \mathbb{C}}^{06} = K_{\kappa, \mathbb{R}}^{06}[i]$  is algebraically closed,  $A_\kappa^{06} = K_\kappa^{06} \cap \mathbf{Oz}$ ,  $\text{Frac}(A_\kappa^{06}) = K_\kappa^{06}$ , and  $A_\kappa^{06}$  is a discretely ordered integer part of  $K_\kappa^{06}$ .*

*Proof.* Inverting a series of support size  $\mu < \kappa$  (factor the leading term and use the geometric series) gives support in a translate of the monoid generated by at most  $\max(\mu, \aleph_0) < \kappa$  elements; sums and products also keep the bound. For closedness, the coefficients of a polynomial use fewer than  $\kappa$  exponents; the rational span  $\Delta$  of these has  $|\Delta| < \kappa$ , the full field  $\mathbb{C}((X^\Delta))$  is algebraically closed [18, Corollary 4] and lies in  $K_{\kappa, \mathbb{C}}^{06}$ ; the real field with its leading-term order has this algebraically closed complexification, so it is real closed. For  $f \in K_\kappa^{06}$ , Lemma 11.7 gives  $h_\beta$  above  $|g|$  for every exponent  $g$ , so  $X^{h_\beta} f \in \Pi_\kappa^{06}$  and  $f$  is a quotient. The floor rule of Section 11.1 gives the integer part, and a nonzero element with positive leading exponent exceeds every real, so there is nothing strictly between 0 and 1.  $\square$

Field closure uses only that  $\kappa$  is uncountable; regularity is used for simultaneous gaps and generation (06). The first-kappa report proves the closedness of  $\{f \in k((t^\Gamma)) : |\text{supp } f| < \kappa\}$  for every uncountable  $\kappa$ , possibly singular (`fkcthm:main` in [29]); 06's closure argument is not a novelty claim, and singular support bounds may preserve field closure while failing the simultaneous gap property used here. The two families  $h_\alpha$  and  $\epsilon_\alpha$  are deliberate: the  $\epsilon_\alpha$  provide smaller positive gaps and the  $h_\alpha$  shift supports upward, and a one-sided family would not automatically give the fraction-field assertion (06).

**Proposition 11.9** (A chain of principal ideals [06]).  *$\Pi_\kappa^{06} = \bigcup_{\alpha < \kappa} X^{\epsilon_\alpha} A_\kappa^{06}$  is a strictly increasing union; the ideal is idempotent and its least number of generators is  $\kappa$ .*

*Proof.* For  $\alpha < \beta$ ,  $\epsilon_\alpha - \epsilon_\beta > 0$  gives the inclusion, strict because the reverse quotient has a negative exponent. A gap  $\epsilon_\alpha < \text{supp } f$  (Lemma 11.7) gives  $f \in X^{\epsilon_\alpha} \Pi_\kappa^{06}$ , hence the union and idempotence. Fewer than  $\kappa$  elements have fewer than  $\kappa$  support exponents in total (regularity), so they lie in some  $X^a A_\kappa^{06}$ , which misses  $X^{a/2}$ .  $\square$

**Lemma 11.10** (Size of the ring and of every smaller-scale field [merge]). *For every  $c > 0$  in  $\Gamma_\kappa^{06}$ , the field  $L_c = \{f \in K_\kappa^{06} : \text{supp } f \subseteq H_c(\Gamma_\kappa^{06})\}$  has cardinality at least  $\kappa^{<\kappa}$ , and*

$$|K_{\kappa, \mathbb{R}}^{06}| = |K_{\kappa, \mathbb{C}}^{06}| = |A_\kappa^{06}| = |\Pi_\kappa^{06}| = \kappa^{<\kappa}.$$

*Proof.* The finite expression of  $c$  uses indices below some  $\alpha_0 < \kappa$ , and its Archimedean class is that of its leading generator; so  $\epsilon_\alpha \ll c$  for every  $\alpha > \alpha_0$ . For  $B \subseteq (\alpha_0, \kappa)$  with  $|B| < \kappa$ , the series  $\sum_{\alpha \in B} X^{\epsilon_\alpha}$  has support  $\{\epsilon_\alpha : \alpha \in B\}$ , reverse well ordered because  $\epsilon_\alpha$  decreases as  $\alpha$  increases, of size  $|B| < \kappa$ , and inside  $H_c$ ; so it lies in  $L_c$ , which is a field by Proposition 10.3(i). Distinct  $B$  give distinct series, and the number of subsets of size  $< \kappa$  of a set of size  $\kappa$  is  $\sum_{\nu < \kappa} \kappa^\nu = \kappa^{<\kappa}$ . Hence  $|L_c| \geq \kappa^{<\kappa}$ , and  $X^c L_c \subseteq \Pi_\kappa^{06}$  gives the same lower bound for the ideal, ring and fields. Conversely a series is given by a support of size  $\nu < \kappa$  and a coefficient function: at most  $\kappa^\nu \cdot (2^{\aleph_0})^\nu$  choices, which is  $\leq \kappa^\nu$  for infinite  $\nu$  and  $\leq \kappa \cdot 2^{\aleph_0} \leq \kappa^{\aleph_0} \leq \kappa^{<\kappa}$  for finite  $\nu$  (as  $\kappa > \aleph_0$ ). Summing over the  $\leq \kappa$  values of  $\nu$  gives at most  $\kappa^{<\kappa}$ .  $\square$

**Theorem 11.11** (The exact threshold of  $A_\kappa^{06}$  [06] [merge]). *Let  $\kappa$  be uncountable and regular. Every homomorphism  $A_\kappa^{06} \rightarrow S$  with  $|S| < \kappa^{<\kappa}$  factors through  $\text{ct}$ , every  $A_\kappa^{06}$ -module  $M$  with  $|M| < \kappa^{<\kappa}$  satisfies  $\Pi_\kappa^{06} M = 0$ , and for each  $0 \neq x \in \Pi_\kappa^{06}$  the least cardinality of a ring target detecting  $x$ , and of a module on which  $x$  acts nontrivially, is exactly  $\kappa^{<\kappa}$ . The same holds for  $A_\kappa^{06}[i]$  with  $\mathbb{Z}[i]$ .*

*Proof.* Theorem 10.10 with Lemmas 11.7 and 11.10: each  $0 \neq x \in \Pi_\kappa^{06}$  has a gap  $\epsilon_\beta < \text{supp } x$ , and with  $c = \epsilon_\beta/2$  the field  $L_c$  has cardinality at least  $\kappa^{<\kappa} = |A_\kappa^{06}|$ .  $\square$

06 proves the lower bound  $\kappa$  (with the explicit telescope family of the  $\kappa$  monomials  $X^{a+\epsilon_\alpha}$ , each witness having countable support) and, under the extra hypothesis  $\kappa^{<\kappa} = \kappa$ , that  $|A_\kappa^{06}| = \kappa$  and the threshold is exactly  $\kappa$ ; that is the special case of the theorem where  $\kappa^{<\kappa} = \kappa$ . The lower bounds of 06 need no cardinal arithmetic, and neither does the exact value  $\kappa^{<\kappa}$ , which comes from 06's own collision lemma (compare 03's field bound). This answers the first half of 06's question "Cardinal arithmetic without the matching hypothesis": the least size of a target detecting a fixed nonzero  $x \in \Pi_\kappa^{06}$  is  $\kappa^{<\kappa}$  in ZFC. The second half ("a different integer-part construction of size exactly  $\kappa$  with the same lower-bound property under weaker hypotheses") is answered by 08 in ZFC for every infinite  $\kappa$  (Theorem 11.16), for a ring of size exactly  $\kappa$  whose fraction field, unlike  $\text{Frac } A_\kappa^{06} = K_\kappa^{06}$ , is not claimed to be its ambient field.

**Example 11.12** (The first uncountable cardinal [06]). Under CH,  $\aleph_1^{<\aleph_1} = \aleph_1^{\aleph_0} = 2^{\aleph_0} = \aleph_1$ , so  $A_{\aleph_1}^{06}$  has size  $\aleph_1$ : every action on a countable abelian group is an ordinary integer action, while the regular module of size  $\aleph_1$  sees all purely infinite elements. Without CH the lower bound at  $\aleph_1$  holds, the ring has size  $\aleph_1^{\aleph_0} = 2^{\aleph_0}$ , and by Theorem 11.11 the exact threshold is  $2^{\aleph_0}$ ; this ring is isomorphic to  $A_{\aleph_1}^{03}$  (Remark 10.11).

### 11.3 A controlled field of size exactly $\kappa$ [08]

Let  $\kappa$  be any infinite cardinal and  $\Gamma_\kappa^{08} = \bigoplus_{\xi < \kappa} \mathbb{Q}e_\xi$ , ordered lexicographically with the least nonzero index deciding the sign, so  $0 < ne_\eta < e_\xi$  for  $\xi < \eta$ ; it is divisible of cardinality  $\kappa$ , and each nonzero element has a finite coordinate support.

**Lemma 11.13** (Scale and support bounds [08]). *For every  $c > 0$  there are  $\kappa$  positive  $a \ll c$  (namely  $e_\xi$  for  $\xi$  beyond the least nonzero index of  $c$ ). If  $S \subseteq (\Gamma_\kappa^{08})_{>0}$  has fewer than  $\text{cf } \kappa$  elements, some  $\beta < \kappa$  has  $e_\beta \ll s$  for all  $s \in S$ .*

*Proof.* A union of fewer than  $\text{cf } \kappa$  finite subsets of  $\kappa$  is bounded; take  $\beta$  above it.  $\square$

Let  $K = \mathbb{R} \cap \overline{\mathbb{Q}}$ ,  $H = K((X^{\Gamma_\kappa^{08}}))$  and  $H_{\text{ctbl}} \subseteq H$  the series with at most countable support.  $H_{\text{ctbl}}$  is a real closed subfield: finitely many series lie in  $K((X^\Delta))$  for the countable divisible subgroup  $\Delta$  generated by their supports, which is real closed [16, Fact 2.1.1] and has only countable supports. A *truncation* of a series is its restriction to an initial segment of its support in decreasing order; a countable support has countably many. Let  $F_0$  be the real closure in  $H_{\text{ctbl}}$  of the field generated by  $K$  and all  $X^g$ ; given  $F_n$ , adjoin all truncations of its elements and let  $F_{n+1}$  be the real closure in  $H_{\text{ctbl}}$  of the resulting field; put  $F = \bigcup_n F_n$ .

**Lemma 11.14** (The controlled field [08]).  *$F$  is real closed and truncation-closed, has cardinality  $\kappa$ , consists of series with countable support, and contains  $K$  and every monomial.*

*Proof.* An algebraic extension of an infinite field of size  $\kappa$  has size  $\leq \kappa$  ( $\kappa$  polynomials, finitely many roots each), and adjoining countably many truncations of each of  $\kappa$  elements adds at most  $\kappa$  elements; so every  $F_n$ , and  $F$ , has size  $\kappa$ , and at least  $\kappa$  because of the monomials. Truncations of an element of  $F_n$  lie in  $F_{n+1}$ , and finitely many coefficients lie in a common real closed stage.  $\square$

**Lemma 11.15** (Finite-coordinate invariant [08]). *Every  $f \in F$  has  $\text{supp } f \subseteq \Gamma_E = \bigoplus_{\xi \in E} \mathbb{Q}e_\xi$  for some finite  $E \subseteq \kappa$  (its support need not be finite).*

*Proof.* For finite  $E$ ,  $H_E = K((X^{\Gamma_E}))$  is real closed and lies in  $H_{\text{ctbl}}$  ( $\Gamma_E$  is countable); the  $H_E$  form a directed family. An element of the field generated by the monomials is a rational expression in finitely many of them, so lies in some  $H_E$ . An element of  $F_0$  is algebraic over that field, hence over some  $H_E$  (finitely many coefficients); as  $H_E$  is real closed and  $H_{\text{ctbl}}$  is an ordered field containing it, it has no proper ordered algebraic extension there, so the element lies in  $H_E$ . Truncations of elements of  $H_E$  stay in  $H_E$ , finite rational expressions stay in  $H_{E \cup E'}$ , and real closure is handled by the same algebraicity argument; induct over  $n$ .  $\square$

This invariant removes regularity: even at a singular  $\kappa$  of countable cofinality, the coordinates in one element are confined to a finite set. The proof uses countable supports to control the number of truncations and finite coordinate sets to find smaller positive scales (08).

The map  $\iota(\sum q_\xi e_\xi) = \sum q_\xi \omega^{-\xi}$  is an increasing additive embedding of  $\Gamma_\kappa^{08}$  into  $\mathbf{No}$  (the least nonzero coordinate decides the sign on both sides), and  $\Phi(\sum a_g X^g) = \sum a_g \omega^{\iota(g)}$  is an ordered field embedding  $H \hookrightarrow \mathbf{No}$  preserving truncations and constant terms. Put  $F_\kappa^{08} = \Phi(F)$ ,  $A_\kappa^{08} = F_\kappa^{08} \cap \mathbf{Oz} = \{f \in F : \text{supp } f \geq 0, \text{ ct}(f) \in \mathbb{Z}\}$  and  $\Pi_\kappa^{08} = \ker(\text{ct}|_{A_\kappa^{08}})$ . Since  $\iota(e_0) = 1$ , the monomial  $X^{e_0}$  is the actual surreal  $\omega$ .  $A_\kappa^{08}$  is a discretely ordered integer part of  $F$  of cardinality  $\kappa$ : truncate  $x$  to its positive part  $P \in F$  and apply the floor rule of Section 11.1 to  $x = P + r + \epsilon$  with  $r \in K$ .

**Theorem 11.16** (Sharp models at every infinite cardinal [08]). *For every infinite cardinal  $\kappa$ :*

- (i)  $F_\kappa^{08}$  is a truncation-closed real closed subfield of  $\mathbf{No}$  with countable supports,  $|F_\kappa^{08}| = |A_\kappa^{08}| = \kappa$ , and  $A_\kappa^{08}$  is an integer part of it;
- (ii) every homomorphism from  $A_\kappa^{08}$  to a ring of cardinality  $< \kappa$  factors through  $\text{ct}$ ;
- (iii) every quotient ring of  $A_\kappa^{08}$  is a quotient  $\mathbb{Z}/n\mathbb{Z}$  ( $n \geq 0$ ) under  $\text{ct}$  or has cardinality exactly  $\kappa$ ;
- (iv)  $\Pi_\kappa^{08}$  needs exactly  $\text{cf } \kappa$  generators;
- (v) every unital  $A_\kappa^{08}$ -module of cardinality  $< \kappa$  has action  $f \cdot m = \text{ct}(f)m$ ;

(vi) the positive-characteristic maximal residue fields are the  $\mathbb{F}_p$ , with maximal ideals  $pA_\kappa^{08}$ ; characteristic-zero maximal residue fields exist and all have cardinality exactly  $\kappa$ .

No hypothesis such as CH, GCH,  $\kappa^{\aleph_0} = \kappa$  or an inaccessible cardinal is used.

*Proof.* (i) is Lemma 11.14 and the paragraph above. Each  $f \in \Pi_\kappa^{08}$  has support in some  $\Gamma_E$  (Lemma 11.15), so  $e_\beta$ ,  $\beta > \max E$ , is a positive strict lower bound; and Lemma 11.13 gives  $\kappa$  scales below every  $c$ . These are the hypotheses of Theorem 10.7, which gives (ii). (iii) If  $|A/J| < \kappa$ , (ii) applied to  $A \rightarrow A/J$  gives  $\Pi \subseteq J$ ; otherwise  $|A/J| \leq |A| = \kappa$ . (iv) With a cofinal  $C \subseteq \kappa$  of size  $\text{cf } \kappa$ , for  $f \in \Pi_\kappa^{08}$  choose  $\beta \in C$  above its coordinate set; then  $X^{-e_\beta} f \in \Pi_\kappa^{08}$ , so  $\{X^{e_\beta} : \beta \in C\}$  generates. Fewer than  $\text{cf } \kappa$  elements have coordinate sets bounded by some  $\beta$ , so they lie in  $(X^{e_\beta})$ , which misses  $X^{e_\beta/2}$ . (v) is Lemma 10.9 with (ii). (vi) Coefficient division gives  $n\Pi_\kappa^{08} = \Pi_\kappa^{08}$ , so  $nA = n\mathbb{Z} + \Pi_\kappa^{08}$  and  $A/pA \cong \mathbb{F}_p$  is already a field; a maximal ideal of characteristic  $p$  contains  $pA$  and equals it. The element  $u = 1 + X^{e_0}$  is a nonunit (Lemma 2.4), so the set-sized ring  $A$  has a maximal ideal  $M \ni u$  (ordinary maximal-ideal theorem); if a nonzero integer  $n$  lay in  $M$ , then  $X^{e_0} \in n\Pi_\kappa^{08} \subseteq M$  and  $1 \in M$ . A characteristic-zero residue field of size  $< \kappa$  would receive a map through  $\mathbb{Z}$  by (ii), with image  $\mathbb{Z}$ , not a field; and it has size  $\leq \kappa$  as a quotient.  $\square$

At  $\kappa = \aleph_0$  the theorem restricts only finite targets, and the characteristic-zero residue fields are countable and detect infinite monomials. For regular  $\kappa$  the generator count is  $\kappa$ ; at a singular  $\kappa$  it is smaller, so a countably generated ideal can be invisible to all ring targets below a much larger singular  $\kappa$  of countable cofinality. Two resources are separated: monomial generators coinital in the support scales ( $\text{cf } \kappa$ ) and pairwise collision witnesses below a fixed scale ( $\kappa$ ) (08).

In the residue field of a maximal ideal containing  $1 + \omega$  the image of  $\omega$  is  $-1$ , not  $0$ , and that field has size  $\kappa$ ; so for every infinite  $\kappa$  the subring  $A_\kappa^{08} \subseteq \mathbf{Oz}$  has a characteristic-zero field quotient of cardinality  $\kappa$  in which  $\omega \mapsto -1$ . This quotient map cannot extend to  $\mathbf{Oz}$ , where every set-sized image sends  $\omega$  to  $0$ : an explicit extension obstruction between a set-sized integer part and the class ring (08). 08 does not identify  $\text{Frac}(A_\kappa^{08})$  with  $F_\kappa^{08}$ , imposes no initial-segment condition or birthday cutoff on the embedding, and designs the construction for truncation, support control and cardinality, not for closure under a derivation or an exponential; the isomorphism types of the residue fields are not classified.

## 11.4 Countable supports over a long one-armed group [09]

Fix a cardinal  $\lambda$  with  $\lambda^{\aleph_0} = \lambda$ ; then automatically  $\lambda \geq 2^{\aleph_0}$ , since  $\lambda^{\aleph_0} \geq 2^{\aleph_0}$  (09 states  $\lambda \geq 2^{\aleph_0}$  as a separate hypothesis; it is redundant). For example  $\lambda = 2^\nu$  for any infinite  $\nu$ . Neither regularity nor CH is assumed. Let  $L_\lambda = \omega_1 \times \lambda$  with the lexicographic order and  $\Gamma_\lambda^{09} = \bigoplus_{\ell \in L_\lambda} \mathbb{Q}e_\ell$ , the least nonzero index deciding the sign.

**Lemma 11.17** (A large small-exponent tail [09]). *For every countable  $S \subseteq (\Gamma_\lambda^{09})_{>0}$  there is  $\alpha_0 < \omega_1$  with  $e_{(\alpha,\beta)} \in U_\Gamma(S)$  for all  $\alpha_0 \leq \alpha < \omega_1$  and  $\beta < \lambda$ ; in particular  $U_\Gamma(S)$  contains  $\lambda$  elements, and every countable positive set has a positive strict lower bound.*

*Proof.* The countable union of the finite index supports has first coordinates bounded below some  $\alpha_0 < \omega_1$ ; each  $s \in S$  has its least index before every  $(\alpha, \beta)$  with  $\alpha \geq \alpha_0$ , with positive coefficient there.  $\square$

For  $k \in \{\mathbb{R}, \mathbb{C}\}$  and unital  $D \subseteq k$  let  $H_{\lambda,k}$  be the countable-support Hahn field over  $\Gamma_\lambda^{09}$ ,  $\Pi_{\lambda,k}^{09}$  its positive-support part and  $A_{\lambda,D}^{09} = D + \Pi_{\lambda,k}^{09}$ . It is a domain with units  $D^\times$ , the quotients of

**Lemma 4.7** of its elements stay in it (countable supports), and  $|A_{\lambda,D}^{09}| = \lambda$ : there are at most  $\lambda^{\aleph_0} = \lambda$  countable supports and  $|k|^{\aleph_0} = 2^{\aleph_0} \leq \lambda$  coefficient choices, and  $\lambda$  positive basis monomials. With  $j(\alpha, \beta) = \lambda \cdot \alpha + \beta$  (ordinal operations),  $\iota(\sum q_\ell e_\ell) = \sum q_\ell \omega^{-j(\ell)}$  is an ordered embedding and  $\sum c_\gamma X^\gamma \mapsto \sum c_\gamma \omega^{\iota(\gamma)}$  embeds  $H_{\lambda,\mathbb{R}}$  in  $\mathbf{No}$  and  $H_{\lambda,\mathbb{C}}$  in  $\mathbf{No}[i]$ , sending  $A_{\lambda,\mathbb{Z}}^{09}$  into  $\mathbf{Oz}$  and  $A_{\lambda,\mathbb{Z}[i]}^{09}$  into  $\mathbf{Oz}[i]$ ; here  $X^{e_\ell} \mapsto \omega^{\omega^{-j(\ell)}}$ , the group coordinates describing exponents.

**Theorem 11.18** (Cardinal barrier [09]). *Let  $A = A_{\lambda,D}^{09}$  and  $\Pi_A = \Pi_{\lambda,k}^{09}$ .*

- (i) *If  $\varphi : A \rightarrow B$  is a unital ring map with  $\varphi(\Pi_A) \neq 0$ , then  $|\varphi(A)| \geq \lambda$ .*
- (ii) *If  $M$  is an  $A$ -module with  $\Pi_A M \neq 0$ , then  $|M| \geq \lambda$ .*
- (iii) *All maps to rings of size  $< \lambda$ , and all modules of size  $< \lambda$ , factor through  $\text{ct}$ .*
- (iv) *For every ideal  $J$ , either  $\Pi_A \subseteq J$  and  $A/J \cong D/\text{ct}(J)$ , or  $|A/J| = \lambda$ .*

*In particular, in ZFC there is an explicit  $A \subseteq \mathbf{Oz}$  of cardinality  $2^{\aleph_0}$  (take  $\lambda = 2^{\aleph_0}$ ,  $D = \mathbb{Z}$ ) whose maps to countable rings factor through  $\text{ct}$ , whose quotients are finite, countably infinite, or of cardinality  $2^{\aleph_0}$  (the last for every quotient not induced from  $\mathbb{Z}$ ), and whose countable modules are abelian groups with constant-term action.*

*Proof.* For  $f \in \Pi_A$  detected by  $\varphi$  or by a vector,  $\text{supp } f$  is countable, Lemma 11.17 supplies  $\lambda$  small exponents, and Lemma 4.8 with Lemma 4.7 injects them into  $\varphi(A)$  (with the module structure given by  $\varphi$  and the vector 1) or into  $M$ . (iii) follows, and (iv) as in Theorem 10.10.  $\square$

When the continuum exceeds  $\aleph_1$  the last statement also excludes all intermediate quotient cardinalities; this is conditional only on that cardinal inequality (09).

**Lemma 11.19** (Recovering the coefficient field [09]). *Let  $\varphi : A \rightarrow F$  be a unital map to a field with  $\varphi(\Pi_A) \neq 0$ . There is a unique embedding  $\sigma : k \hookrightarrow F$  with  $\varphi(rg) = \sigma(r)\varphi(g)$  for  $r \in k$ ,  $g \in \Pi_A$ , and  $\sigma|_D = \varphi|_D$ ; in particular  $F$  has characteristic 0.*

*Proof.* Take  $f \in \Pi_A$  with  $\varphi(f) \neq 0$  and  $\sigma(r) = \varphi(rf)/\varphi(f)$  (note  $rf \in \Pi_A$  even if  $r \notin A$ ). It is additive and unital,  $(rf)(sf) = (rsf)f$  gives multiplicativity, and it is injective as a map of fields. The identity  $(rg)f = (rf)g$  gives the displayed formula and independence of  $f$ .  $\square$

**Theorem 11.20** (Residue fields at the threshold [09]). *Let  $A = A_{\lambda,D}^{09}$ .*

- (i) *For every  $0 \neq f \in \Pi_A$  there is a maximal ideal  $\mathfrak{m}$  with  $f \equiv 1 \pmod{\mathfrak{m}}$  and  $|A/\mathfrak{m}| = \lambda$ .*
- (ii) *Every maximal ideal not containing  $\Pi_A$  has a residue field of cardinality  $\lambda$  with a canonical copy of  $k$ .*
- (iii) *For  $D = \mathbb{Z}$ , the maximal ideals containing  $\Pi_A$  are the  $pA$ , with residue fields  $\mathbb{F}_p$ , and every other maximal ideal has a characteristic-zero residue field of cardinality  $\lambda$ .*
- (iv) *There are at least  $\lambda$  distinct maximal ideals not containing  $\Pi_A$ .*

*Consequently the least cardinality of a field (and of a module) detecting a given  $0 \neq f \in \Pi_A$  is exactly  $\lambda$ ; and for  $D = \mathbb{Z}$  the intersection of all finite-index ideals is  $\Pi_A \neq 0$  while  $\text{Jac}(A) = 0$ .*

*Proof.* (i)  $1 - f$  is a nonconstant nonunit; a maximal ideal of the set-sized ring  $A$  containing it sends  $f$  to 1, so the residue field has size  $\geq \lambda$  by Theorem 11.18 and  $\leq \lambda$  as a quotient. (ii) is Theorem 11.18 with Lemma 11.19; (iii) uses  $pA = p\mathbb{Z} + \Pi_A$ . (iv) For  $\ell < \ell'$  in  $L_\lambda$  put  $\gamma = e_\ell$ ,  $a = e_{\ell'}$ ; then  $na < \gamma$



for all  $n$ , and  $q = -\sum_{n \geq 1} X^{\gamma-na} \in \Pi_A$  satisfies  $(1 - X^a)q = X^\gamma$ , so  $1 = (1 - X^\gamma) + (1 - X^a)q$ : the  $\lambda$  proper ideals  $(1 - X^{e_\ell})$  are pairwise comaximal, and maximal ideals above them (set-sized choice) are distinct and do not contain  $\Pi_A$ . For the consequences, the residue field of (i) is a field target and a module on which  $f$  acts as the identity. Every finite target has size  $< \lambda$ , so the finite residual is  $\Pi_A$ ; if  $0 \neq x = n + f$ , a prime not dividing  $n \neq 0$ , or (i) when  $n = 0$ , gives a maximal ideal avoiding  $x$ .  $\square$

All of this concerns set-sized rings and is not an application of the maximal-ideal theorem to **Oz**. The distinction between finite ring probes and field probes is sharp: the extra information is carried by residue fields of the full workspace cardinality (09). The number of maximal ideals and the isomorphism types of the residue fields are not determined; distinct maximal ideals need not give nonisomorphic fields.  $\Pi_{\lambda,k}^{09} = \bigcup_{\alpha < \omega_1} X^{e_{(\alpha,0)}} A$  is an increasing chain, and the least number of generators of the ideal is exactly  $\aleph_1$  (the chain gives  $\aleph_1$  generators; a countable family lies in  $X^{2h} A$  for a small  $h$  by Lemma 11.17, which misses  $X^h$ ); flatness is in Theorem 12.2.

*Remark 11.21* (Why the support restriction is real [09]). In the unrestricted Hahn field over  $\Gamma_\lambda^{09}$  the series  $f_* = \sum_{\alpha < \omega_1} X^{e_{(\alpha,0)}}$  has reverse well-ordered but uncountable support with no positive lower bound in  $\Gamma_\lambda^{09}$  (a late enough  $e_{(\alpha,0)}$  lies below any given  $h > 0$ ); it is not divisible by any positive monomial within the nonnegative-support ring and is not in  $A_{\lambda,D}^{09}$ . No assertion is made about the quotient classification of that larger unrestricted ring.

## 11.5 Supports of size $< \kappa$ over a one-armed group [11]

Theorem 10.8 (11) applies to any divisible set group  $\Gamma$  with  $\text{coin}(\Gamma_{>0}) \geq \kappa$ . The coinitality condition says that every set of fewer than  $\kappa$  positive elements has a positive strict lower bound.

**Proposition 11.22** (Long lexicographic groups [11]). *For  $\kappa$  uncountable and regular and  $\Gamma = \bigoplus_{\alpha < \kappa} \mathbb{Q}e_\alpha$  with the least nonzero index dominant,  $|\Gamma| = \kappa$  and  $\text{coin}(\Gamma_{>0}) = \kappa$ . The group  $\Gamma_\kappa^{06}$  also has coinitality  $\kappa$ , so  $A_\kappa^{06} = R_{\Gamma_\kappa^{06}, < \kappa}^{11}$  with  $(D, K) = (\mathbb{Z}, \mathbb{R})$  is an instance of Theorem 10.8.*

*Proof.* The  $e_\beta$  with  $\beta$  beyond the least index of a positive vector lie below it, so  $(e_\beta)_{\beta < \kappa}$  is coinital; fewer than  $\kappa$  positive vectors have boundedly many leading indices (regularity), and a later basis vector lies below all of them. For  $\Gamma_\kappa^{06}$  use Lemma 11.7.  $\square$

11 proves sharpness under the displayed extra hypotheses  $|K| \leq \kappa$  and  $\kappa^{<\kappa} = \kappa$ : then  $|R_{\Gamma, < \kappa}^{11}| = \kappa$  and its identity map is a target of size  $\kappa$  not factoring through  $\text{ct}$ . 11 states explicitly that these hypotheses are assumptions for the sharpness example, not assertions about all uncountable regular cardinals, and that regularity enters only the lexicographic example.

**Proposition 11.23** (Exact size and threshold [merge]). *In Proposition 11.22, for every set field  $K$  and every  $c > 0$  the field  $L_c = \{f \in K((X^\Gamma)) : |\text{supp } f| < \kappa, \text{supp } f \subseteq H_c(\Gamma)\}$  has at least  $\kappa^{<\kappa}$  elements. If  $|K| \leq \max(\kappa, 2^{\aleph_0})$  (which includes 11's hypothesis  $|K| \leq \kappa$ ), then  $|R_{\Gamma, < \kappa}^{11}| = \kappa^{<\kappa}$ , and the least cardinality of a ring target or module detecting any nonzero element of the positive-support ideal is exactly  $\kappa^{<\kappa}$ .*

*Proof.* Exactly as in Lemma 11.10: the series  $\sum_{\alpha \in B} X^{e_\alpha}$ ,  $B$  a subset of size  $< \kappa$  of the indices beyond the leading index of  $c$ , lie in  $L_c$  ( $e_\alpha$  decreases with  $\alpha$ ), and the  $< \kappa$ -support series form a field because inverses have support in a monoid generated by fewer than  $\kappa$  elements. The upper bound counts supports of size  $\nu < \kappa$  and coefficient functions: for infinite  $\nu$ ,  $|K|^\nu \leq \max(\kappa^\nu, 2^\nu) = \kappa^\nu$ , and

for finite  $\nu$ ,  $|K|^\nu \leq \max(\kappa, 2^{\aleph_0}) \leq \kappa^{\aleph_0} \leq \kappa^{<\kappa}$ ; also  $|D| \leq |K|$ , and the support choices number  $\kappa^\nu$ . The threshold follows from [Proposition 10.3](#), the coinitality gap, and the identity map.  $\square$

So the sharpness of [Theorem 10.8](#) in the lexicographic example holds without the hypothesis  $\kappa^{<\kappa} = \kappa$  once the bound is stated as  $\kappa^{<\kappa}$ , which equals  $\kappa$  exactly when  $\kappa^{<\kappa} = \kappa$ ; the coefficient field is still assumed to satisfy  $|K| \leq \max(\kappa, 2^{\aleph_0})$ . With  $(D, K) = (\mathbb{Z}, \mathbb{R})$  this ring is a Hahn integer part of the kind treated in this section; 11 does not embed it in **No**, and no embedding is used here.

## 12 Flat ideals and homological reflection

All rings, modules, resolutions, tensor products and Tor, Ext groups in this section are ordinary set-sized objects over the set-sized rings of [Section 11](#). No class-sized resolution is formed, and no directed-colimit theorem is applied to the proper-class union  $\Pi = \bigcup_{a>0} \omega^a \mathbf{Oz}$  (03, 06, 09). The package is 06's, proved there for  $A_\kappa^{06}$ ; 03 and 09 prove its flatness and Tor parts for their rings. The proofs use only the following properties, which all five constructions have:

- (H1)  $A$  is a commutative domain,  $A = D \oplus I$  with  $I = \ker \text{ct}$  an ideal and  $D = A/I$ ;
- (H2)  $A$  is built over a divisible exponent group  $\Gamma$ , and every  $f \in I$  has the form  $f = X^g b$  with  $g \in \Gamma_{>0}$  and  $b \in I$ ; hence  $I = \bigcup_{h \in \Gamma_{>0}} X^h A$ , a directed set-indexed union of principal ideals;
- (H3)  $I$  is not finitely generated.

(H2) is the gap property used in [Theorem 10.10](#) (a positive strict lower bound  $g$  of  $\text{supp } f$  gives  $b = X^{-g} f \in I$ ). (H3) holds because the least number of generators is  $\text{cf } \kappa$ ,  $\kappa$ ,  $\text{cf } \kappa$ ,  $\aleph_1$ , respectively  $\kappa$  ([Theorem 11.5\(iv\)](#), [Proposition 11.9](#), [Theorem 11.16\(iv\)](#), the chain after [Theorem 11.20](#), and for  $R_{\Gamma, <\kappa}^{11}$  the coinital chain  $(X^{e_\alpha})_{\alpha < \kappa}$  of [Proposition 11.22](#): fewer than  $\kappa$  elements have a common monomial divisor  $X^h$ , and  $X^h A$  misses  $X^{h/2}$ ); these counts are infinite.

**Lemma 12.1** (Projective ideals of a domain [06]). *A nonzero projective ideal of a commutative domain is finitely generated.*

*Proof.* Fix  $0 \neq a \in J$ . For  $A$ -linear  $f : J \rightarrow A$  and  $x \in J$ ,  $af(x) = f(ax) = xf(a)$ , so  $f(a) = 0$  forces  $f = 0$ . A dual basis  $(u_s, f_s)$  has  $x = \sum_s f_s(x)u_s$  with finitely many nonzero terms for each  $x$ ; only finitely many  $f_s(a)$  are nonzero, all other  $f_s$  vanish identically, and the finitely many corresponding  $u_s$  generate  $J$ .  $\square$

**Theorem 12.2** (Homological package [03, 06, 09] [merge]). *Let  $(A, I, D)$  satisfy (H1)–(H3). Then:*

- (i)  $I$  is flat,  $I^2 = I$  (each element is a product of two elements of  $I$ ), and  $I$  is not projective;
- (ii) for every  $D$ -module  $N$  (inflated along  $\text{ct}$ ),  $I \otimes_A N = 0$  and  $\text{Tor}_n^A(D, N) = 0$  for  $n > 0$ ; in particular  $\text{Tor}_n^A(D, D) = 0$ ;
- (iii)  $D$  is not flat:  $\text{fd}_A D = 1$ , and  $\text{Tor}_1^A(D, A/xA) \cong D$  for every  $0 \neq x \in I$ ;
- (iv) for all  $D$ -modules  $N, M$  and all  $n \geq 0$ , inflation induces natural isomorphisms  $\text{Ext}_A^n(N, M) \cong \text{Ext}_D^n(N, M)$ ;
- (v)  $\text{pd}_A D \geq 2$ ;
- (vi) the sequence  $0 \rightarrow I \rightarrow A \xrightarrow{\text{ct}} D \rightarrow 0$  does not split, so  $\text{Ext}_A^1(D, I) \neq 0$ ;

(vii) if  $(x_\alpha)_{\alpha \in \Lambda}$  generates  $I$ ,  $P = \bigoplus_\alpha Ae_\alpha \rightarrow I$ ,  $e_\alpha \mapsto x_\alpha$ , and  $J$  is its kernel, then  $\text{Ext}_A^2(D, J) \neq 0$ .

*Proof.* (i) Each  $X^h A$  is free of rank one because  $A$  is a domain, the union in (H2) is directed ( $X^h A \subseteq X^{h'} A$  for  $h' < h$ ), and a filtered colimit of flat modules is flat [19, Lemma 10.39.3]. The factorization  $f = X^g b$  writes  $f$  as a product of two elements of  $I$ , so  $I^2 = I$ . Projectivity would contradict (H3) by Lemma 12.1. (ii) Since  $IN = 0$ ,  $I \otimes_A N \cong (I/I^2) \otimes_D N = 0$  (the map  $x \otimes n \mapsto (x + I^2) \otimes n$  is well defined in reverse because  $(ab) \otimes n = b \otimes an = 0$  for  $a, b \in I$ ); tensoring the flat resolution  $0 \rightarrow I \rightarrow A \rightarrow D \rightarrow 0$  with  $N$  computes all positive Tor groups as 0. (iii) The same resolution gives  $\text{fd}_A D \leq 1$ . For  $0 \neq x \in I$ ,  $0 \rightarrow A \xrightarrow{x} A \rightarrow A/xA \rightarrow 0$  is a free resolution; after  $\otimes_A D$  the first map is 0 because  $x \in I$ , so  $\text{Tor}_1^A(D, A/xA) \cong D \neq 0$  and  $D$  is not flat. Equivalently, the injection  $xA \hookrightarrow A$  becomes the zero map  $D \rightarrow D$  after tensoring (03, 09). (iv) Take a projective  $A$ -resolution  $P_\bullet \rightarrow N$ . By (ii),  $D \otimes_A P_\bullet$  has homology  $N$  in degree 0 only, and its terms are projective  $D$ -modules (direct summands of free ones), so it is a projective  $D$ -resolution; the adjunction  $\text{Hom}_A(P_\bullet, M) \cong \text{Hom}_D(D \otimes_A P_\bullet, M)$  is an isomorphism of complexes, and comparison of resolutions makes the result natural. (v) If  $\text{pd}_A D \leq 1$ , the kernel  $I$  of the free surjection  $A \rightarrow D$  would be projective (Schanuel), against (i). (vi) A section  $s$  gives  $r = s(1)$  with  $\text{ct}(r) = 1$  and  $xr = s(x \cdot 1) = 0$  for  $0 \neq x \in I$ , so  $r = 0$  in the domain  $A$ . (vii)  $P \rightarrow I$  is surjective;  $0 \rightarrow J \rightarrow P \rightarrow I \rightarrow 0$  does not split (else  $I$  would be a summand of a free module, hence projective), giving  $0 \neq \text{Ext}_A^1(I, J)$ ; applying  $\text{Hom}_A(-, J)$  to  $0 \rightarrow I \rightarrow A \rightarrow D \rightarrow 0$  with  $A$  projective gives  $\text{Ext}_A^1(I, J) \cong \text{Ext}_A^2(D, J)$ , represented by the spliced sequence  $0 \rightarrow J \rightarrow P \rightarrow A \rightarrow D \rightarrow 0$ .  $\square$

In standard terms,  $A \rightarrow D$  is a homological ring epimorphism ( $D \otimes_A D = D$ , all higher self-Tor vanish) but not a flat one (03, 06); this terminology is optional. The exact value of  $\text{pd}_A D$  is not determined in general: only  $\text{pd}_A D \geq 2$  and  $\text{fd}_A D = 1$  are proved (06); when  $I$  is countably generated,  $\text{pd}_A D = 2$  (Proposition 13.12(v), from 12's telescope). Hypothesis (H2) is where the totally ordered exponent cone enters: for 12's coordinatewise cones the flat dimension of  $D$  is any  $d \geq 1$ , and the two kinds of statement are compared in Remark 13.16. The nonzero  $\text{Tor}_1^A(D, A/xA)$  does not contradict (ii), because  $A/xA$  is not a  $D$ -module:  $I \not\subseteq xA$ , since a gap below  $\text{supp } x$  exhibits a smaller positive monomial of  $I$  outside  $xA$  (06).  $I/I^2 = 0$  is not a finite-presentation or smoothness statement (09).

**Theorem 12.3** (Exact size of cohomological witnesses [06] [merge]). *Let  $A$  be one of the five rings of Section 10.4, with  $D = \mathbb{Z}$  (or any admissible  $D$ ), and  $\tau = |A|$ . For every  $A$ -module  $M$  with  $|M| < \tau$  and every  $n > 0$ ,  $\text{Ext}_A^n(D, M) = 0$ , and for any two  $A$ -modules of size  $< \tau$  the Ext groups over  $A$  are those of the underlying  $D$ -modules. For  $n \in \{1, 2\}$ ,*

$$\min\{|M| : \text{Ext}_A^n(D, M) \neq 0\} = \tau.$$

*For  $A_\kappa^{06}$  this minimum is  $\kappa^{<\kappa}$  in ZFC; under  $\kappa^{<\kappa} = \kappa$  it is  $\kappa$ , which is 06's statement. For  $A_\kappa^{03}$ ,  $A_\kappa^{08}$  and  $A_{\lambda, D}^{09}$  it is  $\kappa^{\aleph_0}$ ,  $\kappa$  and  $\lambda$ .*

*Proof.* Modules of size  $< \tau$  are annihilated by  $I$  (Theorem 10.10), so they are inflated  $D$ -modules and Theorem 12.2(iv) applies;  $D$  is free over itself. For the upper bounds,  $|I| = \tau$  and  $\text{Ext}_A^1(D, I) \neq 0$ . For  $n = 2$  choose a generating family of cardinality at most  $\tau$  (the generator counts of (H3) are all  $\leq \tau$ ); then  $|P| \leq \tau$ , so  $|J| \leq \tau$ , and  $\text{Ext}_A^2(D, J) \neq 0$  with the lower bound forces  $|J| = \tau$ .  $\square$

The Ext groups are computed in the full category of  $A$ -modules, not only among small modules; this is why Theorem 12.2(iv) is an ambient derived-functor statement and not merely an equivalence of small module categories (06). The two witnesses show that a quotient can look homologically

projective against every small coefficient module while failing projectivity in the ambient category. The whole argument works for the Gaussian versions with  $D = \mathbb{Z}[i]$ , where  $\text{Ext}_D$  means extensions of  $\mathbb{Z}[i]$ -modules (06). Finally, for  $0 \neq x \in \Pi_\kappa^{06}$  the module  $A_\kappa^{06}/xA_\kappa^{06}$  is detected by  $I$  (it is not a  $D$ -module), so its carrier has at least  $\kappa^{<\kappa}$  elements; 06 stated this with the qualifier “under the cardinal-arithmetic hypothesis”, which is not needed.

## 13 Homological and tensor algebra of omnific cores

This section is the part contributed by manuscripts 12 and 14 (Section 1.3). Both study ordinary commutative algebra of rings of the form “constants plus a tail” inside  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , and they share one set-sized model: 14’s finite-support ring  $A_0$  is 12’s rank-one coordinate-cone core  $A_1$ , and 12’s two-generator relation module is the case  $r = 2$  of 14’s lattice syzygy. Each is printed once (Lemma 13.1 and Theorem 13.9). Manuscript 12 computes flat and projective dimensions of set-sized finite-support cores (Sections 13.1 to 13.3 and 13.5); 14 computes tensor products, symmetric and exterior powers, Rees algebras, homomorphisms and duals of coefficient-lattice modules, over the full class ring and over  $A_0$  (Sections 13.4 and 13.6 to 13.8). Labels carry the sub-prefixes **osq:hd:** (12) and **osq:tn:** (14).

**Scope.** The rings of Sections 13.1 to 13.3 and 13.5 and 14’s model  $A_0$  have *finite* supports. They are therefore outside the universal theorem, like the finite-support rings of Section 9: the augmentation that sends every monomial to 1 is a unital map to the coefficient field that detects the tail (Proposition 13.2). What 12 calls arithmetic invisibility is invisibility to *finite* rings and to modules on which the tail acts by zero; these rings are not invisible to set-sized rings in the sense of Theorem 5.1.

**Notation.** 12’s augmentation  $\varepsilon$  is  $\text{ct}$ ; its tail  $M$  is  $\Pi_\Lambda$  (the letter  $M$  is 14’s coefficient lattice); its ordered-cone ring  $T = D \oplus P$  with monomials  $t^\gamma$  is  $T_\Gamma = D \oplus \Pi_\Gamma^{\text{fin}}$  with monomials  $X^\gamma$  (in this report  $t^g$  means  $X^{-g}$ , Section 2.2); its test modules  $N_F$  are  $V_F$ ; its cyclic module  $Q = A/(X_1, cX_1)$  is written  $A/H$ . 14’s ideal  $\mathcal{I}$ , its rings  $B = k + \mathcal{I}$ ,  $A = D + \mathcal{I}$  and its field  $F$  are  $\Pi_k$ ,  $\mathcal{A}_{k,k}$ ,  $\mathcal{A}_{D,k}$  and  $k((X^{\mathbf{No}}))$ ; its set-sized  $P$ ,  $A_0$ ,  $B_0$  are  $\Pi_\mathbb{Q}^{\text{fin}}$ ,  $T_\mathbb{Q}$  and  $k + \Pi_\mathbb{Q}^{\text{fin}}$ ; its multiplication map  $\mu_{M,N}$ , kernel  $K(M, N)$  and carrier  $E(M, N)$  are  $\mathfrak{m}_{M,N}$ ,  $\mathcal{T}(M, N)$  and  $\mathbf{E}(M, N)$ ; its multiplier ring  $\mathcal{O}(M)$  and orders  $\mathcal{O}$  in number fields  $K$  are  $\mathfrak{o}(M)$  and  $\mathfrak{o} \subset K_0$  (the letter  $\mathcal{O}$  denotes the finite surreal rings); the saturated lattice  $L$  of its matrix theorem is  $W_D$ .

### 13.1 Coordinate-cone and ordered-cone cores

Throughout Sections 13.1 to 13.3,  $K$  is a field and  $D \subseteq K$  a unital subring, so  $D$  is a domain. Let  $\Lambda$  be  $[d] = \{1, \dots, d\}$  ( $d \geq 1$ ) or  $\mathbb{N}_{>0}$ , and let  $P_\Lambda = \mathbb{Q}_{\geq 0}^{(\Lambda)}$  be the monoid of finitely supported vectors  $\alpha = (\alpha_i)_{i \in \Lambda}$  with nonnegative rational coordinates. In the monoid algebra  $K[P_\Lambda]$  (finite sums,  $X^\alpha X^\beta = X^{\alpha+\beta}$ ) put

$$\Pi_\Lambda = \bigoplus_{0 \neq \alpha \in P_\Lambda} KX^\alpha, \quad A_\Lambda = D \oplus \Pi_\Lambda, \quad (13.1)$$

and write  $A_d, \Pi_d$  for  $\Lambda = [d]$  and  $X_i = X^{\mathbf{e}_i}$ . The coefficient at  $X^0$  is a split surjective ring homomorphism  $\text{ct} : A_\Lambda \rightarrow D$  with kernel  $\Pi_\Lambda$ , because a sum of two nonnegative vectors is zero only

when both are. For a nonzero divisible subgroup  $\Gamma \subseteq \mathbb{R}$  put

$$\Pi_\Gamma^{\text{fin}} = \bigoplus_{\gamma \in \Gamma_{>0}} KX^\gamma, \quad T_\Gamma = D \oplus \Pi_\Gamma^{\text{fin}}, \quad (13.2)$$

the finite-support ring of Section 9 over the fixed ordered group  $\Gamma$ , again with  $\text{ct} : T_\Gamma \rightarrow D$ . The difference between the two families is the exponent cone: coordinatewise nonnegative vectors in  $A_\Lambda$ , the positive cone of a totally ordered group in  $T_\Gamma$ .

**Lemma 13.1** (Domains and embeddings [12; 14]).

- (i)  $K[P_\Lambda]$ ,  $A_\Lambda$  and  $T_\Gamma$  are domains, and  $\Pi_\Lambda$  is a nonzero prime ideal.
- (ii) Let  $(\gamma_i)_{i \in \Lambda}$  be positive surreals linearly independent over  $\mathbb{Q}$ . Then  $X^\alpha \mapsto \omega^{\sum_i \alpha_i \gamma_i}$  defines an injective ring homomorphism  $\mathbb{R}[P_\Lambda] \rightarrow \mathbf{No}$ , and, applied to real and imaginary parts,  $\mathbb{C}[P_\Lambda] \rightarrow \mathbf{No}[i]$ ; it maps  $\mathbb{Z} + \Pi_\Lambda$  into  $\mathbf{Oz}$  and  $\mathbb{Z}[i] + \Pi_\Lambda$  into  $\mathbf{Oz}[i]$ . The  $\gamma_i$  may be taken real:  $\gamma_i = \theta^{i-1}$  for a transcendental real  $\theta > 1$ . Likewise  $X^\gamma \mapsto \omega^\gamma$  embeds  $T_\Gamma$  in  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  for  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ .
- (iii) If the  $\gamma_i$  are positive reals linearly independent over  $\mathbb{Q}$  and  $\Gamma = \sum_i \mathbb{Q}\gamma_i$ , then  $X_i \mapsto X^{\gamma_i}$  is an injective ring homomorphism  $A_\Lambda \hookrightarrow T_\Gamma$  compatible with  $\text{ct}$ . For  $d = 1$  and  $\gamma_1 = 1$  it is an isomorphism  $A_1 \cong T_\mathbb{Q}$ ; this ring, with  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , is 14's set-sized model  $A_0$ .

*Proof.* (i) Finitely many elements of  $K[P_\Lambda]$  involve finitely many coordinates and a common denominator  $n$  of their exponents, so they lie in the polynomial ring  $K[X_{i_1}^{1/n}, \dots, X_{i_s}^{1/n}]$ , a domain. In  $T_\Gamma$  the largest exponents of two nonzero finite sums add and their leading coefficients multiply.  $A_\Lambda/\Pi_\Lambda = D$  is a domain and  $X_1 \in \Pi_\Lambda$ . (ii) The map  $\ell(\alpha) = \sum_i \alpha_i \gamma_i$  is additive, injective by linear independence, and positive on nonzero  $\alpha \in P_\Lambda$ ; so the monomials go to distinct Conway monomials,  $\omega^u \omega^v = \omega^{u+v}$  and finite distributivity give a ring homomorphism, and uniqueness of finite normal forms gives injectivity. Nonconstant terms go to positive exponents and the constant coefficient is an integer (Gaussian integer), which is membership in  $\mathbf{Oz}$  ( $\mathbf{Oz}[i]$ ). A rational relation among the  $\theta^{i-1}$  would give a nonzero rational polynomial vanishing at  $\theta$ . The case of  $T_\Gamma$  is the same with  $\ell = \text{id}$ . (iii)  $\ell$  is an injective monoid map  $P_\Lambda \rightarrow \Gamma_{\geq 0}$  with  $\ell^{-1}(0) = 0$ ; for  $d = 1$  it is  $\mathbb{Q}_{\geq 0} \rightarrow \mathbb{Q}_{\geq 0}$ , the identity.  $\square$

Thus  $d$  counts independent monomial directions, not Archimedean scales: with  $\gamma_i = \theta^{i-1}$  all exponents lie in the single Archimedean class of 1 (12). Every element of every ring in this subsection is a finite sum; no topology, convergence or infinite summation is involved.

**Proposition 13.2** (Finite targets only [12]). *Let  $A$  be  $A_\Lambda$  or  $T_\Gamma$  with tail  $\Pi$ , and  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ .*

- (i) For  $0 \neq a \in D$ ,  $a\Pi = \Pi$  and  $A/aA \cong D/aD$ . Every unital map from  $A$  to a finite ring, and every finite  $A$ -module, factors through  $\text{ct}$ ;  $\hat{A} \cong \hat{D}$  for the profinite completions,  $\varprojlim_k A/p^k A \cong \varprojlim_k D/p^k D$  for every ordinary prime  $p$ ,  $\bigcap_k p^k A = \Pi$ , and  $\text{Tor}_{>0}^A(D, E) = 0$  for every finite  $A$ -module  $E$ .
- (ii) The augmentation  $\sum_\alpha c_\alpha X^\alpha \mapsto \sum_\alpha c_\alpha$  is a unital ring homomorphism  $A \rightarrow K$  sending every monomial to 1; it does not factor through  $\text{ct}$ .

*Proof.* (i) Divide the coefficients of a tail by  $a$ ; so  $aA = aD \oplus \Pi$ . A finite ring has a positive characteristic  $n$ , so a map to it kills  $n\Pi = \Pi$ ; a finite module is killed by some  $n \geq 1$ , and

$m = n(m/n)$  for  $m \in \Pi$ . The ideals  $nA = nD \oplus \Pi$  are cofinal among ideals of finite index, and  $\bigcap_n nD = 0 = \bigcap_k p^k D$  (coordinatewise for  $\mathbb{Z}[i]$ ). A finite module is therefore a  $D$ -module, and the Tor assertion is [Corollary 13.5](#) below for  $A_\Lambda$  and [Theorem 12.2\(ii\)](#) with [Theorem 13.13](#) for  $T_\Gamma$ . (ii) is the computation of [Theorem 9.1\(ii\)](#): for finite supports, the coefficient sum of a product is the product of the coefficient sums.  $\square$

Hence all the cores, and 14's  $A_0$ , have the same finite quotients and profinite completion as  $D$ , while [Corollary 13.8](#) below shows that their homological dimensions differ; this limits any reconstruction of an omnific subring from its finite arithmetic shadows (12). Item (ii) is the reason the word “invisible” must be read with (i)’s restriction.

### 13.2 Flat face ideals and a Boolean resolution

For a finite nonempty  $S \subseteq \Lambda$  put  $p_S = \prod_{i \in S} X_i$  and let

$$I_S = \bigoplus_{\alpha \in P_\Lambda, \alpha_i > 0 \ (i \in S)} KX^\alpha, \quad (13.3)$$

an ideal of  $A_\Lambda$  with  $I_S = \bigcap_{i \in S} I_{\{i\}}$  and  $\Pi_\Lambda = \sum_i I_{\{i\}}$ .

**Lemma 13.3** (Face ideals [12]). *Let  $A = A_\Lambda$  and  $\Pi = \Pi_\Lambda$ .*

- (i)  $I_S = \bigcup_{n \geq 0} p_S^{2^{-n}} A$ , an increasing chain of free rank-one  $A$ -modules; hence  $I_S$  is flat.
- (ii)  $\Pi^2 = \Pi$ ,  $I_S^2 = I_S$  and  $\Pi I_S = I_S$ ; for every  $D$ -module  $E$ , regarded as an  $A$ -module through  $\text{ct}$ ,  $I_S \otimes_A E = 0$ .
- (iii) Neither  $\Pi$  nor any  $I_S$  is finitely generated.

*Proof.* (i) Multiplication by  $p_S^{2^{-n}}$  is injective ([Lemma 13.1](#)), and  $p_S^{2^{-n}} = p_S^{2^{-n-1}} p_S^{2^{-n-1}}$  gives the chain. Given  $f \in I_S$ , choose  $n$  with  $2^{-n}$  below every coordinate indexed by  $S$  of the finitely many exponents of  $f$ ; then  $f/p_S^{2^{-n}}$  has nonnegative exponents, positive on  $S$ , so it lies in  $\Pi \subseteq A$  although its coefficients are arbitrary in  $K$ . A filtered colimit of flat modules is flat [[19](#), Lemma 10.39.3]. (ii)  $kX^\alpha = (kX^{\alpha/2})X^{\alpha/2}$ , both factors in  $I_S$  when  $\alpha$  is positive on  $S$ . Since  $\Pi E = 0$ ,  $I_S \otimes_A E \cong (I_S/\Pi I_S) \otimes_D E = 0$ . (iii) Let  $|\alpha| = \sum_i \alpha_i$ . If finitely many elements generated  $\Pi$  (or  $I_S$ ), let  $\delta > 0$  be the least total degree of their monomials; every element of the generated ideal has only monomials of total degree at least  $\delta$ , since multiplication by  $A$  adds nonnegative degrees and cancellation creates no new monomial. But  $X_1^q \in \Pi$  (respectively  $p_S^q \in I_S$ ) with  $q$  small has smaller total degree.  $\square$

Flatness of every  $I_S$  does not imply flatness of their sum  $\Pi$ ; the failure begins at  $d = 2$  ([Corollary 13.8](#)). For  $p \geq 1$  put  $C_p = \bigoplus_{S \subseteq \Lambda, |S|=p} I_S$  and  $C_0 = A$ ; for  $S = \{s_1 < \dots < s_p\}$  let  $d_p$  send  $f$  in the summand  $I_S$  to  $\sum_{r=1}^p (-1)^{r-1} f$  in the summands  $I_{S \setminus \{s_r\}}$  (for  $p = 1$  the target is  $A$  and the map is the inclusion). Deleting two indices in either order gives opposite signs, so  $d_{p-1} d_p = 0$ .

**Theorem 13.4** (Coordinate-intersection resolution [12]). *The augmented complex*

$$\dots \longrightarrow C_2 \xrightarrow{d_2} C_1 \xrightarrow{d_1} A_\Lambda \xrightarrow{\text{ct}} D \longrightarrow 0$$

*is exact and every  $C_p$  is flat. For  $\Lambda = [d]$ ,  $C_p = 0$  for  $p > d$ , and this is a flat resolution of  $D$  of length  $d$ ; also for  $\Lambda = \mathbb{N}_{>0}$  the complex is exact.*



*Proof.* All maps preserve the exponent grading. In degree  $\mathbf{0}$  only  $A$  and  $D$  contribute, and  $ct$  is the identity of  $D$ . Fix  $\alpha \neq \mathbf{0}$  with support  $T = \{i : \alpha_i > 0\}$ . The degree- $\alpha$  part of  $I_S$  is  $K$  exactly when  $\emptyset \neq S \subseteq T$ , so the degree- $\alpha$  part of the complex is the augmented chain complex of the full simplex on  $T$  with coefficients  $K$ , a  $p$ -subset in degree  $p$ . With  $t = \min T$ , the homotopy that inserts  $t$  (with the sign placing it in the ordered list) into subsets not containing  $t$ , is zero on subsets containing  $t$ , and sends the augmentation vector to the vertex  $\{t\}$ , satisfies  $dh + hd = 1$ . Each element involves finitely many degrees and summands, so degreewise exactness is exactness. Flatness is [Lemma 13.3\(i\)](#) and closure under direct sums.  $\square$

**Corollary 13.5** (Arithmetic modules are Tor-acyclic [12]). *For every  $D$ -module  $E$ ,  $D \otimes_{A_\Lambda} E \cong E$  and  $\mathrm{Tor}_j^{A_\Lambda}(D, E) = 0$  for  $j > 0$ ; in particular  $A_\Lambda \rightarrow D$  is a homological ring epimorphism.*

*Proof.* Tensor the resolution with  $E$ : every positive term vanishes by [Lemma 13.3\(ii\)](#), and flat resolutions compute  $\mathrm{Tor}$ .  $\square$

This concerns every module on which the tail acts by zero; it is why such modules cannot see the flat dimension computed next (12).

**Proposition 13.6** (Formal lifting and differentials [12; 13]). *Let  $A$  be  $A_\Lambda$  or  $T_\Gamma$  with tail  $\Pi$ . Then  $A \rightarrow D$  is formally étale for nilpotent lifting problems, but neither flat nor finitely presented as a ring map, and  $\Omega_{A/D} \otimes_A D = 0$ : every  $D$ -derivation of  $A$  into a module killed by  $\Pi$  vanishes.*

*Proof.* If  $f : A \rightarrow R$  is an  $A$ -algebra structure map and  $D \rightarrow R/J$  an  $A$ -algebra map with  $J^n = 0$ , then  $f(\Pi) \subseteq J$ , so  $f(\Pi) = f(\Pi^n) \subseteq J^n = 0$  ( $\Pi^2 = \Pi$  by halving exponents) and  $f$  factors uniquely through  $D$ ; this is the idempotent-ideal mechanism of [20, Tag 060H]. Multiplication by a nonzero monomial of  $\Pi$  is injective on  $A$  but becomes 0 on  $D = D \otimes_A A$ , so  $D$  is not flat;  $A/\Pi$  is finitely presented only if  $\Pi$  is finitely generated, which it is not ([Lemma 13.3\(iii\)](#)); for  $T_\Gamma$ , a finite family of tails has a least exponent among its monomials). A derivation  $\partial$  as stated satisfies  $\partial(uv) = u\partial v + v\partial u = 0$  for  $u, v \in \Pi$ , and  $\Pi = \Pi^2$ .  $\square$

The mechanism is classical and not a novelty claim (12). Manuscript 13 proves the same statements for the full omnific ring in a two-universe reading ([Remark 13.16](#)), and [Theorem 7.4](#) is the class-ring analogue for derivations. “Nilpotent” is a test condition on  $R$ : the cores are domains.

### 13.3 The multigraded Tor and exact flat dimensions

For a finite nonempty  $F \subseteq \Lambda$  let  $J_F = (X_i : i \in F)A_\Lambda$  and  $V_F = A_\Lambda/J_F$ . For finite  $S$  put  $\mathbf{e}_S = \sum_{i \in S} \mathbf{e}_i$ ; the notation  $K\langle \mathbf{e}_S \rangle$  means a copy of the  $D$ -module  $K$  placed in multidegree  $\mathbf{e}_S$  (not a subring of  $A_\Lambda$ ).

**Theorem 13.7** (Boolean-face Tor formula [12]). *For  $\Lambda$  finite or countably infinite and  $F$  as above,*

$$\begin{aligned} \mathrm{Tor}_0^{A_\Lambda}(D, V_F) &\cong D\langle \mathbf{0} \rangle, & \mathrm{Tor}_1^{A_\Lambda}(D, V_F) &\cong \bigoplus_{i \in F} D\langle \mathbf{e}_i \rangle, \\ \mathrm{Tor}_p^{A_\Lambda}(D, V_F) &\cong \bigoplus_{S \subseteq F, |S|=p} K\langle \mathbf{e}_S \rangle & (p \geq 2). \end{aligned}$$

*In particular  $\mathrm{Tor}_p = 0$  for  $p > |F|$ .*

*Proof.* Tensor [Theorem 13.4](#) with  $V_F$ ; the positive terms become  $I_S/J_F I_S$  and the degree-zero term is  $V_F$ . Fix  $\alpha \neq \mathbf{0}$  with support  $T$ . If  $\alpha_j > 1$  for some  $j \in F$ , then  $\alpha - \mathbf{e}_j$  is nonnegative, nonzero and positive wherever  $\alpha$  is, so the whole coefficient space at  $\alpha$  lies in  $X_j I_S$  and in  $X_j A$ ; every term vanishes in this degree. So assume  $\alpha_j \leq 1$  for  $j \in F$  and put  $U = \{j \in F : \alpha_j = 1\}$ . For  $\emptyset \neq S \subseteq T$  the degree- $\alpha$  part of  $I_S/J_F I_S$  is a full copy of  $K$  exactly when  $U \subseteq S$  (multiplication by  $X_j$  reaches this degree from  $I_S$  only when  $\alpha_j = 1$  and  $j \notin S$ ), and zero otherwise.

*Case*  $U = \emptyset$ .  $(V_F)_\alpha = K$  and all faces survive; the augmented simplex is exact. *Case*  $U \neq \emptyset \neq T \setminus U$ .  $(V_F)_\alpha = 0$ , since subtracting  $\mathbf{e}_j$  ( $j \in U$ ) leaves a nonzero vector; inserting some  $t \in T \setminus U$  contracts the complex of faces  $U \subseteq S \subseteq T$ . *Case*  $T = U$ ,  $|U| \geq 2$ . Only  $K$  in degree  $|U|$  survives, with no incoming or outgoing maps ( $(V_F)_{\mathbf{e}_U} = 0$ ); this gives  $K\langle \mathbf{e}_U \rangle$ . *Case*  $T = U = \{j\}$ . In degree  $\mathbf{e}_j$  the complex is  $K \rightarrow (V_F)_{\mathbf{e}_j} = K/D$ , because  $J_F$  has coefficient space  $D$  (not  $K$ ) at  $\mathbf{e}_j$ : the cofactor of  $X_j$  is a constant. The map is the quotient map, with kernel  $D$  and zero cokernel; this gives  $D\langle \mathbf{e}_j \rangle$  in degree one. In degree  $\mathbf{0}$  only  $D$  in degree zero remains.  $\square$

For  $|S| = p \geq 2$  the class of  $k \in K$  is represented by  $kX^{\mathbf{e}_S}$  in  $I_S/J_F I_S$ ; in degree one only coefficients  $d \in D$  give cycles  $dX_i$ . Treating the constant coefficient as if it lay in  $K$  gives a wrong answer (12). Scaling coordinates by positive rationals  $b_i$  ( $i \in F$ ) is an automorphism of  $A_\Lambda$  fixing the coefficients, so the same formula holds for  $A_\Lambda/(X_i^{b_i} : i \in F)$  with  $\mathbf{e}_S$  replaced by  $\sum_{i \in S} b_i \mathbf{e}_i$  (12).

**Corollary 13.8** (Exact flat dimensions [12]). *For  $\Lambda = [d]$ ,  $\text{fd}_{A_d} D = d$  and  $\text{fd}_{A_d} \Pi_d = d - 1$ ; for  $\Lambda = \mathbb{N}_{>0}$  both are infinite. All positive self-Tor groups  $\text{Tor}_{>0}^{A_d}(D, D)$  vanish.*

*Proof.* The resolution gives  $\text{fd } D \leq d$ , and  $\text{Tor}_d(D, V_{[d]})$  is  $D$  ( $d = 1$ ) or  $K$  ( $d \geq 2$ ), nonzero. Truncation gives  $\text{fd } \Pi_d \leq d - 1$ ; for  $d \geq 2$ ,  $0 \rightarrow \Pi_d \rightarrow A_d \rightarrow D \rightarrow 0$  gives  $\text{Tor}_{d-1}(\Pi_d, V_{[d]}) \cong \text{Tor}_d(D, V_{[d]}) \neq 0$ , and for  $d = 1$ ,  $\Pi_1 = I_{\{1\}}$  is flat. In infinite rank take  $|F| = n$  for every  $n$ . The last assertion is [Corollary 13.5](#).  $\square$

For  $d = 2$  the groups are  $D^2$  and  $K$  (the latter represented by  $kX_1X_2$ ), and for  $d = 3$  they are  $D^3, K^3, K$ : vertices, edges and the full face (12). The finite audit of 12 checks this formula for  $D = \mathbb{Q} \subset K = \mathbb{Q}(\sqrt{2})$  ([Section 19.1](#)).

### 13.4 Coefficient lattices and their syzygies

The next results need only a ring of the following shape, which covers 12's cores, 12's ordered-cone rings and 14's rings.

**Setting (L).**  $K$  is a field,  $D \subseteq K$  a unital subring, and  $A = D \oplus \Pi$  is one of: (a)  $A_\Lambda$  with  $\Pi = \Pi_\Lambda$ ; (b)  $T_\Gamma$  with  $\Pi = \Pi_\Gamma^{\text{fin}}$ ; (c) the class ring  $\mathcal{A}_{D,K}$  with  $\Pi = \Pi_K$  (full surreal exponents,  $K$  a set field). In each case  $B = K \oplus \Pi$  is a ring containing  $A$ ,  $\Pi$  is an ideal of  $B$  and a  $K$ -vector space,  $K \cap \Pi = 0$ , all three lie in a field, and  $\Pi$  is not finitely generated in (a), (b) and not generated by a set in (c) ([Lemma 13.3\(iii\)](#), [Proposition 13.6](#), [Proposition 3.3\(iii\)](#)). A *coefficient lattice* is  $M = Dm_1 + \cdots + Dm_r \subseteq K$  with  $m_1, \dots, m_r$  linearly independent over  $D$  (equivalently over  $\text{Frac } D$ );  $r = \text{rank } M$ . Put

$$L(M) = M + \Pi \subseteq B, \quad J_g(M) = X^g L(M) \subseteq A$$

for a nonzero monomial  $X^g \in \Pi$  (a vector  $g \neq \mathbf{0}$  in (a),  $g \in \Gamma_{>0}$  in (b),  $g > 0$  in (c)). For  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ,  $D$  is a principal ideal domain and every nonzero finitely generated  $D$ -submodule of  $K$  is free, so these are exactly 14's coefficient lattices.

**Theorem 13.9** (Generators and the first syzygy [14; 12] [merge]). *In Setting (L), let  $M$  have basis  $m_1, \dots, m_r$ .*

(i)  $L(M) = \sum_j A m_j$ ,  $\Pi L(M) = \Pi$  and  $L(M)/\Pi L(M) \cong M$ ; the least number of generators of  $L(M)$  is  $r$ , although  $L(M)$  has rank one over the fraction field.

(ii) The sequence

$$0 \longrightarrow \Pi^{r-1} \xrightarrow{\sigma} A^r \xrightarrow{\pi} L(M) \longrightarrow 0, \quad \pi(a) = \sum_j a_j m_j,$$

$$\sigma(u_2, \dots, u_r) = \left( -\sum_{j \geq 2} \frac{m_j}{m_1} u_j, u_2, \dots, u_r \right),$$

is exact. For  $r = 1$ ,  $L(M) \cong A$ ; for  $r \geq 2$ ,  $L(M)$  is not finitely presented in (a), (b), and in (c) the kernel is not generated by any set.

(iii) (12's certificate.) Let  $c \in K \setminus \text{Frac}(D)$  and  $H = (X^g, cX^g)A$ . Then  $X^g A \cap cX^g A = X^g \Pi$  and  $0 \rightarrow \Pi \xrightarrow{m \mapsto (-cm, m)} A^2 \xrightarrow{(a, b) \mapsto X^g a + cX^g b} H \rightarrow 0$  is exact; in (a) and (b),  $A$  is not coherent.

*Proof.* (i) Since  $0 \neq m_1 \in K$  and  $\Pi$  is a  $K$ -space,  $m_1 \Pi = \Pi$ ; so  $\sum_j A m_j \supseteq M + m_1 \Pi = L(M)$ , and the reverse inclusion is clear.  $\Pi L(M) \subseteq \Pi$  because  $\Pi$  is an ideal of  $B$ , and  $\Pi L(M) \supseteq \Pi m_1 = \Pi$ . As  $M \cap \Pi \subseteq K \cap \Pi = 0$ ,  $L(M)/\Pi \cong M \cong D^r$ . Generators of  $L(M)$  give generators of the  $D$ -module  $L(M)/\Pi L(M)$ , and a free module of rank  $r$  over a nonzero commutative ring needs at least  $r$  generators (reduce modulo a maximal ideal). (ii) If  $\sum_j a_j m_j = 0$  with  $a_j = d_j + u_j$ , the components in  $K$  and  $\Pi$  give  $\sum_j d_j m_j = 0$ , so every  $d_j = 0$  by independence, and  $u_1 = -\sum_{j \geq 2} (m_j/m_1) u_j$ ; conversely these tuples lie in  $\ker \pi$ , and  $(m_j/m_1) u_j \in \Pi$ .  $\sigma$  is injective. For  $r \geq 2$  the kernel projects onto  $\Pi$ , which is not finitely (set-) generated, and a finitely presented module has finitely generated kernels for every surjection from a finite free module. (iii) Apply (ii) to  $M = D + Dc$  (the elements  $1, c$  are independent over  $D$  because  $c \notin \text{Frac}(D)$ ) and multiply by  $X^g$ , which is injective:  $H = X^g L(M)$ . If  $X^g(d + u) = cX^g(e + v)$  then  $d = ce$  forces  $d = e = 0$ , so  $X^g A \cap cX^g A = X^g(\Pi \cap c\Pi) = X^g \Pi$ , not finitely generated; in a coherent ring the intersection of two finitely generated ideals is finitely generated [21, Tag 05CU].  $\square$

Theorem 13.9(iii) is 12's noncoherence certificate for  $A_\Lambda$  and  $T_\Gamma$  with any unital  $D \subseteq K$ ; (i) and (ii) are 14's theorem for  $\mathbf{Oz}$ ,  $\mathbf{Oz}[i]$  and  $A_0$  with  $D$  a principal ideal domain. Stating both at once in Setting (L) is the merge's; the proofs are the sources'. All compatible relations between two generators with the same monomial and incompatible constant lines are forced into the infinitely divisible tail (12). No novelty is attached to noncoherence as such: rings of the form  $D + M$  and their coherence have a long history, including Dobbs–Papick [10]; 14 uses no valuation-ring hypothesis on  $B$ , and the useful information is the exact syzygy (14).

**Proposition 13.10** (Antisymmetric torsion and nonflatness [14] [merge]). *In Setting (L), let  $M, N$  be coefficient lattices and  $\mathbf{m}_{M,N} : M \otimes_D N \rightarrow K$  the multiplication map. On  $\mathbf{E}(M, N) = \Pi \oplus (M \otimes_D N)$  the rule*

$$(d + h) \cdot (u, t) = (du + hu + h \mathbf{m}_{M,N}(t), dt) \quad (d \in D, h \in \Pi) \quad (13.4)$$

is an  $A$ -module structure, and  $(m + u) \otimes (n + v) \mapsto (mv + un + uv, m \otimes n)$  is a well-defined  $A$ -linear map  $\beta : L(M) \otimes_A L(N) \rightarrow \mathbf{E}(M, N)$ . If  $\text{rank } M \geq 2$  and  $m_1, m_2$  are basis vectors, then  $\tau = m_1 \otimes m_2 - m_2 \otimes m_1$  is a nonzero element of  $L(M) \otimes_A L(M)$  with product 0. Consequently, in (a) and (b),  $L(M)$  and every ideal  $J_g(M)$  are not flat, hence neither projective nor invertible.

*Proof.* For  $a = d + h$ ,  $b = e + g'$  in  $A$  one computes that both  $a \cdot (b \cdot (u, t))$  and  $(ab) \cdot (u, t)$  have first coordinate  $abu + (dg' + he + hg')m_{M,N}(t)$  and second coordinate  $det$ ; additivity and  $1 \cdot (u, t) = (u, t)$  are clear, and  $hm_{M,N}(t) \in \Pi$  because  $\Pi$  is a  $K$ -space. Write  $x = m + u \in L(M)$ ,  $y = n + v \in L(N)$  (uniquely, since  $M \cap \Pi = 0$ ); the formula is  $(xy - mn, m \otimes n)$ , which is biadditive. For  $a = d + h \in A$ , the constant part of  $ax$  is  $dm$ , so the value at  $(ax, y)$  is  $(axy - dm n, dm \otimes n)$ , which equals  $a \cdot (xy - mn, m \otimes n)$  by (13.4), and the value at  $(x, ay)$  is the same. Hence  $\beta$  exists.  $\beta(\tau) = (0, \tau)$ , and  $\tau \neq 0$  in the free  $D$ -module  $M \otimes_D M$  with basis  $m_i \otimes m_j$ . For flatness,  $J = J_g(M) \cong L(M)$  is an ideal of  $A$  ( $X^g M \subseteq \Pi$  and  $X^g \Pi \subseteq \Pi$ ); tensoring the injection  $J \hookrightarrow A$  with  $J$  gives the multiplication  $J \otimes_A J \rightarrow J$ , which kills the nonzero element corresponding to  $\tau$ . Projective modules and invertible ideals are flat.  $\square$

In (c) the tensor product is 14's explicit carrier (Section 13.6); there the same element shows that tensoring  $J_g(M) \hookrightarrow A$  with  $J_g(M)$  is not injective, which is 14's statement that  $L(M)$  is not flat over  $\mathbf{Oz}$ . Over  $A_0$  this is 14's corollary; allowing any unital  $D \subseteq K$  and 12's rings is the merge's (the proof is 14's, and no principal-ideal hypothesis is used in it).

### 13.5 Flat and projective dimensions; the ordered cone

**Theorem 13.11** (Lattice modules and cyclic quotients [12, 14] [merge]). *Let  $A$  be  $A_d$  ( $d \geq 1$ ) or  $T_\Gamma$ , and put  $\delta = d$ , respectively  $\delta = 1$ . Let  $M$  be a coefficient lattice of rank  $r \geq 2$  and  $J = J_g(M)$ . Then*

$$\text{fd}_A L(M) = \delta, \quad \text{fd}_A(A/J) = \delta + 1,$$

*and for every  $A$ -module  $N$  and  $j \geq 2$ ,  $\text{Tor}_j^A(L(M), N) \cong \text{Tor}_j^A(D, N)^{r-1} \cong \text{Tor}_{j+1}^A(A/J, N)$ . Over  $A_{\mathbb{N}_{>0}}$  both dimensions are infinite. In particular, for  $c \in K \setminus \text{Frac}(D)$  and  $H = (X_1, cX_1)$ ,  $\text{fd}_{A_d}(A_d/H) = d + 1$  and  $\text{fd}_{T_\Gamma}(T_\Gamma/(X^g, cX^g)) = 2$ , and over  $A_0 = T_\mathbb{Q}$ ,  $\text{fd} L(M) = 1$  and  $\text{fd}(A_0/J_e(M)) = 2$ . If moreover  $c^2 \in D$ , then  $\text{Tor}_2^A(A/H, A/H) \cong D + cD \cong D \oplus D$ .*

*Proof.* By Theorem 13.9(ii) there is an exact sequence  $0 \rightarrow \Pi^{r-1} \rightarrow A^r \rightarrow L(M) \rightarrow 0$ , and  $\text{fd} \Pi = \delta - 1$  (Corollary 13.8; for  $T_\Gamma$ ,  $\Pi$  is flat by Theorem 13.13 below). Hence  $\text{fd} L(M) \leq \delta$  and, from  $0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0$  with  $J \cong L(M)$ ,  $\text{fd}(A/J) \leq \delta + 1$ . The long exact sequences give, for  $j \geq 2$ ,  $\text{Tor}_j(L(M), N) \cong \text{Tor}_{j-1}(\Pi, N)^{r-1} \cong \text{Tor}_j(D, N)^{r-1}$  and  $\text{Tor}_{j+1}(A/J, N) \cong \text{Tor}_j(J, N)$ . If  $\delta = d \geq 2$ ,  $\text{Tor}_d(D, V_{[d]}) \cong K \neq 0$  (Theorem 13.7) gives both lower bounds. If  $\delta = 1$ ,  $L(M)$  is not flat (Proposition 13.10), so  $\text{fd} L(M) = 1$  and  $\text{Tor}_2(A/J, N) \cong \text{Tor}_1(J, N) \neq 0$  for some  $N$ . In infinite rank,  $\text{Tor}_n(D, V_F) \neq 0$  for  $|F| = n$ . The special cases are  $M = D + Dc$  with  $J = H$ , and  $M$  of rank at least two over  $T_\mathbb{Q}$ . For the last assertion (12), with  $U = D + cD$  one has  $H = UX^g \oplus X^g \Pi \subseteq \Pi$ ,  $H\Pi = X^g \Pi$  (as  $\Pi^2 = \Pi$  and  $c\Pi = \Pi$ ) and  $cH \subseteq H$  (as  $c^2 \in D$ ). The two sequences give  $\text{Tor}_2(A/H, A/H) \cong \text{Tor}_1(H, A/H) \cong \ker(\Pi/X^g \Pi \rightarrow (A/H)^2, m \mapsto (-cm, m))$ , and the kernel consists of the classes of  $m \in H$ , that is  $H/X^g \Pi \cong U$ .  $\square$

**Correction to 12 and the relation to 14.** 12 states  $\text{fd}(A_d/H) = d + 1$  and  $\text{fd}(T_\Gamma/H) = 2$  under the extra hypothesis  $c^2 \in D$ , which it uses only to compute  $\text{Tor}_2(A/H, A/H)$ ; for  $d \geq 2$  its own shift argument never uses it, and for  $d = 1$  and  $T_\Gamma$  the lower bound 2 follows from 14's antisymmetric torsion tensor. The hypothesis is needed only for the formula  $\text{Tor}_2(A/H, A/H) \cong D \oplus D$ . 14's flat-dimension proposition for  $A_0$  is the case  $A = T_\mathbb{Q} = A_1$  of the theorem. Since a nonzero Tor in degree  $n$  forces weak global dimension at least  $n$ : if  $K \neq \text{Frac } D$  (for instance  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$  with  $c = \sqrt{2}$ ), then  $\text{w.gl.dim } A_d \geq d + 1$ ,  $\text{w.gl.dim } T_\Gamma \geq 2$ , and the countable-rank core has infinite weak global dimension (12, 14). Neither source claims the exact weak global dimension of any of these rings.

**Proposition 13.12** (Projective dimensions [12] [merge]).

- (i) (Telescope, 12.) Let  $A$  be a domain and  $I = \bigcup_{n \geq 0} x_n A$  for nonzero  $x_n$  with  $x_n = x_{n+1} u_n$ ,  $u_n \in A$ . With  $F = \bigoplus_{n \geq 0} A e_n$ , the sequence  $0 \rightarrow F \xrightarrow{\Delta} F \xrightarrow{\pi} I \rightarrow 0$ ,  $\Delta(e_n a) = e_n a - e_{n+1} u_n a$ ,  $\pi(e_n a) = x_n a$ , is exact; so  $\text{pd}_A I \leq 1$ . This applies to every  $I_S$  ( $x_n = p_S^{2^{-n}}$ ) and to  $\Pi_\Gamma^{\text{fin}}$  ( $x_n = X^{2^{-n}g}$ ,  $g \in \Gamma_{>0}$ ).
- (ii) [merge]  $\text{pd}_{A_\Lambda} I_S = 1$  and  $\text{pd}_{T_\Gamma} \Pi_\Gamma^{\text{fin}} = 1$ .
- (iii)  $d \leq \text{pd}_{A_d} D \leq d+1$  (12), and [merge]  $\text{pd}_{A_1} D = \text{pd}_{T_\Gamma} D = 2$ .
- (iv) [merge] Over  $T_\Gamma$  (in particular over  $A_0$ ), for a lattice of rank at least two,  $\text{pd } L(M) = 2$  and  $\text{pd}(T_\Gamma/J_g(M)) = 3$ . Over  $A_d$ ,  $d \leq \text{pd } L(M) \leq d+1$  and  $d+1 \leq \text{pd}(A_d/J_g(M)) \leq d+2$  (12, for  $M = D + Dc$ ).
- (v) [merge] Let  $(A, I, D)$  satisfy (H1)–(H3) of Section 12, with  $A$  consisting of series of nonnegative support (as do all rings of Section 11 and  $T_\Gamma$ ). If  $I$  is countably generated, equivalently if  $\Gamma_{>0}$  has a countable coinital subset, then  $\text{pd}_A I = 1$  and  $\text{pd}_A D = 2$ .

*Proof.* (i)  $\pi\Delta = 0$  since  $x_{n+1}u_n = x_n$ ;  $\pi$  is onto.  $\Delta$  is injective: at the least index  $n_0$  of a nonzero vector the coefficient is unchanged. If  $\pi(v) = 0$ , use  $e_n a \equiv e_{n+1} u_n a$  modulo  $\text{im } \Delta$  to move  $v$  to a single index  $N$ :  $v \equiv e_N a$  with  $x_N a = \pi(v) = 0$ , so  $a = 0$ . For  $T_\Gamma$ , every tail has a positive exponent below its finitely many exponents, hence below some  $2^{-n}g$ . (ii) These ideals are nonzero and not finitely generated (Lemma 13.3(iii) and its analogue for  $T_\Gamma$ ), so they are not projective by Lemma 12.1. (iii) Every  $C_p$  of Theorem 13.4 has projective dimension at most 1 by (i); splicing (or dimension shifting along the resolution) gives  $\text{pd } D \leq d+1$ , and  $\text{pd} \geq \text{fd} = d$ . For  $T_\Gamma$  (and  $A_1 \cong T_\mathbb{Q}$ ),  $0 \rightarrow \Pi \rightarrow T_\Gamma \rightarrow D \rightarrow 0$  gives  $\text{pd } D \leq \text{pd } \Pi + 1 = 2$ ; if  $\text{pd } D \leq 1$ ,  $\Pi$  would be projective (Schanuel), against (ii). (iv)  $0 \rightarrow \Pi^{r-1} \rightarrow T_\Gamma^r \rightarrow L(M) \rightarrow 0$  gives  $\text{pd } L(M) \leq 2$ ; if  $\text{pd } L(M) \leq 1$ , the kernel  $\Pi^{r-1}$ , hence its summand  $\Pi$ , would be projective. So  $\text{pd } L(M) = 2$ , and since  $J = J_g(M) \cong L(M)$  is not projective,  $\text{pd}(T_\Gamma/J) = \text{pd } J + 1 = 3$ . Over  $A_d$  the upper bounds come from  $\text{pd } \Pi_d \leq d$  (the truncated resolution, by (i)), and the lower bounds from Theorem 13.11, since  $\text{pd} \geq \text{fd}$ . (v) If  $I$  is generated by countably many  $f_n = X^{g_n} b_n$  ((H2)), then for  $h \in \Gamma_{>0}$  the element  $X^h$  lies in  $\sum_{n \leq N} f_n A \subseteq X^g A$  with  $g = \min_{n \leq N} g_n$ , and  $X^{h-g} \in A$  forces  $g \leq h$  because  $A$  has no negative exponents: the  $g_n$  are coinital. Conversely, for a coinital decreasing sequence  $h_n$ , (H2) gives  $I = \bigcup_n X^{h_n} A$  with  $X^{h_n} = X^{h_{n+1}} X^{h_n - h_{n+1}}$ , a countable chain as in (i). So  $\text{pd } I \leq 1$ ;  $I$  is not projective by (H3) and Lemma 12.1; hence  $\text{pd } I = 1$  and, as in (iii),  $\text{pd } D = 2$ .  $\square$

Items (ii)–(v) sharpen 12’s bounds  $1 \leq \text{pd}_{A_1} D \leq 2$  and are consistent with 06’s  $\text{pd} \geq 2$  (Theorem 12.2(v)). For  $d \geq 2$  the value  $\text{pd}_{A_d} D \in \{d, d+1\}$  is not determined. Among the five constructions of Section 11, (v) applies exactly to  $A_\kappa^{08}$  with  $\text{cf } \kappa = \aleph_0$ : the tail of each construction needs as many generators as the length of its coinital chain (Remark 10.11), namely  $\text{cf } \kappa$  for 03 and 08,  $\kappa$  for 06 and 11,  $\aleph_1$  for 09, and only 08 allows countable cofinality. Question 18.3 is re-scoped accordingly.

**Theorem 13.13** (Ordered-cone cores [12]).  $(T_\Gamma, \Pi_\Gamma^{\text{fin}}, D)$  satisfies (H1)–(H3) of Section 12. Consequently  $\Pi_\Gamma^{\text{fin}} = \bigcup_{\delta \in \Gamma_{>0}} X^\delta T_\Gamma$  is flat and idempotent but not projective,  $\text{Tor}_{>0}^{\Pi_\Gamma^{\text{fin}}}(D, N) = 0$  for every  $D$ -module  $N$ ,  $\text{fd}_{T_\Gamma} D = 1$  with  $\text{Tor}_1(D, T_\Gamma/X^\gamma T_\Gamma) \cong D$ , and the other items of Theorem 12.2 hold;  $T_\Gamma \rightarrow D$  is formally étale and not flat,  $\text{pd}_{T_\Gamma} D = 2$ , and  $T_\Gamma$  is not coherent.

*Proof.* (H1) is Lemma 13.1(i). For (H2), a nonzero tail  $f$  has finitely many positive exponents; with  $g$  half the least of them,  $f = X^g(X^{-g}f)$  and  $X^{-g}f$  is again a tail, its coefficients being



arbitrary in  $K$ . (H3) holds because a finite family of tails has a least exponent among its monomials and cannot generate a monomial of smaller exponent. Apply [Theorem 12.2](#), [Proposition 13.6](#), [Proposition 13.12\(iii\)](#) and [Theorem 13.9\(iii\)](#).  $\square$

**Theorem 13.14** (Same exponent group, different dimension [12]). *Let  $(\gamma_i)_{i \in \Lambda}$  be positive reals linearly independent over  $\mathbb{Q}$ ,  $\Gamma = \sum_i \mathbb{Q}\gamma_i$ , and  $A_\Lambda \hookrightarrow T_\Gamma$  the inclusion of [Lemma 13.1\(iii\)](#). Then  $\Pi_\Lambda T_\Gamma = \Pi_\Gamma^{\text{fin}}$  and  $T_\Gamma \otimes_{A_\Lambda} D \cong D$ . If  $|\Lambda| \geq 2$  the extension is not flat:  $(X_1 A_\Lambda \cap X_2 A_\Lambda) T_\Gamma \neq X_1 T_\Gamma \cap X_2 T_\Gamma$ . The flat dimension of  $D$  drops from  $d$  to 1 in rank  $d$ , and from  $\infty$  to 1 in countable rank.*

*Proof.* For  $\gamma \in \Gamma_{>0}$  choose a positive rational  $q$  with  $q\gamma_1 < \gamma$ ; then  $kX^\gamma = X_1^q \cdot kX^{\gamma-q\gamma_1} \in \Pi_\Lambda T_\Gamma$ , so  $\Pi_\Lambda T_\Gamma = \Pi_\Gamma^{\text{fin}}$  and  $T_\Gamma \otimes_{A_\Lambda} A_\Lambda / \Pi_\Lambda \cong T_\Gamma / \Pi_\Gamma^{\text{fin}} = D$ . Relabel so that  $\gamma_1 < \gamma_2$ . Every monomial of an element of  $X_1 A_\Lambda \cap X_2 A_\Lambda$  has first and second coordinate at least 1, hence real exponent at least  $\gamma_1 + \gamma_2$ , and multiplying by  $T_\Gamma$  only adds nonnegative exponents; but  $X^{\gamma_2} = X_1 X^{\gamma_2-\gamma_1} \in X_1 T_\Gamma \cap X_2 T_\Gamma$ . A flat extension preserves finite intersections of ideals: tensor the exact sequence  $0 \rightarrow I \cap I' \rightarrow A \rightarrow A/I \oplus A/I'$  with it. The dimensions are [Corollary 13.8](#) and [Theorem 13.13](#).  $\square$

**Example 13.15** (Two directions inside  $\mathbf{Oz}$  [12]). With  $\gamma_1 = 1$ ,  $\gamma_2 = \sqrt{2}$  and  $(D, K) = (\mathbb{Z}, \mathbb{R})$ , the core  $A_2 = \mathbb{Z} + \bigoplus_{a,b \in \mathbb{Q}_{\geq 0}, a+b>0} \mathbb{R} \omega^{a+b\sqrt{2}}$  lies in the ring  $T_{\mathbb{Q}+\mathbb{Q}\sqrt{2}} \subseteq \mathbf{Oz}$ , which also contains  $\omega^{\sqrt{2}-1} \notin A_2$ ; the identity  $\omega^{\sqrt{2}} = \omega \cdot \omega^{\sqrt{2}-1}$  is the extra intersection element, and the two arithmetic quotients have flat dimensions 2 and 1. Replacing  $(\mathbb{Z}, \mathbb{R})$  by  $(\mathbb{Z}[i], \mathbb{C})$  gives the same example in  $\mathbf{Oz}[i]$ .

*Remark 13.16* (Flat dimension one is a property of the cone [merge]). The report now contains two kinds of statements about the flat dimension of the arithmetic quotient, and they concern different rings.

- $\text{fd}_A D = 1$  holds whenever (H1)–(H3) hold: the tail is a directed union of principal monomial ideals, because the positive exponents form the positive cone of a totally ordered group and every family of tails of the relevant size (finite for finite supports, of fewer than  $\kappa$  or countably many elements in the constructions, a set for the class ring) has a common monomial divisor in the ring. This covers the five constructions of [Section 11](#) ([Theorem 12.2](#)), including 06’s ring  $A_\kappa^{06}$ ; the full omnific ring in the two-universe reading, where 13 proves that the tail is flat, idempotent and not generated by a family from the smaller universe, that  $\text{Tor}_{>0}(D, D) = 0$  and that  $D$  is not flat, so  $\text{fd } D = 1$ ; 14’s  $A_0$ ; and 12’s  $T_\Gamma$  ([Theorem 13.13](#)).
- For the coordinatewise cone  $\mathbb{Q}_{\geq 0}^d$  the flat dimension is  $d$  ([Corollary 13.8](#)): under the embedding, with  $\gamma_1 < \gamma_2$ , the quotient  $\omega^{\epsilon(\gamma_2-\gamma_1)}$  of  $X_2^\epsilon$  by  $X_1^\epsilon$  lies in  $\mathbf{Oz}$  but not in  $A_d$ , so the tail of  $A_d$  is not a directed union of principal ideals; it is a sum of  $d$  flat ideals that is not flat for  $d \geq 2$ .

So “flat dimension one” must never be asserted for an arbitrary subring of  $\mathbf{Oz}$ :  $A_d \subseteq T_\Gamma \subseteq \mathbf{Oz}$ , with all exponents in one Archimedean class, have flat dimensions  $d$  and 1, and  $A_d \rightarrow T_\Gamma$  is not flat. Being a subring of a ring with small quotient flat dimension imposes no bound on the subring, and conversely the cores say nothing about the homological dimensions of the full class ring  $\mathbf{Oz}$ , which none of the sources computes (12, 14). The projective dimension of  $D$  behaves differently again: under (H1)–(H3) it is 2 when the tail is countably generated, in particular for  $T_\Gamma$ , and is otherwise known only to be at least 2; for  $A_d$  with  $d \geq 2$  it is  $d$  or  $d+1$  ([Proposition 13.12](#)).



### 13.6 The tensor normal form

For the rest of this section (14) fix  $(D, k) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , and let  $(A, \Pi, B)$  be either the class rings  $(\mathcal{A}_{D,k}, \Pi_k, \mathcal{A}_{k,k})$ , that is  $A = \mathbf{Oz}$  or  $\mathbf{Oz}[i]$ , or the set-sized model  $(T_{\mathbb{Q}}, \Pi_{\mathbb{Q}}^{\text{fin}}, k + \Pi_{\mathbb{Q}}^{\text{fin}})$  with  $T_{\mathbb{Q}} = A_0$ . Coefficient lattices are those of Setting (L). Every result below holds in both cases (14); its proofs use only that  $D$  is a principal ideal domain in  $k$ , that  $A = D \oplus \Pi$  with  $\Pi$  a  $k$ -linear ideal of  $B$ , and that every  $u \in \Pi$  is a product  $u = ca$  with  $c, a \in \Pi$  (Proposition 3.3(ii); in  $T_{\mathbb{Q}}$  split off a smaller monomial). This interface is 14's proposal for a formalization (Section 19.2). For the class rings a tensor product means an explicit carrier with the balanced universal property proved by finite identities (for two lattices,  $\Pi \times (M \otimes_D N)$ ); no quotient of a proper class by a class relation and no class colimit is used (14).

**Lemma 13.17** (Absorption [14]). *For every lattice  $N$  and  $0 \neq n_0 \in N$ , multiplication is an isomorphism  $\Pi \otimes_A L(N) \cong \Pi$  with inverse  $u \mapsto (u/n_0) \otimes n_0$ ; indeed  $u \otimes v = (uv/n_0) \otimes n_0$  for  $u \in \Pi$ ,  $v \in L(N)$ .*

*Proof.* Write  $u = ca$  with  $c, a \in \Pi$ . Since  $v/n_0 \in B$ ,  $b = a(v/n_0) \in \Pi \subseteq A$ , and  $u \otimes v = c \otimes av = c \otimes bn_0 = cb \otimes n_0 = (uv/n_0) \otimes n_0$ . So every finite tensor sum is the image of its product, multiplication is onto ( $u = (u/n_0)n_0$ ), and the two maps are mutually inverse. For a balanced bilinear map  $b'$  into a class module,  $u \mapsto b'(u/n_0, n_0)$  is the required factorization, unique by the same identity.  $\square$

**Theorem 13.18** (Exact tensor normal form [14]). *The map  $\beta$  of Proposition 13.10 is an  $A$ -module isomorphism  $L(M) \otimes_A L(N) \cong \mathbf{E}(M, N)$ . Multiplication corresponds to  $(u, t) \mapsto u + \mathbf{m}_{M,N}(t)$ , and there is an exact sequence*

$$0 \longrightarrow \mathcal{T}(M, N) \longrightarrow L(M) \otimes_A L(N) \longrightarrow L(MN) \longrightarrow 0, \quad \mathcal{T}(M, N) = \ker(M \otimes_D N \rightarrow MN),$$

*with  $\mathcal{T}(M, N)$  embedded as  $(0, \mathcal{T}(M, N))$ . This is exactly the  $A$ -torsion submodule; if it is nonzero its annihilator is exactly  $\Pi$ , and  $\text{rank}_D \mathcal{T}(M, N) = \text{rank } M \cdot \text{rank } N - \text{rank}(MN)$ .*

*Proof.* Fix  $0 \neq n_0 \in N$ , let  $j(w) = (w/n_0) \otimes n_0$  on  $\Pi$  and let  $s$  be induced by the constant inclusions on  $M \otimes_D N$ . Then  $(u, t) \mapsto j(u) + s(t)$  is inverse to  $\beta$ : in an expanded pure tensor the terms with an infinite first factor reduce by Lemma 13.17, a term  $m \otimes v$  with  $v = ca \in \Pi$  equals  $cm \otimes a$ , and all mixed terms become  $j(mv + un + uv)$ , while the constant term is  $s(m \otimes n)$ . The same finite identities give the universal property for balanced maps into class modules, so this carrier is the tensor product in the class case. The image of multiplication is  $MN + \Pi = L(MN)$ , and since  $\Pi \cap k = 0$  its kernel is  $(0, \mathcal{T}(M, N))$ , killed by  $\Pi$  by (13.4); the quotient lies in a field and is torsion free, so the kernel is all the torsion. It is a submodule of a finite free  $D$ -module, hence free, and a nonzero  $d \in D$  kills none of its nonzero elements. Ranks:  $M \otimes_D N \rightarrow MN$  is onto.  $\square$

For  $M = D + D\alpha$  with  $\alpha \notin \text{Frac } D$ ,  $\tau = 1 \otimes \alpha - \alpha \otimes 1$  is a nonzero tensor with  $h\tau = 0$  for all  $h \in \Pi$ ; if  $\alpha^2 = c \in D$ , then  $\tau$  and  $\alpha \otimes \alpha - c(1 \otimes 1)$  form a  $D$ -basis of the multiplication kernel (14). Here “torsion” means  $A$ -torsion:  $\mathcal{T}(M, N)$  is a free abelian group (14).

**Proposition 13.19** (Several factors [14]). *For lattices  $M_1, \dots, M_n$ ,  $\bigotimes_{j,A} L(M_j) \cong \Pi \oplus \bigotimes_{j,D} M_j$  with the action (13.4) for the  $n$ -fold product map; the torsion is its kernel, killed by  $\Pi$ . For equal factors, permutations act trivially on the first coordinate and by permuting tensor factors on the second.*

*Proof.* Expand a pure tensor: the all-constant term keeps its  $D$ -tensor coordinate, and every other term contains a factor from  $\Pi$  and reduces, by repeated absorption (moving a tail scalar to an adjacent factor), to its product in one copy of  $\Pi$ . So the multilinear map has coordinates  $(\prod_j (m_j + u_j) - \prod_j m_j, \otimes_j m_j)$ , with inverse as in [Theorem 13.18](#).  $\square$

**Theorem 13.20** (Exterior and symmetric powers [14]). *Let  $r = \text{rank } M$ .*

- (i) *For  $j \geq 2$ ,  $\bigwedge_A^j L(M) \cong \bigwedge_D^j M$  with  $A$  acting through  $\text{ct}$ : a free  $D$ -module of rank  $\binom{r}{j}$  with annihilator  $\Pi$  when nonzero.*
- (ii) *For  $n \geq 1$ ,  $\text{Sym}_A^n L(M) \cong \Pi \oplus \text{Sym}_D^n M$  with the action (13.4) for the product map  $\mathbf{m}_n : \text{Sym}_D^n M \rightarrow k$ ; the product map has image  $L(M^n)$  ( $M^n$  the  $D$ -span of  $n$ -fold products) and torsion kernel  $\ker \mathbf{m}_n$ , of rank  $\binom{r+n-1}{n} - \text{rank}(M^n)$ , annihilated by  $\Pi$ .*

*Proof.* (i) For  $u \in \Pi$  and  $0 \neq m \in M$ ,  $u/m \in A$  and  $u \wedge m = (u/m)m \wedge m = 0$ ; for  $u, v \in \Pi$  both are multiples of one  $m_0$ , so  $u \wedge v = 0$ . Hence every wedge with a tail factor vanishes,  $h(m_1 \wedge \cdots \wedge m_j) = 0$  for  $h \in \Pi$ , the constant inclusion induces a surjection  $\bigwedge_D^j M \rightarrow \bigwedge_A^j L(M)$ , and taking constants in all factors is an alternating  $A$ -multilinear left inverse. (ii) By [Proposition 13.19](#) a permutation difference has the form  $(0, t - \sigma t)$ , with product zero, so multiplying it by an element of  $\Pi$  gives zero and imposes no relation on the first coordinate; the quotient is  $\Pi \oplus \text{Sym}_D^n M$ , and the kernel argument of [Theorem 13.18](#) applies.  $\square$

A rank-one ideal of  $\mathbf{Oz}$  can thus have a nonzero  $r$ th exterior power for every  $r \geq 2$ ; these powers vanish after passing to the fraction field, in agreement with the field-valued rank (14).

### 13.7 Rees equations

For  $e > 0$  the Rees algebra of  $L(M)$  is  $\mathcal{R}(L(M)) = A \oplus \bigoplus_{n \geq 1} L(M)^n T^n \subseteq F[T]$  ( $F$  the fraction field); degreewise multiplication by  $X^{ne}$  identifies it, and the symmetric algebra, with those of the ideal  $J_e(M)$  [23, Tag 052P]. An ideal is *of linear type* if  $\text{Sym} \rightarrow \mathcal{R}$  is an isomorphism, and the *first nonlinear degree* is the least degree of a nonzero element of the kernel after the linear relations are imposed (14).

**Theorem 13.21** (All nonlinear Rees equations are arithmetic [14]). *Let  $m_1, \dots, m_r$  be a basis of  $M$  and  $\mathfrak{q}_M = \ker(D[X_1, \dots, X_r] \rightarrow k[T], X_i \mapsto m_i T)$ . The constant-coefficient map  $D[X_1, \dots, X_r] \rightarrow \text{Sym}_A L(M)$  is injective, and under it  $\ker(\text{Sym}_A L(M) \rightarrow \mathcal{R}(L(M))) = \mathfrak{q}_M$ ; this is all the  $A$ -torsion of the symmetric algebra, it is killed by  $\Pi$ , and it is generated by any set of homogeneous generators of  $\mathfrak{q}_M$ , finitely many of which suffice. With  $\mathfrak{l}_M = (hX_j - \frac{m_j}{m_1} hX_1 : h \in \Pi, 2 \leq j \leq r) \subseteq A[X_1, \dots, X_r]$ ,*

$$\text{Sym}_A L(M) \cong A[\mathbf{X}]/\mathfrak{l}_M, \quad \mathcal{R}(L(M)) \cong A[\mathbf{X}]/(\mathfrak{l}_M + \mathfrak{q}_M A[\mathbf{X}]).$$

*Proof.* The symmetric algebra is presented by the linear relations of the generator map [22, Tag 00DM], which are those of [Theorem 13.9\(ii\)](#). In degree  $n \geq 1$ , [Theorem 13.20\(ii\)](#) identifies the constant polynomials with  $\text{Sym}_D^n M$  and the kernel with  $\ker \mathbf{m}_n$ , the degree- $n$  part of  $\mathfrak{q}_M$ ; in degree 0 the map is the identity of  $A$ . The quotient embeds in the domain  $F[T]$ , so the kernel is all the torsion. A polynomial in  $A[\mathbf{X}]$  acts on the kernel through its coefficientwise constant term, so generators over  $D[\mathbf{X}]$  generate the kernel, and  $D[\mathbf{X}]$  is Noetherian.  $\square$

The nonlinear kernel is finitely generated while, for  $r \geq 2$ , the linear relation module is not finitely (set-) generated ([Theorem 13.9](#)); the two statements are compatible (14).

**Theorem 13.22** (The algebraic degree is the first nonlinear degree [14]). *Let  $\alpha \in k \setminus \text{Frac}(D)$ ,  $M = D + D\alpha$  and  $J = (X^e, \alpha X^e) \subseteq A$ .*

- (i) *If  $\alpha$  is transcendental over  $\text{Frac}(D)$ , then  $J$  is of linear type, although it is not flat and not finitely presented.*
- (ii) *If  $\alpha$  is algebraic of degree  $d$  (necessarily  $d \geq 2$ ), let  $f \in D[Z]$  be primitive irreducible with  $f(\alpha) = 0$  and  $F_f(X, Y) = X^d f(Y/X)$ . Then  $\mathcal{R}(J) \cong A[X, Y]/(hY - \alpha hX \ (h \in \Pi), F_f(X, Y))$ ; the kernel of  $\text{Sym}_A J \rightarrow \mathcal{R}(J)$  is generated by  $F_f$ , and the first nonlinear degree is exactly  $d$ .*

*Proof.* A homogeneous  $G$  of degree  $n$  lies in  $\mathfrak{q}_M$  iff  $G(1, \alpha) = 0$ . In the transcendental case  $G = 0$ . In the algebraic case the minimal polynomial of  $\alpha$  divides  $g(Z) = G(1, Z)$  over  $\text{Frac } D$ , and Gauss's lemma for the unique factorization domain  $D$  and the primitivity of  $f$  give  $g = fh$  with  $h \in D[Z]$ ,  $\deg h \leq n - d$ ; homogenizing,  $G = F_f \cdot X^{n-d} h(Y/X)$ . Conversely  $F_f(1, \alpha) = 0$ . Apply Theorem 13.21; nonflatness and failure of finite presentation are Proposition 13.10 and Theorem 13.9.  $\square$

**Corollary 13.23** (A family of every degree [14]). *For  $d \geq 2$  let  $J_d = (\omega, 2^{1/d}\omega) \subseteq \mathbf{Oz}$ . Then  $\mathcal{R}_{\mathbf{Oz}}(J_d) \cong \mathbf{Oz}[X, Y]/(hY - 2^{1/d}hX \ (h \in \Pi), Y^d - 2X^d)$ , there is no nonlinear equation in degrees  $2, \dots, d - 1$ , and for  $n \geq 1$*

$$\begin{aligned} \mu_{\mathbf{Oz}}(J_d^n) &= \min(n + 1, d), & \text{rank}_{\mathbb{Z}} \ker(\text{Sym}_{\mathbf{Oz}}^n J_d \rightarrow J_d^n) &= \max(0, n - d + 1), \\ \text{rank}_{\mathbb{Z}} \ker(J_d^{\otimes n} \rightarrow J_d^n) &= 2^n - \min(n + 1, d), \end{aligned}$$

where  $\mu$  is the least number of generators. All these kernels are killed by  $\Pi$ .

*Proof.*  $Z^d - 2$  is irreducible by Eisenstein at 2, so  $1, \alpha, \dots, \alpha^{d-1}$  are independent and higher powers reduce by  $\alpha^d = 2$ ; the lattice spanned by  $1, \alpha, \dots, \alpha^n$  has rank  $\min(n + 1, d)$ , and  $J_d^n$  is the monomial scaling of its lattice module. Apply Theorem 13.9(i), Theorem 13.20(ii) and Proposition 13.19.  $\square$

For  $d = 5$  the two-generated ideal  $(\omega, 2^{1/5}\omega)$  has powers needing 2, 3, 4, 5, 5,  $\dots$  generators, a symmetric kernel of ranks 0, 0, 0, 0, 1, 2,  $\dots$ , and a nonzero antisymmetric tensor already in its tensor square: symmetric and tensor obstructions occur at different stages (14).

### 13.8 Homomorphisms, ideal classes and duals

For lattices put  $(N : M) = \{c \in k : cM \subseteq N\}$  and  $\mathfrak{o}(M) = (M : M)$ .

**Theorem 13.24** (Scalar homomorphisms and isomorphism types [14]). *Identifying a scalar with multiplication by it,  $\text{Hom}_A(L(M), L(N)) = \Pi + (N : M)$ , composition is multiplication, and  $L(M) \cong L(N)$  if and only if  $N = cM$  for some  $c \in k^\times$ . Hence  $\text{End}_A L(M) = \Pi + \mathfrak{o}(M)$  and  $\text{Aut}_A L(M) = \mathfrak{o}(M)^\times$ . The ring  $\mathfrak{o}(M)$  is finite free over  $D$  and integral over  $D$ ; in the real case  $\mathbb{Q}\mathfrak{o}(M) \subseteq \mathbb{R}$  is a number field of degree at most  $\text{rank } M$  (not necessarily totally real).*

*Proof.* An  $A$ -linear map  $\varphi$  between nonzero  $A$ -submodules of the fraction field is multiplication by a scalar: for nonzero  $a, b$  in a common ideal (after a monomial denominator),  $a\varphi(b) = b\varphi(a)$ . If  $qL(M) \subseteq L(N)$  and  $0 \neq m_0 \in M$ , then  $q = (qm_0)/m_0 \in B$ ; write  $q = c + h$  with  $c \in k$ ,  $h \in \Pi$ ; the inclusion is equivalent to  $cM \subseteq N$ . An isomorphism has  $q$  and  $q^{-1}$  in  $B$ , so  $q \in B^\times = k^\times$  (Lemma 2.4), and nonzero constants preserve  $\Pi$ . Multiplication by  $m_0$  embeds  $\mathfrak{o}(M)$  in  $M$ ; Cayley–Hamilton for the matrix of multiplication by  $c$  on a basis of  $M$  gives integrality; in the real case the rational span is a finite-dimensional domain, hence a field, of dimension at most  $\text{rank } M$ .  $\square$

Let  $K_0 \subset \mathbb{R}$  be a number field and  $\mathfrak{o} \subset K_0$  an order. An invertible fractional  $\mathfrak{o}$ -ideal is a lattice, and for these modules put  $U \odot V = (U \otimes_A V)/(\text{elements killed by a nonzero element of } A)$ ; by [Theorem 13.18](#),  $L(M) \odot L(N) \cong L(MN)$ .

**Theorem 13.25** (Ideal classes survive torsion-free multiplication [14]).  *$P \mapsto L(P)$  distinguishes exactly the ideal classes of invertible fractional  $\mathfrak{o}$ -ideals, and under  $\odot$  these isomorphism types form a group isomorphic to  $\text{Pic}(\mathfrak{o})$  with identity  $[L(\mathfrak{o})]$ . If  $[K_0 : \mathbb{Q}] \geq 2$  this is not a subgroup of  $\text{Pic}(A)$  and its identity is not  $[A]$ .*

*Proof.* If  $L(P) \cong L(Q)$  then  $Q = cP$  with  $c \in \mathbb{R}^\times$  ([Theorem 13.24](#)), and  $c = (cp)/p \in K_0^\times$  for  $0 \neq p \in P$ ; the converse is clear.  $\odot$  corresponds to ideal multiplication,  $\mathfrak{o}P = P$  and  $PP^{-1} = \mathfrak{o}$ . For  $[K_0 : \mathbb{Q}] \geq 2$ ,  $L(P)$  needs  $[K_0 : \mathbb{Q}]$  generators and is not flat ([Proposition 13.10](#)), so it is not invertible, and  $L(\mathfrak{o})$  needs two or more generators while  $A = L(\mathbb{Z})$  needs one.  $\square$

The group has its own idempotent identity inside a larger multiplicative semigroup; it is not a unital homomorphism into the Picard group of  $A$ . In the class setting the isomorphism is read as an explicit set-indexed family of representatives with its equivalence criterion (14).

**Example 13.26** (A class of order two [14]). In  $K_0 = \mathbb{Q}(\sqrt{10})$  with  $\mathfrak{o} = \mathbb{Z} + \mathbb{Z}\sqrt{10}$  let  $P = 2\mathbb{Z} + \mathbb{Z}\sqrt{10}$ . It is an  $\mathfrak{o}$ -ideal,  $P^2 = 4\mathbb{Z} + 2\sqrt{10}\mathbb{Z} + 10\mathbb{Z} = 2\mathfrak{o}$ , so  $P$  is invertible. It is not principal:  $P = c\mathfrak{o}$  with  $c = a + b\sqrt{10} \in P$  would give  $|a^2 - 10b^2| = [\mathfrak{o} : P] = 2$ , impossible modulo 5. Hence  $L(P) \not\cong L(\mathfrak{o})$  and  $L(P) \odot L(P) \cong L(\mathfrak{o})$ ; both have rank one and need two generators, and the tensor square of  $L(P)$  has torsion of rank  $4 - 2 = 2$ . As ideals,  $\omega L(P) = (2\omega, \sqrt{10}\omega)$  and  $\omega L(\mathfrak{o}) = (\omega, \sqrt{10}\omega)$ .

**Theorem 13.27** (Uniform duals [14]).  *$(A : \Pi) = B$  and  $(A : B) = \Pi$  (in both cases of this subsection). For a lattice  $M$  of rank at least two,  $L(M)^* = \text{Hom}_A(L(M), A) = \Pi$ ,  $L(M)^{**} = B$ , the canonical bidual map is the inclusion  $L(M) \hookrightarrow B$ , and  $B/L(M) \cong k/M$  with  $A$  acting through  $\text{ct}$ . Moreover  $\Pi^* = B$ ,  $B^* = \Pi$ ,  $\Pi^{**} = \Pi$ ,  $B^{**} = B$ , and  $J_e(M)^* = X^{-e}\Pi$ ,  $J_e(M)^{**} = X^e B$ .*

*Proof.* If  $q\Pi \subseteq A$  and  $q$  had a negative exponent  $\gamma$ , then for  $0 < \delta < -\gamma$  the coefficient of  $X^{\gamma+\delta}$  in  $qX^\delta$  would be the same nonzero coefficient; so  $q \in B$  (in  $T_{\mathbb{Q}}$ , first  $qX^1 \in T_{\mathbb{Q}}$  shows that  $q$  has finite rational support). If  $qB \subseteq A$  then  $q = d + u \in A$  and  $dk \subseteq D$  forces  $d = 0$ . By [Theorem 13.24](#),  $L(M)^* = \Pi + (D : M)$ , and  $(D : M) = 0$  because a nonzero  $c$  would embed  $M$  in  $D$ . Evaluation at  $x$  sends  $u \in \Pi$  to  $ux$ , so the bidual scalar of  $x$  is  $x$ . Scaling by  $X^e$  reverses in the first dual and restores in the second.  $\square$

So the two nonisomorphic modules of [Example 13.26](#) have the same dual and reflexive hull: their difference lies below the hull and is seen by coefficient arithmetic, not by rank or double dualization (14).

**Theorem 13.28** (Common-scale constant matrices [14]). *Let  $C \in k^{m \times n}$ ,  $e > 0$ ,  $W = \ker_k C$ ,  $q = \dim_k W$ ,  $W_D = W \cap D^n$  and  $\ell = \text{rank}_D W_D$ . Then  $\ker_A(X^e C) \cong A^\ell \oplus \Pi^{q-\ell}$ . The image of  $X^e C : A^n \rightarrow A^m$  is finitely presented if and only if  $\ell = q$ , that is, if and only if  $W$  is defined over  $\text{Frac}(D)$ , and it is then free of rank  $n - q$ .*

*Proof.* For  $x = d + u$ ,  $Cx = 0$  splits into  $Cd = 0$  and  $Cu = 0$  (constants against tails).  $W_D$  is saturated in  $D^n$ , hence a direct summand; a  $D$ -basis of it is  $k$ -independent and extends to a  $k$ -basis of  $W$ ; in these coordinates the kernel is  $A^\ell \oplus \Pi^{q-\ell}$  (Gaussian elimination with constants preserves  $\Pi$ ). If  $q > \ell$  the kernel projects onto  $\Pi$  and is not finitely (set-) generated. If  $q = \ell$ , a constant change of basis over  $D$  makes the first  $q$  columns zero and the others independent over the fraction field.  $\square$

The matrix theorem concerns constant matrices at one common monomial scale; several support levels need more analysis (14). Its companion, 14's criterion that a point  $[r_1 : \cdots : r_n] \in \mathbb{P}^{n-1}(k)$  has a unimodular representative in  $A^n$  if and only if it is  $\text{Frac}(D)$ -rational, is in the real case the rationality theorem for primitive constant directions of the Diophantine report [30], which 14 does not credit (it credits only that report's example  $[1 : \sqrt{2}]$ , to a companion manuscript); the Gaussian and principal-ideal form is printed there with that theorem, not here.

**The Gaussian case and the reach of the results (14).** All proofs above use only a principal ideal domain  $D \subset k$  and the common-factor property, so the Gaussian versions over  $\mathbf{Oz}[i] = \mathbb{Z}[i] + \Pi_{\mathbb{C}}$  are proved, not merely analogous: ranks are  $\mathbb{Z}[i]$ -ranks, multiplication kernels are killed by  $\Pi_{\mathbb{C}}$ , and in Theorem 13.22 the degree is taken over  $\mathbb{Q}(i)$ . For real algebraic  $\alpha$  it equals the degree over  $\mathbb{Q}$  ( $[K_0(i) : \mathbb{Q}(i)] = [K_0 : \mathbb{Q}]$  for  $K_0 \subset \mathbb{R}$ ), so Corollary 13.23 holds over  $\mathbf{Oz}[i]$  with the same ranks over  $\mathbb{Z}[i]$ ; for  $\sqrt{2} \notin \mathbb{Q}(i)$  the ideal  $(\omega, \sqrt{2}\omega) \subseteq \mathbf{Oz}[i]$  has the single quadratic equation  $Y^2 - 2X^2$ . Complex conjugation transports the construction for  $M$  to that for  $\overline{M}$ . These results concern the omnific *rings*: over the surcomplex field itself all these module calculations are vector-space calculations and the arithmetic torsion vanishes. They do not classify arbitrary finitely generated omnific ideals, whose generators may lie at several support levels.

## 14 Large internal fields and the binomial quotient

The universal theorem (Theorem 5.1) describes what a set-sized target can see of  $\mathbf{Oz}$ : every purely infinite element dies, and only the constant term survives. This section, taken from manuscript 07 together with one proposition of manuscript 06, shows how much survives in a large target once a single purely infinite element becomes a *unit*. The answer is a great deal: the target then contains a real closed Hahn field into which all of  $\mathbf{No}$  embeds as an ordered field. In the nonzero binomial quotient  $\mathcal{Q}_a = \mathbf{Oz}/(1 + \omega^a)$ , which has no nonzero set-sized ring image at all, the same mechanism produces an algebraically closed class field containing a copy of  $\mathbf{No}[i]$ , exact finite cyclotomic algebras over it, and an atomless Boolean algebra of idempotents.

The dependence of the results is as follows (07). An ordered additive compression of  $\mathbf{No}$  and the closedness of set-sized Hahn fields give a copy of  $\mathbf{No}$  inside the compressed field  $\mathcal{F}_a$  (Proposition 14.1 and Theorem 14.4); localization transfers  $\mathcal{F}_a$  into every quotient in which a suitable purely infinite element is invertible (Theorem 14.7); a support argument gives exact binomial kernels (Theorem 14.15); and finite Chinese remainder decompositions give the internal field  $\mathcal{E}_a$  and the Cantor algebra (Corollary 14.17 and Theorem 14.19). The absence of small images of  $\mathcal{Q}_a$  comes from the other branch, Theorems 5.1 and 6.1. No factorization conjecture enters either chain. What manuscript 07 calls its Main Theorem B is Theorem 14.7 together with Theorem 14.4, and its Main Theorem C is the combination of Lemma 14.14, Corollary 14.17, and Theorem 14.19 with the absence of small images of  $\mathcal{Q}_a$  recorded before Lemma 14.14.

### 14.1 Compressed exponent fields

Fix  $a > 0$ . Recall from (2.4) and Lemma 4.2 that  $H_a = \{h \in \mathbf{No} : n|h| < a \text{ for all ordinary } n \geq 1\}$  is a convex divisible subgroup of  $(\mathbf{No}, +)$  containing a proper class of positive elements, and that  $\mathcal{F}_a(k) = k((X^{H_a}))$  is a field for every set field  $k$ . For  $k = \mathbb{R}$  and  $k = \mathbb{C}$  we realize these fields inside



the surreals and surcomplex numbers,

$$\begin{aligned}\mathcal{F}_a &= \mathcal{F}_a(\mathbb{R}) = \{f \in \mathbf{No} : \text{supp } f \subseteq H_a\}, \\ \mathcal{F}_a(i) &= \mathcal{F}_a(\mathbb{C}) = \{f \in \mathbf{No}[i] : \text{supp } f \subseteq H_a\}.\end{aligned}\tag{14.1}$$

Taking real and imaginary parts of coefficients shows  $\mathcal{F}_a(i) = \mathcal{F}_a + i\mathcal{F}_a$ . Manuscript 07 calls them *compressed exponent fields*; the phrase refers to the restriction on exponents, not to finiteness of supports. Both are proper classes. Manuscript 07 checks directly that  $H_a$  is a subgroup by the inequality  $n|h + h'| \leq n|h| + n|h'| < a/2 + a/2$ , using the defining inequalities at  $2n$ ; this is part of Lemma 4.2.

**Proposition 14.1** (Closedness of the compressed fields [07]). *The class  $\mathcal{F}_a$  is a real closed subfield of  $\mathbf{No}$ , and  $\mathcal{F}_a(i)$  is an algebraically closed subfield of  $\mathbf{No}[i]$ .*

*Proof.* Since  $H_a$  is a subgroup, sums and products of series supported in  $H_a$  are supported in  $H_a$ ; the inverse of a nonzero Hahn series has support in the subgroup generated by its own support (factor out the leading monomial and use the Hahn geometric series), so it stays in  $H_a$  as well. Hence  $\mathcal{F}_a$  and  $\mathcal{F}_a(i)$  are subfields (Lemma 4.2).

Let  $P$  be a nonconstant polynomial with coefficients in  $\mathcal{F}_a(i)$ . The union of the supports of its finitely many coefficients is a set; its rational span is a set-sized divisible ordered subgroup  $G \subseteq H_a$ , because  $H_a$  is a divisible subgroup. The series with support in  $G$  form the set-sized Hahn fields  $\mathbb{R}((\omega^G)) \subseteq \mathcal{F}_a$  and  $\mathbb{C}((\omega^G)) \subseteq \mathcal{F}_a(i)$ , whose operations are those of  $\mathbf{No}$  and  $\mathbf{No}[i]$ , and  $P$  has coefficients in  $\mathbb{C}((\omega^G))$ . For divisible  $G$  this field is algebraically closed by the Hahn-field theorem [18, Corollary 4] (stated there for well-ordered supports; the substitution  $t^g = \omega^{-g}$  converts). So  $P$  has a root in  $\mathbb{C}((\omega^G)) \subseteq \mathcal{F}_a(i)$ , and  $\mathcal{F}_a(i)$  is algebraically closed.

For the real field, note first that  $\mathbb{R}((\omega^G))$  is real closed. If  $0 < f \in \mathbb{R}((\omega^G))$ , write a square root of  $f$  in  $\mathbb{C}((\omega^G))$  as  $u + iv$  with  $u, v \in \mathbb{R}((\omega^G))$ ; then  $u^2 - v^2 = f$  and  $2uv = 0$ . If  $u = 0$  then  $f = -v^2 \leq 0$ , which is impossible; so  $v = 0$  and  $u^2 = f$ . If  $Q$  has odd degree and coefficients in  $\mathbb{R}((\omega^G))$ , coefficientwise conjugation is an automorphism of  $\mathbb{C}((\omega^G))$  fixing  $Q$ , so the nonreal roots of  $Q$  occur in conjugate pairs with equal multiplicities, and an odd number of roots counted with multiplicity must be real. An ordered field in which positive elements are squares and odd-degree polynomials have roots is real closed. Now let  $0 < f \in \mathcal{F}_a$ , or let  $Q$  be of odd degree over  $\mathcal{F}_a$ ; choosing  $G$  as above for  $f$  or for the coefficients of  $Q$ , the required square root or root lies in  $\mathbb{R}((\omega^G)) \subseteq \mathcal{F}_a$ . Hence  $\mathcal{F}_a$  is real closed. Only set-sized instances of the classical theorem have been used.  $\square$

*Remark 14.2* (Bounded in the ambient field, yet a proper class [07]). Every positive  $f \in \mathcal{F}_a$  satisfies  $\omega^{-a} < f < \omega^a$ : its leading exponent  $h$  satisfies  $|h| < a/2$  and its leading coefficient is a positive real. A field contained in such an ambient interval need not be small, since the interval is itself a proper class;  $\mathcal{F}_a$  is a proper class, and by Theorem 14.4 it even contains an ordered-field copy of  $\mathbf{No}$ . Bounds in the surrounding field do not give bounds inside an isomorphic copy of  $\mathbf{No}$ .

## 14.2 An ordered additive compression of the surreals

Knowing that  $\mathcal{F}_a$  contains Hahn fields of every set size (Lemma 4.2) would already show considerable largeness. Manuscript 07 does more: it embeds all of  $\mathbf{No}$  into  $\mathcal{F}_a$  as an ordered field. Define

$$\vartheta(x) = \frac{1}{2} \left( \frac{x}{1 + |x|} - 3 \right), \quad \vartheta : \mathbf{No} \longrightarrow (-2, -1).\tag{14.2}$$



For  $x \geq 0$  the inner fraction is  $x/(1+x) = 1 - 1/(1+x) \in [0, 1)$ , and for  $x < 0$  it is  $x/(1-x) = -1 + 1/(1-x) \in (-1, 0)$ . Each branch is strictly increasing and they agree at 0, so  $\vartheta$  is strictly increasing with values in  $(-2, -1)$ . For a real-coefficient normal form put

$$\Psi\left(\sum_{\gamma} r_{\gamma} \omega^{\gamma}\right) = \sum_{\gamma} r_{\gamma} \omega^{\vartheta(\gamma)}, \quad \Psi_a(x) = a\Psi(x). \quad (14.3)$$

(Manuscript 07 writes  $E_a$  for  $\Psi_a$ ; the letter  $\mathcal{E}_a$  is reserved here for the internal field of [Corollary 14.17](#).)

**Lemma 14.3** (Ordered additive compression [07]). *The map  $\Psi_a : (\mathbf{No}, +) \rightarrow (H_a, +)$  is an injective, order-preserving,  $\mathbb{R}$ -linear homomorphism of additive groups.*

*Proof.* A strictly increasing map sends a reverse well-ordered set to a reverse well-ordered set, so the support  $\vartheta(\text{supp } x)$  is a reverse well-ordered set and  $\Psi(x)$  exists. Since  $\vartheta$  is injective, distinct exponents stay distinct, and the coefficient of  $\Psi(x)$  at  $\vartheta(\gamma)$  is  $x_{\gamma}$ ; hence coefficientwise addition, with all cancellations, and multiplication by real scalars are preserved. If  $x \neq 0$ , the greatest exponent  $\gamma_0$  of  $\text{supp } x$  is sent to the greatest exponent  $\vartheta(\gamma_0)$  of  $\text{supp } \Psi(x)$  with the same coefficient; since the sign of a nonzero normal form is the sign of its leading coefficient,  $\Psi$  maps positive elements to positive elements, and being additive it is injective and order preserving. The leading exponent of a nonzero  $\Psi(x)$  lies in  $(-2, -1)$ , so  $\Psi(x)$  is infinitesimal and  $n|\Psi(x)| < 1$  for every ordinary  $n \geq 1$ . Multiplying by  $a > 0$  gives  $n|\Psi_a(x)| < a$ , that is,  $\Psi_a(x) \in H_a$ , and multiplication by the positive surreal  $a$  preserves additivity,  $\mathbb{R}$ -linearity and order.  $\square$

The map  $\vartheta$  is not additive, and  $\Psi$  is not asserted to be multiplicative. The construction deliberately has two levels: first an additive exponent group is compressed, and only then is the additive map used as a substitution on monomials.

**Theorem 14.4** (A full surreal field in every compressed field [07]). *The formula*

$$\Theta_a\left(\sum_{\gamma} r_{\gamma} \omega^{\gamma}\right) = \sum_{\gamma} r_{\gamma} \omega^{\Psi_a(\gamma)} \quad (14.4)$$

*defines an injective ordered-field homomorphism  $\Theta_a : \mathbf{No} \hookrightarrow \mathcal{F}_a$  fixing  $\mathbb{R}$ . It preserves every set-indexed strongly summable family. Extending coefficients,  $x + iy \mapsto \Theta_a(x) + i\Theta_a(y)$ , gives an injective field homomorphism  $\mathbf{No}[i] \hookrightarrow \mathcal{F}_a(i)$  fixing  $\mathbb{C}$ .*

*Proof.* The support of the right side of (14.4) is  $\Psi_a(\text{supp } x)$ , the image of a reverse well-ordered set under a strictly increasing map into  $H_a$  ([Lemma 14.3](#)); so the output exists and lies in  $\mathcal{F}_a$ . If  $(x_{\lambda})$  is a strongly summable set-indexed family, the union of the supports is reverse well ordered and each exponent receives finitely many contributions; since  $\Psi_a$  is injective and strictly increasing, the same holds for the family  $(\Theta_a(x_{\lambda}))$  with the same contributions at  $\Psi_a(\gamma)$  as at  $\gamma$ , so  $\Theta_a(\sum_{\lambda} x_{\lambda}) = \sum_{\lambda} \Theta_a(x_{\lambda})$ .

For multiplication, write  $x = \sum r_{\gamma} \omega^{\gamma}$  and  $y = \sum s_{\delta} \omega^{\delta}$ . Because  $\Psi_a$  is additive and injective,  $\Psi_a(\gamma) + \Psi_a(\delta) = \Psi_a(\varepsilon)$  holds exactly when  $\gamma + \delta = \varepsilon$ , and every sum  $\Psi_a(\gamma) + \Psi_a(\delta)$  lies in the image of  $\Psi_a$ . Hence the coefficient of  $\Theta_a(x)\Theta_a(y)$  at  $\Psi_a(\varepsilon)$  is  $\sum_{\gamma+\delta=\varepsilon} r_{\gamma}s_{\delta} = (xy)_{\varepsilon}$ , and its coefficients at exponents outside the image vanish; so  $\Theta_a(xy) = \Theta_a(x)\Theta_a(y)$ . Additivity is clear, and constants are fixed because  $\Psi_a(0) = 0$ . A nonzero  $x$  has a nonzero image whose leading coefficient is that of  $x$ , so  $\Theta_a$  is injective and preserves signs, hence order. It is an injective unital ring map between fields, hence a field embedding. The complex extension is the same coefficient-and-support argument with complex coefficients.  $\square$

**Example 14.5** ([07]). Since  $\vartheta(0) = -3/2$ , one has  $\Psi_a(1) = a\omega^{-3/2}$  and

$$\Theta_a(\omega) = \omega^{a\omega^{-3/2}}.$$

No preservation of the ordinal embedding, of birthdays or simplicity, or of the surreal exponential is asserted or needed:  $\Theta_a$  is an explicit ordered-field embedding with a controlled normal-form support image. Constructions of self-similar surreal subclasses and strongly linear maps have a substantial prior literature, notably the surreal substructures of Bagayoko and van der Hoeven [2]. Manuscript 07 uses the elementary construction above to make the transfer theorem independent of a general classification of surreal substructures, and does not claim that surreal self-similarity itself is new.

### 14.3 Transfer by inverting a purely infinite element

**Lemma 14.6** (Multiplication by a purely infinite denominator [07]). *Let  $0 \neq g \in \Pi$  and choose  $a > 0$  with  $a < \gamma$  for every  $\gamma \in \text{supp } g$  (possible by Lemma 3.1). If  $f \in \mathcal{F}_a$ , then  $gf \in \Pi$ . The same holds for  $0 \neq g \in \Pi_{\mathbb{C}}$  and  $f \in \mathcal{F}_a(i)$ .*

*Proof.* Every  $h \in \text{supp } f$  satisfies  $|h| < a/2$ . Every exponent occurring in the product before cancellation has the form  $\gamma + h$  with  $\gamma \in \text{supp } g$ , and  $\gamma + h > a - a/2 > 0$ . Hahn multiplication is defined and cancellation cannot create an exponent outside the sum of the supports, so  $gf$  has strictly positive support and real (respectively complex) coefficients.  $\square$

**Theorem 14.7** (Localization and quotient transfer [07]). *Under the hypotheses of Lemma 14.6, the subfield  $\mathcal{F}_a$  of  $\mathbf{No}$  lies in the localization  $\mathbf{Oz}[1/g] \subseteq \mathbf{No}$ . If  $q : \mathbf{Oz} \rightarrow B$  is a unital homomorphism to a nonzero class ring and  $q(g)$  is invertible in  $B$ , then*

$$\iota_g(f) = q(gf)q(g)^{-1} \quad (f \in \mathcal{F}_a) \quad (14.5)$$

*is an injective field homomorphism  $\mathcal{F}_a \hookrightarrow B$ , and  $\iota_g \circ \Theta_a$  is an injective field homomorphism  $\mathbf{No} \hookrightarrow B$ . For  $q : \mathbf{Oz}[i] \rightarrow B$  and  $0 \neq g \in \Pi_{\mathbb{C}}$  with  $q(g)$  invertible, the same formula gives an injective field homomorphism  $\mathcal{F}_a(i) \hookrightarrow B$ , and hence a copy of  $\mathbf{No}[i]$  in  $B$ . In either case  $B$  is a proper class.*

*Proof.* Inside  $\mathbf{No}$ ,  $f = (gf)/g$  with  $gf \in \Pi \subseteq \mathbf{Oz}$  by Lemma 14.6, so  $f \in \mathbf{Oz}[1/g]$ . The elements of  $q(\mathbf{Oz})$  commute pairwise, and  $q(g)^{-1}$  commutes with every element  $y$  commuting with  $q(g)$  (multiply  $q(g)y = yq(g)$  on both sides by  $q(g)^{-1}$ ). Hence  $q(\mathbf{Oz})$  and  $q(g)^{-1}$  generate a commutative subring of  $B$ , and all computations below take place there; no commutativity of  $B$  is needed. The map (14.5) is additive and  $\iota_g(1) = 1$ . For  $f, f' \in \mathcal{F}_a$  the elements  $gf, gf', gff'$  lie in  $\mathbf{Oz}$  and  $(gf)(gf') = g \cdot (gff')$ , so

$$\iota_g(f)\iota_g(f') = q(gf)q(gf')q(g)^{-2} = q(g)q(gff')q(g)^{-2} = \iota_g(ff').$$

(Equivalently,  $\iota_g$  is the restriction to  $\mathcal{F}_a$  of the extension of  $q$  to  $\mathbf{Oz}[1/g]$  given by the universal property of localization.) If  $f \neq 0$  and  $\iota_g(f) = 0$ , then  $1_B = \iota_g(f)\iota_g(f^{-1}) = 0$ , contradicting  $B \neq 0$ ; so  $\iota_g$  is injective. Composing with Theorem 14.4 gives the copy of  $\mathbf{No}$ . The complex case is identical, with Lemma 14.6 for  $\Pi_{\mathbb{C}}$  and the complex extension of  $\Theta_a$ . Finally, if  $B$  were set-sized, Theorem 5.1 would give  $q(g) = 0$ , which is not invertible in a nonzero ring.  $\square$

*Remark 14.8* ([07]). If the target  $B$  is noncommutative, the image of the localization still lies in a commutative subring, as the proof shows; no commutativity condition on all of  $B$  is needed. The hypothesis on  $a$  is always available in the full surreal exponent class, even when the exponents of  $g$  approach zero inside a smaller ordered group (Lemma 3.1). No inference of this kind is valid for an arbitrary fixed set-sized exponent group without an additional lower-bound hypothesis.

**Corollary 14.9** (Quotients by a unit-constant perturbation [07]). *If  $0 \neq g \in \Pi$ , then  $\mathbf{Oz}/(1+g)$  is nonzero and contains a class copy of  $\mathbf{No}$ . If  $0 \neq g \in \Pi_{\mathbb{C}}$ , then  $\mathbf{Oz}[i]/(1+g)$  is nonzero and contains a class copy of  $\mathbf{No}[i]$ . Neither quotient has a unital homomorphism to a nonzero set-sized ring.*

*Proof.* The element  $1+g$  is nonconstant, hence a nonunit by Lemma 2.4, so the quotients are nonzero. In the quotient  $g = -1$  is a unit, so Theorem 14.7 applies. Since  $\text{ct}(1+g) = 1$ , the ideal  $(1+g)$  has constant-term image all of  $\mathbb{Z}$  (respectively  $\mathbb{Z}[i]$ ), and Theorem 6.1 makes the universal set-sized image the zero ring; a unital map from the zero ring exists only to the zero ring.  $\square$

**Corollary 14.10** (A conditional statement about nonarithmetic primes [07]). *Let  $P$  be a prime ideal of  $\mathbf{Oz}$  such that  $P \cap \mathbb{Z} = 0$  and  $P \not\subseteq \Pi$ . Then  $\mathbf{Oz}/P$  contains a class copy of  $\mathbf{No}$  and is a  $\mathbb{Q}$ -algebra. Likewise, if  $P$  is a prime ideal of  $\mathbf{Oz}[i]$  with  $P \cap \mathbb{Z}[i] = 0$  and  $P \not\subseteq \Pi_{\mathbb{C}}$ , then  $\mathbf{Oz}[i]/P$  contains a class copy of  $\mathbf{No}[i]$  and is a  $\mathbb{Q}(i)$ -algebra.*

*Proof.* Choose  $f \in P$  with nonzero constant term  $c$ . The series  $f/c$  still lies in the omnific ring: its constant term is 1 and its other coefficients are real (respectively complex). Since  $f = c \cdot (f/c)$ ,  $P$  is prime and  $c \notin P$ , we get  $f/c \in P$ . Write  $f/c = 1+g$  with  $g$  purely infinite;  $g \neq 0$  because  $c \notin P$ . In the quotient  $g = -1$  is a unit, and Theorem 14.7 applies. For every nonzero ordinary coefficient  $d$  ( $d \in \mathbb{Z}$ , respectively  $d \in \mathbb{Z}[i]$ ),  $g/d$  lies in the ring, and in the quotient  $d \cdot (-g/d) = -g = 1$ ; so  $d$  is invertible, which gives the  $\mathbb{Q}$ -, respectively  $\mathbb{Q}(i)$ -, algebra structure.  $\square$

The corollary is deliberately conditional. Manuscript 07 does not assert that prime ideals with these properties exist; see Section 14.8.

**Example 14.11** (Invertibility cannot be replaced by nonvanishing [07]). The quotient  $\mathbf{Oz}/(\omega^a)$  is nonzero, and the class of  $\omega^{a/2}$  is a nonzero element of square zero: it is nonzero because  $\omega^{a/2}/\omega^a = \omega^{-a/2} \notin \mathbf{Oz}$ . Since  $(\omega^a) \subseteq \Pi$ , this quotient still maps onto  $\mathbb{Z}$  by constant terms, and so it cannot contain a unital copy of  $\mathbb{Q}$ : composing such a copy with that map would give a unital map  $\mathbb{Q} \rightarrow \mathbb{Z}$ . A surviving nilpotent and a surviving unit are therefore substantively different.

The same mechanism, with a fraction field in place of the quotient, applies to every prime ideal not containing the purely infinite ideal. Here  $X^g$  is the monomial of  $k((X^{\mathbf{No}}))$ , which is  $\omega^g$  when  $k = \mathbb{R}$  or  $k = \mathbb{C}$ .

**Proposition 14.12** (A scaled field in every non-arithmetic residue field [06]). *Let  $k$  be a set field,  $D \subseteq k$  a unital subring,  $A = \mathcal{A}_{D,k}$ , and let  $P$  be a prime class ideal of  $A$  with  $\Pi_k \not\subseteq P$ . There exists  $a > 0$  for which the class field  $\mathcal{F}_a(k) = k((X^{H_a}))$  embeds in  $\text{Frac}(A/P)$ .*

*Proof.* Choose  $f \in \Pi_k \setminus P$  and write  $f = X^g h$  with  $g > 0$  and  $h \in \Pi_k$  (Lemma 3.1 applied to  $\text{supp } f$ ). Then  $X^g \notin P$ , since otherwise  $f \in P$ . Put  $a = g/2$  and  $m = X^a$ . As  $m^2 = X^g \notin P$  and  $P$  is prime,  $\bar{m} \neq 0$  in the domain  $A/P$ . Since  $m\mathcal{F}_a(k) \subseteq \Pi_k \subseteq A$  (Lemma 4.2), we may define

$$\iota(x) = \frac{\overline{mx}}{\bar{m}} \in \text{Frac}(A/P) \quad (x \in \mathcal{F}_a(k)).$$

Additivity and  $\iota(1) = 1$  are immediate. Multiplicativity follows by reducing the identity  $(mx)(my) = m(mxy)$  in  $A$  and dividing by  $\bar{m}^2$ . For  $x \neq 0$ , the identity  $(mx)(m/x) = m^2$  in  $A$  shows  $\bar{m}x \neq 0$ , because  $\bar{m}^2 \neq 0$ . Thus the unital field homomorphism  $\iota$  is injective.  $\square$

This is the scaled-field collision mechanism of [Lemma 4.1](#) in a fraction field. The conclusion says more than that the residue field is large: it contains a specific Hahn field built on all exponent scales infinitesimal relative to one positive  $a$ . It is a necessary structural condition on such residue fields; manuscript 06 states that it does not classify those prime ideals, assert that they exist, or assert that every field of this form is a residue field.

*Remark 14.13* (The composed embedding [merge]). Let  $P$  be a prime class ideal of  $\mathbf{Oz}$  with  $\Pi \not\subseteq P$ . Then there is an injective field homomorphism  $\mathbf{No} \hookrightarrow \text{Frac}(\mathbf{Oz}/P)$  whose first factor is an ordered-field embedding,

$$\mathbf{No} \xrightarrow{\Theta_a} \mathcal{F}_a \xrightarrow{\iota} \text{Frac}(\mathbf{Oz}/P).$$

Indeed, [Proposition 14.12](#) with  $(D, k) = (\mathbb{Z}, \mathbb{R})$  gives  $a > 0$  and an injective field homomorphism  $\iota : \mathcal{F}_a \rightarrow \text{Frac}(\mathbf{Oz}/P)$ , and [Theorem 14.4](#) gives  $\Theta_a$ ; a composite of injective field homomorphisms is one. With  $(D, k) = (\mathbb{Z}[i], \mathbb{C})$  and the complex extension of  $\Theta_a$ , every prime  $P$  of  $\mathbf{Oz}[i]$  with  $\Pi_{\mathbb{C}} \not\subseteq P$  gives  $\mathbf{No}[i] \hookrightarrow \text{Frac}(\mathbf{Oz}[i]/P)$ . Neither source states this composition.

The hypotheses of [Proposition 14.12](#) and [Corollary 14.10](#) are different. If  $P \cap \mathbb{Z} = 0$  and  $P \not\subseteq \Pi$ , then  $\Pi \not\subseteq P$ : otherwise, for  $x \in P$  we would have  $x - \text{ct}(x) \in \Pi \subseteq P$ , hence  $\text{ct}(x) \in P \cap \mathbb{Z} = 0$ , so  $P \subseteq \Pi$ . Conversely,  $\Pi \not\subseteq P$  forces  $P \cap \mathbb{Z} = 0$ , since a nonzero  $n \in P$  would give  $\Pi = n\Pi \subseteq P$  ([Proposition 3.3](#)), but it allows  $P \subseteq \Pi$ , for instance  $P = 0$ , where the statement is the trivial  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  ([Proposition 3.4](#)). Thus 07's hypothesis is the stronger one and yields the stronger conclusion (a copy of  $\mathbf{No}$  inside  $\mathbf{Oz}/P$  itself, and a  $\mathbb{Q}$ -algebra structure), while 06's hypothesis also covers primes contained in  $\Pi$ , with the conclusion only in the fraction field.

## 14.4 The binomial quotient and its localization

Fix  $a > 0$  and write

$$\mathcal{Q}_a = \mathbf{Oz}/(1 + \omega^a), \quad q_a : \mathbf{Oz} \longrightarrow \mathcal{Q}_a$$

for the quotient map (07's  $B_a$ ). The element  $1 + \omega^a$  is a nonunit ([Lemma 2.4](#)), so  $\mathcal{Q}_a \neq 0$ ; and since  $\text{ct}(1 + \omega^a) = 1$ , [Theorem 6.1](#) shows that  $\mathcal{Q}_a$  has no unital homomorphism to a nonzero set-sized ring. A purely arithmetic calculation shows why: in a set-sized target  $\omega^a$  becomes 0 ([Theorem 5.1](#)), so  $1 + \omega^a$  becomes 1, and a target in which it vanishes has  $1 = 0$ . Its failure to have small ring images is therefore not caused by its being zero, a field, or a domain; it is none of these by [Corollary 14.22](#). By [Corollary 14.9](#) it contains a copy of  $\mathbf{No}$ ; the rest of this section identifies much more of its internal structure.

The support description of the localization

$$\mathbf{Oz}[\omega^{-a}] = \{f \in \mathbf{No} : \text{supp } f \subseteq [-Na, \infty) \text{ for some ordinary } N \geq 0\}$$

is [Proposition 6.11](#); manuscripts 04 and 07 both prove it. (07 calls this ring  $L_a$ .)

**Lemma 14.14** (Localization of the binomial quotient [07]).

- (i)  $\omega^a \mathcal{F}_a \subseteq \Pi$ ; in particular  $\mathcal{F}_a \subseteq \mathbf{Oz}[\omega^{-a}]$ .
- (ii) There is a unique ring homomorphism  $\tilde{q}_a : \mathbf{Oz}[\omega^{-a}] \rightarrow \mathcal{Q}_a$  extending  $q_a$ . It is surjective with kernel  $(1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$ , so

$$\mathcal{Q}_a \cong \mathbf{Oz}[\omega^{-a}]/(1 + \omega^a)\mathbf{Oz}[\omega^{-a}]. \tag{14.6}$$

(iii) The restriction of  $\tilde{q}_a$  to  $\mathcal{F}_a$  is the injective field homomorphism

$$\iota_a(f) = -q_a(\omega^a f) \quad (f \in \mathcal{F}_a). \quad (14.7)$$

*Proof.* (i) Every  $h \in H_a$  satisfies  $a + h > 0$ , so translating a support in  $H_a$  by  $a$  gives a strictly positive support (Lemma 4.2); then  $f = \omega^{-a}(\omega^a f)$ .

(ii) The subring of  $\mathbf{No}$  generated by  $\mathbf{Oz}$  and  $\omega^{-a}$  consists of the finite sums  $\sum_{j \leq N} h_j \omega^{-ja} = \omega^{-Na} \sum_{j \leq N} h_j \omega^{(N-j)a}$ , so every element has the form  $\omega^{-Na} h$  with  $N \geq 0$  and  $h \in \mathbf{Oz}$ . Any extension of  $q_a$  sends  $\omega^{-a}$  to  $q_a(\omega^a)^{-1} = -1$ , which proves uniqueness; define  $\tilde{q}_a(\omega^{-Na} h) = (-1)^N q_a(h)$ . This is well defined: if  $\omega^{-Na} h = \omega^{-Ma} h'$  with  $N \leq M$ , then  $h' = \omega^{(M-N)a} h$  and  $(-1)^M q_a(h') = (-1)^M (-1)^{M-N} q_a(h) = (-1)^N q_a(h)$ . Writing two elements over a common power  $\omega^{-Na}$  shows that  $\tilde{q}_a$  is a ring homomorphism; it extends  $q_a$  and is surjective. If  $\tilde{q}_a(\omega^{-Na} h) = 0$ , then  $q_a(h) = 0$ , so  $h = (1 + \omega^a)y$  with  $y \in \mathbf{Oz}$  and  $\omega^{-Na} h = (1 + \omega^a)\omega^{-Na} y \in (1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$ . Conversely  $\tilde{q}_a((1 + \omega^a)\omega^{-Na} y) = (-1)^N q_a((1 + \omega^a)y) = 0$  for  $y \in \mathbf{Oz}$ .

(iii) By (i) and (ii),  $\tilde{q}_a(f) = \tilde{q}_a(\omega^{-a} \cdot \omega^a f) = -q_a(\omega^a f)$ . A unital ring homomorphism from a field to a nonzero ring is injective.  $\square$

Henceforth  $\mathcal{F}_a$  is identified with its image  $\iota_a(\mathcal{F}_a) \subseteq \mathcal{Q}_a$  whenever we work inside  $\mathcal{Q}_a$ ; all coefficient fields below are embedded with the identity of  $\mathcal{Q}_a$ . This identification is not constant-term extraction: a nonintegral real scalar  $r$  is represented in  $\mathcal{Q}_a$  by the omnific integer  $-r\omega^a$ .

## 14.5 Exact binomial kernels

For an ordinary integer  $n \geq 1$  put  $x_n = q_a(\omega^{a/n}) \in \mathcal{Q}_a$ . Certainly  $x_n^n = q_a(\omega^a) = -1$ . The substantive question is whether the full ambient omnific ring, with its arbitrary surreal exponents, creates additional polynomial relations over  $\mathcal{F}_a$ .

**Theorem 14.15** (Exact binomial kernel [07]). *For every  $n \geq 1$ , evaluation at  $x_n$  induces an injective ring homomorphism*

$$\mathcal{F}_a[T]/(T^n + 1) \hookrightarrow \mathcal{Q}_a. \quad (14.8)$$

*Equivalently, the kernel of  $\mathcal{F}_a[T] \rightarrow \mathcal{Q}_a$ ,  $T \mapsto x_n$ , is exactly  $(T^n + 1)$ .*

*Proof.* The polynomial  $T^n + 1$  lies in the kernel. Dividing by this monic polynomial over the field  $\mathcal{F}_a$ , it suffices to show that no nonzero remainder

$$R(T) = \sum_{j=0}^{n-1} r_j T^j \quad (r_j \in \mathcal{F}_a)$$

vanishes at  $x_n$ . Put  $\xi = \omega^{a/n} \in \mathbf{Oz}$ ; then  $R(\xi) \in \mathbf{Oz}[\omega^{-a}]$  by Lemma 14.14(i), and  $\tilde{q}_a(R(\xi)) = R(x_n)$ .

*The quotient in  $\mathbf{No}$ .* The series  $U = \sum_{k \geq 0} (-1)^k \omega^{-(k+1)a}$  has the decreasing support  $\{-(k+1)a\}$ , and  $(1 + \omega^a)U = \sum_{k \geq 0} (-1)^k \omega^{-(k+1)a} + \sum_{k \geq 0} (-1)^k \omega^{-ka} = 1$ , each exponent receiving at most two contributions that cancel except at exponent 0. Hence  $U = (1 + \omega^a)^{-1}$  in  $\mathbf{No}$ , and multiplying the finite sum  $R(\xi) = \sum_j r_j \omega^{ja/n}$  by  $U$  gives

$$\frac{R(\xi)}{1 + \omega^a} = \sum_{k \geq 0} \sum_{j=0}^{n-1} (-1)^k r_j \omega^{(j/n - k - 1)a}, \quad (14.9)$$

provided the right side is a strongly summable family, which we now check together with the absence of cancellation between blocks.

*Block cosets.* The support of the block indexed by  $(j, k)$  lies in the coset  $c_{j,k} + H_a$  with centre  $c_{j,k} = (j/n - k - 1)a$ , because  $\text{supp } r_j \subseteq H_a$ . Distinct pairs give distinct cosets. Indeed,  $c_{j,k} - c_{j',k'} = (m/n)a$  with  $m = j - j' - n(k - k') \in \mathbb{Z}$ , and  $(m/n)a \in H_a$  only for  $m = 0$ , since for  $m \neq 0$  one has  $n \cdot |(m/n)a| = |m|a \geq a$ . From  $j - j' = n(k - k')$  and  $|j - j'| < n$  we get  $j = j'$  and  $k = k'$ . Cosets of  $H_a$  are disjoint, so distinct blocks have disjoint supports, and each exponent receives a contribution from at most one block.

*Reverse well-ordering.* As  $(j, k)$  runs over  $\{0, \dots, n-1\} \times \mathbb{N}$ , the integer  $j - n(k+1)$  runs bijectively over the integers  $m \leq -1$ , so the centres are the points  $(m/n)a$ ,  $m \leq -1$ , a strictly descending discrete progression with step  $a/n$ . Every offset  $h \in H_a$  satisfies  $4n|h| < a$ , that is,  $|h| < a/(4n)$ , so the block with centre  $(m/n)a$  lies in the open interval of radius  $a/(4n)$  about its centre; these intervals are pairwise disjoint and ordered as their centres. Given a nonempty subset  $S$  of the union of the supports, the set of  $m$  whose block meets  $S$  is a nonempty set of integers bounded above by  $-1$ , so it has a largest element  $m^*$ ; the elements of  $S$  in block  $m^*$  exceed all other elements of  $S$ , and since that block has reverse well-ordered support,  $S$  has a greatest element. All supports and the index family are sets. Hence the right side of (14.9) is a legitimate strongly summable Hahn expression, and (14.9) holds because Hahn multiplication distributes over strongly summable families. (Equivalently, multiplying the right side by  $1 + \omega^a$  cancels consecutive  $k$ -blocks and leaves exactly  $R(\xi)$ .)

*Unbounded-below support.* Choose  $j$  with  $r_j \neq 0$  and  $h \in \text{supp } r_j$ . For every  $k \geq 0$  the exponent  $h + (j/n - k - 1)a$  lies only in block  $(j, k)$ , so it occurs in (14.9) with the nonzero coefficient  $(-1)^k (r_j)_h$ . Since  $h < a$  and  $j/n < 1$ , this exponent is less than  $a + a - (k+1)a = (1-k)a$ , which is below  $-Na$  as soon as  $k \geq N+1$ . Thus the support of (14.9) is not contained in  $[-Na, \infty)$  for any ordinary  $N$ , and by Proposition 6.11 the quotient  $R(\xi)/(1 + \omega^a)$  does not belong to  $\mathbf{Oz}[\omega^{-a}]$ .

*Conclusion.* If  $R(x_n) = 0$ , then  $R(\xi) \in \ker \tilde{q}_a = (1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$  by Lemma 14.14(ii), so  $R(\xi) = (1 + \omega^a)u$  with  $u \in \mathbf{Oz}[\omega^{-a}]$ . Division in the field  $\mathbf{No}$  identifies  $u$  with (14.9), a contradiction. Hence  $R(x_n) \neq 0$ .  $\square$

*Remark 14.16 ([07]).* The proof does not restrict the exponents of an alleged ambient factor  $u$  to rational multiples of  $a$ ; the full support characterization of  $\mathbf{Oz}[\omega^{-a}]$  is exactly what rules out such hidden factors. A calculation solely in a finite group algebra (the span of the monomials  $\omega^{ja/n}$ ) would not establish the theorem.

**Corollary 14.17** (Internal algebraically closed field [07]). *Let  $u = x_2 = q_a(\omega^{a/2})$ . The subring  $\mathcal{E}_a = \mathcal{F}_a[u] \subseteq \mathcal{Q}_a$  is a field, isomorphic over  $\mathcal{F}_a$  to  $\mathcal{F}_a(i)$  by  $i \mapsto u$ . It is algebraically closed and contains a class copy of  $\mathbf{No}[i]$ .*

*Proof.* By Theorem 14.15 with  $n = 2$ ,  $\mathcal{E}_a \cong \mathcal{F}_a[T]/(T^2+1)$ . Since  $\mathcal{F}_a$  is real closed (Proposition 14.1),  $-1$  is not a square in it,  $T^2 + 1$  is irreducible, and the quotient is the field  $\mathcal{F}_a + i\mathcal{F}_a = \mathcal{F}_a(i)$ , which is algebraically closed by Proposition 14.1. The complex extension of  $\Theta_a$  (Theorem 14.4) embeds  $\mathbf{No}[i]$  in it.  $\square$

There is no order contradiction. Although  $\omega^{a/2}$  is positive in  $\mathbf{Oz}$ , its square becomes  $-1$  in  $\mathcal{Q}_a$ ; the order of  $\mathbf{Oz}$  does not descend to  $\mathcal{Q}_a$ . The corollary is a statement about rings and fields, not about ordered quotients.



More generally, finite real factorization and the Chinese remainder theorem give

$$\mathcal{F}_a[T]/(T^n + 1) \cong \begin{cases} \mathcal{E}_a^m, & n = 2m, \\ \mathcal{F}_a \times \mathcal{E}_a^m, & n = 2m + 1. \end{cases} \quad (14.10)$$

The roots of  $T^n + 1$  are the  $n$  distinct ordinary complex numbers  $\exp(i\pi(2k+1)/n)$ , which lie in  $\mathbb{C} \subseteq \mathcal{F}_a(i)$ . For even  $n$  none is real, and the  $m$  conjugate pairs give  $m$  distinct monic quadratic factors with real coefficients, irreducible over  $\mathcal{F}_a$  because their roots are not in  $\mathcal{F}_a$ . For odd  $n$  the only real root is  $-1$ , and the remaining roots give  $m$  such quadratics. All roots are simple, so the factors are pairwise coprime, and each quadratic quotient is isomorphic to  $\mathcal{F}_a(i) \cong \mathcal{E}_a$ . The isomorphisms in (14.10) need not identify a chosen imaginary unit in the same way at different stages; the dyadic tower below makes coherent choices.

## 14.6 The dyadic tower and its Cantor algebra

For  $r \geq 1$  define

$$\xi_r = q_a(\omega^{a/2^r}) = x_{2^r}, \quad \mathcal{S}_r = \mathcal{F}_a[\xi_r] \subseteq \mathcal{Q}_a$$

(07's  $\mathcal{S}_r$ ). Then  $\xi_r^2 = \xi_{r-1}$  for  $r \geq 2$ ,  $\mathcal{S}_r \subseteq \mathcal{S}_{r+1}$ , and  $\mathcal{S}_1 = \mathcal{E}_a$ , so each  $\mathcal{S}_r$  is an  $\mathcal{E}_a$ -algebra. In this subsection  $i$  denotes the element  $u = \xi_1$  of  $\mathcal{E}_a$ ; under  $\mathcal{E}_a \cong \mathcal{F}_a(i)$  the ordinary complex field  $\mathbb{C} \subseteq \mathcal{F}_a(i)$  corresponds to  $\mathbb{R} + \mathbb{R}u \subseteq \mathcal{E}_a$ , and ordinary complex numbers are read in  $\mathcal{E}_a$  through this identification.

**Lemma 14.18** (A single stage over the fixed complex field [07]). *Let  $m = 2^{r-1}$ . As  $\mathcal{E}_a$ -algebras,*

$$\mathcal{S}_r \cong \mathcal{E}_a[T]/(T^m - i), \quad T \mapsto \xi_r. \quad (14.11)$$

*Proof.* The  $\mathcal{E}_a$ -algebra map  $\mathcal{E}_a[T] \rightarrow \mathcal{S}_r$ ,  $T \mapsto \xi_r$ , is surjective, since  $\mathcal{E}_a[\xi_r] = \mathcal{F}_a[\xi_r^m, \xi_r] = \mathcal{S}_r$ , and it kills  $T^m - i$  because  $\xi_r^m = \xi_1 = i$ . For injectivity on the quotient, suppose  $\sum_{j < m} (a_j + b_j i) \xi_r^j = 0$  with  $a_j, b_j \in \mathcal{F}_a$ . Substituting  $i = \xi_r^m$  gives  $\sum_{j < m} a_j \xi_r^j + \sum_{j < m} b_j \xi_r^{j+m} = 0$ , a polynomial relation over  $\mathcal{F}_a$  of degree less than  $2m = 2^r$  satisfied by  $\xi_r = x_{2^r}$ . By Theorem 14.15 with  $n = 2^r$  all its coefficients vanish, so all  $a_j$  and  $b_j$  are zero.  $\square$

The roots of  $T^m - i$  can be chosen in the ordinary complex subfield of  $\mathcal{E}_a$ :

$$\zeta_{r,k} = \exp\left(i \frac{\pi/2 + 2\pi k}{2^{r-1}}\right) \quad (k \in \mathbb{Z}/2^{r-1}\mathbb{Z}). \quad (14.12)$$

The expression is evaluated in  $\mathbb{C}$  and then embedded in  $\mathcal{E}_a$ ; no exponential function on surcomplex arguments is assumed. These  $m$  roots are distinct, and

$$\zeta_{r+1,k}^2 = \zeta_{r,k \bmod 2^{r-1}} \quad (k \in \mathbb{Z}/2^r\mathbb{Z}).$$

Since the roots are distinct, the Chinese remainder theorem and (14.11) give an isomorphism of  $\mathcal{E}_a$ -algebras

$$\Phi_r : \mathcal{S}_r \longrightarrow \mathcal{E}_a^{\mathbb{Z}/2^{r-1}\mathbb{Z}}, \quad P(\xi_r) \longmapsto (P(\zeta_{r,k}))_k \quad (P \in \mathcal{E}_a[T]). \quad (14.13)$$

The inclusion  $\mathcal{S}_r \subseteq \mathcal{S}_{r+1}$  replaces  $\xi_r$  by  $\xi_{r+1}^2$ , so  $P(\xi_r) = P(\xi_{r+1}^2)$  has stage- $(r+1)$  coordinates  $P(\zeta_{r+1,k}^2) = P(\zeta_{r,k \bmod m})$ . In coordinates the transition map is therefore exactly

$$(c_k)_{k \bmod m} \longmapsto (c_{k \bmod m})_{k \bmod 2m},$$

which repeats each coordinate on its two lifts. This compatibility, and not merely the existence of separate finite product decompositions, is what identifies the direct limit.

**Theorem 14.19** (Cantor algebra theorem [07]). *Let  $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a)$  be the class of functions from the 2-adic integers  $\mathbb{Z}_2$  to  $\mathcal{E}_a$  that factor through some finite quotient  $\mathbb{Z}/2^N\mathbb{Z}$ , with pointwise operations; thus  $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a) = \bigcup_{N \geq 0} \mathcal{E}_a^{\mathbb{Z}/2^N\mathbb{Z}}$  with the transition maps repeating each coordinate on its two lifts. There is an explicit  $\mathcal{E}_a$ -algebra isomorphism*

$$\bigcup_{r \geq 1} \mathcal{S}_r \cong \text{LC}(\mathbb{Z}_2, \mathcal{E}_a), \quad (14.14)$$

under which  $\xi_r$  corresponds to the function  $z \mapsto \zeta_{r, z \bmod 2^{r-1}}$ . The left side is the subalgebra of  $\mathcal{Q}_a$  generated by  $\mathcal{F}_a$  and the classes of  $\omega^{a/2^r}$ ,  $r \geq 1$ .

*Proof.* Identify  $\mathcal{E}_a^{\mathbb{Z}/2^{r-1}\mathbb{Z}}$  with the functions  $\mathbb{Z}_2 \rightarrow \mathcal{E}_a$  factoring through  $\mathbb{Z}/2^{r-1}\mathbb{Z}$ , via  $c \mapsto (z \mapsto c_{z \bmod 2^{r-1}})$ . By the computation of the transition maps, the function attached to an element of  $\mathcal{S}_r$  by  $\Phi_r$  is the same as the one attached by  $\Phi_{r+1}$ , so the  $\Phi_r$  combine into a map  $\Phi$  on the increasing union; it is an  $\mathcal{E}_a$ -algebra homomorphism, and it is injective because each  $\Phi_r$  is. Conversely, every function factoring through  $\mathbb{Z}/2^N\mathbb{Z}$  lies in the image of the bijection  $\Phi_{N+1}$ , so  $\Phi$  is surjective. Evaluating  $T$  at the roots gives the function attached to  $\xi_r$ . Finally,  $\xi_{r'} = \xi_r^{2^{r-r'}}$   $\in \mathcal{S}_r$  for  $r' \leq r$ , so the union of the  $\mathcal{S}_r$  is the subalgebra generated by  $\mathcal{F}_a$  and all  $\xi_r$ .  $\square$

The isomorphism is an algebraic statement about a specified subalgebra; it is not a classification of all of  $\mathcal{Q}_a$ .

*Remark 14.20* (Locally constant functions [07]). The topological terminology is justified: a locally constant function on the ordinary compact space  $\mathbb{Z}_2$  has a finite subcover of neighbourhoods of constancy and hence factors through one finite quotient, and conversely such a factorization is locally constant. This uses only the ordinary Cantor space; no topology is imposed on the proper-class field  $\mathcal{E}_a$ , and one can equally take the finite-factorization condition as the definition.

*Remark 14.21* (Evaluation at a point [07]). Evaluation at a point  $z \in \mathbb{Z}_2$  is a homomorphism from the subalgebra (14.14) onto  $\mathcal{E}_a$ . Its target is a proper-class field, not a set-sized ring, and no assertion is made that it extends to all of  $\mathcal{Q}_a$ . The construction is therefore fully compatible with the absence of small ring images of  $\mathcal{Q}_a$ .

For  $m = 2^{r-1}$  and  $k \in \mathbb{Z}/m\mathbb{Z}$  define, in  $\mathcal{S}_r$ ,

$$e_{r,k} = \frac{1}{m} \sum_{\ell=0}^{m-1} \left( \frac{\xi_r}{\zeta_{r,k}} \right)^\ell. \quad (14.15)$$

At the root  $\zeta_{r,j}$  the sum has value 1 for  $j = k$  and 0 otherwise, by the finite geometric-sum identity for the  $m$ th root of unity  $\zeta_{r,j}/\zeta_{r,k}$ . Thus  $\Phi_r(e_{r,k})$  is the indicator of the coordinate  $k$ , and

$$e_{r,k}^2 = e_{r,k}, \quad e_{r,k}e_{r,j} = 0 \ (k \neq j), \quad \sum_{k \bmod m} e_{r,k} = 1, \quad (14.16)$$

$$e_{r,k} = e_{r+1,k} + e_{r+1,k+m}. \quad (14.17)$$

Each  $e_{r,k}$  is nonzero because  $\Phi_r$  is injective, which rests on Theorem 14.15. The refinement formula (14.17) follows from the coordinate transition, or by checking values at all roots of stage  $r+1$ .

The first nontrivial instance can be written without trigonometric notation. Put

$$y = \xi_2, \quad u = \xi_1 = y^2, \quad \zeta = \frac{1+u}{\sqrt{2}} \in \mathcal{E}_a.$$

Then  $u^2 = -1$  and  $\zeta^2 = (1 + 2u + u^2)/2 = u = y^2$ , so  $(y/\zeta)^2 = 1$ , and

$$e_+ = \frac{1}{2}\left(1 + \frac{y}{\zeta}\right), \quad e_- = \frac{1}{2}\left(1 - \frac{y}{\zeta}\right) \quad (14.18)$$

satisfy  $e_{\pm}^2 = e_{\pm}$ ,  $e_+e_- = \frac{1}{4}(1 - (y/\zeta)^2) = 0$  and  $e_+ + e_- = 1$ . Nonvanishing is not inferred from the suggestive formula: the roots at stage 2 are  $\zeta_{2,0} = \zeta$  and  $\zeta_{2,1} = -\zeta$ , so  $e_+ = e_{2,0}$ ,  $e_- = e_{2,1}$ , and  $\Phi_2$  identifies  $\mathcal{S}_2 \cong \mathcal{E}_a \times \mathcal{E}_a$  and sends  $e_+, e_-$  to  $(1, 0)$  and  $(0, 1)$ . This is why the exact-kernel theorem is an essential part of the argument.

**Corollary 14.22** (Boolean algebra and failure of finiteness conditions [07]). *The idempotents of  $\bigcup_r \mathcal{S}_r$  form precisely the Boolean algebra of clopen subsets of  $\mathbb{Z}_2$ , and this Boolean algebra is atomless. The ring  $\mathcal{Q}_a$  has an infinite sequence of pairwise orthogonal nonzero idempotents, is neither a domain nor a local ring, and fails the ascending chain condition on ideals.*

*Proof.* A field has only the idempotents 0 and 1, so an idempotent of  $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a)$  is the indicator of a set that is a union of residue classes modulo some  $2^N$ , that is, of a clopen subset of  $\mathbb{Z}_2$ ; conversely, by compactness every clopen subset is a finite union of such residue classes, and its indicator is idempotent. Every nonempty clopen set contains a residue class modulo some  $2^N$ , which splits into two nonempty residue classes modulo  $2^{N+1}$ ; so there are no atoms.

For  $j \geq 0$  the sets  $C_j = \{z \in \mathbb{Z}_2 : z \equiv 2^j \pmod{2^{j+1}}\}$  (the  $z$  of 2-adic valuation exactly  $j$ ) are nonempty, clopen and pairwise disjoint. Their indicators give pairwise orthogonal nonzero idempotents  $f_j \in \mathcal{Q}_a$ . The ideals

$$(f_0) \subsetneq (f_0, f_1) \subsetneq (f_0, f_1, f_2) \subsetneq \cdots$$

of  $\mathcal{Q}_a$  strictly increase: if  $f_{n+1} = \sum_{j \leq n} c_j f_j$  with  $c_j \in \mathcal{Q}_a$ , multiplication by  $f_{n+1}$  would give  $f_{n+1} = 0$ . This proves the failure of the ascending chain condition without forming any class-sized generating family. A nontrivial idempotent  $e$  satisfies  $e(1 - e) = 0$ , so  $\mathcal{Q}_a$  is not a domain; and in a local ring (for every  $x$ , one of  $x$  and  $1 - x$  is a unit) either  $e$  or  $1 - e$  is a unit, which forces  $e = 1$  or  $e = 0$ .  $\square$

Grouping a sufficiently fine dyadic partition of  $\mathbb{Z}_2$  into any prescribed number  $d \geq 1$  of nonempty blocks yields a unital subalgebra of  $\mathcal{Q}_a$  isomorphic to  $\mathcal{E}_a^d$ . Manuscript 07 claims neither that all idempotents of  $\mathcal{Q}_a$  lie in the Cantor algebra (14.14) nor that  $\mathcal{Q}_a$  itself is a ring of locally constant functions. No infinite sum of the orthogonal sequence  $(f_j)$  is used.

## 14.7 Gaussian binomial quotients

Let

$$\mathcal{Q}_a^{\mathbb{C}} = \mathbf{Oz}[i]/(1 + \omega^a), \quad q_a^{\mathbb{C}} : \mathbf{Oz}[i] \rightarrow \mathcal{Q}_a^{\mathbb{C}}.$$

Since  $\mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz}$  and  $\omega^{-a}$  is real,  $\mathbf{Oz}[i][\omega^{-a}] = \mathbf{Oz}[\omega^{-a}] + i\mathbf{Oz}[\omega^{-a}]$ , so by Proposition 6.11 it consists of the surcomplex normal forms whose supports are bounded below by some  $-Na$ . It contains  $\mathcal{F}_a(i)$ , and the proof of Lemma 14.14 applies verbatim with  $\mathbf{Oz}[i]$  in place of  $\mathbf{Oz}$ : the extension  $\tilde{q}_a^{\mathbb{C}}$  of  $q_a^{\mathbb{C}}$  has kernel  $(1 + \omega^a)\mathbf{Oz}[i][\omega^{-a}]$ , and it embeds the coefficient field by

$$f \mapsto -q_a^{\mathbb{C}}(\omega^a f) \quad (f \in \mathcal{F}_a(i)).$$

For  $f = i$  this is the Gaussian imaginary unit of the quotient, since  $-q_a^{\mathbb{C}}(\omega^a i) = -q_a^{\mathbb{C}}(\omega^a)q_a^{\mathbb{C}}(i) = q_a^{\mathbb{C}}(i)$ .

**Theorem 14.23** (Complex exact kernels and dyadic algebra [07]). *For every  $n \geq 1$  there is an injective ring homomorphism*

$$\mathcal{F}_a(i)[T]/(T^n + 1) \hookrightarrow \mathcal{Q}_a^{\mathbb{C}}, \quad T \mapsto q_a^{\mathbb{C}}(\omega^{a/n}),$$

*and its source is isomorphic to  $\mathcal{F}_a(i)^n$ . The increasing union of the dyadic stages  $n = 2^r$ ,  $r \geq 0$ , is isomorphic to  $\text{LC}(\mathbb{Z}_2, \mathcal{F}_a(i))$ . The ring  $\mathcal{Q}_a^{\mathbb{C}}$  is nonzero, has no unital homomorphism to a nonzero set-sized ring, and contains a class copy of  $\mathbf{No}[i]$ .*

*Proof.* The proof of Theorem 14.15 uses only supports and the nonvanishing of coefficients, not their reality, so it applies verbatim over  $\mathcal{F}_a(i)$  and  $\mathbf{Oz}[i][\omega^{-a}]$ : a nonzero remainder  $R$  with coefficients in  $\mathcal{F}_a(i)$  gives the expansion (14.9), whose blocks lie in the distinct cosets  $(j/n - k - 1)a + H_a$ , whose union of supports is reverse well ordered, and whose support is not bounded below by any  $-Na$ ; so  $R(\omega^{a/n}) \notin (1 + \omega^a)\mathbf{Oz}[i][\omega^{-a}]$ . The polynomial  $T^n + 1$  has  $n$  distinct roots  $\exp(i\pi(2k+1)/n)$  in  $\mathbb{C} \subseteq \mathcal{F}_a(i)$ , so the Chinese remainder theorem gives  $\mathcal{F}_a(i)[T]/(T^n + 1) \cong \mathcal{F}_a(i)^n$  by evaluation at the roots.

For  $n = 2^r$  label the roots  $\eta_{r,k} = \exp(i(\pi + 2\pi k)/2^r)$ ,  $k \in \mathbb{Z}/2^r\mathbb{Z}$ . Then  $\eta_{r,k}^2 = \eta_{r-1, k \bmod 2^{r-1}}$ , and the stage- $(r-1)$  generator  $q_a^{\mathbb{C}}(\omega^{a/2^{r-1}})$  is the square of the stage- $r$  generator; so, exactly as in Theorem 14.19, the coordinate transitions repeat each entry on its two lifts, and the union is  $\text{LC}(\mathbb{Z}_2, \mathcal{F}_a(i))$ . The ring is nonzero because  $1 + \omega^a$  is not among the units  $\pm 1, \pm i$  of  $\mathbf{Oz}[i]$  (Lemma 2.4); it has no nonzero set-sized image by Theorem 6.1, since  $\text{ct}(1 + \omega^a) = 1$  generates  $\mathbb{Z}[i]$ ; and it contains a copy of  $\mathbf{No}[i]$  by the complex extension of  $\Theta_a$  (Theorem 14.4).  $\square$

Coefficientwise complex conjugation preserves  $\mathbf{Oz}[i]$  and, because  $1 + \omega^a$  is real, the ideal  $(1 + \omega^a)$ ; hence it descends to an involution of  $\mathcal{Q}_a^{\mathbb{C}}$ . It commutes with the coefficient-field embedding  $f \mapsto -q_a^{\mathbb{C}}(\omega^a f)$  because the multiplier  $-\omega^a$  is real. This supplies a compatible involution without requiring an order or an analytic interpretation of the quotient.

## 14.8 Boundaries of the large-field results

The small-image theorem and the large-field results concern different hypotheses and reinforce one another (07). A homomorphism to a set-sized ring kills every purely infinite element (Theorem 5.1); a homomorphism to a nonzero ring that makes even one purely infinite element invertible has a huge field in its target (Theorem 14.7). That target is necessarily a proper class, but the explicit field embedding gives considerably more than this size conclusion, and the Cantor algebra describes zero-divisor structure that no cardinality argument captures. Manuscript 07 insists on the following distinctions.

**A nonzero element versus a unit.** A nonzero nilpotent, as in Example 14.11, does not produce a field. Invertibility is what allows the formula  $f = (gf)/g$  to be transported to a quotient. No cancellation in a ring with zero divisors has been used.

**A specified subalgebra versus the entire quotient.** The exact kernel theorem identifies the subalgebras  $\mathcal{F}_a[x_n]$ , and their dyadic union is a Cantor algebra. Surreal normal forms with other supports can still create further elements of  $\mathcal{Q}_a$ . Neither a classification of all idempotents of  $\mathcal{Q}_a$  nor a description of its prime ideals has been proved.

**Class ideals versus set-theoretic Zorn arguments.** Zorn’s lemma applies to sets of candidates. The ring  $\mathcal{Q}_a$  is a proper class, so the usual existence argument for maximal ideals cannot simply be repeated over its class of ideals. Statements such as [Corollary 14.10](#) are deliberately conditional: it is not concluded that a proper-class nonarithmetic maximal ideal exists merely because  $(1 + \omega^a)$  is proper. Likewise (06) [Proposition 14.12](#) is a necessary condition on residue fields, not an existence theorem.

**Strong sums versus limits.** Every infinite sum used above, namely the geometric series for  $(1 + \omega^a)^{-1}$  and the block expansion [\(14.9\)](#), has a set-sized reverse well-ordered support and finitely many contributions at each exponent, and is verified coefficientwise. Finite geometric remainders of the kind in [Lemma 4.3](#) do not tend to zero, and no limit argument is used.

**Finite checks.** The program accompanying manuscript 07 checks exact finite geometric identities, finite binomial remainders, the first nontrivial idempotents [\(14.18\)](#) and finite coordinate refinement maps (07 records that all 621 named checks passed). It does not represent arbitrary surreal normal forms, and it does not verify the ambient support argument, namely that the infinite expansion [\(14.9\)](#) lies outside  $\mathbf{Oz}[\omega^{-a}]$ ; that is a mathematical proof in the text. For a formalization, 07 recommends keeping the support characterization of  $\mathbf{Oz}[\omega^{-a}]$  and the exact binomial kernel separate from the routine finite Chinese remainder calculations; no corresponding Lean declarations are claimed.

## 15 Integral closure, complete integral closure, and invisible equations

The universal theorem uses exponent scales *below* the supports of purely infinite elements. This section uses a scale *above* all negative exponents, which yields one denominator for a whole set-generated algebra (03). The consequences separate three notions that coincide for Noetherian domains: membership in  $\mathbf{Oz}$ , integrality over  $\mathbf{Oz}$ , and almost integrality over  $\mathbf{Oz}$ . The complete integral closure of  $\mathbf{Oz}$  is all of  $\mathbf{No}$ , while its integral closure  $\mathcal{N}$  is a proper, fine-dense subring that no set of generators produces (03). The section ends with a monic equation of 08 that has no root in  $\mathbf{Oz}[i]$  although its image in every set-sized ring has the root 1; its positive root is an explicit element of  $\mathcal{N} \setminus \mathbf{Oz}$ .

**Standing notation.** As in [Section 11.1](#),  $\kappa$  is an infinite cardinal with  $\mu = \text{cf}(\kappa) > \aleph_0$ ,  $\theta = \kappa^{\aleph_0}$ ,  $G_\kappa$  is the two-armed lexicographic group of 03,  $F_\kappa^{03} = H_\omega(\mathbb{R}, G_\kappa)$  is the field of countably supported Hahn series over  $G_\kappa$ , identified with its image in  $\mathbf{No}$ , and

$$A_\kappa^{03} = F_\kappa^{03} \cap \mathbf{Oz} = \mathbb{Z} + \Pi_\kappa^{03},$$

where  $\Pi_\kappa^{03}$  consists of the elements of  $F_\kappa^{03}$  with strictly positive support. We write  $X^g$  ( $g \in G_\kappa$ ) for the monomials of  $F_\kappa^{03}$ ; under the identification they are surreal monomials. The Gaussian versions are  $F_\kappa^{03}[i] = H_\omega(\mathbb{C}, G_\kappa)$ ,  $A_\kappa^{03}[i] = \mathbb{Z}[i] + \Pi_\kappa^{03} + i\Pi_\kappa^{03}$  and the ideal  $\Pi_\kappa^{03} + i\Pi_\kappa^{03}$ . The field  $F_\kappa^{03}$  is real closed and  $F_\kappa^{03}[i]$  is algebraically closed ([Section 11.1](#)). We write  $\overline{\mathbb{Z}}$  for the ring of complex algebraic integers and  $\overline{\mathbb{Z}}_\mathbb{R} = \overline{\mathbb{Z}} \cap \mathbb{R}$  for the real ones. For an inclusion of rings  $A \subseteq R$  the *conductor* is  $(A : R) = \{a \in A : aR \subseteq A\}$ .

## 15.1 Uniform denominators

The clearing statement [Proposition 3.4\(i\)](#) clears one set of surreals. Applied to a well-chosen set it clears a whole algebra.

**Theorem 15.1** (One denominator for a set-generated algebra [03]). *For every set  $S \subseteq \mathbf{No}$  there is a positive monomial  $d = \omega^h \in \Pi$  with*

$$d \mathbf{Oz}[S] \subseteq \Pi. \quad (15.1)$$

*For every set  $S \subseteq \mathbf{No}[i]$  there is such a  $d$  with*

$$d \mathbf{Oz}[i][S] \subseteq \Pi_{\mathbb{C}}. \quad (15.2)$$

*In particular  $d$  is a nonzero element of the conductor  $(\mathbf{Oz} : \mathbf{Oz}[S])$ , respectively  $(\mathbf{Oz}[i] : \mathbf{Oz}[i][S])$ .*

*Proof.* Although  $\mathbf{Oz}[S]$  is in general a proper class, the ring  $\mathbb{Z}[S]$  is a set: it consists of the values of ordinary integer polynomials in finitely many members of  $S$ . By [Proposition 3.4\(i\)](#) there is  $h > 0$  with  $\omega^h \mathbb{Z}[S] \subseteq \Pi$ . Every element of  $\mathbf{Oz}[S]$  is a finite sum  $\sum_j a_j b_j$  with  $a_j \in \mathbf{Oz}$  and  $b_j \in \mathbb{Z}[S]$ , and

$$\omega^h \sum_j a_j b_j = \sum_j a_j (\omega^h b_j) \in \Pi$$

because  $\Pi$  is an ideal of  $\mathbf{Oz}$  ([Lemma 2.2](#)). The complex statement is the same argument with the set  $\mathbb{Z}[i][S]$ , the complex clause of [Proposition 3.4\(i\)](#) and the ideal  $\Pi_{\mathbb{C}}$  of  $\mathbf{Oz}[i]$ . Finally  $\Pi \subseteq \mathbf{Oz}$  and  $\Pi_{\mathbb{C}} \subseteq \mathbf{Oz}[i]$ .  $\square$

*Remark 15.2* (Conductors [03]). The theorem gives more than a nonzero element of the conductor  $(\mathbf{Oz} : \mathbf{Oz}[S])$ : it gives a monomial whose products with the whole algebra land in the strictly positive-support ideal  $\Pi$ . The real clearing statement and its proof are due to manuscript 05 (now in the sibling report [30]); the algebra formulation and the closure consequences below are 03's.

**Theorem 15.3** (The bounded-size version [03]). *Let  $A$  be  $A_{\kappa}^{03}$  or  $A_{\kappa}^{03}[i]$ , with ambient field  $F = F_{\kappa}^{03}$ , respectively  $F_{\kappa}^{03}[i]$ , constant ring  $D = \mathbb{Z}$ , respectively  $\mathbb{Z}[i]$ , and ideal  $P = \Pi_{\kappa}^{03}$ , respectively  $\Pi_{\kappa}^{03} + i\Pi_{\kappa}^{03}$ . For every set  $S \subseteq F$  with  $|S| < \mu = \text{cf}(\kappa)$  there is  $h > 0$  in  $G_{\kappa}$  with*

$$X^h A[S] \subseteq P. \quad (15.3)$$

*In particular  $\text{Frac}(A_{\kappa}^{03}) = F_{\kappa}^{03}$  and  $\text{Frac}(A_{\kappa}^{03}[i]) = F_{\kappa}^{03}[i]$ .*

*Proof.* The ring  $D[S]$  has cardinality at most  $\max(|S|, \aleph_0) < \mu$ , since  $\mu$  is uncountable. Each of its elements has countable support (in the complex case, take the union of the supports of the real and imaginary coordinates). The union  $U$  of these supports is a union of fewer than  $\mu$  countable sets, so  $|U| < \mu$  because  $\mu$  is regular and uncountable. By [Lemma 11.2\(i\)](#) the family  $\{-g : g \in U\} \cup \{0\}$ , which has fewer than  $\mu$  members, has an upper bound  $b \in G_{\kappa}$ ; put  $h = b + e$  for any  $e > 0$  in  $G_{\kappa}$ . Then  $h > 0$  and  $h > -g$  for every  $g \in U$ . Multiplication by  $X^h$  translates each support by  $h$ , so it maps every element of  $D[S]$  to a countably supported series with strictly positive support, that is, into  $P$ . Every element of  $A[S]$  is a finite sum  $\sum_j a_j b_j$  with  $a_j \in A$  and  $b_j \in D[S]$ , and  $X^h \sum_j a_j b_j = \sum_j a_j (X^h b_j) \in P$  because  $P$  is an ideal of  $A$ .

For  $x \in F$  apply this to  $S = \{x\}$ : then  $x = (X^h x)/X^h$  is a quotient of elements of  $A$ , and  $X^h \in P \subseteq A$  is nonzero.  $\square$

This supplies the fraction-field clause  $\text{Frac}(A_{\kappa}^{03}) = F_{\kappa}^{03}$  of 03's cardinal realization (compare [Theorem 11.5](#)). Clearing one element is weaker than clearing *all powers* of that element by the same denominator; both theorems supply the latter as well, since  $x^n \in A[x]$  for every  $n$ .



## 15.2 Almost integrality and complete integral closure

**Definition 15.4** ([03]). Let  $A$  be a domain contained in a field  $F$ . An element  $x \in F$  is *almost integral* over  $A$  if there is  $0 \neq d \in A$  with  $dx^n \in A$  for every ordinary  $n \geq 0$ . When  $F = \text{Frac}(A)$ , the set  $A^*$  of these elements is the *complete integral closure* of  $A$ . The *integral closure*  $\text{IC}_F(A)$  consists instead of the roots in  $F$  of monic polynomials in  $A[T]$ .

These are the standard notions of commutative algebra; they differ in non-Noetherian settings, and we take the terminology from [3] without importing any result of that paper as a proof. The quantifier over powers ranges over ordinary finite powers, not over ordinal iterations or surreal exponentiation. Each individual witness  $d$  and each monic polynomial is set-sized data, so the definitions apply verbatim to the class rings  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ .

**Theorem 15.5** (Every ambient element is almost integral [03]). *Relative to their fraction fields,*

$$\mathbf{Oz}^* = \mathbf{No}, \quad \mathbf{Oz}[i]^* = \mathbf{No}[i], \quad (15.4)$$

$$(A_\kappa^{03})^* = F_\kappa^{03}, \quad (A_\kappa^{03}[i])^* = F_\kappa^{03}[i]. \quad (15.5)$$

More generally, every ring  $R$  with  $A \subseteq R \subseteq F$ , where  $(A, F)$  is one of the four pairs  $(\mathbf{Oz}, \mathbf{No})$ ,  $(\mathbf{Oz}[i], \mathbf{No}[i])$ ,  $(A_\kappa^{03}, F_\kappa^{03})$ ,  $(A_\kappa^{03}[i], F_\kappa^{03}[i])$ , has  $\text{Frac}(R) = F$  and  $R^* = F$ .

*Proof.* The fraction fields are the indicated ambient fields by Proposition 3.4(ii) and Theorem 15.3, so there are no further elements to consider. For  $x$  in the ambient field apply Theorem 15.1 or Theorem 15.3 to  $S = \{x\}$ . The single nonzero denominator  $d$  supplied there satisfies  $dx^n \in dA[x] \subseteq A$  for every ordinary  $n \geq 0$ , so  $x$  is almost integral. If  $A \subseteq R \subseteq F$ , then  $F = \text{Frac}(A) \subseteq \text{Frac}(R) \subseteq F$ , and the same  $d \in A \subseteq R$  gives  $dx^n \in A \subseteq R$ .  $\square$

**Example 15.6** (Two explicit denominators [03]). The rational  $1/2$  is almost integral over  $\mathbf{Oz}$ , witnessed by  $d = \omega$ : indeed  $\omega/2^n \in \Pi$  for all  $n \geq 0$ . The monomial  $\omega^{-1}$  is almost integral, witnessed by  $d = \omega^\omega$ : indeed  $\omega^{\omega-n} \in \Pi$  for every ordinary  $n$ , since  $\omega - n > 0$ . Neither claim uses a limit as  $n$  tends to an infinite ordinal. Neither element is integral (Lemma 15.7 below).

## 15.3 Integral closure: reciprocals, small units and density

The geometric statements are proved first for an arbitrary integer part of a real closed field; this isolates what is a consequence of integer-part geometry rather than of the proper-class size of  $\mathbf{No}$  (03).

**Lemma 15.7** (An integral reciprocal was already present [03]). *Let  $A$  be a domain in a field  $F$  and  $0 \neq a \in A$ . If  $a^{-1}$  is integral over  $A$ , then  $a^{-1} \in A$ . Hence the integral closure of  $A$  in  $F$  contains the inverse of no nonunit of  $A$ .*

*Proof.* Suppose  $(a^{-1})^n + c_{n-1}(a^{-1})^{n-1} + \dots + c_0 = 0$  with  $c_j \in A$  and  $n \geq 1$ . Multiplying by  $a^{n-1}$  gives

$$a^{-1} = -c_{n-1} - c_{n-2}a - \dots - c_0a^{n-1} \in A. \quad \square$$

Consequently  $1/2$  is not integral over  $\mathbf{Oz}$ ,  $A_\kappa^{03}$ ,  $\mathbf{Oz}[i]$  or  $A_\kappa^{03}[i]$ , despite Example 15.6:  $2$  is a nonunit of each of these rings by Lemma 2.4 (the units of  $A_\kappa^{03} \subseteq \mathbf{Oz}$  are among  $\mathbf{Oz}^\times = \{\pm 1\}$ , and similarly in the Gaussian case). More generally, for  $h > 0$  and  $0 \neq c \in \mathbb{R}$  the negative monomial  $c\omega^{-h}$  is not integral over  $\mathbf{Oz}$ , and for  $0 \neq c \in \mathbb{C}$  it is not integral over  $\mathbf{Oz}[i]$ : its reciprocal  $c^{-1}\omega^h$

lies in the purely infinite ideal  $\Pi$ , respectively  $\Pi_{\mathbb{C}}$ , and is a nonunit. The same holds for  $cX^{-h}$  over  $A_{\kappa}^{03}$  and  $A_{\kappa}^{03}[i]$  when  $h > 0$  in  $G_{\kappa}$ .

**Lemma 15.8** (Integer parts [03]). *Recall that a subring  $A$  of an ordered field  $F$  is an integer part if for every  $x \in F$  there is a unique  $\lfloor x \rfloor \in A$  with  $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$ . Then  $\mathbf{Oz}$  is an integer part of  $\mathbf{No}$  and  $A_{\kappa}^{03}$  is an integer part of  $F_{\kappa}^{03}$ . Every nonzero element of  $\mathbf{Oz}$  has absolute value at least 1, and every nonzero element of  $\mathbf{Oz}[i]$  has surreal modulus  $|a + ib| = \sqrt{a^2 + b^2}$  at least 1.*

*Proof.* A nonzero element of  $\mathbf{Oz}$  is either a nonzero ordinary integer or has positive degree (Lemma 2.4); in the second case its absolute value exceeds every real number. Hence  $|a| \geq 1$  for  $0 \neq a \in \mathbf{Oz}$ , and for  $a + ib \in \mathbf{Oz}[i]$  with  $a, b \in \mathbf{Oz}$  not both zero,  $\sqrt{a^2 + b^2} \geq \max(|a|, |b|) \geq 1$ . This lower bound also gives uniqueness of floors: two elements of  $\mathbf{Oz}$  in a half-open interval of length 1 differ by an element of absolute value less than 1. For existence, write the normal form of  $x \in \mathbf{No}$  uniquely as  $x = P + r + \epsilon$  with  $P \in \Pi$ ,  $r \in \mathbb{R}$  and  $\text{supp } \epsilon$  strictly negative, so that  $\epsilon$  is zero or infinitesimal. Take  $\lfloor x \rfloor = P + \lfloor r \rfloor$ , except when  $r \in \mathbb{Z}$  and  $\epsilon < 0$ , where  $\lfloor x \rfloor = P + r - 1$ . In every case  $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$ . For  $x \in F_{\kappa}^{03}$  the same decomposition has  $P \in \Pi_{\kappa}^{03}$ , so the floor lies in  $A_{\kappa}^{03}$ .  $\square$

The exceptional case in the floor formula matters: discarding all negative exponents sends  $-\omega^{-1}$  to 0, which is not its floor.

**Why integral elements form a ring.** That the integral elements over a commutative ring  $A$  form a ring, and that integrality is transitive, is finite algebra, which we recall because it is applied to class rings. If  $x$  satisfies a monic equation of degree  $n$  over  $A$ , then  $A[x] = A + Ax + \dots + Ax^{n-1}$  is a finitely generated  $A$ -module, by reducing powers with that equation. If  $x$  and  $y$  are integral,  $A[x, y]$  is generated by finitely many monomials  $w_1 = 1, w_2, \dots, w_N$  in  $x$  and  $y$ . For  $z \in A[x, y]$  write  $zw_k = \sum_l a_{kl}w_l$  with  $a_{kl} \in A$ ; the determinant trick shows that  $\det(zI - (a_{kl}))$  annihilates every  $w_l$ , in particular  $w_1 = 1$ , so it is zero, and this is a monic equation for  $z$ . The same argument, applied to the finitely many coefficients of a monic equation and to the element, proves transitivity. Only finitely many elements are involved at each step, so the arguments apply unchanged to  $\mathbf{Oz}$ ,  $\mathbf{Oz}[i]$  and their overrings in  $\mathbf{No}$  and  $\mathbf{No}[i]$  (03).

**Theorem 15.9** (Small integral units [03]). *Let  $F$  be a real closed ordered field,  $A \subseteq F$  an integer part, and  $N$  the integral closure of  $A$  in  $F$ . For every  $H \in A_{>0}$  the element  $u_H = \sqrt{H^2 + 1} - H$  is a positive unit of  $N$  satisfying*

$$u_H^2 + 2Hu_H - 1 = 0, \quad u_H^{-1} = u_H + 2H, \quad 0 < u_H < \frac{1}{2H}. \quad (15.6)$$

*For every  $\epsilon \in F_{>0}$  there is such a unit with  $0 < u_H < \epsilon$ .*

*Proof.* Since  $\sqrt{H^2 + 1} > H > 0$ , the element  $u_H$  is positive. Squaring  $u_H + H = \sqrt{H^2 + 1}$  gives the first equation, which is monic over  $A$ ; so  $u_H \in N$ . The same equation says  $u_H(u_H + 2H) = 1$ , and  $u_H + 2H \in N$  because  $N$  is a ring containing  $A$ ; so  $u_H$  is a unit of  $N$ . Rationalizing,

$$u_H = \frac{1}{\sqrt{H^2 + 1} + H} < \frac{1}{2H}.$$

An integer part is cofinal in  $F$ :  $\lfloor 1/(2\epsilon) \rfloor + 1 \in A$  is positive and exceeds  $1/(2\epsilon)$ . Taking  $H$  to be this element gives  $u_H < 1/(2H) < \epsilon$ .  $\square$

**Theorem 15.10** (Dense normalization of an integer part [03]). *Under the hypotheses of Theorem 15.9,  $N$  is dense in  $F$  for the order topology. More precisely, for every  $x \in F$  and every  $\epsilon \in F_{>0}$  there is  $y \in N$  with*

$$0 \leq x - y < \epsilon. \quad (15.7)$$

*Proof.* Choose a positive unit  $u < \epsilon$  of  $N$  by Theorem 15.9 and let  $n = \lfloor x/u \rfloor \in A$ . Multiplying  $n \leq x/u < n+1$  by  $u > 0$  gives  $0 \leq x - un < u < \epsilon$ ; put  $y = un \in N$ . To meet a prescribed open interval  $(a, b)$ , apply this with  $x = (a+b)/2$  and  $\epsilon < (b-a)/2$ ; then  $y \in (x - \epsilon, x] \subseteq (a, b)$ .  $\square$

No completion and no sequential argument is used. The same proof is valid for the order topology of **No** (the fine topology), with all positive surreal radii. Results on dense subfields and integer parts in other settings provide context [15]; the object here is the integral closure of a specified integer part, not an arbitrary dense subfield, and the normalization theorem is not attributed to that work. The small-unit argument is a general elementary lemma about integer parts, and 03 makes no priority claim for it.

**The normalizations.** We write

$$\mathcal{N} = \text{IC}_{\mathbf{No}}(\mathbf{Oz}), \quad \mathcal{N}_\kappa = \text{IC}_{F_\kappa^{03}}(A_\kappa^{03}). \quad (15.8)$$

These are 03's  $B$  and  $B_\kappa$ . Manuscript 03 wrote  $\text{Int}_F(A)$  for the integral closure of  $A$  in  $F$ ; we write  $\text{IC}_F(A)$  because  $\text{Int}$  denotes rings of integer-valued polynomials, such as  $\text{Int}(\mathbb{Z}^r)$ , elsewhere in this report. Note that  $\mathcal{N}_\kappa$  is defined relative to  $F_\kappa^{03}$ ; only  $\mathcal{N}_\kappa \subseteq \mathcal{N} \cap F_\kappa^{03}$  is automatic, and no descent equality is claimed.

**Theorem 15.11** (Set-generated intermediate algebras are uniformly discrete [03]). *For every set  $S \subseteq \mathbf{No}$  the algebra  $C = \mathbf{Oz}[S]$  has a positive surreal separation bound  $r$ :*

$$|x - y| \geq r \quad \text{for all distinct } x, y \in C.$$

*It is a closed, uniformly discrete subclass of **No** in the fine order topology. The same holds for  $A_\kappa^{03}[S] \subseteq F_\kappa^{03}$ , in the order topology of  $F_\kappa^{03}$ , whenever  $|S| < \mu$ .*

*Proof.* Choose the positive monomial  $d$  of Theorem 15.1, respectively  $d = X^h$  of Theorem 15.3. For distinct  $x, y \in C$  the product  $d(x - y)$  is a nonzero element of  $\Pi$  (respectively  $\Pi_\kappa^{03}$ ), hence of the integer part, so  $|d(x - y)| \geq 1$  by Lemma 15.8. Thus  $|x - y| \geq 1/d =: r$ .

Uniform discreteness gives an isolating interval of radius  $r/3$  around each point. For closedness, let  $z \notin C$ . If no point of  $C$  lies within  $r/3$  of  $z$ , that neighbourhood misses  $C$ . Otherwise let  $c \in C$  be such a point; every other point of  $C$  is at distance greater than  $2r/3$  from  $z$  by the triangle inequality, so a radius smaller than both  $r/3$  and  $|z - c|$  gives a neighbourhood of  $z$  disjoint from  $C$ .  $\square$

This is a class-size statement about a proper-class base ring; it is not the familiar assertion that a finite extension of a discrete ring is discrete in an Archimedean topology (03).

**Theorem 15.12** (Proper dense normalization, zero conductor, no set of generators [03]). *For the full omnific ring,*

$$\mathbf{Oz} \subsetneq \mathcal{N} \subsetneq \mathbf{No}, \quad \overline{\mathcal{N}}^{\text{fine}} = \mathbf{No}, \quad (\mathbf{Oz} : \mathcal{N}) = 0, \quad (15.9)$$

*and  $\mathcal{N}$  is not generated as an  $\mathbf{Oz}$ -algebra by any set of its elements. For the set-sized ring  $A_\kappa^{03}$ , the normalization  $\mathcal{N}_\kappa$  satisfies  $A_\kappa^{03} \subsetneq \mathcal{N}_\kappa \subsetneq F_\kappa^{03}$ , is dense in  $F_\kappa^{03}$ , has conductor  $(A_\kappa^{03} : \mathcal{N}_\kappa) = 0$ , and requires at least  $\mu$  algebra generators over  $A_\kappa^{03}$ .*

*Proof.* The element  $\sqrt{2} \in \mathbb{R} \subseteq F_\kappa^{03}$  is integral, being a root of  $T^2 - 2$ , and it is in neither integer part, whose real elements are the ordinary integers; so the first inclusions are strict. The element  $1/2$  is not integral by Lemma 15.7, so the second inclusions are strict. The fields  $\mathbf{No}$  and  $F_\kappa^{03}$  are real closed and  $\mathbf{Oz}$ ,  $A_\kappa^{03}$  are integer parts (Lemma 15.8); density is Theorem 15.10.

For the conductor, fix  $a \neq 0$  in the integer part. Choose a positive unit  $u$  of the normalization with  $u < 1/|a|$  (Theorem 15.9). Then  $au$  lies in the normalization and  $0 < |au| < 1$ , so  $au$  is not in the integer part by Lemma 15.8. Hence  $a$  is not in the conductor, and this holds for every  $a \neq 0$ .

If  $\mathcal{N} = \mathbf{Oz}[S]$  for a set  $S$ , then Theorem 15.1 would give a nonzero element  $d$  of the conductor, a contradiction. (Equivalently,  $\mathcal{N}$  would be uniformly discrete by Theorem 15.11, contradicting density.) The same argument with Theorem 15.3 shows that  $\mathcal{N}_\kappa \neq A_\kappa^{03}[S]$  whenever  $|S| < \mu$ .  $\square$

Integral closure therefore changes the fine geometry of the ring, although  $\mathcal{N}$  still contains neither  $1/2$  nor any nonzero negative monomial  $c\omega^{-h}$ . For  $\kappa = \aleph_1$  the theorem says, without the continuum hypothesis, that the normalization cannot be obtained by adjoining countably many elements to the continuum-sized ring  $A_{\aleph_1}^{03}$  (here  $\theta = \aleph_1^{\aleph_0} = 2^{\aleph_0}$ ). For general  $\kappa$  the theorem gives only the *lower bound*  $\mu$  on the number of algebra generators of  $\mathcal{N}_\kappa$ , not an exact generation number; the exact count  $\mu$  of 03's cardinal realization (Theorem 11.5) concerns the ideal  $\Pi_\kappa^{03}$ , not the normalization.

*Remark 15.13* (Dense does not mean sequentially approximable [03]). The fine topology of  $\mathbf{No}$  has the size obstructions discussed in the foundations report [26]. The density statement chooses a point separately for each positive surreal radius; it does not produce a nonconstant convergent set-indexed net. Indeed, every fixed set of elements of  $\mathcal{N}$  lies in a single uniformly discrete algebra  $\mathbf{Oz}[S]$  by Theorem 15.11. The dense normalization requires access to more than any fixed set of its additional generators.

## 15.4 Constant slices, truncation, and a unit equation

The results above do not give a complete normal-form test for integrality. They do give exact tests on the nonnegative-support slice, and they show why leading-term tests fail (03). Recall  $\mathcal{A}_{\mathbb{R},\mathbb{R}} = \mathbb{R} + \Pi$  (03's  $P_{\mathbb{R}}$ ), on which  $\text{ct}$  is a ring homomorphism onto  $\mathbb{R}$  by Lemma 2.2, although  $\mathcal{A}_{\mathbb{R},\mathbb{R}}$  is not an integer part; its countable version is  $\mathbb{R} + \Pi_\kappa^{03}$ .

**Theorem 15.14** (The real constant and support slices [03]).

$$\mathcal{N} \cap \mathcal{A}_{\mathbb{R},\mathbb{R}} = \overline{\mathbb{Z}}_{\mathbb{R}} + \Pi = \mathcal{A}_{\overline{\mathbb{Z}}_{\mathbb{R}},\mathbb{R}}, \quad \mathcal{N} \cap \mathbb{R} = \overline{\mathbb{Z}}_{\mathbb{R}}, \quad \mathcal{N} \cap \mathbb{Q} = \mathbb{Z}. \quad (15.10)$$

The same formulas hold for  $\mathcal{N}_\kappa$ , with  $\mathcal{A}_{\mathbb{R},\mathbb{R}}$  and  $\Pi$  replaced by  $\mathbb{R} + \Pi_\kappa^{03}$  and  $\Pi_\kappa^{03}$  in the first formula.

*Proof.* Let  $x = r + P$  with  $r \in \mathbb{R}$  and  $P \in \Pi$ , and suppose  $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$  with  $a_j \in \mathbf{Oz}$ . All terms lie in  $\mathcal{A}_{\mathbb{R},\mathbb{R}}$ , and applying the ring homomorphism  $\text{ct}$  gives  $r^n + \text{ct}(a_{n-1})r^{n-1} + \cdots + \text{ct}(a_0) = 0$ , a monic equation over  $\mathbb{Z}$ . So  $r \in \overline{\mathbb{Z}}_{\mathbb{R}}$ . Conversely, a real algebraic integer  $r$  is integral over  $\mathbb{Z} \subseteq \mathbf{Oz}$ , and  $P \in \Pi \subseteq \mathbf{Oz}$ , so  $r + P$  is integral over  $\mathbf{Oz}$  because integral elements form a ring. This is the first equality; intersecting with  $\mathbb{R}$  gives the second, and the third holds because a rational root of a monic integer polynomial is an integer. For  $\mathcal{N}_\kappa$  the coefficients lie in  $A_\kappa^{03} \subseteq \mathbf{Oz}$ , the element lies in  $\mathbb{R} + \Pi_\kappa^{03}$ , and the same argument applies; conversely  $\overline{\mathbb{Z}}_{\mathbb{R}} \subseteq \mathbb{R} \subseteq F_\kappa^{03}$  and  $\Pi_\kappa^{03} \subseteq A_\kappa^{03}$ .  $\square$

**Example 15.15** ([03]). The element  $\pi\omega + \sqrt{2}$  is integral over  $\mathbf{Oz}$ , whereas  $\pi\omega + \pi$  is not. The nature of a coefficient at a *positive* exponent is irrelevant to the test; only the ordinary constant coefficient is constrained. The element  $\omega + 1/2$  is almost integral (Theorem 15.5) but not integral, since  $1/2 \notin \overline{\mathbb{Z}}_{\mathbb{R}}$ .

**Proposition 15.16** (Normalization is not truncation closed [03]). *Let  $h > 0$  and  $H = \omega^h$  (or  $H = X^h$  with  $h > 0$  in  $G_\kappa$ ). The integral unit  $u_H$  of Theorem 15.9 has the normal form*

$$u_H = \sum_{n \geq 1} \binom{1/2}{n} \omega^{-(2n-1)h} = \frac{1}{2}\omega^{-h} - \frac{1}{8}\omega^{-3h} + \frac{1}{16}\omega^{-5h} - \frac{5}{128}\omega^{-7h} + \frac{7}{256}\omega^{-9h} - \dots \quad (15.11)$$

*Its leading monomial  $\frac{1}{2}\omega^{-h}$  is not integral over the integer part. Therefore neither  $\mathcal{N}$  nor  $\mathcal{N}_\kappa$  is closed under leading normal-form truncation.*

*Proof.* By Lemma 15.24 below with  $m = 2$  and  $\gamma = 2h$ , the series  $s = \omega^h \mathbf{b}_2(\omega^{-2h})$  satisfies  $s^2 = \omega^{2h}(1 + \omega^{-2h}) = H^2 + 1$ . Its leading term is  $\omega^h > 0$ , so  $s$  is the positive square root  $\sqrt{H^2 + 1}$  (an ordered field has at most one positive square root of a given element). Hence

$$u_H = s - \omega^h = \omega^h (\mathbf{b}_2(\omega^{-2h}) - 1) = \sum_{n \geq 1} \binom{1/2}{n} \omega^{h-2nh},$$

a series with countable, reverse well-ordered support  $\{-(2n-1)h : n \geq 1\}$ . The first five binomial coefficients are  $\binom{1/2}{1} = \frac{1}{2}$ ,  $\binom{1/2}{2} = -\frac{1}{8}$ ,  $\binom{1/2}{3} = \frac{1}{16}$ ,  $\binom{1/2}{4} = -\frac{5}{128}$  and  $\binom{1/2}{5} = \frac{7}{256}$ . In the countable model the same computation takes place in  $F_\kappa^{03}$ . The leading monomial  $\frac{1}{2}\omega^{-h}$  is the reciprocal of the nonunit  $2\omega^h$  of the integer part, so it is not integral by Lemma 15.7.  $\square$

This is stronger than the statement that some integral elements have negative exponents: their negative tails can be essential, since deleting all but the leading term destroys integrality. In the opposite direction, Theorem 15.14 gives a complete constant-term criterion when negative exponents are absent (03).

**Proposition 15.17** (A unit equation acquires solutions only after normalization [03]). *Let  $A$  be  $\mathbf{Oz}$  or  $A_\kappa^{03}$  and  $0 \neq H \in A$ . The equation*

$$U^2 + 2HU - 1 = 0 \quad (15.12)$$

*has no solution in  $A$ . For  $H > 0$  its two roots in the real closed ambient field are  $u_H$  and  $-2H - u_H$ ; both are integral units and belong to the normalization.*

*Proof.* A solution in  $A$  would satisfy  $U(U + 2H) = 1$ , so  $U$  would be a unit of  $A$ , hence  $U = \pm 1$  (Lemma 2.4;  $A_\kappa^{03} \subseteq \mathbf{Oz}$ ). Substituting  $U = 1$  gives  $2H = 0$  and substituting  $U = -1$  gives  $-2H = 0$ , both contrary to  $H \neq 0$ . The two roots are  $u_H$  and  $-2H - u_H$  because their sum is  $-2H$ . Each is integral by the monic equation, and their product is  $-1$ , so each is a unit of the normalization.  $\square$

Thus membership, integrality and almost integrality give three different arithmetic environments even for a single quadratic equation with a parameter. This does not contradict constant-term transfer for systems with ordinary integer coefficients: here a possibly infinite coefficient  $H$  is fixed, and changing the ambient ring changes the allowed witnesses (03).

## 15.5 The surcomplex and Gaussian versions

The Gaussian omnific ring  $\mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz} = \mathbb{Z}[i] \oplus \Pi_\mathbb{C}$  is the natural Gaussian analogue of  $\mathbf{Oz}$ ; it is not discretely ordered. It has the split Gaussian constant term  $\text{ct} : \mathbf{Oz}[i] \rightarrow \mathbb{Z}[i]$  (Lemma 2.2), its

units are  $\pm 1, \pm i$  (Lemma 2.4), and each nonzero element has surreal modulus at least 1 (Lemma 15.8). Put

$$\mathcal{N}_{\mathbb{C}} = \text{IC}_{\mathbf{No}[i]}(\mathbf{Oz}[i]), \quad \mathcal{N}_{\kappa, \mathbb{C}} = \text{IC}_{F_{\kappa}^{03}[i]}(A_{\kappa}^{03}[i]), \quad (15.13)$$

03's  $C$  and  $C_{\kappa}$ . Since  $i$  is integral over  $\mathbf{Oz}$ , transitivity shows that these are also the integral closures of  $\mathbf{Oz}$  in  $\mathbf{No}[i]$  and of  $A_{\kappa}^{03}$  in  $F_{\kappa}^{03}[i]$ . Complex conjugation is a ring automorphism of  $\mathbf{No}[i]$  preserving  $\mathbf{Oz}[i]$  (and of  $F_{\kappa}^{03}[i]$  preserving  $A_{\kappa}^{03}[i]$ ), so it preserves  $\mathcal{N}_{\mathbb{C}}$  and  $\mathcal{N}_{\kappa, \mathbb{C}}$ . Consequently

$$\mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}, \quad \mathcal{N}_{\kappa, \mathbb{C}} \cap F_{\kappa}^{03} = \mathcal{N}_{\kappa}. \quad (15.14)$$

Complexification nevertheless does not commute with normalization, and the failure is located at the prime 2.

**Proposition 15.18** (The complexification defect is killed by 2 [03]). *There are strict inclusions of additive subclasses*

$$\mathcal{N}[i] \subsetneq \mathcal{N}_{\mathbb{C}} \subsetneq \tfrac{1}{2}\mathcal{N}[i], \quad \mathcal{N}_{\mathbb{C}}[1/2] = \mathcal{N}[1/2][i]. \quad (15.15)$$

The last object of the chain is a fractional module, not asserted to be a ring. In particular  $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i]$  although  $\mathcal{N}_{\mathbb{C}} \neq \mathcal{N}[i]$ . The same inclusions hold for  $\mathcal{N}_{\kappa}$  and  $\mathcal{N}_{\kappa, \mathbb{C}}$ , and the set-sized quotient  $\mathcal{N}_{\kappa, \mathbb{C}}/\mathcal{N}_{\kappa}[i]$  is a nonzero  $\mathcal{N}_{\kappa}$ -module annihilated by 2.

*Proof.* Every element of  $\mathcal{N}[i]$  is integral over  $\mathbf{Oz}[i]$ , so  $\mathcal{N}[i] \subseteq \mathcal{N}_{\mathbb{C}}$ . If  $z = x + iy \in \mathcal{N}_{\mathbb{C}}$  with  $x, y \in \mathbf{No}$ , then  $\bar{z} \in \mathcal{N}_{\mathbb{C}}$ , so

$$2x = z + \bar{z} \in \mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}, \quad 2y = -i(z - \bar{z}) \in \mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}$$

by (15.14). Hence  $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i]$ , that is,  $\mathcal{N}_{\mathbb{C}} \subseteq \tfrac{1}{2}\mathcal{N}[i]$ . Inverting 2 in  $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i] \subseteq \mathcal{N}_{\mathbb{C}}$  gives  $\mathcal{N}_{\mathbb{C}}[1/2] = \mathcal{N}[i][1/2] = \mathcal{N}[1/2][i]$ .

The root of unity  $\zeta = (-1 + i\sqrt{3})/2$  satisfies  $\zeta^2 + \zeta + 1 = 0$ , so  $\zeta \in \mathcal{N}_{\mathbb{C}}$ ; its real coordinate  $-1/2$  is not in  $\mathcal{N}$  by Lemma 15.7, so  $\zeta \notin \mathcal{N}[i]$ . Finally  $1/2 \in \tfrac{1}{2}\mathcal{N}[i]$  is not in  $\mathcal{N}_{\mathbb{C}}$ , again by the reciprocal lemma, since 2 is a nonunit of  $\mathbf{Oz}[i]$ . The proof is unchanged in  $F_{\kappa}^{03}[i]$ , which contains  $\zeta$ ; there  $\zeta$  represents a nonzero class of  $\mathcal{N}_{\kappa, \mathbb{C}}/\mathcal{N}_{\kappa}[i]$ , and this quotient of  $\mathcal{N}_{\kappa}$ -modules is annihilated by 2 because  $2\mathcal{N}_{\kappa, \mathbb{C}} \subseteq \mathcal{N}_{\kappa}[i]$ .  $\square$

**Theorem 15.19** (Dense Gaussian normalization and its constant slice [03]). *The ring  $\mathcal{N}_{\mathbb{C}}$  is a proper dense subring of  $\mathbf{No}[i]$  for the surreal-modulus topology, its conductor  $(\mathbf{Oz}[i] : \mathcal{N}_{\mathbb{C}})$  is zero, and it cannot be generated over  $\mathbf{Oz}[i]$  by a set. Moreover, with  $\mathcal{A}_{\mathbb{C}, \mathbb{C}} = \mathbb{C} + \Pi_{\mathbb{C}}$ ,*

$$\mathcal{N}_{\mathbb{C}} \cap \mathcal{A}_{\mathbb{C}, \mathbb{C}} = \overline{\mathbb{Z}} + \Pi_{\mathbb{C}} = \mathcal{A}_{\overline{\mathbb{Z}}, \mathbb{C}}, \quad \mathcal{N}_{\mathbb{C}} \cap \mathbb{C} = \overline{\mathbb{Z}}. \quad (15.16)$$

For  $\mathcal{N}_{\kappa, \mathbb{C}}$  the analogous properness, density, slice (with  $\Pi_{\kappa}^{03} + i\Pi_{\kappa}^{03}$  in place of  $\Pi_{\mathbb{C}}$ ) and zero-conductor statements hold, and fewer than  $\mu$  algebra generators over  $A_{\kappa}^{03}[i]$  do not suffice.

*Proof.* Let  $z = x + iy \in \mathbf{No}[i]$  and  $\epsilon > 0$ . Choose a positive real integral unit  $u < \epsilon/2$  of  $\mathcal{N}$  (Theorem 15.9) and put

$$n = \lfloor x/u \rfloor, \quad m = \lfloor y/u \rfloor, \quad w = u(n + im).$$

The floors lie in  $\mathbf{Oz}$  and  $u \in \mathcal{N} \subseteq \mathcal{N}_{\mathbb{C}}$ , so  $w \in \mathcal{N}_{\mathbb{C}}$ . Each coordinate of  $z - w$  lies in  $[0, u)$ , hence  $|z - w| < \sqrt{2}u < \epsilon$ . This proves density. The element  $1/2$  is excluded by Lemma 15.7, which proves properness.



For  $0 \neq a \in \mathbf{Oz}[i]$  choose such a  $u$  with  $0 < u < 1/|a|$ . Then  $au \in \mathcal{N}_{\mathbb{C}}$  and  $0 < |au| < 1$ , so  $au \notin \mathbf{Oz}[i]$  by Lemma 15.8; thus the conductor is zero. If  $\mathcal{N}_{\mathbb{C}} = \mathbf{Oz}[i][S]$  for a set  $S$ , Theorem 15.1 would give a nonzero element of the conductor; for  $\mathcal{N}_{\kappa, \mathbb{C}}$  and fewer than  $\mu$  generators, Theorem 15.3 does the same. The two absorption theorems also show directly that every such small-generated algebra has a uniform modulus separation bound, as in Theorem 15.11.

Finally, let  $z = c + P$  with  $c \in \mathbb{C}$  and  $P \in \Pi_{\mathbb{C}}$ , integral over  $\mathbf{Oz}[i]$ . All terms of its monic equation lie in  $\mathcal{A}_{\mathbb{C}, \mathbb{C}}$ , and applying the ring homomorphism  $\text{ct} : \mathcal{A}_{\mathbb{C}, \mathbb{C}} \rightarrow \mathbb{C}$  shows that  $c$  is integral over  $\mathbb{Z}[i]$ ; since  $\mathbb{Z}[i]$  is integral over  $\mathbb{Z}$ ,  $c \in \overline{\mathbb{Z}}$ . Conversely such a  $c$  is integral over the base, and  $P$  already belongs to it. Restricting to constants gives the second equality. The same arguments work in  $F_{\kappa}^{03}[i]$ , using the real integer part  $A_{\kappa}^{03}$  of  $F_{\kappa}^{03}$  for the floors.  $\square$

The original integer-part rings have many finite quotients:  $\mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}$  for ordinary  $n \geq 1$  by (3.2), and likewise  $A_{\kappa}^{03}/nA_{\kappa}^{03} \cong \mathbb{Z}/n\mathbb{Z}$ , because the purely infinite ideal is divisible by every nonzero ordinary integer (compare Theorem 6.2). Their normalizations behave oppositely.

**Theorem 15.20** (No nonzero finite ring targets [03]). *None of  $\mathcal{N}$ ,  $\mathcal{N}_{\kappa}$ ,  $\mathcal{N}_{\mathbb{C}}$ ,  $\mathcal{N}_{\kappa, \mathbb{C}}$  admits a unital homomorphism to a nonzero finite ring. In particular none of them has a proper ideal of finite index.*

*Proof.* Suppose such a homomorphism exists. Its image is a nonzero finite commutative ring, even if the target is not commutative, and a maximal ideal of the image gives a unital homomorphism onto a finite field  $\mathbb{F}_q$ . The ordinary monic polynomial  $T^q - T - 1$  is negative at 0 and positive at 2, so it has a real root  $\alpha \in (0, 2)$ . This  $\alpha$  is a real algebraic integer and belongs to all four normalizations by Theorems 15.14 and 15.19. Its image  $\bar{\alpha}$  would satisfy  $\bar{\alpha}^q - \bar{\alpha} - 1 = 0$  in  $\mathbb{F}_q$ , whereas every element of  $\mathbb{F}_q$  satisfies  $\bar{\alpha}^q - \bar{\alpha} = 0$ ; so  $1 = 0$  in  $\mathbb{F}_q$ , which is impossible. A proper ideal of finite index would give a unital homomorphism onto a nonzero finite ring.  $\square$

The proof uses only the real algebraic integer constants. Accordingly this finite-quotient obstruction is a general consequence of containing those constants, not a phenomenon specific to surreal supports. It says nothing about infinite set-sized homomorphic images of the normalizations; none are classified here (03).

## 15.6 Set-valued valuations cannot contain the omnific ring

A *valuation* of a field  $F$  (possibly a proper class, such as  $\mathbf{No}$  or  $\mathbf{No}[i]$ ) is a map  $v : F^{\times} \rightarrow \Gamma$  to an ordered abelian group with  $v(xy) = v(x) + v(y)$  and  $v(x + y) \geq \min(v(x), v(y))$  whenever  $x + y \neq 0$ ; its valuation ring consists of 0 and the elements of nonnegative value. No compatibility with the ordering of  $F$  is assumed. The value group is *set-sized* if  $\Gamma$  is a set.

**Lemma 15.21** (No nonzero set-valued ordered quotient of  $(\mathbf{No}, +)$  [03]). *Every order-preserving additive class map  $w : (\mathbf{No}, +) \rightarrow \Gamma$  into a set-sized ordered abelian group is zero.*

*Proof.* Suppose  $w(x) \neq 0$  for some  $x$ . Replacing  $x$  by  $-x$  if necessary,  $w(x) > 0$ ; then  $x \neq 0$ , and  $x < 0$  would give  $w(x) \leq w(0) = 0$ , so  $h := x > 0$  and  $w(h) > 0$ . Choose a cardinal  $\lambda > |\Gamma|$  and consider the set of surreals  $\alpha h$  (surreal products) for ordinals  $\alpha < \lambda$ . If  $\alpha < \beta$ , then  $\beta \geq \alpha + 1$  in the surreal order, the ordinal successor being the surreal sum with 1. Hence  $\beta h - \alpha h \geq h$  and, by additivity and monotonicity,

$$w(\beta h) - w(\alpha h) = w(\beta h - \alpha h) \geq w(h) > 0.$$

So the  $\lambda$  elements  $\alpha h$  have pairwise distinct images in  $\Gamma$ , contradicting  $\lambda > |\Gamma|$ . Only a set of ordinals was used.  $\square$

**Theorem 15.22** (Full-class valuation obstruction [03]). *Let  $v : \mathbf{No}^\times \rightarrow \Gamma$  be a valuation with set-sized value group. If  $v(a) \geq 0$  for every  $0 \neq a \in \mathbf{Oz}$ , then  $v$  is trivial. Likewise, a valuation of  $\mathbf{No}[i]$  with set-sized value group whose valuation ring contains  $\mathbf{Oz}[i]$  is trivial.*

*Proof.* Define  $w(g) = v(\omega^g)$  for  $g \in \mathbf{No}$ . The monomial law  $\omega^{g+g'} = \omega^g \omega^{g'}$  makes  $w$  additive. If  $g > 0$  then  $\omega^g \in \Pi \subseteq \mathbf{Oz}$ , so  $w(g) \geq 0$ ; hence  $g \leq g'$  implies  $w(g') - w(g) = w(g' - g) \geq 0$ , and  $w$  is order preserving. By Lemma 15.21,  $w = 0$ .

For  $x \neq 0$ , clear the set  $\{x, x^{-1}\}$  by Proposition 3.4(i): there is a monomial  $d = \omega^h$  with  $dx, dx^{-1} \in \Pi \subseteq \mathbf{Oz}$ . Since  $v(d) = w(h) = 0$ ,

$$v(x) = v(dx) \geq 0, \quad -v(x) = v(x^{-1}) = v(dx^{-1}) \geq 0,$$

so  $v(x) = 0$ . The complex proof uses the same real monomials and the complex clause of Proposition 3.4(i), with  $dx, dx^{-1} \in \Pi_{\mathbb{C}} \subseteq \mathbf{Oz}[i]$ .  $\square$

*Remark 15.23* (Why the size restriction is essential [03]). The theorem does not say that  $\mathbf{No}$  has no interesting valuations. Its normal-form valuation  $x \mapsto -\max \text{supp } x$  takes values in the proper-class group  $(\mathbf{No}, +)$  and is negative on positive-exponent monomials, so it does not satisfy the containment hypothesis anyway. Nor is the theorem a consequence of complete integral closure alone: in the lexicographically ordered group  $\mathbb{Z} \oplus \mathbb{Z}$ , the values  $v(d) = (1, 0)$  and  $v(x) = (0, -1)$  would give  $v(dx^n) = (1, -n) > 0$  for every ordinary  $n$  although  $v(x) < 0$ . Almost integrality by itself cannot eliminate valuations of higher rank.

For ordinary set-sized domains the integral closure is characterized by valuation overrings with set-sized value groups [24, Lemmas 10.50.3 and 10.50.11]. That valuative criterion of integrality cannot be imported to a class ring while retaining only set-sized value groups:  $1/2$  is not integral over  $\mathbf{Oz}$ , yet Theorem 15.22 rules out every nontrivial valuation of that restricted kind whose ring contains  $\mathbf{Oz}$ . No assertion is made about the existence or classification of *proper-class-valued* valuation overrings separating nonintegral surreal elements.

## 15.7 An equation invisible to every set-sized image

Manuscript 08 writes  $B_m(t)$  for the binomial series; we write  $\mathbf{b}_m(t)$ .

**Lemma 15.24** (The binomial identity [08]). *For an ordinary integer  $m \geq 2$  the formal power series*

$$\mathbf{b}_m(t) = \sum_{n \geq 0} \binom{1/m}{n} t^n \in \mathbb{Q}[[t]]$$

*satisfies  $\mathbf{b}_m(t)^m = 1 + t$ . For every  $\gamma \in \mathbf{No}_{>0}$  the substitution  $t = \omega^{-\gamma}$  is a valid Hahn substitution, and  $\mathbf{b}_m(\omega^{-\gamma})^m = 1 + \omega^{-\gamma}$  in  $\mathbf{No}$ . The same holds for  $t = X^{-\gamma}$  with  $\gamma > 0$  in any ordered abelian group, in particular in  $F_{\kappa}^{03}$  (this last sentence is 03's use of the binomial expansion in its Hahn fields, not 08's).*

*Proof.* Write  $b_n = \binom{1/m}{n}$ ; then  $b_0 = 1$  and  $(n+1)b_{n+1} = (1/m - n)b_n$ . The coefficient of  $t^n$  in  $(1+t)\mathbf{b}'_m$  is  $(n+1)b_{n+1} + nb_n = b_n/m$ , so  $(1+t)\mathbf{b}'_m = \frac{1}{m}\mathbf{b}_m$ . Put  $c = \mathbf{b}_m^m = \sum_n c_n t^n$ . Then

$(1+t)c' = m\mathbf{b}_m^{m-1}(1+t)\mathbf{b}'_m = c$  and  $c_0 = 1$ . Comparing coefficients gives  $(n+1)c_{n+1} = (1-n)c_n$  for  $n \geq 0$ , so  $c_1 = 1$ ,  $c_2 = 0$  and hence  $c_n = 0$  for all  $n \geq 2$ . Thus  $c = 1+t$ .

After substitution the support lies in  $\{-n\gamma : n \geq 0\}$ , which is reverse well ordered. In any fixed finite product each exponent  $-n\gamma$  receives finitely many contributions, because an ordinary integer  $n \geq 0$  has only finitely many ordered decompositions into a fixed number of nonnegative integers. Hence  $\sum_n c_n t^n \mapsto \sum_n c_n \omega^{-n\gamma}$  is a ring homomorphism  $\mathbb{Q}[[t]] \rightarrow \mathbf{No}$ , and the formal identity remains a Hahn identity.  $\square$

**Theorem 15.25** (Universal small-target failure of root detection [08]). *Fix  $\gamma \in \mathbf{No}_{>0}$  and an ordinary integer  $m \geq 2$ . The equation*

$$T^m = \omega^\gamma + 1 \tag{15.17}$$

*has no root in  $\mathbf{Oz}[i]$ , and therefore none in  $\mathbf{Oz}$ . Nevertheless, for every unital ring homomorphism  $\varphi : \mathbf{Oz} \rightarrow R$  with  $R$  set-sized, the image equation  $T^m = \varphi(\omega^\gamma) + 1_R$  has the solution  $T = 1_R$ . The same holds for unital homomorphisms with domain  $\mathbf{Oz}[i]$ .*

*Proof.* Put  $r = \omega^{\gamma/m} \mathbf{b}_m(\omega^{-\gamma}) \in \mathbf{No}$ . By Lemma 15.24,  $r^m = \omega^\gamma(1 + \omega^{-\gamma}) = \omega^\gamma + 1$ . For every ordinary complex  $m$ th root of unity  $\zeta$ , the element  $\zeta r \in \mathbf{No}[i]$  is a root. These are  $m$  distinct roots (as  $r \neq 0$ ), so they exhaust the roots of the degree- $m$  polynomial  $T^m - \omega^\gamma - 1$  in the field  $\mathbf{No}[i]$ ; no algebraic-closedness argument is needed. The complex normal form of each root begins

$$\zeta r = \zeta \omega^{\gamma/m} + \frac{\zeta}{m} \omega^{\gamma/m-\gamma} + \sum_{n \geq 2} \zeta \binom{1/m}{n} \omega^{\gamma/m-n\gamma}. \tag{15.18}$$

The exponent  $\gamma/m - \gamma$  is strictly negative because  $m \geq 2$ , its coefficient  $\zeta/m$  is nonzero, and it differs from every other exponent  $\gamma/m - n\gamma$  ( $n \neq 1$ ) of the expansion. So this negative-exponent term cannot cancel, and no root lies in  $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$ , whose elements have nonnegative support.

On the other hand, by Theorem 5.1 every unital homomorphism  $\varphi$  from  $\mathbf{Oz}$  to a set-sized ring kills  $\Pi$ , so  $\varphi(\omega^\gamma) = 0$  and the image equation is  $T^m = 1_R$ , solved by  $1_R$ . For a unital homomorphism with domain  $\mathbf{Oz}[i]$ , the universal theorem with  $D = \mathbb{Z}[i]$  and  $k = \mathbb{C}$  (see Corollary 5.3 for the description of these maps) again gives  $\varphi(\omega^\gamma) = 0$ .  $\square$

The nonexistence of a root in  $\mathbf{Oz}[i]$  is also a case of 16's image-gap theorem (Remark 16.37(c)). This is not merely a failure of a local-global test by congruences modulo ordinary primes. Even allowing *every set-sized ring* as a target does not reveal the missing negative tail in (15.18). The obstruction belongs to the full normal-form arithmetic and survives in the proper-class ring itself (08).

**Corollary 15.26** ([08; 03]). *The ring  $\mathbf{Oz}$  is not integrally closed in its fraction field  $\mathbf{No}$ .*

*Proof.* The positive square root of  $\omega + 1$  lies in  $\mathbf{No} = \text{Frac}(\mathbf{Oz})$  (Proposition 3.4(ii)). It is integral over  $\mathbf{Oz}$ , being a root of the monic polynomial  $T^2 - (\omega + 1)$ , but it is not in  $\mathbf{Oz}$  by Theorem 15.25 with  $m = 2$  and  $\gamma = 1$ .  $\square$

The same corollary is contained in 03's Theorem 15.12, with the witness  $\sqrt{2}$ , and manuscript 05 of the sibling report [30] proves it with the same witness  $\sqrt{2}$ . This elementary corollary is included to clarify the distinction between an integer part and an integral closure; it is not offered as a new factorization theorem (08).

*Remark 15.27* (Explicit elements of  $\mathcal{N} \setminus \mathbf{Oz}$  [merge]). Let  $r$  be the positive root of  $T^m - (\omega^\gamma + 1)$  in  $\mathbf{No}$ , for  $\gamma > 0$  and  $m \geq 2$ ; it is the element  $r = \omega^{\gamma/m} \mathbf{b}_m(\omega^{-\gamma})$  of the proof of [Theorem 15.25](#). Then

$$r \in \mathcal{N} \setminus \mathbf{Oz}, \quad \zeta r \in \mathcal{N}_{\mathbb{C}} \setminus \mathbf{Oz}[i] \quad (\zeta^m = 1).$$

Indeed  $r$  is a root of a monic polynomial with coefficients in  $\mathbf{Oz}$ , so it lies in  $\mathcal{N}$  by definition, and  $\zeta r$  lies in  $\mathcal{N}_{\mathbb{C}}$  for the same reason; neither lies in the respective integer ring by [Theorem 15.25](#). Thus 08's equation exhibits explicit elements of the normalization whose absence from  $\mathbf{Oz}$  is invisible to every set-sized image: under every unital homomorphism from  $\mathbf{Oz}$  to a set-sized ring the defining equation of  $r$  becomes  $T^m = 1$ , which already has the root 1. Equivalently, if  $\psi$  is a unital homomorphism from  $\mathcal{N}$  (or from  $\mathbf{Oz}[r]$ ) to a set-sized ring, then its restriction to  $\mathbf{Oz}$  kills  $\omega^\gamma$  by [Theorem 5.1](#), so  $\psi(r)^m = 1$ . Unlike  $\sqrt{2}$ , the elements  $r$  are not constants: by (15.18) they have infinite leading term  $\omega^{\gamma/m}$  and a nonzero negative tail. Nothing is claimed here about the set-sized images of  $\mathcal{N}$  themselves.

## 16 Arithmetic of polynomial maps on the omnific lattices

This section is manuscript 10's arithmetic of finite-degree polynomial and rational maps on the real lattice  $\mathbf{Oz} \subset \mathbf{No}$  and the Gaussian lattice  $\mathbf{Oz}[i] \subset \mathbf{No}[i]$ , followed by manuscript 16's fresh-scale results for  $D + \Pi_k$  with any unital constant ring  $D$  ([Sections 16.6](#) and [16.7](#)). It uses from the core only the universal theorem [Theorem 5.1](#), the quotient list [Theorem 6.2](#) with the identity (3.2), the constant-term lemma [Lemma 2.2](#), the clearing of supports [Proposition 3.4\(i\)](#) and the unit lists of [Lemma 2.4](#).

**Renamings.** Manuscript 10's symbols are renamed as follows. Its ideals  $\mathfrak{P}$  and  $\mathfrak{P}_{\mathbb{C}}$  are  $\Pi$  and  $\Pi_{\mathbb{C}}$ ; its constant-term map  $\pi$  is  $\text{ct}$ ; its projection  $\Pi$  on polynomial rings (the *shadow*) is  $\text{ct}_*$ , the coefficientwise constant term; its Gaussian lattice is  $\mathbf{Oz}[i]$  inside  $\mathbf{No}[i]$ ; its ring  $R(D, K)$  is  $\mathcal{A}_{D,K}$ ; its binomial polynomials  $B_k(X)$  are written  $\binom{X}{k}$ ; its  $L_k = \text{lcm}(1, \dots, k)$  is written  $\text{lcm}_k$ ; and its rings  $S_r$  and  $I_r$  are written  $\mathcal{N}_r(\mathbf{Oz})$  and  $\text{Int}(\mathbb{Z}^r)$  (the letter  $I_r$  would collide with ideals). The numbered results keep 10's content.

**Notation.** Throughout,  $r \geq 1$  is an ordinary integer,  $\mathbf{X} = (X_1, \dots, X_r)$ , and bold letters denote  $r$ -tuples. For an ordinary  $k \geq 0$ ,

$$\binom{X}{k} = \frac{X(X-1) \cdots (X-k+1)}{k!}, \quad \binom{X}{0} = 1,$$

and for a multi-index  $\mathbf{k} = (k_1, \dots, k_r)$  we put  $\binom{\mathbf{X}}{\mathbf{k}} = \prod_{j=1}^r \binom{X_j}{k_j}$ . We write  $\Delta_j F(\mathbf{X}) = F(\mathbf{X} + \mathbf{e}_j) - F(\mathbf{X})$ , and  $\Delta = \Delta_1$  when  $r = 1$ . Put  $\text{lcm}_0 = 1$  and  $\text{lcm}_k = \text{lcm}(1, \dots, k)$  for  $k \geq 1$ . Recall  $\mathcal{A}_{\mathbb{Q}, \mathbb{R}} = \mathbb{Q} + \Pi$  and  $\mathcal{A}_{\mathbb{Q}(i), \mathbb{C}} = \mathbb{Q}(i) + \Pi_{\mathbb{C}}$ ; by [Lemma 2.2](#) the constant term  $\text{ct}$  is a ring homomorphism on both, with kernels  $\Pi$  and  $\Pi_{\mathbb{C}}$ , so  $\Pi$  is an ideal of  $\mathbb{Q} + \Pi$  and  $\Pi_{\mathbb{C}}$  is an ideal of  $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$ . Its coefficientwise extension

$$\text{ct}_* : (\mathbb{Q} + \Pi)[\mathbf{X}] \rightarrow \mathbb{Q}[\mathbf{X}], \quad \text{ct}_* : (\mathbb{Q}(i) + \Pi_{\mathbb{C}})[\mathbf{X}] \rightarrow \mathbb{Q}(i)[\mathbf{X}]$$

is therefore a ring homomorphism with kernel  $\Pi[\mathbf{X}]$ , respectively  $\Pi_{\mathbb{C}}[\mathbf{X}]$ . Define

$$\begin{aligned} \mathcal{N}_r(\mathbf{Oz}) &= \{F \in \mathbf{No}[\mathbf{X}] : F(\mathbf{Oz}^r) \subseteq \mathbf{Oz}\}, & \text{Int}(\mathbb{Z}^r) &= \{f \in \mathbb{Q}[\mathbf{X}] : f(\mathbb{Z}^r) \subseteq \mathbb{Z}\}, \\ \mathcal{N}_r(\mathbf{Oz}[i]) &= \{F \in \mathbf{No}[i][\mathbf{X}] : F(\mathbf{Oz}[i]^r) \subseteq \mathbf{Oz}[i]\}, & \text{Int}(\mathbb{Z}[i]^r) &= \{f \in \mathbb{Q}(i)[\mathbf{X}] : f(\mathbb{Z}[i]^r) \subseteq \mathbb{Z}[i]\}. \end{aligned}$$

These are rings under pointwise operations. All degrees, finite sums and factorials are ordinary.

## 16.1 Lattice separation and an omitted congruence type

**Lemma 16.1** (Lattice separation and cofinality [10]). *For  $\Lambda = \mathbf{Oz}$  and for  $\Lambda = \mathbf{Oz}[i]$ , every nonzero  $x \in \Lambda$  satisfies  $|x| \geq 1$ . The finite elements of  $\mathbf{Oz}$  are exactly the ordinary integers. The positive elements of  $\mathbf{Oz}$  are cofinal in  $\mathbf{No}$ . The units are  $\mathbf{Oz}^\times = \{\pm 1\}$  and  $\mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}$  (Lemma 2.4).*

*Proof.* A nonzero real purely infinite series has a positive leading exponent, so its absolute value exceeds every positive real. Hence  $n + u$  with  $n \in \mathbb{Z}$  and  $0 \neq u \in \Pi$  is infinite, the finite elements of  $\mathbf{Oz}$  are the ordinary integers, and every nonzero element of  $\mathbf{Oz}$  has absolute value at least 1. In the Gaussian case write  $x = a + ib$  with  $a, b \in \mathbf{Oz}$ , not both zero; the surcomplex modulus dominates either coordinate,  $|x| = \sqrt{a^2 + b^2} \geq \max(|a|, |b|) \geq 1$ .

Given  $y \in \mathbf{No}$ , choose an exponent  $g > 0$  larger than the leading exponent of  $y$  (any  $g > 0$  if  $y = 0$ ). Then  $\omega^g \in \mathbf{Oz}$  is positive and exceeds  $n|y|$  for every ordinary  $n$ , in particular  $\omega^g > |y|$ . This proves cofinality. The unit lists are Lemma 2.4.  $\square$

*Remark 16.2* (Modulus and standard part [10]). The surcomplex modulus needs only real closedness of  $\mathbf{No}$ :  $|a + ib| = \sqrt{a^2 + b^2}$  for  $a, b \in \mathbf{No}$ , and it is multiplicative and satisfies the triangle inequality. The standard part of a finite surcomplex number is taken coordinatewise in  $\mathbb{C}$ ; on the finite rings  $\mathcal{O}_{\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}}$  it is the map  $\text{st}$  of Lemma 2.2.

**Proposition 16.3** (A finitely satisfiable countable congruence type [10]). *Fix an ordinary prime  $p$  and  $\alpha \in \mathbb{Z}_p \setminus \mathbb{Z}$ . For  $e \geq 1$  let  $a_e$  be the representative of  $\alpha \bmod p^e$  in  $\{0, \dots, p^e - 1\}$ . The countable family of ring formulas*

$$\exists y (x - a_e = p^e y), \quad e = 1, 2, \dots, \quad (16.1)$$

*is finitely satisfiable in  $\mathbb{Z} \subset \mathbf{Oz}$  but has no realization in  $\mathbf{Oz}$ . An explicit choice is*

$$\alpha = \sum_{j \geq 0} p^{2^j} \in \mathbb{Z}_p. \quad (16.2)$$

*Proof.* For a finite subfamily let  $E$  be its largest index and take  $x = a_E$ . Since the residues are compatible,  $a_E \equiv a_e \pmod{p^e}$  for  $e \leq E$ , which gives witnesses  $y \in \mathbb{Z}$  for every formula of the subfamily. If  $x \in \mathbf{Oz}$  realized all formulas, applying the ring homomorphism  $\text{ct}$  would give an ordinary integer  $\text{ct}(x)$  with  $\text{ct}(x) \equiv a_e \equiv \alpha \pmod{p^e}$  for every  $e$  (by (3.2),  $x - a_e \in p^e \mathbf{Oz}$  means  $\text{ct}(x) - a_e \in p^e \mathbb{Z}$ ). In  $\mathbb{Z}_p$  this forces  $\text{ct}(x) = \alpha$ , contradicting  $\alpha \notin \mathbb{Z}$ .

For (16.2): the powers  $p^{2^j}$  are distinct, so the base- $p$  digits of  $\alpha$  are 1 at the positions  $2^j$  and 0 elsewhere. They are not eventually zero, so  $\alpha$  is not a nonnegative integer; and they are not eventually  $p - 1$  (for  $p = 2$  the positions  $2^j + 1 < 2^{j+1}$  with  $j \geq 1$  carry the digit 0), so  $\alpha$  is not a negative integer.  $\square$

*Remark 16.4* (Ring types, not saturation [10]). This elementary omitted type concerns the ring structure. It shows that the order-theoretic richness of the full surreal class does not give a corresponding realization property for countable ring types. No use of the term “saturated model” for a proper-class structure is needed, and none is made.

## 16.2 Polynomials preserving the omnific lattices

**Lemma 16.5** (Omnific binomial closure [10]). *For every  $x \in \mathbf{Oz}$  and every ordinary  $k \geq 0$ ,  $\binom{x}{k} \in \mathbf{Oz}$ . More generally, if  $f \in \mathbb{Q}[\mathbf{X}]$  and  $f(\mathbb{Z}^r) \subseteq \mathbb{Z}$ , then  $f(\mathbf{Oz}^r) \subseteq \mathbf{Oz}$ .*

*Proof.* Write  $x_j = n_j + u_j$  with  $n_j \in \mathbb{Z}$  and  $u_j \in \Pi$ . Expanding each monomial of  $f$  shows that  $f(\mathbf{n} + \mathbf{u}) - f(\mathbf{n})$  is a finite sum of terms, each a rational multiple of a product of ordinary integers and of powers of the  $u_j$  containing at least one factor  $u_j$ . Since  $\Pi$  is an ideal of  $\mathbb{Q} + \Pi$ , every such term lies in  $\Pi$ ; thus

$$f(\mathbf{n} + \mathbf{u}) - f(\mathbf{n}) \in \Pi.$$

As  $f(\mathbf{n}) \in \mathbb{Z}$ , we get  $f(\mathbf{x}) \in \mathbb{Z} + \Pi = \mathbf{Oz}$ . The binomial case is the case  $f = \binom{X}{k}$ , which takes integer values at all ordinary integers, including negative ones:  $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$ .  $\square$

Thus  $\mathbf{Oz}$  is a binomial ring in the usual torsion-free sense: it is  $\mathbb{Z}$ -torsion free and closed under  $x \mapsto \binom{x}{k}$  [11]. The terminology is classical; the support argument above explains its application here without importing an abstract free-ring theorem.

**Theorem 16.6** (Exact finite-grid test [10; 16]). *Let  $F \in \mathbf{No}[\mathbf{X}]$  with  $\deg_{X_j} F \leq d_j$  for  $j = 1, \dots, r$ . The following are equivalent:*

- (i)  $F(\mathbf{Oz}^r) \subseteq \mathbf{Oz}$ ;
- (ii)  $F(n_1, \dots, n_r) \in \mathbf{Oz}$  for all ordinary integers  $0 \leq n_j \leq d_j$ ;
- (iii) in the unique expansion

$$F(\mathbf{X}) = \sum_{\mathbf{k} \leq \mathbf{d}} c_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}} \quad (16.3)$$

all coefficients  $c_{\mathbf{k}}$  belong to  $\mathbf{Oz}$ .

Explicitly,

$$c_{\mathbf{k}} = (\Delta_1^{k_1} \cdots \Delta_r^{k_r} F)(\mathbf{0}) = \sum_{\ell \leq \mathbf{k}} (-1)^{\sum_j (k_j - \ell_j)} \left( \prod_{j=1}^r \binom{k_j}{\ell_j} \right) F(\ell). \quad (16.4)$$

*Proof.* The polynomials  $\binom{X}{0}, \dots, \binom{X}{d}$  form a basis of the univariate polynomials of degree at most  $d$  over the field  $\mathbf{No}$ : their degrees are  $0, \dots, d$  and their leading coefficients are nonzero. Taking products, the  $\binom{\mathbf{X}}{\mathbf{k}}$  with  $\mathbf{k} \leq \mathbf{d}$  form a basis of the polynomials with  $\deg_{X_j} \leq d_j$ , which gives existence and uniqueness of (16.3). Since  $\Delta_m^{(X)} = \binom{X}{m-1}$  for  $m \geq 1$  and  $\Delta_0^{(X)} = 0$ , one has  $\Delta_j^{k_j} \binom{X_j}{m_j} = \binom{X_j}{m_j - k_j}$  for  $m_j \geq k_j$  and 0 otherwise; as  $\binom{0}{m} = 0$  for  $m > 0$ , applying  $\Delta_1^{k_1} \cdots \Delta_r^{k_r}$  to (16.3) and evaluating at  $\mathbf{0}$  leaves exactly  $c_{\mathbf{k}}$ . Writing  $\Delta_j = \tau_j - 1$  with the shift  $\tau_j$  and expanding by the binomial theorem gives the last expression in (16.4).

(i) $\Rightarrow$ (ii) is restriction to ordinary grid points. Under (ii), the last expression in (16.4) is an integral linear combination of grid values in  $\mathbf{Oz}$ , which proves (iii). Under (iii), each factor  $\binom{x_j}{k_j}$  with  $x_j \in \mathbf{Oz}$  lies in  $\mathbf{Oz}$  by Lemma 16.5, and evaluating (16.3) proves (i).  $\square$

The test is finite for a given finite-degree polynomial, although the preservation property quantifies over the entire class  $\mathbf{Oz}^r$ . It is a mathematical test with exact surreal input; no claim is made that arbitrary surreal coefficients have an effective finite encoding.

**Theorem 16.7** (The real numerical-polynomial ring [10; 16]). *Every  $F \in \mathcal{N}_r(\mathbf{Oz})$  has a unique decomposition*

$$F = U + f, \quad U \in \Pi[\mathbf{X}], \quad f \in \text{Int}(\mathbb{Z}^r), \quad (16.5)$$



so that  $\mathcal{N}_r(\mathbf{Oz}) = \Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$  additively. The coefficientwise constant term  $\text{ct}_*(F) = f$  is a surjective ring homomorphism  $\mathcal{N}_r(\mathbf{Oz}) \rightarrow \text{Int}(\mathbb{Z}^r)$  with kernel  $\Pi[\mathbf{X}]$ , split by the inclusion  $\text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$ . For  $\mathbf{x} \in \mathbf{Oz}^r$ ,

$$\text{ct}(F(\mathbf{x})) = \text{ct}_*(F)(\text{ct}(x_1), \dots, \text{ct}(x_r)). \quad (16.6)$$

*Proof.* Let  $F \in \mathcal{N}_r(\mathbf{Oz})$ . By Theorem 16.6 its Newton coefficients  $c_{\mathbf{k}}$  lie in  $\mathbf{Oz}$ ; write  $c_{\mathbf{k}} = m_{\mathbf{k}} + u_{\mathbf{k}}$  with  $m_{\mathbf{k}} \in \mathbb{Z}$  and  $u_{\mathbf{k}} \in \Pi$ , and put

$$U = \sum_{\mathbf{k}} u_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}}, \quad f = \sum_{\mathbf{k}} m_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}}.$$

The binomials have rational coefficients and  $\Pi$  is a  $\mathbb{Q}$ -vector space, so  $U \in \Pi[\mathbf{X}]$ ; products of binomials are integer valued on  $\mathbb{Z}^r$ , so  $f \in \text{Int}(\mathbb{Z}^r)$ . Conversely, a polynomial in  $\Pi[\mathbf{X}]$  takes values in  $\Pi$  on  $\mathbf{Oz}^r$ , because  $\Pi$  is an ideal of  $\mathbb{Q} + \Pi \supseteq \mathbf{Oz}$ , and  $\text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$  by Lemma 16.5; hence  $\Pi[\mathbf{X}] + \text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$ . The sum is direct because a coefficient that is both rational and purely infinite is zero.

By (16.5) all coefficients of  $F$  lie in  $\mathbb{Q} + \Pi$ , so  $\mathcal{N}_r(\mathbf{Oz})$  is a subring of  $(\mathbb{Q} + \Pi)[\mathbf{X}]$  and  $\text{ct}_*$  restricts to a ring homomorphism on it. It sends  $U$  to 0 and fixes  $f$ , so its restriction is the projection onto the second summand; its kernel is  $\Pi[\mathbf{X}]$  and the inclusion of  $\text{Int}(\mathbb{Z}^r)$  is a ring section. Finally write  $x_j = n_j + u_j$  with  $n_j = \text{ct}(x_j)$ . Then  $U(\mathbf{x}) \in \Pi$  and  $f(\mathbf{x}) - f(\mathbf{n}) \in \Pi$  by the proof of Lemma 16.5, while  $f(\mathbf{n}) \in \mathbb{Z}$ . Hence  $\text{ct}(F(\mathbf{x})) = f(\mathbf{n})$ , which is (16.6).  $\square$

The decomposition is an additive direct sum whose first summand is an ideal; it is *not* a direct product of rings.

**Example 16.8** ([10]). The polynomial

$$F(X) = 3 \binom{X}{2} + \omega^{\sqrt{2}} X^3 + \left( \sum_{m \geq 1} \omega^{1/m} \right) (X - 1)$$

is omnific-integer valued and has shadow  $\text{ct}_*(F) = 3 \binom{X}{2}$ . By contrast  $\omega^{-1}X$  fails already at  $X = 1$ . No negative-exponent coefficient can be hidden in a finite-degree lattice-preserving polynomial: by Theorem 16.7 every coefficient of an element of  $\mathcal{N}_r(\mathbf{Oz})$  lies in  $\mathbb{Q} + \Pi$ .

**Theorem 16.9** (Gaussian numerical polynomials [10; 16]). *For every ordinary  $r \geq 1$ ,*

$$\mathcal{N}_r(\mathbf{Oz}[i]) = \Pi_{\mathbb{C}}[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}[i]^r) \quad (16.7)$$

*additively. The coefficientwise constant term  $\text{ct}_*$  is a surjective ring homomorphism  $\mathcal{N}_r(\mathbf{Oz}[i]) \rightarrow \text{Int}(\mathbb{Z}[i]^r)$  with kernel  $\Pi_{\mathbb{C}}[\mathbf{X}]$ , split by inclusion, and for  $\mathbf{x} \in \mathbf{Oz}[i]^r$ , with  $\text{ct} : \mathbf{Oz}[i] \rightarrow \mathbb{Z}[i]$  the Gaussian constant term,*

$$\text{ct}(F(\mathbf{x})) = \text{ct}_*(F)(\text{ct}(x_1), \dots, \text{ct}(x_r)). \quad (16.8)$$

*Proof.* Let  $F \in \mathcal{N}_r(\mathbf{Oz}[i])$  with  $\deg_{X_j} F \leq d_j$ . The Newton expansion (16.3) and formula (16.4) hold over the field  $\mathbf{No}[i]$  by the same basis argument. The ordinary grid points lie in  $\mathbf{Oz}[i]^r$ , so (16.4) shows  $c_{\mathbf{k}} \in \mathbf{Oz}[i] = \mathbb{Z}[i] + \Pi_{\mathbb{C}}$ . Expanding the binomials, each monomial coefficient of  $F$  is a  $\mathbb{Q}$ -linear combination of the  $c_{\mathbf{k}}$ , hence lies in  $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$ . Put  $f = \text{ct}_*(F) \in \mathbb{Q}(i)[\mathbf{X}]$  and  $U = F - f \in \Pi_{\mathbb{C}}[\mathbf{X}]$ . For an ordinary Gaussian tuple  $\mathbf{z} \in \mathbb{Z}[i]^r$  we have  $U(\mathbf{z}) \in \Pi_{\mathbb{C}}$ , because  $\Pi_{\mathbb{C}}$  is an ideal of  $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$ ,

and  $F(\mathbf{z}) = w + v$  with  $w \in \mathbb{Z}[i]$ ,  $v \in \Pi_{\mathbb{C}}$ . Then  $f(\mathbf{z}) - w = v - U(\mathbf{z})$  lies in  $\mathbb{Q}(i) \cap \Pi_{\mathbb{C}} = 0$ , so  $f(\mathbf{z}) = w \in \mathbb{Z}[i]$ . Thus  $f \in \text{Int}(\mathbb{Z}[i]^r)$ .

Conversely write a Gaussian omnific input as  $\mathbf{z} + \mathbf{u}$  with  $\mathbf{z} \in \mathbb{Z}[i]^r$  and  $u_j \in \Pi_{\mathbb{C}}$ . For  $f \in \text{Int}(\mathbb{Z}[i]^r)$  the expansion argument of Lemma 16.5 gives  $f(\mathbf{z} + \mathbf{u}) - f(\mathbf{z}) \in \Pi_{\mathbb{C}}$ , so  $f(\mathbf{z} + \mathbf{u}) \in \mathbb{Z}[i] + \Pi_{\mathbb{C}} = \mathbf{Oz}[i]$ ; and a polynomial in  $\Pi_{\mathbb{C}}[\mathbf{X}]$  takes values in  $\Pi_{\mathbb{C}}$  on  $\mathbf{Oz}[i]^r$ . This proves (16.7); the sum is direct since  $\mathbb{Q}(i) \cap \Pi_{\mathbb{C}} = 0$ . As in Theorem 16.7,  $\mathcal{N}_r(\mathbf{Oz}[i])$  is a subring of  $(\mathbb{Q}(i) + \Pi_{\mathbb{C}})[\mathbf{X}]$ , on which  $\text{ct}_*$  is a ring homomorphism; this gives multiplicativity, the kernel and the section. The same computation,  $\text{ct}(F(\mathbf{z} + \mathbf{u})) = f(\mathbf{z})$ , proves (16.8).  $\square$

*Remark 16.10* (Why the real Newton test cannot simply be copied [10]). The ordinary binomial polynomials need not be Gaussian-integer valued:

$$\binom{i}{2} = \frac{i(i-1)}{2} = \frac{-1-i}{2} \notin \mathbb{Z}[i].$$

Consequently values on an ordinary integer grid do not by themselves certify preservation of  $\mathbf{Oz}[i]$  (the grid values are necessary, not sufficient). The Gaussian theorem retains the ordinary ring  $\text{Int}(\mathbb{Z}[i]^r)$  rather than imposing a false binomial basis on it; the arithmetic of  $\text{Int}(\mathbb{Z}[i]^r)$  is not replaced by the simpler real Newton basis.

**Theorem 16.11** (Small-target reflection for numerical polynomials [10]). *Every class ring homomorphism  $\varphi$  from  $\mathcal{N}_r(\mathbf{Oz})$  to a set-sized ring  $A$  factors uniquely through*

$$\text{ct}_* : \mathcal{N}_r(\mathbf{Oz}) \longrightarrow \text{Int}(\mathbb{Z}^r).$$

*The analogous assertion holds for  $\mathcal{N}_r(\mathbf{Oz}[i])$  with universal image  $\text{Int}(\mathbb{Z}[i]^r)$ . Therefore the set-sized quotient presentations of either numerical-polynomial ring are precisely the quotient presentations of its ordinary shadow pulled back along  $\text{ct}_*$ : their kernels are the class ideals  $\text{ct}_*^{-1}(\mathfrak{a})$  for ideals  $\mathfrak{a}$  of  $\text{Int}(\mathbb{Z}^r)$ , respectively  $\text{Int}(\mathbb{Z}[i]^r)$ , with quotients  $\text{Int}(\mathbb{Z}^r)/\mathfrak{a}$ , respectively  $\text{Int}(\mathbb{Z}[i]^r)/\mathfrak{a}$ .*

*Proof.* The restriction of  $\varphi$  to the ideal  $\Pi$  of constant polynomials is a (possibly nonunital) class ring homomorphism  $\Pi \rightarrow A$ , hence zero by Theorem 5.1. Every element of  $\Pi[\mathbf{X}]$  is a finite sum of products  $u\mathbf{X}^{\mathbf{m}}$  with  $u \in \Pi$  and  $\mathbf{X}^{\mathbf{m}} \in \mathcal{N}_r(\mathbf{Oz})$ , so  $\varphi(u\mathbf{X}^{\mathbf{m}}) = \varphi(u)\varphi(\mathbf{X}^{\mathbf{m}}) = 0$  and  $\varphi$  kills  $\Pi[\mathbf{X}] = \ker \text{ct}_*$  (Theorem 16.7). Hence  $\varphi = \psi \circ \text{ct}_*$  with  $\psi$  the restriction of  $\varphi$  to the section  $\text{Int}(\mathbb{Z}^r)$ ;  $\psi$  is unique because  $\text{ct}_*$  is surjective. The Gaussian case is identical, with  $\Pi_{\mathbb{C}}$  and Theorem 16.9.

If  $\varphi$  is surjective onto a set-sized ring with kernel  $J$ , then  $\psi$  is surjective and  $J = \text{ct}_*^{-1}(\ker \psi)$ . Conversely, for an ideal  $\mathfrak{a}$  of the shadow ring, the composite of  $\text{ct}_*$  with the quotient map onto the set  $\text{Int}(\mathbb{Z}^r)/\mathfrak{a}$  is a set-sized quotient presentation with kernel  $\text{ct}_*^{-1}(\mathfrak{a})$ . This is the kernel argument of Theorem 6.2.  $\square$

### 16.3 Congruences modulo every omnific integer

**Definition 16.12** ([10]). A function  $F : \mathbf{Oz} \rightarrow \mathbf{Oz}$  is *congruence preserving* if  $a - b$  divides  $F(a) - F(b)$  in  $\mathbf{Oz}$  for all  $a, b \in \mathbf{Oz}$ ; equivalently, for every  $m \in \mathbf{Oz}$ ,  $a \equiv b \pmod{m\mathbf{Oz}}$  implies  $F(a) \equiv F(b) \pmod{m\mathbf{Oz}}$ . We write  $\text{CP}(\mathbf{Oz})$  for the congruence-preserving elements of  $\mathcal{N}_1(\mathbf{Oz})$  and  $\text{CP}(\mathbb{Z})$  for the elements  $f \in \text{Int}(\mathbb{Z})$  such that  $a - b$  divides  $f(a) - f(b)$  in  $\mathbb{Z}$  for all  $a, b \in \mathbb{Z}$ .

The two formulations agree: the forward implication is transitivity of divisibility, and the reverse one takes  $m = a - b$ . When  $a = b$  the condition reads  $0 \mid 0$ , which holds. Differences and finite sums of congruence-preserving functions are congruence preserving.

**Lemma 16.13** (Integral divided differences of weighted binomials [10]). *For every ordinary  $k \geq 1$ , the identity*

$$\frac{\binom{X+T}{k} - \binom{X}{k}}{T} = \sum_{j=1}^k \binom{X}{k-j} \frac{1}{j} \binom{T-1}{j-1} \quad (16.9)$$

*holds in  $\mathbb{Q}[X, T]$ . In particular  $\text{lcm}_k \binom{X}{k}$  is congruence preserving on  $\mathbf{Oz}$ .*

*Proof.* Vandermonde's identity gives  $\binom{X+T}{k} = \sum_{j=0}^k \binom{X}{k-j} \binom{T}{j}$ . For  $j \geq 1$  the falling-factorial formula gives  $\binom{T}{j} = \frac{T}{j} \binom{T-1}{j-1}$ . Subtracting the term  $j = 0$  and cancelling the polynomial factor  $T$  gives (16.9).

After multiplication by  $\text{lcm}_k$ , every scalar  $\text{lcm}_k / j$  with  $1 \leq j \leq k$  is an ordinary integer, so by Lemma 16.5 the right side evaluates into  $\mathbf{Oz}$  for all  $X, T \in \mathbf{Oz}$ . Taking  $X = b$  and  $T = a - b$  gives  $\text{lcm}_k \binom{a}{k} - \text{lcm}_k \binom{b}{k} = (a - b)w$  with  $w \in \mathbf{Oz}$ , including for nonzero infinite  $a - b$ ; the case  $a = b$  is trivial.  $\square$

**Theorem 16.14** (All-modulus Newton criterion [10]). *Let  $F \in \mathcal{N}_1(\mathbf{Oz})$  have Newton expansion  $F = \sum_{k=0}^d c_k \binom{X}{k}$ ,  $c_k \in \mathbf{Oz}$ . The following are equivalent:*

- (i)  $F$  is congruence preserving on  $\mathbf{Oz}$ ;
- (ii)  $c_k \in \text{lcm}_k \mathbf{Oz}$  for every  $0 \leq k \leq d$ ;
- (iii)  $\text{ct}(c_k) \in \text{lcm}_k \mathbb{Z}$  for every  $0 \leq k \leq d$ ;
- (iv) the ordinary numerical polynomial  $\text{ct}_*(F)$  is congruence preserving on  $\mathbb{Z}$ .

*Consequently, within the respective numerical-polynomial rings,*

$$\text{CP}(\mathbf{Oz}) = \Pi[X] \oplus \text{CP}(\mathbb{Z}). \quad (16.10)$$

*Proof.* (ii) $\Rightarrow$ (i). Write  $c_k = \text{lcm}_k d_k$  with  $d_k \in \mathbf{Oz}$ . Each summand  $d_k \text{lcm}_k \binom{X}{k}$  is congruence preserving by Lemma 16.13 (multiply the divisibility by  $d_k$ ), and a finite sum of congruence-preserving functions is congruence preserving.

(i) $\Rightarrow$ (ii). We show  $c_k \in \text{lcm}_k \mathbf{Oz}$  by induction on  $k$ . For  $k = 0$  it is automatic since  $\text{lcm}_0 = 1$ . Suppose it holds for all  $j < k$ . Then  $\sum_{j < k} c_j \binom{X}{j}$  is congruence preserving by the previous paragraph, so

$$G(X) = F(X) - \sum_{j < k} c_j \binom{X}{j} = \sum_{j \geq k} c_j \binom{X}{j}$$

is congruence preserving. Since  $\binom{n}{j} = 0$  for ordinary  $0 \leq n < j$  and  $\binom{k}{k} = 1$ , we get  $G(0) = \dots = G(k-1) = 0$  and  $G(k) = c_k$ . For each  $1 \leq m \leq k$ , comparing the inputs  $k$  and  $k - m$  gives  $m \mid G(k) - G(k - m) = c_k$  in  $\mathbf{Oz}$ . By (3.2), the ordinary integer  $\text{ct}(c_k)$  is divisible by every  $m \leq k$ , hence by  $\text{lcm}_k$ , and again by (3.2),  $c_k \in \text{lcm}_k \mathbf{Oz}$ .

(ii) $\Leftrightarrow$ (iii) is (3.2). For (iii) $\Leftrightarrow$ (iv), note that  $\text{ct}_*(F) = \sum_k \text{ct}(c_k) \binom{X}{k}$  is the Newton expansion of  $\text{ct}_*(F)$ , since the binomials have rational coefficients. The same two arguments over  $\mathbb{Z}$  (the identity (16.9) with  $X, T \in \mathbb{Z}$ , and the same induction with divisibility in  $\mathbb{Z}$ ) show that an  $f \in \text{Int}(\mathbb{Z})$  with Newton coefficients  $m_k \in \mathbb{Z}$  is congruence preserving on  $\mathbb{Z}$  if and only if  $m_k \in \text{lcm}_k \mathbb{Z}$  for all  $k$ . Applied to  $m_k = \text{ct}(c_k)$ , this is (iii) $\Leftrightarrow$ (iv).

For (16.10), decompose  $F = U + f$  by Theorem 16.7. The Newton coefficients of  $U \in \Pi[X]$  are integral combinations of values  $U(\ell) \in \Pi$ , and  $\Pi \subseteq \text{lcm}_k \mathbf{Oz}$  by (3.2); so  $U \in \text{CP}(\mathbf{Oz})$  by (ii). An  $f \in \text{Int}(\mathbb{Z})$  lies in  $\text{CP}(\mathbf{Oz})$  exactly when it lies in  $\text{CP}(\mathbb{Z})$ , by (i) $\Leftrightarrow$ (iv) applied to  $f = \text{ct}_*(f)$ . Since  $\text{ct}_*(F) = f$ , the equivalence (i) $\Leftrightarrow$ (iv) for  $F$  now gives  $F \in \text{CP}(\mathbf{Oz})$  if and only if  $f \in \text{CP}(\mathbb{Z})$ .  $\square$

The divisibility condition in (i) includes infinite differences and therefore every infinite omnific modulus; no extra coefficient condition appears at those moduli. The ordinary least-common-multiple pattern is classical, and related finite-ring formulations are proved in [7]; Newton bases and integer-valued polynomials are treated in [6]. The content specific to the present domain is that the pattern suffices for differences at every surreal scale, not only for congruences modulo ordinary integers.

**Example 16.15** (Two kinds of denominators [10]). The polynomial  $12\binom{X}{4}$  is congruence preserving ( $\text{lcm}_4 = 12$ ), although its leading coefficient  $1/2$  is not an ordinary integer. On the other hand

$$F(X) = \binom{X}{2} + \omega^{\sqrt{2}}X^3$$

is integer valued on all of  $\mathbf{Oz}$  but not congruence preserving:  $F(2) - F(0) = 1 + 8\omega^{\sqrt{2}}$  has odd constant term and so is not divisible by 2 in  $\mathbf{Oz}$ , by (3.2). The purely infinite term does not repair the odd constant term.

*Remark 16.16* (What Theorem 16.14 does not say [10]). The theorem does not say that reduction modulo an infinite omnific integer is an ordinary finite congruence; such a quotient need not have any set-sized presentation (by Theorem 6.2 it has one only when it contains  $\Pi$ ). It says that the polynomial *operation* respects those congruences, with an explicit quotient supplied by (16.9). It also does not classify all congruence-preserving class functions  $\mathbf{Oz} \rightarrow \mathbf{Oz}$ : the finite Newton expansion is essential to the statement and the proof, and no infinite interpolation formula or full-class summation is implicit.

## 16.4 $p$ -adic characters and completions

In this subsection  $p$  is an ordinary prime,  $e \geq 1$  and  $r \geq 1$  are ordinary integers, and  $C(\mathbb{Z}_p^r, M)$  denotes the continuous functions from  $\mathbb{Z}_p^r$  with its  $p$ -adic topology to  $M$  ( $M = \mathbb{Z}/p^e\mathbb{Z}$  or  $\mathbb{F}_p$  discrete, or  $M = \mathbb{Z}_p$ ). All  $p$ -adic fields, rings and function spaces here are ordinary set-sized structures.

**Lemma 16.17** (Continuous extension [10]). *Every  $f \in \text{Int}(\mathbb{Z}^r)$  extends, by its rational-coefficient formula, to a continuous function  $\mathbb{Z}_p^r \rightarrow \mathbb{Z}_p$ , and this continuous extension is unique.*

*Proof.* A rational polynomial defines a continuous function  $\mathbb{Q}_p^r \rightarrow \mathbb{Q}_p$ . Ordinary integer tuples are dense in  $\mathbb{Z}_p^r$  and take values in  $\mathbb{Z} \subseteq \mathbb{Z}_p$ ; since  $\mathbb{Z}_p$  is closed in  $\mathbb{Q}_p$ , all values on  $\mathbb{Z}_p^r$  lie in  $\mathbb{Z}_p$ . Density also gives uniqueness.  $\square$

Reduction of values modulo  $p^e$  is therefore a ring homomorphism  $\text{Int}(\mathbb{Z}^r) \rightarrow C(\mathbb{Z}_p^r, \mathbb{Z}/p^e\mathbb{Z})$ . Its surjectivity needs an approximation argument; we prove it explicitly by indicator polynomials rather than assume it, so no unproved approximation assertion is used. For  $0 \leq a < p^k$  write  $a = a_0 + a_1p + \cdots + a_{k-1}p^{k-1}$  with digits  $0 \leq a_j < p$ , and put

$$E_{a,k}(X) = \prod_{j=0}^{k-1} \left[ 1 - \left( \binom{X}{p^j} - a_j \right)^{p-1} \right] \in \text{Int}(\mathbb{Z}), \quad I_{a,k,e}(X) = E_{a,k}(X)^{p^{e-1}}. \quad (16.11)$$

**Lemma 16.18** (Digit and indicator congruences [10]). *For every  $x \in \mathbb{Z}_p$ , with  $\mathbf{1}_{a,k}(x) = 1$  if  $x \equiv a \pmod{p^k}$  and  $\mathbf{1}_{a,k}(x) = 0$  otherwise,*

$$E_{a,k}(x) \equiv \mathbf{1}_{a,k}(x) \pmod{p}, \quad I_{a,k,e}(x) \equiv \mathbf{1}_{a,k}(x) \pmod{p^e}.$$

*Proof.* First let  $x$  be a nonnegative ordinary integer with base- $p$  digits  $x_j$ . In  $\mathbb{F}_p[T]$  the Frobenius identity gives

$$(1+T)^x = \prod_{j \geq 0} (1+T)^{x_j p^j} = \prod_{j \geq 0} (1+T^{p^j})^{x_j}.$$

The coefficient of  $T^{p^j}$  on the right is  $x_j$ : the factors of index below  $j$  have combined degree at most  $\sum_{i < j} (p-1)p^i = p^j - 1$ , and factors of index above  $j$  contribute only exponents 0 or at least  $p^{j+1}$ . Hence  $\binom{x}{p^j} \equiv x_j \pmod{p}$ . Both sides are continuous functions  $\mathbb{Z}_p \rightarrow \mathbb{F}_p$  of  $x$  (Lemma 16.17; the  $j$ th digit depends only on  $x \bmod p^{j+1}$ ), and the nonnegative integers are dense in  $\mathbb{Z}_p$ , so the congruence holds for all  $x \in \mathbb{Z}_p$  with  $x_j$  the  $j$ th  $p$ -adic digit. By Fermat,  $t^{p-1} = 1$  for  $0 \neq t \in \mathbb{F}_p$ , so the  $j$ th factor of  $E_{a,k}(x)$  is  $\equiv 1$  if  $x_j = a_j$  and  $\equiv 0$  otherwise. The product is the indicator of  $x_j = a_j$  for all  $j < k$ , that is, of  $x \equiv a \pmod{p^k}$ .

For the lift, let  $z \in \mathbb{Z}_p$ . If  $z \equiv 0 \pmod{p}$  then  $z^{p^{e-1}} \equiv 0 \pmod{p^e}$ , because  $p^{e-1} \geq e$ . If  $z = 1 + pw$ , then  $(1 + pw)^{p^{e-1}} \equiv 1 \pmod{p^e}$  by induction from  $(1 + p^s v)^p \equiv 1 \pmod{p^{s+1}}$  for  $s \geq 1$ : in  $(1 + p^s v)^p = 1 + p^{s+1} v + \sum_{i=2}^p \binom{p}{i} p^{si} v^i$  the middle binomial coefficients are divisible by  $p$  and  $sp \geq s+1$ ; this is also valid for  $p = 2$ . Applying these two cases to  $z = E_{a,k}(x)$  gives the second congruence.  $\square$

**Theorem 16.19** (Exact finite-level function rings [10]). *The coefficientwise constant term and  $p$ -adic evaluation induce natural ring isomorphisms*

$$\mathcal{N}_r(\mathbf{Oz})/p^e \mathcal{N}_r(\mathbf{Oz}) \simeq \text{Int}(\mathbb{Z}^r)/p^e \text{Int}(\mathbb{Z}^r) \simeq C(\mathbb{Z}_p^r, \mathbb{Z}/p^e \mathbb{Z}). \quad (16.12)$$

*Proof.* By Theorem 16.7,  $\mathcal{N}_r(\mathbf{Oz}) = \Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$  additively, and  $p^e \Pi[\mathbf{X}] = \Pi[\mathbf{X}]$  because  $\Pi$  is a  $\mathbb{Q}$ -vector space. Hence  $p^e \mathcal{N}_r(\mathbf{Oz}) = \Pi[\mathbf{X}] \oplus p^e \text{Int}(\mathbb{Z}^r)$ , and the ring homomorphism  $\text{ct}_*$  induces the first isomorphism.

For the second, the map sends  $f$  to  $x \mapsto f(x) \bmod p^e$  (Lemma 16.17). If  $f = p^e g$  with  $g \in \text{Int}(\mathbb{Z}^r)$ , its reduction is zero. Conversely, if  $f$  vanishes modulo  $p^e$  at every  $p$ -adic tuple, its values at ordinary integer tuples are divisible by  $p^e$ , so  $f/p^e \in \text{Int}(\mathbb{Z}^r)$ . The kernel is exactly  $p^e \text{Int}(\mathbb{Z}^r)$ .

Let  $g \in C(\mathbb{Z}_p^r, \mathbb{Z}/p^e \mathbb{Z})$ . The target is discrete, so every point has a residue box  $\mathbf{y} + p^k \mathbb{Z}_p^r$  on which  $g$  is constant; by compactness of  $\mathbb{Z}_p^r$  finitely many of these boxes cover it. Let  $k$  be the largest of their levels. Every level- $k$  box  $\mathbf{y} + p^k \mathbb{Z}_p^r$  lies in the covering box that contains  $\mathbf{y}$ , so  $g$  is constant on every residue box modulo  $p^k$ . For  $\mathbf{a} \in \{0, \dots, p^k - 1\}^r$  choose an ordinary integer  $c_{\mathbf{a}}$  lifting the value of  $g$  on the box of  $\mathbf{a}$ , and put

$$f(\mathbf{X}) = \sum_{\mathbf{a}} c_{\mathbf{a}} \prod_{j=1}^r I_{a_j, k, e}(X_j) \in \text{Int}(\mathbb{Z}^r).$$

By Lemma 16.18, at  $\mathbf{x}$  in the box of  $\mathbf{a}$  the product indexed by  $\mathbf{a}$  is  $\equiv 1$  and all others are  $\equiv 0$  modulo  $p^e$ , so  $f(\mathbf{x}) \equiv g(\mathbf{x}) \pmod{p^e}$ . This proves surjectivity.  $\square$

**Corollary 16.20** (The  $p$ -adic completion [10]). *There is a canonical isomorphism of topological rings*

$$\varprojlim_{e \geq 1} \mathcal{N}_r(\mathbf{Oz})/p^e \mathcal{N}_r(\mathbf{Oz}) \simeq C(\mathbb{Z}_p^r, \mathbb{Z}_p), \quad (16.13)$$

where the left side carries the inverse-limit topology of the discrete quotients and the right side the uniform  $p$ -adic topology. The canonical map  $\mathcal{N}_r(\mathbf{Oz}) \rightarrow C(\mathbb{Z}_p^r, \mathbb{Z}_p)$  sends  $F$  to the continuous extension of  $\text{ct}_*(F)$ , has dense image, and has kernel  $\Pi[\mathbf{X}]$ .

*Proof.* The isomorphisms of [Theorem 16.19](#) commute with reduction of values from level  $e + 1$  to level  $e$ . A coherent sequence of functions  $g_e : \mathbb{Z}_p^r \rightarrow \mathbb{Z}/p^e\mathbb{Z}$  determines a unique function  $g : \mathbb{Z}_p^r \rightarrow \mathbb{Z}_p$  pointwise, and  $g$  is continuous exactly when every  $g_e = g \bmod p^e$  is continuous, since the sets  $y + p^e\mathbb{Z}_p$  form a base of the topology of  $\mathbb{Z}_p$ . This identifies the inverse limit with  $C(\mathbb{Z}_p^r, \mathbb{Z}_p)$  as rings. The kernel of the projection to level  $e$  corresponds to the functions that vanish modulo  $p^e$  everywhere, which are the basic neighbourhoods of 0 in the uniform  $p$ -adic topology; so the topologies agree. Surjectivity at every finite level gives density of the image.

If the continuous extension of  $f \in \text{Int}(\mathbb{Z}^r)$  is zero, then  $f$  vanishes on  $\mathbb{Z}^r$ , and a rational polynomial vanishing on  $\mathbb{Z}^r$  is zero (write it as a polynomial in  $X_r$  with coefficients in  $\mathbb{Q}[X_1, \dots, X_{r-1}]$ ; for each ordinary tuple  $(n_1, \dots, n_{r-1})$  the resulting univariate polynomial has infinitely many roots, so all coefficients vanish on  $\mathbb{Z}^{r-1}$ ; induct on  $r$ ). With [Theorem 16.7](#) the kernel on  $\mathcal{N}_r(\mathbf{Oz})$  is exactly  $\Pi[\mathbf{X}]$ . Equivalently  $\bigcap_e p^e \text{Int}(\mathbb{Z}^r) = 0$ , because an ordinary integer divisible by every  $p^e$  is zero.  $\square$

*Remark 16.21* (Not a profinite ring completion [10]). For  $r \geq 1$  the characteristic functions of residue boxes at increasing depths are infinitely many distinct elements of  $C(\mathbb{Z}_p^r, \mathbb{F}_p)$ . Therefore  $\mathcal{N}_r(\mathbf{Oz})/p\mathcal{N}_r(\mathbf{Oz})$  is infinite (and discrete in its quotient topology), as are all the rings in (16.12). The completion (16.13) is a  $p$ -adic completion, not an inverse limit of finite rings. This differs from the genuine profinite completion of  $\mathbf{Oz}$ : its finite quotient presentations are the rings  $\mathbb{Z}/n\mathbb{Z}$  with kernels  $n\mathbf{Oz}$  ([Theorem 6.2](#)), so  $\varprojlim_n \mathbf{Oz}/n\mathbf{Oz} \simeq \widehat{\mathbb{Z}}$ .

The ordinary polynomial-approximation content is part of the classical theory of integer-valued polynomials [6]. The explicit proof above also identifies exactly which omnific coefficients disappear, rather than only invoking density for the ordinary subring. The compactness of  $\mathbb{Z}_p^r$  used here is ordinary  $p$ -adic compactness, not compactness of an omnific interval.

**Theorem 16.22** ( $p$ -adic classification of characters [10]). *Every unital class ring homomorphism  $\chi : \mathcal{N}_r(\mathbf{Oz}) \rightarrow \mathbb{F}_p$  has the form*

$$\chi_\alpha(F) = \text{ct}_*(F)(\alpha) \bmod p \quad (16.14)$$

*for a unique  $\alpha \in \mathbb{Z}_p^r$ , and every  $\alpha \in \mathbb{Z}_p^r$  defines such a character. The prime ideals of  $\mathcal{N}_r(\mathbf{Oz})$  containing  $p\mathcal{N}_r(\mathbf{Oz})$  are exactly the kernels  $\ker \chi_\alpha$ , with distinct  $\alpha$  giving distinct kernels; every such prime ideal is maximal with quotient  $\mathbb{F}_p$ .*

*Proof.* The map  $\chi_\alpha$  is a composite of ring homomorphisms ([Theorem 16.7](#) and [Lemma 16.17](#)), so it is a character. Conversely, a unital  $\chi$  kills  $p\mathcal{N}_r(\mathbf{Oz})$ , so by [Theorem 16.19](#) with  $e = 1$  it is  $\psi \circ \rho$ , where  $\rho(F) = \text{ct}_*(F) \bmod p$  and  $\psi : C(\mathbb{Z}_p^r, \mathbb{F}_p) \rightarrow \mathbb{F}_p$  is a unital ring homomorphism. At level  $k$ , the indicators of the  $p^{kr}$  residue boxes modulo  $p^k$  are pairwise orthogonal idempotents with sum 1. Their images are idempotents of  $\mathbb{F}_p$ , hence 0 or 1, pairwise orthogonal with sum 1, so exactly one box of level  $k$  is selected. A box indicator is the sum of the indicators of its children at level  $k + 1$ , and a child indicator  $c$  satisfies  $c = c \cdot b$  for its parent  $b$ ; so the selected boxes are nested. They are nonempty compact sets with diameters tending to 0, so by completeness of  $\mathbb{Z}_p^r$  they have exactly one common point  $\alpha$ . Every  $h \in C(\mathbb{Z}_p^r, \mathbb{F}_p)$  is constant on the residue boxes of some level  $k$  (as in the proof of [Theorem 16.19](#)), so  $h$  is an  $\mathbb{F}_p$ -linear combination of box indicators and  $\psi(h)$  is the value of  $h$  on the selected box, that is,  $h(\alpha)$ . Hence  $\chi = \chi_\alpha$ . Distinct points are separated by a residue-box indicator, which is  $\rho(F)$  for some  $F$  by surjectivity; so  $\alpha$  is unique.

The prime ideals of  $\mathcal{N}_r(\mathbf{Oz})$  containing  $p\mathcal{N}_r(\mathbf{Oz})$  correspond, through the explicit set-sized presentation  $\rho$ , to the prime ideals of  $C(\mathbb{Z}_p^r, \mathbb{F}_p)$ . Every element  $z$  of this ring satisfies  $z^p = z$  pointwise. The quotient by a prime ideal is a commutative domain in which every element is a root of  $X^p - X$ ; such a domain has at most  $p$  elements, and it contains the  $p$  distinct images of the



constants, so it is  $\mathbb{F}_p$ . Thus every such prime is maximal with quotient  $\mathbb{F}_p$ , and the quotient map is a character, hence some  $\chi_\alpha$ . A unital surjection onto  $\mathbb{F}_p$  is determined by its kernel, because  $\mathbb{F}_p$  has only the identity automorphism; with the uniqueness of  $\alpha$  this shows that distinct parameters give distinct kernels.  $\square$

**Corollary 16.23** (Most characters are not omnific evaluations [10]). *A character  $\mathcal{N}_r(\mathbf{Oz}) \rightarrow \mathbb{F}_p$  obtained by evaluating at a tuple  $\mathbf{x} \in \mathbf{Oz}^r$  and then applying a unital ring homomorphism  $\mathbf{Oz} \rightarrow \mathbb{F}_p$  has parameter  $\alpha = (\text{ct}(x_1), \dots, \text{ct}(x_r)) \in \mathbb{Z}^r \subset \mathbb{Z}_p^r$ . Conversely every parameter in  $\mathbb{Z}^r$  arises in this way. Parameters in  $\mathbb{Z}_p^r \setminus \mathbb{Z}^r$  give characters that are not evaluation at any omnific tuple.*

*Proof.* By Theorem 5.1 the unital homomorphism  $\mathbf{Oz} \rightarrow \mathbb{F}_p$  is uniquely  $\text{ct}$  followed by reduction modulo  $p$ . By (16.6), its composite with evaluation at  $\mathbf{x}$  is  $F \mapsto \text{ct}_*(F)(\text{ct}(\mathbf{x})) \bmod p$ , the character with parameter  $\text{ct}(\mathbf{x})$ . An ordinary integer tuple is itself an allowed input, which gives the converse. The last assertion follows from the uniqueness of the parameter in Theorem 16.22.  $\square$

For  $r = 1$  the lacunary  $p$ -adic integer (16.2) of the omitted type in Proposition 16.3 gives a concrete nonevaluation character. There are uncountably many parameters and only countably many integer ones. The enormous class of omnific inputs does not enlarge the evaluation parameters beyond the ordinary integers, because the constant term of every omnific input is an ordinary integer.

## 16.5 Rational self-maps and gaps at arbitrary surreal scales

Throughout this subsection  $(E, \Lambda)$  is one of the pairs  $(\mathbf{No}, \mathbf{Oz})$  or  $(\mathbf{No}[i], \mathbf{Oz}[i])$ ; in the second the absolute value is the surreal-valued modulus. The input and output lattices are the full classes, not a fixed set-sized truncation. We write  $\mathcal{N}_1(\Lambda)$  for  $\mathcal{N}_1(\mathbf{Oz})$ , respectively  $\mathcal{N}_1(\mathbf{Oz}[i])$ , and  $E(X)$  for the rational functions in one variable over the class field  $E$ . We use repeatedly the following estimate, valid in  $E$  by the properties of the modulus: if  $Q(X) = \sum_{j \leq n} q_j X^j \in E[X]$  has  $q_n \neq 0$  and  $|y| \geq R_Q := 1 + 2 \sum_{j < n} |q_j/q_n|$ , then

$$\frac{1}{2}|q_n||y|^n \leq |Q(y)| \leq \frac{3}{2}|q_n||y|^n, \quad (16.15)$$

because  $|\sum_{j < n} q_j y^j| \leq \sum_{j < n} |q_j||y|^{n-1} \leq \frac{1}{2}|q_n||y|^n$ .

**Lemma 16.24** (Finite differences of a proper rational function [10]). *Let  $r \in E(X)$  be nonzero and proper, and let  $\Delta r(X) = r(X+1) - r(X)$ . For every ordinary  $k \geq 0$ ,  $\Delta^k r$  is nonzero and proper. If the leading Laurent term of  $r$  at infinity is  $cX^{-m}$  with  $m \geq 1$ , then the leading term of  $\Delta^k r$  is*

$$(-1)^k c m(m+1) \cdots (m+k-1) X^{-m-k}, \quad (16.16)$$

*the empty product for  $k = 0$  being 1.*

*Proof.* Compute in the field of formal Laurent series in  $X^{-1}$  over a set-sized subfield of  $E$  containing the finitely many coefficients involved; the rational functions over that subfield embed in it, a rational function is proper exactly when its expansion has only negative exponents, and the shift  $X \mapsto X+1$  becomes the substitution  $X^{-1} \mapsto X^{-1}(1+X^{-1})^{-1}$ . The binomial series gives

$$(X+1)^{-n} - X^{-n} = X^{-n}((1+X^{-1})^{-n} - 1) = -nX^{-n-1} + \text{terms of lower degree in } X.$$

Since  $r = cX^{-m} + (\text{terms } X^{-n}, n > m)$ , and  $\Delta$  sends each such lower term to terms of degree at most  $-n-1 < -m-1$ , the leading term of  $\Delta r$  is  $-cmX^{-m-1}$ . Induction on  $k$  gives (16.16).

The coefficient is nonzero in characteristic zero and the exponent is negative, which proves both assertions.  $\square$

**Theorem 16.25** (Polynomiality of rational lattice self-maps [10; 16]). *Let  $f \in E(X)$  satisfy  $f(x) \in \Lambda$  for every  $x \in \Lambda$  at which  $f$  is defined. Then  $f \in E[X]$ . In particular its reduced rational presentation has no finite poles anywhere in  $E$ .*

*Proof.* Divide the numerator by the denominator to write  $f = P + r$  with  $P \in E[X]$  and  $r$  proper, and suppose  $r \neq 0$ . Choose an ordinary  $k \geq 1$  larger than  $\deg P$  (any  $k \geq 1$  if  $P = 0$ ). Then  $\Delta^k P = 0$ , so  $g := \Delta^k f = \Delta^k r$  is nonzero and proper by Lemma 16.24. Write  $f$  and  $g$  in reduced form,  $f = N_f/D_f$  and  $g = N_g/D_g$ , with  $\deg N_g < \deg D_g$ .

We find  $B \in \mathbf{No}$  such that for every  $x \in \mathbf{No}$  with  $x > B$  all of  $f(x), \dots, f(x+k)$  are defined,  $g(x)$  is defined, and

$$0 < |g(x)| < 1. \quad (16.17)$$

By (16.15),  $D_f(y) \neq 0$  for  $|y| \geq R_{D_f}$ ,  $N_g(y) \neq 0$  for  $|y| \geq R_{N_g}$ , and for  $|y| \geq \max(1, R_{N_g}, R_{D_g})$ ,

$$|g(y)| \leq \frac{\frac{3}{2}|\ell(N_g)||y|^{\deg N_g}}{\frac{1}{2}|\ell(D_g)||y|^{\deg D_g}} \leq 3 \frac{|\ell(N_g)|}{|\ell(D_g)|} |y|^{-1},$$

where  $\ell(\cdot)$  is the leading coefficient. Take  $B$  larger than 1, than  $R_{D_f}, R_{N_g}, R_{D_g}$  and than  $3|\ell(N_g)/\ell(D_g)|$ ; for real  $x > B$  the points  $x, \dots, x+k$  are real and larger than  $B$ , so every requirement holds. These are finitely many inequalities in the ordered field  $\mathbf{No}$ .

By Lemma 16.1 choose  $x \in \mathbf{Oz}$  with  $x > B$ . Then  $x+j \in \mathbf{Oz} \subseteq \Lambda$  for  $0 \leq j \leq k$ ,  $f$  is defined at these points, and

$$g(x) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(x+j) \in \Lambda.$$

This contradicts (16.17) and the lattice separation of Lemma 16.1. Therefore  $r = 0$  and  $f = P$ .  $\square$

The finite-difference argument is a general discrete-lattice method. It is included as a supporting result, not asserted to be a new ordinary theorem about rational functions on  $\mathbb{Z}$ . 16 proves the theorem for every unital constant ring with one fresh input (Theorem 16.42); the two routes are kept.

Recall that  $y \in E$  is finite if  $|y|$  is bounded by an ordinary integer and infinitesimal if  $|y| < 1/n$  for every ordinary  $n \geq 1$ ; the standard part of a finite element is the unique real (respectively complex) number at infinitesimal distance, taken coordinatewise in  $\mathbf{No}[i]$  (Remark 16.2).

**Lemma 16.26** (Finite roots and reduction [10]). *A root in  $E$  of a monic polynomial with finite coefficients is finite. The standard part is a ring homomorphism on the finite elements of  $E$  and carries such a root to a root of the coefficientwise standard-part polynomial. For finite  $z \in \mathbf{No}[i]$ ,*

$$\text{st}(|z|) = |\text{st}(z)|.$$

*Proof.* Let  $z^d + \sum_{j<d} c_j z^j = 0$  with  $|c_j| \leq C$  for an ordinary integer  $C$ . If  $|z| > 1 + dC$ , then

$$\left| \sum_{j<d} c_j z^j \right| \leq dC|z|^{d-1} < |z|^d,$$

which excludes a root; so every root is finite. Sums and products of finite elements with infinitesimal differences have infinitesimal differences, so  $\text{st}$  is a ring homomorphism on the finite elements (Lemma 2.2 for  $\mathcal{O}_{\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}}$ ) and commutes with evaluation of polynomials; applying it to the equation gives the second assertion. For finite  $z \in \mathbf{No}[i]$ ,  $|z| \leq |\Re z| + |\Im z|$  is finite; taking standard parts in  $|z|^2 = (\Re z)^2 + (\Im z)^2$  gives  $\text{st}(|z|)^2 = |\text{st}(z)|^2$ , and  $\text{st}(|z|) \geq 0$ , so uniqueness of nonnegative real square roots gives the identity.  $\square$

For an ordinary  $d \geq 2$  define the ordinary integer

$$q_{E,d} = \begin{cases} 1, & E = \mathbf{No} \text{ and } d \text{ odd,} \\ 2, & E = \mathbf{No} \text{ and } d \text{ even,} \\ d, & E = \mathbf{No}[i]. \end{cases} \quad (16.18)$$

**Theorem 16.27** (A bounded number of lattice preimages at every scale [10]). *Let  $F \in \mathcal{N}_1(\Lambda)$  have degree  $d \geq 2$ , and let  $H \in \mathbf{No}_{>0}$ . There is a positive surreal  $T_0$  such that, for every  $T \in \mathbf{Oz}$  with  $T > T_0$ , the class*

$$\{u \in \Lambda : |F(u) - F(T)| \leq H\} \quad (16.19)$$

*has at most  $q_{E,d}$  elements. In particular  $F(\Lambda)$  has at most  $q_{E,d}$  points in the closed ball of radius  $H$  about  $F(T)$ .*

*Proof.* Write  $F(X) = a_d X^d + \sum_{j < d} a_j X^j$  with  $a_d \neq 0$ , and put  $T_0 = \omega \cdot \max(1, H/|a_d|, |a_j/a_d| \ (j < d))$ . For  $T > T_0$  we have  $T > 1$  and, for every ordinary  $n \geq 1$ ,  $T > n|a_j/a_d|$  and  $T > nH/|a_d|$ . Since  $d - j \geq 1$  and  $d - 1 \geq 1$ , it follows that

$$b_j := \frac{a_j}{a_d T^{d-j}} \ (j < d) \quad \text{and} \quad \frac{H}{|a_d| T^{d-1}} \quad \text{are infinitesimal.} \quad (16.20)$$

Fix such a  $T$  and an element  $u$  of (16.19), and put  $z = u/T$ . Dividing  $F(u) - F(T)$  by  $a_d T^d$  gives

$$z^d + \sum_{j < d} b_j z^j = 1 + \sum_{j < d} b_j + \delta, \quad |\delta| \leq \frac{H}{|a_d| T^d} \leq \frac{H}{|a_d| T^{d-1}}, \quad (16.21)$$

so  $\delta$  is infinitesimal. Thus  $z$  is a root of a monic polynomial with finite coefficients, hence finite by Lemma 16.26, and taking standard parts gives  $\text{st}(z)^d = 1$ . There are at most  $d$  such residues in  $\mathbb{C}$ ; among real residues only 1 is possible for odd  $d$ , and only  $\pm 1$  for even  $d$ . So there are at most  $q_{E,d}$  possible residues  $\text{st}(u/T)$ .

It remains to show that each residue belongs to at most one lattice input. Let  $u, v$  be elements of (16.19) with  $\text{st}(u/T) = \text{st}(v/T) = \zeta$ , and put  $z = u/T$ ,  $w = v/T$ . Write  $F(u) - F(v) = (u - v)Q_F(u, v)$  with  $Q_F(u, v) = \sum_{j=1}^d a_j \sum_{\ell=0}^{j-1} u^{j-1-\ell} v^\ell$ . Direct expansion gives

$$\frac{Q_F(u, v)}{a_d T^{d-1}} = \sum_{\ell=0}^{d-1} z^{d-1-\ell} w^\ell + \sum_{j=1}^{d-1} b_j \sum_{\ell=0}^{j-1} z^{j-1-\ell} w^\ell. \quad (16.22)$$

Its standard part is  $d\zeta^{d-1}$ , of modulus  $d$  because  $\zeta^d = 1$ ; by Lemma 16.26 the standard part of its modulus is  $d$ , so

$$|Q_F(u, v)| > \frac{d}{2} |a_d| T^{d-1}.$$

If  $u \neq v$ , the triangle inequality gives  $|F(u) - F(v)| \leq 2H$ , hence

$$|u - v| = \frac{|F(u) - F(v)|}{|Q_F(u, v)|} < \frac{4H}{d|a_d|T^{d-1}} < 1,$$

the middle quantity being infinitesimal by (16.20). This contradicts lattice separation for  $0 \neq u - v \in \Lambda$  (Lemma 16.1). So  $u = v$ , and (16.19) has at most  $q_{E,d}$  elements. Every image point in the ball is the image of an element of (16.19), which gives the last assertion.  $\square$

The proof uses the finite residue polynomial  $Z^d - 1$  only as a label for possible clusters. It assumes no convergence of an iterative root-finding scheme, no compactness of an omnific interval, and no existence theorem for new roots in the full field.

**Theorem 16.28** (Arbitrary-scale holes in nonlinear polynomial images [10]). *Let  $F \in \mathcal{N}_1(\Lambda)$  have degree  $d \geq 2$ . For every  $H > 0$  and every  $B \geq 0$  in  $\mathbf{No}$  there is  $c \in \Lambda$  with  $|c| > B$  such that*

$$\{y \in E : |y - c| \leq H\} \cap F(\Lambda) = \emptyset. \quad (16.23)$$

*For the real lattice this is an empty closed interval; for the Gaussian lattice it is an empty closed surcomplex disk.*

*Proof.* Put  $q = q_{E,d}$  and choose, by Lemma 16.1, a positive  $M \in \mathbf{Oz}$  with  $M > H$ . Let  $L = M + 1$  and  $R = 3qL + M$ . Apply Theorem 16.27 with radius  $R$  to get  $T_0$ , and choose  $T \in \mathbf{Oz}$  with  $T > T_0$  and also

$$|F(T)| > B + 3qL.$$

This is possible: by (16.15), for  $T \geq \max(1, R_F)$  one has  $|F(T)| \geq \frac{1}{2}|a_d|T^d \geq \frac{1}{2}|a_d|T$ , so any  $T \in \mathbf{Oz}$  exceeding  $T_0$ ,  $1$ ,  $R_F$  and  $2(B + 3qL)/|a_d|$  works (Lemma 16.1).

Set  $y = F(T) \in \Lambda$  and consider the  $q + 1$  lattice points  $c_j = y + 3jL$ ,  $j = 0, \dots, q$ . The closed balls of radius  $M$  about them are pairwise disjoint, since distinct centres are at distance at least  $3L > 2M$ . Each lies in the closed ball of radius  $R$  about  $y$ , since  $|x - y| \leq |x - c_j| + 3jL \leq M + 3qL$ . That ball contains at most  $q$  points of  $F(\Lambda)$  by Theorem 16.27, so one of the  $q + 1$  smaller balls contains none. Its centre  $c = c_j$  satisfies  $|c| \geq |y| - 3qL > B$ , and since  $H < M$  the ball of radius  $H$  about  $c$  is disjoint from  $F(\Lambda)$  as well.  $\square$

Failure of surjectivity is therefore not a single small defect near zero: the gaps exceed any prescribed infinite surreal radius and occur beyond any prescribed bound in modulus, that is, at arbitrarily large moduli.

**Corollary 16.29** (All rational bijections of the two lattices [10; 16]). *A rational function  $F \in E(X)$  defining a bijection  $\Lambda \rightarrow \Lambda$  has the form*

$$F(X) = uX + b, \quad b \in \Lambda, \quad u \in \Lambda^\times,$$

*and every such map is a bijection. Thus the rational bijections of  $\mathbf{Oz}$  are  $X \mapsto \pm X + b$  with  $b \in \mathbf{Oz}$ , and those of  $\mathbf{Oz}[i]$  are  $X \mapsto uX + b$  with  $b \in \mathbf{Oz}[i]$  and  $u \in \{\pm 1, \pm i\}$ .*

*Proof.* By Theorem 16.25,  $F$  is a polynomial, so  $F \in \mathcal{N}_1(\Lambda)$ . A constant is not a bijection of  $\Lambda$ . Degree at least two is excluded by Theorem 16.28, because the centre  $c$  there is itself a lattice point missing from the image. Hence  $F(X) = aX + b$  with  $b = F(0) \in \Lambda$  and  $a = F(1) - F(0) \in \Lambda$ . Surjectivity gives  $x \in \Lambda$  with  $F(x) = b + 1$ , so  $ax = 1$  and  $a$  is a unit of  $\Lambda$ . Conversely, an affine map with slope  $u \in \Lambda^\times$  has the lattice-preserving inverse  $X \mapsto u^{-1}(X - b)$ . The unit lists are those of Lemma 2.4.  $\square$

Here “rational” means a rational function in one variable over the ambient field  $E$ . Arbitrary functions  $\Lambda \rightarrow \Lambda$ , and maps such as coordinatewise conjugation on  $\mathbf{Oz}[i]$ , are not being classified. 16 shows that surjectivity alone already forces the affine form, for every unital constant ring (Theorem 16.43).

## 16.6 Fresh scales: image gaps, fibers and surjective self-maps

This subsection and the next are manuscript 16’s. They work over an arbitrary constant ring and replace the lattice-separation mechanism of Section 16.5 by a *fresh scale*: an exponent above an entire set-sized group of old exponents. Throughout,  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $E_k = k((X^{\mathbf{No}}))$  (so  $E_{\mathbb{R}} = \mathbf{No}$ ,  $E_{\mathbb{C}} = \mathbf{No}[i]$ ),  $D \subseteq k$  is a unital subring, and  $\mathcal{A}_{D,k} = D + \Pi_k$  as in Section 2.3;  $\mathcal{A}_{\mathbb{Z},\mathbb{R}} = \mathbf{Oz}$  and  $\mathcal{A}_{\mathbb{Z}[i],\mathbb{C}} = \mathbf{Oz}[i]$ . The ring  $\mathcal{A}_{k,k} = k + \Pi_k$  of normal forms without negative exponents (16’s  $\mathcal{I}_k$ ) is *not* the ring  $\mathcal{O}_k$  of finite surreals:  $\omega \in \mathcal{A}_{k,k}$  and  $\omega^{-1} \notin \mathcal{A}_{k,k}$ . 16’s  $\mathcal{P}_k$  and  $\mathcal{A}_D$  are  $\Pi_k$  and  $\mathcal{A}_{D,k}$ ; its defect polynomial  $D_{\zeta,c}$  is  $d_{\zeta,c}$  (the letter  $D$  is the constant ring), its centre  $\mu$  is  $s$ , its affine part  $L_{\zeta}$  is  $a_{\zeta}$  and its forbidden coefficient  $A_{\zeta}$  is  $c_{\zeta}$ . For a set-sized additive subgroup  $G \subseteq \mathbf{No}$ ,  $K_G = k((X^G))$  is the Hahn field with supports in  $G$ , identified with its normal-form image in  $E_k$  ( $X^g \mapsto \omega^g$ ).

**Lemma 16.30** (Fresh exponents [16]). *Every set-sized family of elements of  $E_k$  lies in  $K_G$  for some set-sized subgroup  $G \subseteq \mathbf{No}$ . There is  $b > 0$  with  $b > g$  for all  $g \in G$  (a fresh exponent), and then  $jb + g > 0$  for  $j > 0$  and  $jb + g < 0$  for  $j < 0$  ( $j \in \mathbb{Z}$ ,  $g \in G$ ); in particular  $G \cap \mathbb{Z}b = \{0\}$ .*

*Proof.* Let  $G$  be the subgroup generated by the union of the supports; it is a set, and  $K_G$  is a set-sized Hahn field. Choose  $b$  strictly above the set  $G \cup \{0\}$  (Lemma 3.1); since  $G = -G$ ,  $b + g > 0 > -b + g$ , and the same follows for all nonzero integral multiples of  $b$ .  $\square$

Freshness dominates the whole group, not only finitely many leading exponents: to dominate  $G = \mathbb{Q}$  one may take  $b = \omega$ , while no ordinary real suffices (16).

**Theorem 16.31** (Laurent blocks [16]). *Let  $b > G$  be fresh and  $t = \omega^b$ . The map  $\iota_b : K_G((T^{-1})) \rightarrow E_k$ ,  $\sum_{j \leq m} c_j T^j \mapsto \sum_{j \leq m} c_j \omega^{jb}$ , is an injective field homomorphism, the identity on  $K_G$ , with  $\iota_b(T) = t$ . Each nonzero  $c_j$  occupies the support block  $jb + \text{supp } c_j$ ; blocks with different  $j$  are disjoint and ordered by  $j$ . In particular  $t$  is transcendental over  $K_G$ .*

*Proof.* For  $j < \ell$  and  $g, h \in G$ ,  $(\ell b + h) - (jb + g) = (\ell - j)b + (h - g) > 0$ . A nonempty subset of the union of the blocks has a greatest element: take the greatest outer index of a block meeting it (the indices are bounded above), then the greatest element in that block. So the image is a legitimate Hahn sum even when the  $c_j$  have infinite supports. Addition is coefficientwise; in a product a fixed outer index receives contributions from finitely many pairs of outer indices, each an ordinary Hahn product. A nonzero coefficient gives a nonempty block that cannot cancel, so  $\iota_b$  is injective; a unital injective ring map from a field preserves inverses, and injectivity on  $K_G[T]$  is transcendence.  $\square$

**Corollary 16.32** (Exact support test [16]). *For  $L = \sum_{j \leq m} c_j T^j$ ,  $\iota_b(L) \in \mathcal{A}_{D,k}$  if and only if  $c_j = 0$  for all  $j < 0$  and  $c_0 \in \mathcal{A}_{D,k}$ ; the same holds with  $\mathcal{A}_{k,k}$  in place of  $\mathcal{A}_{D,k}$ . Positive-index blocks are purely infinite.*

*Proof.* Negative-index blocks consist of negative exponents and cannot cancel; the block of index 0 is  $c_0$ ; positive-index blocks do not affect the negative terms or the constant term.  $\square$

**Theorem 16.33** (One fresh value forces polynomiality [16]). *Let  $F \in K_G(T)$ . Its denominator does not vanish at  $t$ , and  $F(t) \in \mathcal{A}_{D,k}$  if and only if  $F \in K_G[T]$  and the constant coefficient of  $F$  lies in  $\mathcal{A}_{D,k}$ . The same holds with  $t$  replaced by  $ut$  for any  $u \in K_G^\times$ ; these inputs lie in  $\Pi_k$ . If the proper part of  $F$  begins  $aT^{-n} + O(T^{-n-1})$  with  $a \neq 0$ ,  $n \geq 1$ , then  $F(t)$  has the nonzero block  $-nb + \text{supp } a$ , and if the constant coefficient of the polynomial part has no negative terms, the leading negative exponent of  $F(t)$  is  $\max \text{supp } a - nb$ , with the leading coefficient of  $a$ .*

*Proof.* Transcendence gives the first assertion. Expand  $F$  at infinity by polynomial division and a geometric inverse; its image under  $\iota_b$  is  $F(t)$ , and a rational function has no negative Laurent coefficients exactly when it is a polynomial. Apply Corollary 16.32. Substituting  $uT$  multiplies  $c_j$  by  $u^j$  and fixes  $c_0$ ;  $ut$  occupies the positive block  $b + G$ . The last assertion is the block order of Theorem 16.31.  $\square$

One fresh value detects negative Laurent tails, not integer-valuedness:  $F(X) = X/2$  passes the test but fails at  $X = 1$  (16).

**Lemma 16.34** (Branches of a shifted value [16]). *Let  $d \geq 2$ , let  $K$  be a field of characteristic 0 containing the  $d$ th roots of unity, and  $f = a_d X^d + \dots + a_0 \in K[X]$  with  $a_d \neq 0$ . For every  $c \in K$  the equation  $f(Y) = f(T) + c$  has exactly  $d$  distinct roots in  $K((T^{-1}))$ , indexed by  $\zeta^d = 1$ , of the form*

$$Y_{\zeta,c} = \zeta T + (\zeta - 1)s + O(T^{-1}), \quad s = \frac{a_{d-1}}{da_d}.$$

*Proof.* With  $u = T^{-1}$  and  $W = uY$ , multiplying by  $u^d$  gives  $\sum_j a_j u^{d-j}(W^j - 1) - cu^d = 0$ , which at  $u = 0$  is  $a_d(W^d - 1) = 0$  with simple roots  $\zeta$  (the  $W$ -derivative there is  $da_d \zeta^{d-1} \neq 0$ ). The coefficient of  $u^n$  is then solved uniquely step by step, giving  $W_\zeta \in K[[u]]$  with  $W_\zeta(0) = \zeta$ , and  $Y = TW_\zeta(T^{-1})$ ; the  $d$  distinct leading coefficients give  $d$  distinct roots, which exhaust the roots of a polynomial of degree  $d$ . Comparing coefficients of  $T^{d-1}$  in  $f(\zeta T + w + \dots) = f(T) + c$  gives  $da_d \zeta^{d-1} w + a_{d-1} \zeta^{d-1} = a_{d-1}$ , that is  $w = (\zeta - 1)s$ .  $\square$

Put  $\mathbf{a}_\zeta(T) = \zeta T + (\zeta - 1)s$  and  $\mathbf{d}_{\zeta,c}(T) = f(\mathbf{a}_\zeta(T)) - f(T) - c$ . The choices of  $\zeta$  and  $s$  cancel the terms of degrees  $d$  and  $d - 1$ , so  $\deg \mathbf{d}_{\zeta,c} \leq d - 2$  unless  $\mathbf{d}_{\zeta,c} = 0$ . Since  $\mathbf{a}_\zeta$  fixes  $-s$ ,

$$\mathbf{d}_{\zeta,c}(-s) = -c : \quad (16.24)$$

a nonzero output shift cannot be produced by an affine change of input, even if the coefficients of  $f$  are infinite or infinitesimal (16). Degrees and leading coefficients lc below refer to the formal variable  $T$ .

**Theorem 16.35** (Forbidden layers [16]). *In Lemma 16.34 let  $c \neq 0$ . For each  $\zeta^d = 1$  put  $q_\zeta = \deg \mathbf{d}_{\zeta,c}$ ,  $r_\zeta = d - 1 - q_\zeta$  and  $\mathbf{c}_\zeta = -\text{lc}(\mathbf{d}_{\zeta,c})/(da_d \zeta^{d-1})$ . Then  $\mathbf{d}_{\zeta,c} \neq 0$ ,  $1 \leq r_\zeta \leq d - 1$ ,  $\mathbf{c}_\zeta \neq 0$  and*

$$Y_{\zeta,c} = \mathbf{a}_\zeta(T) + \mathbf{c}_\zeta T^{-r_\zeta} + O(T^{-r_\zeta-1}); \quad \text{in particular} \quad Y_{1,c} = T + \frac{c}{da_d} T^{1-d} + O(T^{-d}),$$

so the bound  $d - 1$  is attained for every  $c \neq 0$ .

*Proof.* By (16.24) the defect is nonzero, of degree between 0 and  $d - 2$ . The remainder  $R = Y_{\zeta,c} - \mathbf{a}_\zeta$  lies in  $T^{-1}K[[T^{-1}]]$  and is nonzero; let  $\chi T^m$  ( $m \leq -1$ ) be its first term. Taylor expansion gives  $0 = \mathbf{d}_{\zeta,c} + f'(\mathbf{a}_\zeta)R + \sum_{j \geq 2} f^{(j)}(\mathbf{a}_\zeta)R^j/j!$ . The linear term has degree  $d - 1 + m$  and leading coefficient  $da_d \zeta^{d-1} \chi$ , while the  $j$ th term has degree at most  $d - j + jm < d - 1 + m$  because  $(j - 1)(1 - m) > 0$ . Hence  $q_\zeta = d - 1 + m$  and  $\text{lc}(\mathbf{d}_{\zeta,c}) + da_d \zeta^{d-1} \chi = 0$ . For  $\zeta = 1$ ,  $\mathbf{a}_1 = T$  and  $\mathbf{d}_{1,c} = -c$ .  $\square$



**Theorem 16.36** (Uniform image gap [16]). *Let  $f \in K_G[X]$  have degree  $d \geq 2$ , let  $b > G$  be fresh and  $t = \omega^b$ . Then*

$$f(\mathcal{A}_{k,k}) \cap (f(t) + K_G) = \{f(t)\}, \quad \text{hence} \quad f(\mathcal{A}_{D,k}) \cap (f(t) + K_G) = \{f(t)\}.$$

*For every  $0 \neq c \in K_G$ , every root in  $\mathbf{No}[i]$  of  $f(Y) = f(t) + c$  has a nonzero block  $-r_\zeta b + \text{supp } c_\zeta$  of negative exponents, with  $1 \leq r_\zeta \leq d - 1$  as in Theorem 16.35.*

*Proof.* Work over  $\mathbb{C}((X^G))$  if  $k = \mathbb{R}$  (the roots of unity are ordinary constants, so no new exponents arise). Lemma 16.34 and Theorem 16.31 give  $d$  distinct roots of  $f(Y) - f(t) - c$  in  $\mathbf{No}[i]$ , hence all of them. For  $c \neq 0$  each has, by Theorem 16.35, a nonzero coefficient in a block of negative index, which lies below exponent 0 and cannot cancel against the affine constant or another block; so no root lies in  $\mathcal{A}_{\mathbb{C},\mathbb{C}}$ , and none in  $\mathcal{A}_{k,k} \supseteq \mathcal{A}_{D,k}$ . For  $c = 0$ ,  $t \in \Pi_k$  gives the value  $f(t)$ .  $\square$

The same  $b$  works for every  $c \in K_G$  at once, and  $G$  need not be Archimedean, countable or divisible (16).

*Remark 16.37* (Special cases printed elsewhere [16; 08]). Three nonsolvability statements proved separately are instances of Theorem 16.36 with  $G = \{0\}$ , where every  $b > 0$  is fresh and  $K_G = k$ . (a)  $x^2 = \omega^2 + 1$  has no solution in  $\mathbf{Oz}$ : take  $f = X^2$ ,  $t = \omega$ ,  $c = 1$ ; this is an example of the Diophantine report [30]. (b) For  $a \in \mathbb{R}_{>0}$ ,  $\gamma > 0$ ,  $0 \neq \beta \in \mathbb{Z}$  and  $n \geq 2$ ,  $X^n = a\omega^\gamma + \beta$  has no solution in  $\mathbf{Oz}$ : put  $X = a^{1/n}Y$  with  $Y \in \mathcal{A}_{\mathbb{R},\mathbb{R}}$ ,  $f = aY^n$ ,  $t = \omega^{\gamma/n}$ ,  $c = \beta$ ; this is the two-term root obstruction of the Diophantine report [30], whose short proof by the binomial expansion is kept there. (c) For  $\gamma > 0$  and  $m \geq 2$ ,  $X^m = \omega^\gamma + 1$  has no root in  $\mathbf{Oz}[i]$ : take  $k = \mathbb{C}$ ,  $f = X^m$ ,  $t = \omega^{\gamma/m}$ ,  $c = 1$ ; this is 08's invisible equation (Theorem 15.25), whose explicit expansion (15.18) exhibits the forbidden block of layer  $r_1 = m - 1$ . Theorem 16.36 is also consistent with 10's local count (Theorem 16.27): for fresh  $t$  the fiber of  $f(t)$  has at most  $e(f) \leq d$  points in the complex case and at most one ( $d$  odd) or two ( $d$  even) in the real case (Theorem 16.40 below), within 10's bound  $q_{E,d}$ .

**Corollary 16.38** (Omnific non-solvability [16]). *If  $f$  maps  $\mathcal{A}_{D,k}$  into itself and has degree at least two, then with  $K_G$  and  $t$  as above,  $f(Y) = f(t) + c$  has no solution  $Y \in \mathcal{A}_{D,k}$  for any  $0 \neq c \in K_G \cap \mathcal{A}_{D,k}$ , although its right side lies in  $\mathcal{A}_{D,k}$ . For  $f \in k[X]$  one may take  $G = \{0\}$  and any  $b > 0$ .*

**Definition 16.39** (Centered polynomial [16]). For  $f$  of degree  $d \geq 2$  with centre  $s = a_{d-1}/(da_d)$  write  $h(U) = f(U - s) = \sum_{j=0}^d h_j U^j$  (so  $h_{d-1} = 0$ ) and  $e(f) = \gcd\{j \geq 1 : h_j \neq 0\}$ .

**Theorem 16.40** (Exact fresh fibers [16]). *For  $\zeta^d = 1$ ,  $f(a_\zeta(T)) = f(T)$  if and only if  $h(\zeta U) = h(U)$  if and only if  $\zeta^{e(f)} = 1$ ; if this fails, the branch  $Y_{\zeta,0}$  has a nonzero negative Laurent term, and if it holds,  $Y_{\zeta,0} = a_\zeta$ . Consequently, for  $f \in K_G[X]$  of degree at least two and fresh  $t$ ,*

$$\{y \in \mathcal{A}_{D,k} : f(y) = f(t)\} = \{\zeta t + (\zeta - 1)s : \zeta \in k, \zeta^{e(f)} = 1, (\zeta - 1)s \in \mathcal{A}_{D,k}\},$$

*distinct  $\zeta$  giving distinct points. If  $s$  has a nonzero negative-exponent part the fiber is  $\{t\}$ ; if  $s \in \Pi_k$ , all  $e(f)$  points occur when  $k = \mathbb{C}$ , and one or two when  $k = \mathbb{R}$ , according as  $e(f)$  is odd or even.*

*Proof.*  $a_\zeta(T) + s = \zeta(T + s)$  gives the first equivalence; the second is  $\zeta^j = 1$  for all supported  $j$ , that is  $\zeta^{e(f)} = 1$  by Bézout. If the defect vanishes,  $a_\zeta$  is a root with leading coefficient  $\zeta$ , hence the branch by uniqueness of the simple-root lift; otherwise the remainder is nonzero and the argument of Theorem 16.35 applies. Every root is a branch; nonsymmetric branches have a forbidden block; a symmetric one is  $\zeta t + (\zeta - 1)s$ , whose fresh term cannot affect the old negative terms or the

constant, so it lies in  $\mathcal{A}_{D,k}$  exactly when  $(\zeta - 1)s$  does (and  $\zeta \in k$ : a nonreal  $\zeta$  gives a nonreal leading coefficient). Distinct  $\zeta$  have distinct fresh leading coefficients; multiplying a nonzero negative part of  $s$  by  $\zeta - 1 \neq 0$  keeps it; the count is that of roots of unity in  $k$ .  $\square$

**Example 16.41** (Worked fibers and defects [16]). (a) For  $f = X^d$ ,  $s = 0$  and  $e(f) = d$ : the fresh fiber of  $t^d$  in  $\mathbf{Oz}[i]$  is  $\{\zeta t : \zeta^d = 1\}$  (the constant term of  $\zeta t$  is 0, so  $\zeta$  need not be a Gaussian integer), and for  $0 \neq c \in k$  the shifted roots are  $\zeta t(1 + ct^{-d})^{1/d} = \zeta t + \frac{\zeta c}{d} t^{1-d} + \frac{\zeta(1-d)c^2}{2d^2} t^{1-2d} + \dots$ , with first negative layer exactly  $d - 1$  (a formal binomial expansion followed by the block embedding). (b)  $f = X^3 + X^2$  has  $s = \frac{1}{3}$ ,  $h = U^3 - \frac{1}{3}U + \frac{2}{27}$  and  $e(f) = 1$ , so  $f(t)$  has the single preimage  $t$  in  $\mathbf{Oz}$  and in  $\mathbf{Oz}[i]$ ; this does not make  $f$  injective ( $f(0) = f(-1)$ ). (c)  $f = \omega X^2 + X$  has the infinitesimal centre  $1/(2\omega)$ ; at  $t = \omega^\omega$  the other algebraic preimage of  $f(t)$  is  $-t - \omega^{-1}$ , and  $f$  is in fact injective on  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , since  $f(x) = f(y)$ ,  $x \neq y$ , forces  $x + y = -\omega^{-1}$ . (d) For  $f = X^3 + \lambda X + \nu$  ( $s = 0$ ),  $d_{\zeta,c} = \lambda(\zeta - 1)T - c$ : the identity branch first detects the shift at layer two ( $\frac{c}{3}T^{-2}$ ), a nontrivial branch with  $\lambda \neq 0$  at layer one ( $-\lambda(\zeta - 1)T^{-1}/(3\zeta^2)$ ). (e) For  $f = X^4 + pX^2 + qX + r$ ,  $d_{\zeta,c} = p(\zeta^2 - 1)T^2 + q(\zeta - 1)T - c$ ; the first negative terms are  $cT^{-3}/4$  at  $\zeta = 1$ ,  $-qT^{-2}/2$  at  $\zeta = -1$  ( $q \neq 0$ ) and  $ipT^{-1}/2$  at  $\zeta = i$  ( $p \neq 0$ ); when a displayed coefficient vanishes the next term of the finite defect decides. (f) Freshness cannot be weakened casually: with  $G = \mathbb{Q}$ ,  $F(T) = \omega/T$  takes the value 1 at the non-fresh  $T = \omega$ ; and  $f = X^2$ ,  $t = \omega$  with the shift  $2\omega + 1$ , which is not below a fresh exponent, gives  $(t + 1)^2$ , an omnific square.

**Theorem 16.42** (Rational self-maps over any constant ring [16]; second route to Theorem 16.25). *Let  $F \in E_k(X)$  be a rational function, in lowest terms, whose values lie in  $\mathcal{A}_{D,k}$  at every point of  $\mathcal{A}_{D,k}$  where it is defined. Then  $F$  is a polynomial. Hence the rational self-maps of  $\mathcal{A}_{D,k}$  are exactly its integer-valued polynomials.*

*Proof.* Put the finitely many coefficients in some  $K_G$  (Lemma 16.30) and take a fresh  $t = \omega^b \in \Pi_k \subseteq \mathcal{A}_{D,k}$ . The denominator does not vanish at  $t$ , and  $F(t) \in \mathcal{A}_{D,k}$ , so  $F$  is a polynomial by Theorem 16.33.  $\square$

For  $D = \mathbb{Z}, \mathbb{Z}[i]$  this is 10's Theorem 16.25; the two proofs are different routes. 10's uses finite differences and lattice separation ( $|x| \geq 1$  for  $0 \neq x \in \mathbf{Oz}$ ), which fails for  $D = \mathbb{Q}$  or  $D = k$ ; 16's uses one fresh input and works for every  $D$ . The bare statement that integer-valued rational functions are polynomials is not new in principle: it is the D-ring phenomenon of Gunji–McQuillan [14], as recalled by Liu [17, Proposition 1.12]; for  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , whose unit groups are finite, a nonconstant polynomial cannot take all its values in the unit group, which is the link with that criterion. 16 avoids importing a theorem about set-sized domains into a proper-class ring and supplies an explicit fresh witness (16).

**Theorem 16.43** (Surjective rational self-maps [16]; second route to Corollary 16.29). *A rational function over  $E_k$  that is defined at every point of  $\mathcal{A}_{D,k}$  and maps  $\mathcal{A}_{D,k}$  into itself is surjective onto  $\mathcal{A}_{D,k}$  if and only if it is  $X \mapsto uX + a$  with  $a \in \mathcal{A}_{D,k}$  and  $u \in D^\times$ ; every such map is bijective, and under composition they form the affine group  $\mathcal{A}_{D,k} \rtimes D^\times$ ,  $(a, u)(a', u') = (a + ua', uu')$ . For  $\mathbf{Oz}$  these are  $X \mapsto \pm X + a$ , and for  $\mathbf{Oz}[i]$  they are  $X \mapsto uX + a$  with  $u \in \{\pm 1, \pm i\}$ .*

*Proof.* By Theorem 16.42 the map is a polynomial; constants are not surjective. If its degree is at least two, then for a fresh  $t$  the element  $F(t) + 1 \in \mathcal{A}_{D,k}$  is not a value, by Theorem 16.36. So  $F = uX + a$  with  $a = F(0)$  and  $u = F(1) - F(0)$  in  $\mathcal{A}_{D,k}$ , and surjectivity at  $a + 1$  gives  $ux = 1$  with  $x \in \mathcal{A}_{D,k}$ , so  $u \in D^\times$  (Lemma 2.4). The inverse is  $X \mapsto u^{-1}(X - a)$ .  $\square$

For  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  this contains 10's classification of the rational bijections; 16 assumes only surjectivity and allows every unital  $D \subseteq k$ , and its proof uses the image gap where 10's uses holes of every radius ([Theorem 16.28](#)). Both are kept.

**Corollary 16.44** (Omitted values form a proper class [16]). *If  $F$  is an integer-valued polynomial of  $\mathcal{A}_{D,k}$  of degree at least two, then  $\mathcal{A}_{D,k} \setminus F(\mathcal{A}_{D,k})$  is a proper class; changing  $F$  on a set of inputs cannot make it surjective.*

*Proof.* Given a set  $S \subseteq \mathcal{A}_{D,k}$ , enlarge  $G$  to contain the supports of  $S$  and of the coefficients of  $F$  and take  $b > G$ . The omitted value  $F(\omega^b) + 1$  has leading exponent  $db + g_d$  ( $g_d$  the leading exponent of the leading coefficient), which is not in  $G$ , so it is not in  $S$ . Changing  $F$  on a set adds at most a set of values.  $\square$

Complex conjugation bijects  $\mathbf{Oz}[i]$  but is not a rational function of one surcomplex variable (a rational function agreeing with  $X$  on the ordinary integers is  $X$ ); admitting  $F(\overline{X})$  adds exactly the maps  $u\overline{X} + a$ . The theorem is one-variable:  $(x, y) \mapsto (x + y^2, y)$  is a nonlinear polynomial bijection of  $\mathcal{A}_{D,k}^2$  (16).

## 16.7 Finite tests, fresh probes and sampling

For a unital  $D \subseteq k$  and  $r \geq 1$  put  $\text{Int}(D^r) = \{g \in \text{Frac}(D)[\mathbf{X}] : g(D^r) \subseteq D\}$  and  $\mathcal{N}_r(\mathcal{A}_{D,k}) = \{F \in E_k[\mathbf{X}] : F(\mathcal{A}_{D,k}^r) \subseteq \mathcal{A}_{D,k}\}$ , so that  $\mathcal{N}_r(\mathbf{Oz})$  and  $\mathcal{N}_r(\mathbf{Oz}[i])$  are those of [Section 16](#).

**Proposition 16.45** (Constant-ring transfer [16]; 10 for  $D = \mathbb{Z}, \mathbb{Z}[i]$ ). *For every unital  $D \subseteq k$  and  $r \geq 1$ ,  $\mathcal{N}_r(\mathcal{A}_{D,k}) = \text{Int}(D^r) \oplus \Pi_k[\mathbf{X}]$  additively, the second summand an ideal, and the coefficientwise constant term is a ring homomorphism onto  $\text{Int}(D^r)$ , split by inclusion.*

*Proof.* Let  $F \in \mathcal{N}_r(\mathcal{A}_{D,k})$  have degree at most  $d$  in each variable. The Newton coefficients (16.4) are integral combinations of values at  $\{0, \dots, d\}^r$ , hence lie in  $\mathcal{A}_{D,k}$ , and  $(d!)^r \binom{\mathbf{X}}{\mathbf{k}} \in \mathbb{Z}[\mathbf{X}]$ , so every monomial coefficient of  $F$  lies in  $\text{Frac}(D) + \Pi_k$ . Split  $F = g + p$  with  $g \in \text{Frac}(D)[\mathbf{X}]$ ,  $p \in \Pi_k[\mathbf{X}]$ . The constant term is a ring homomorphism on  $\text{Frac}(D) + \Pi_k$  ([Lemma 2.2](#)), so for  $\mathbf{a} \in D^r$ ,  $g(\mathbf{a}) = \text{ct}(F(\mathbf{a})) \in D$ ; thus  $g \in \text{Int}(D^r)$ . Conversely, for  $\mathbf{x} = \mathbf{a} + \mathbf{z}$  with  $\mathbf{a} \in D^r$ ,  $\mathbf{z} \in \Pi_k^r$ , every term of  $g(\mathbf{a} + \mathbf{z}) - g(\mathbf{a})$  contains a factor  $z_j$ , so it lies in the ideal  $\Pi_k$  of  $\mathcal{A}_{k,k}$ , and  $p(\mathbf{x}) \in \Pi_k$ ; hence  $F(\mathbf{x}) \in g(\mathbf{a}) + \Pi_k$ . Uniqueness is  $k \cap \Pi_k = 0$ , and the section is the inclusion.  $\square$

For  $D = \mathbb{Z}$  and  $\mathbb{Z}[i]$  this is [Theorems 16.7](#) and [16.9](#) (10); 16 proves it for  $r = 1$  and for several variables with any  $D$ . The finite-difference identity needs no convergence argument: both sides agree at the grid points (16).

**Theorem 16.46** (Optimal point tests [16]; 10). *Let  $L \subseteq \mathbb{N}^n$  be a finite nonempty lower set ( $\alpha \in L$  and  $\beta \leq \alpha$  coordinatewise imply  $\beta \in L$ ) and  $V_L = \text{span}_{\mathbf{No}}\{\mathbf{X}^\alpha : \alpha \in L\}$ . For  $F \in V_L$  the following are equivalent:  $F(\mathbf{Oz}^n) \subseteq \mathbf{Oz}$ ;  $F(\alpha) \in \mathbf{Oz}$  for all  $\alpha \in L$ ; all coefficients of the expansion  $F = \sum_{\alpha \in L} c_\alpha \binom{\mathbf{X}}{\alpha}$  lie in  $\mathbf{Oz}$ . Moreover  $|L|$  is the least size of a set of points of  $\mathbf{Oz}^n$ , arbitrary omnific points allowed, whose values test membership for all of  $V_L$ . In particular, for one variable and degree at most  $d$ , the  $d + 1$  points  $0, \dots, d$  suffice and no  $d$  points do.*

*Proof.* The  $\binom{\mathbf{X}}{\alpha}$ ,  $\alpha \in L$ , form a basis of  $V_L$ : each has leading monomial  $\mathbf{X}^\alpha / \prod_j \alpha_j!$  and its other monomials have smaller indices, which stay in  $L$ . In a linear extension of the coordinatewise order the evaluation matrix  $((\binom{\beta}{\alpha}))$  on  $L$  is integral and unitriangular, so  $c_\alpha = \sum_{\beta \leq \alpha} (-1)^{|\alpha| - |\beta|} \prod_j \binom{\alpha_j}{\beta_j} F(\beta)$ .

Each  $(X_j)_{\alpha_j}$  preserves  $\mathbf{Oz}$  (Lemma 16.5). If fewer than  $|L|$  points are proposed, their evaluation maps have a common nonzero kernel element  $h \in V_L$ ; a polynomial vanishing on  $\mathbb{Z}^n$  is zero, so  $h(\mathbf{n}) \neq 0$  for some  $\mathbf{n} \in \mathbb{Z}^n$ , and  $h/(2h(\mathbf{n}))$  passes all proposed tests but takes the value  $\frac{1}{2}$  at  $\mathbf{n}$ .  $\square$

For rectangular  $L$  the equivalences are 10's Theorem 16.6; general lower sets (for instance total degree at most  $d$ , with  $\binom{n+d}{n}$  points) and the optimality statement are 16's. So the binomial polynomials form a basis of the  $\mathbf{Oz}$ -module  $\mathcal{N}_1(\mathbf{Oz})$ , each element having finitely many nonzero coordinates, and the countable set  $\mathbb{Z}$  tests all of  $\mathcal{N}_1(\mathbf{Oz})$ ; no topological density is asserted (16).

**Proposition 16.47** (A finite Gaussian test [16]). *Let  $F \in \mathbf{No}[i][\mathbf{X}]$  have total degree at most  $d$ ,  $M = d!$  and  $C_M = \{a + bi : a, b \in \mathbb{Z}, 0 \leq a, b < M\}$ . Then  $F$  preserves  $\mathbf{Oz}[i]$  if and only if its values lie in  $\mathbf{Oz}[i]$  on  $L_d \cup C_M^n$ , where  $L_d = \{\alpha \in \mathbb{N}^n : |\alpha| \leq d\}$ . This is an explicit finite certificate, not a claim of minimal size.*

*Proof.* The values on  $L_d$  give  $F = g + p$  with  $g \in M^{-1}\mathbb{Z}[i][\mathbf{X}]$  and  $p \in \Pi_{\mathbb{C}}[\mathbf{X}]$  (the denominators  $\prod_j \alpha_j!$  divide  $d!$ ). With  $h = Mg \in \mathbb{Z}[i][\mathbf{X}]$ ,  $g(\mathbf{z}) \in \mathbb{Z}[i]$  means  $h(\mathbf{z}) \equiv 0 \pmod{M\mathbb{Z}[i]}$ , which depends only on the residues of the coordinates, and  $C_M$  is a complete residue system. So  $g \in \text{Int}(\mathbb{Z}[i]^n)$ ; apply Proposition 16.45.  $\square$

The ordinary grid alone is not enough over  $\mathbf{Oz}[i]$ :  $\binom{i}{2} = (-1 - i)/2$  (Remark 16.10).

**Lemma 16.48** (Independent fresh coordinates [16]). *For a set-sized subgroup  $G$  there are positive  $b_1, \dots, b_n$  whose classes are linearly independent over  $\mathbb{Q}$  modulo  $G$ , and the  $t_j = \omega^{b_j}$  are algebraically independent over  $K_G$ .*

*Proof.* Choose  $b_j$  above the set-sized rational span of  $G, b_1, \dots, b_{j-1}$ ; a nontrivial relation would contradict the freshness of the last participating  $b_j$ . Distinct monomials  $\prod_j t_j^{\alpha_j}$  have exponents in distinct cosets of  $G$ , so their  $K_G$ -multiples have disjoint supports.  $\square$

With such  $b_j$ , let  $H = G + \sum_j \mathbb{Q}b_j$ , choose  $B > H$ , and let  $\xi^{(i)} \in \mathcal{A}_{D,k}^n$  have  $i$ th coordinate  $\omega^B$  and  $j$ th coordinate  $\omega^{b_j}$  for  $j \neq i$ .

**Theorem 16.49** (Coordinate-dominant probes [16]). *If  $F \in K_G(X_1, \dots, X_n)$  and  $F(\xi^{(i)}) \in \mathcal{A}_{D,k}$  for  $1 \leq i \leq n$ , then  $F \in K_G[X_1, \dots, X_n]$ . Consequently every rational function mapping  $\mathcal{A}_{D,k}^n$  into  $\mathcal{A}_{D,k}$  at its points of definition is a polynomial.*

*Proof.* Fix  $i$  and view  $F$  as a rational function of  $X_i$  over  $E_i = K_G(X_j : j \neq i)$ . Substituting  $X_j = t_j$  ( $j \neq i$ ) embeds  $E_i$  in  $K_H$  (Lemma 16.48); a reduced numerator and denominator stay coprime (their Bézout identity survives) and keep their degrees. As  $\omega^B$  is fresh over  $K_H$ , Theorem 16.33 makes the denominator constant in  $X_i$ , so  $F \in E_i[X_i]$  for each  $i$ . If  $F = P/Q$  in lowest terms in the unique factorization domain  $K_G[\mathbf{X}]$  had nonconstant  $Q$ , an irreducible factor of positive degree in some  $X_i$  would remain a denominator over  $E_i$  (Gauss's lemma).  $\square$

**Corollary 16.50** (Workspace-uniform rational certificates [16]). *Fix  $K_G$  and  $d \geq 0$ . For rational functions over  $K_G$  with numerator of total degree at most  $d$  in lowest terms, being a self-map of  $\mathbf{Oz}^n \rightarrow \mathbf{Oz}$  is tested by  $L_d$  together with the  $n$  probes  $\xi^{(i)}$  (at most  $\binom{n+d}{n} + n$  values;  $d+2$  for  $n=1$ ), and of  $\mathbf{Oz}[i]^n \rightarrow \mathbf{Oz}[i]$  by  $L_d \cup C_d^n$  together with the probes.*

*Proof.* The probes force a polynomial, of total degree at most  $d$ ; then apply Theorem 16.46 or Proposition 16.47. Necessity is clear since all test points lie in the domain.  $\square$

These certificates are uniform for a whole coefficient workspace  $K_G$ ; they are relative to exact coefficient data, zero tests and a certificate of freshness, and are not an algorithm for arbitrary presentations of surreal coefficients (16).

**Theorem 16.51** (No set-sized sample tests all rational maps [16]). *Let  $\Lambda$  be  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  and  $S \subseteq \Lambda$  a set. There is  $\beta > 0$  such that  $R_S(X) = \omega^\beta / (X - \frac{1}{2})$  is defined on all of  $\Lambda$ , satisfies  $R_S(S) \subseteq \Pi_k$ , and  $R_S(\omega^{2\beta}) = \sum_{m \geq 0} 2^{-m} \omega^{-(2m+1)\beta} \notin \Lambda$ .*

*Proof.*  $\frac{1}{2} \notin \Lambda$ . Clear the set of supports of the  $(s - \frac{1}{2})^{-1}$ ,  $s \in S$  (Proposition 3.4(i)) to get  $\beta$ . At  $x = \omega^{2\beta}$ ,  $R_S(x) = \omega^{-\beta}(1 - \frac{1}{2}\omega^{-2\beta})^{-1}$  is a nonzero infinitesimal.  $\square$

Thus no set of points, with all values known, detects all rational self-maps, even among functions of numerator degree 0 and denominator degree 1 with no pole on  $\Lambda$ : the single function  $\omega / (X - \frac{1}{2})$  takes purely infinite values at every ordinary and Gaussian integer, yet  $\omega / (\omega^2 - \frac{1}{2}) = \omega^{-1} + \frac{1}{2}\omega^{-3} + \dots \notin \mathbf{Oz}$ . For polynomials, by contrast, the ordinary integers suffice in every degree (Theorem 16.46) (16).

**Theorem 16.52** (Co-small integrality [16]). *If  $F \in E_k(X)$  is defined and  $\mathcal{A}_{D,k}$ -valued outside a set-sized subset of  $\mathcal{A}_{D,k}$ , then  $F$  is an integer-valued polynomial of  $\mathcal{A}_{D,k}$ : a rational function that is not a self-map fails on a proper class of inputs (poles counted as failures).*

*Proof.* Let  $S$  contain all failures and put its supports and the coefficients of  $F$  into  $K_G$ . If  $F$  is not a polynomial, a fresh  $t \notin S$  gives  $F(t) \notin \mathcal{A}_{D,k}$  (Theorem 16.33). If  $F$  is a polynomial with  $F(a) \notin \mathcal{A}_{D,k}$ , enlarge  $G$  by the supports of  $a$ ; the constant coefficient of  $F(a + T)$  is  $F(a)$ , so  $F(a + t) \notin \mathcal{A}_{D,k}$  by Corollary 16.32, and  $a + t \notin S$ .  $\square$

Theorems 16.51 and 16.52 do not conflict: the first builds a bad function adapted to a set of successful tests, the second says its failures form a proper class; neither provides one set meeting the failures of every function (16).

**Proposition 16.53** (Hulls [16]). *For  $S \subseteq \mathcal{A}_{D,k}$  let  $\text{Hull}_{\text{rat}}(S)$  be the set of  $a \in \mathcal{A}_{D,k}$  such that every rational function over  $E_k$  defined and  $\mathcal{A}_{D,k}$ -valued on  $S$  is defined and  $\mathcal{A}_{D,k}$ -valued at  $a$  (poles off  $S$  allowed), and define  $\text{Hull}_{\text{pol}}$  with polynomials. Then  $\text{Hull}_{\text{rat}}(S) = S$  for every set  $S$ , while  $\text{Hull}_{\text{pol}}(\mathbb{Z}) = \mathbf{Oz}$  and  $\text{Hull}_{\text{pol}}(\mathbb{Z}[i]) = \mathbf{Oz}[i]$ .*

*Proof.* For  $a \notin S$ , clearing the supports of the  $(s - a)^{-1}$  gives  $\beta$  with  $\omega^\beta / (X - a)$  purely infinite on  $S$  and a pole at  $a$ . The polynomial statements follow from Proposition 16.45: values on  $D$  force the ordinary part into  $\text{Int}(D)$ , and the purely infinite part is harmless.  $\square$

The pole allowed in the hull is not what drives Theorem 16.51, whose functions have no pole on the ring (16). In a fixed smaller Hahn field whose value group has no exponent above the coefficient group, the fresh-scale argument cannot be reused; this is one of 16's questions (Question 18.14).

## 16.8 Scope of the polynomial arithmetic

Two mechanisms act in this section and have different hypotheses. The reflection theorem Theorem 16.11 transports the size-sensitive divisibility mechanism of Theorem 5.1 through finite polynomial algebra; that mechanism applies to arbitrary set coefficient fields in the class Hahn construction. The results of Section 16.5 use instead *finite reduction plus lattice separation*: at a large scale  $T$ , polynomial preimages of a prescribed output ball reduce to roots of  $Z^d - 1$ , and a



nonzero reduced divided difference would force two inputs of one cluster to lie less than 1 apart, which the lattice forbids. This gives [Theorems 16.27](#) and [16.28](#) without a count of points in a surreal interval or a real-valued asymptotic density; it uses the real or surcomplex modulus, the standard part and the unit-spaced lattices. Neither mechanism should be used as a substitute for the other. 16 adds a third: *fresh scales* ([Sections 16.6](#) and [16.7](#)). A monomial above an entire set-sized exponent group separates formal Laurent powers into disjoint support blocks, so negative Laurent tails cannot be hidden by cancellation; this needs no lattice separation, works for every unital  $D \subseteq k$ , and is a support argument, not a convergence argument in the fine topology. It needs the full surreal exponent class: in a fixed Hahn field whose value group has no exponent above the coefficient group it cannot be reused (16).

Manuscript 10 proposes [Theorem 16.11](#) and the arbitrary-scale bound [Theorem 16.27](#) as new; the other results of this section are explicit consequences or transfers of classical arithmetic facts (Newton interpolation, binomial-ring facts [11], the least-common-multiple pattern [7],  $p$ -adic approximation by integer-valued polynomials [6]), proved here to make the scope complete. The results do not classify all class ideals of  $\mathbf{Oz}$  or of  $\mathcal{N}_r(\mathbf{Oz})$ , all automorphisms of  $\mathbf{Oz}$ , or the factorization of arbitrary omnific integers, and they neither construct nor use a global surcomplex exponential. They do not assert a general transfer principle from  $\mathbb{Z}$  to  $\mathbf{Oz}$ ; the omitted type of [Proposition 16.3](#) is a concrete warning against such a principle. The Gaussian theorem [Theorem 16.9](#) does not replace the arithmetic of  $\text{Int}(\mathbb{Z}[i]^r)$  by a binomial basis, and no Gaussian analogue of the congruence criterion [Theorem 16.14](#) is given. For  $r \geq 1$  the completion of [Corollary 16.20](#) is a  $p$ -adic, not a profinite, completion. The classification of [Corollary 16.29](#) concerns rational functions only. Manuscript 10 records a question suggested by this material: for the full omnific integers, determine how much an automorphism can do inside the ideal  $\Pi$  while fixing the ordinary shadow  $\text{ct}$  (every ring endomorphism of  $\mathbf{Oz}$  preserves  $\text{ct}$ , by the uniqueness in [Theorem 5.1](#)). Manuscript 16 presents the image-gap, forbidden-layer and fresh-fiber package and the two sampling statements as its proposed contributions, credits rational collapse to the D-ring theory and Newton interpolation to the classical literature, and proves only one-variable surjectivity rigidity.

## 17 Natural ordinal arithmetic is not a quotient of the omnific integers

This section is manuscript 11's. Let  $\mathcal{G}_{\text{nat}}$  be the Grothendieck ring of the ordinals under Hessenberg's natural sum and natural product. Every ordinal has a finite Cantor normal form  $\sum_j n_j \omega^{\gamma_j}$ , each  $\gamma_j$  has its own finite Cantor normal form  $\sum_\alpha m_{j,\alpha} \omega^\alpha$ , and under natural multiplication  $\omega^{\gamma_j} = \prod_\alpha (\omega^{\omega^\alpha})^{m_{j,\alpha}}$ . Uniqueness of these forms gives the free commutative semiring  $\mathbb{N}[T_\alpha : \alpha \in \mathbf{On}]$ , and adjoining additive inverses gives

$$\mathcal{G}_{\text{nat}} \cong \mathbb{Z}[T_\alpha : \alpha \in \mathbf{On}], \quad (17.1)$$

with  $T_\alpha$  represented by  $\omega^{\omega^\alpha}$  under the natural embedding in  $\mathbf{No}$ . Natural operations, not the usual noncommutative ordinal operations, are essential. The description (17.1) appears in the question [12]; the argument above fixes the operations.

**Theorem 17.1** (The ordinal quotient question [11, 13]). *Every unital ring map  $h : \mathbf{Oz} \rightarrow \mathcal{G}_{\text{nat}}$  has image exactly the ordinary integral constants. In particular  $\mathcal{G}_{\text{nat}}$  is not isomorphic to a quotient ring of  $\mathbf{Oz}$ . More generally, no ring map  $\mathbf{Oz} \rightarrow \mathbb{Z}[X]$  with  $X \neq \emptyset$  is surjective.*



*Proof. First proof.*  $\mathbb{Z}[X]$  is residually set-sized by [Theorem 5.12](#) (for any class  $X$  of indeterminates), so  $h$  kills  $\Pi$  and factors through  $\text{ct}$ ; its image is  $\mathbb{Z}$ , a proper subring since  $T_0 \notin \mathbb{Z}$ .

*Second proof, specific to this target.* The additive group of  $\Pi$  is divisible ( $s/n \in \Pi$  for every ordinary  $n \geq 1$ ), while no nonzero element of  $\mathbb{Z}[T_\alpha]$  is divisible by every  $n$  (each of its finitely many integral coefficients would be). So every additive map from  $\Pi$  to  $\mathbb{Z}[X]$  is zero, and a unital ring map from  $\mathbf{Oz} = \mathbb{Z} + \Pi$  has image  $\mathbb{Z}$ .  $\square$

The question was posted by Jesse Elliott as the final addition to MathOverflow question 188430, “Transcendence degree of the surreals over the subfield generated by the ordinals” (30 November 2014) [\[12\]](#); the displayed answer, by Eric Wofsey, concerns the transcendence degree, a different question. The answer to the quotient question is negative. 11 gives the short additive proof precisely so that the answer is not mistaken for a difficult new factorization theorem: no claim is made that the elementary obstruction was unknown to specialists or absent from every other source. The first proof illustrates the general residual theorem, and the second isolates the weaker mechanism that suffices for this target. The obstruction is not a cardinal comparison, since both rings may be proper classes; it is their divisibility and representation properties (11).

*Remark 17.2* (An independent answer [\[13\]](#)). Manuscript 13, dated a day after 11 and pinned to a repository commit that precedes 11’s placement, answers the same addendum negatively with the same two proofs: its lemma that every unital map from  $\mathbf{Oz}$  into a ring  $B$  with  $\bigcap_n nB = 0$  factors through  $\text{ct}$  is the second proof above, and its route through the maps  $q_F : \mathcal{G}_{\text{nat}} \rightarrow \mathbb{Z}[T_\alpha : \alpha \in F]$  ( $F$  finite) and the universal theorem is the first (its corollary for rings “separated by set-sized ring targets” is [Theorem 5.12](#)). 13 adds that the inclusion  $\mathcal{G}_{\text{nat}} \subset \mathbf{Oz}$  has no unital ring retraction, which is immediate since every unital map has image  $\mathbb{Z}$ , and derives (17.1) from Cantor normal forms in the same way. 13 records that the question writes the ring as  $K[\Omega]$ , with  $K$  a Grothendieck construction; that notation was not rechecked for this merge. 13 does not claim to answer the original transcendence-degree question, and states that its priority as a first answer is not established.

## 18 Boundaries, non-claims and open questions

### 18.1 What the results do not say

The universal theorem is a statement about homomorphisms into set-sized rings and about set-sized modules. It is not an isomorphism  $\mathbf{Oz} \cong \mathbb{Z}$ ; it does not classify class ideals, class modules, automorphisms or factorizations of  $\mathbf{Oz}$ ; it concerns multiplicative maps and says nothing against additive or valuation-theoretic observations; and it fails verbatim for every fixed set-sized exponent group, whose identity map is a set-sized target. Size alone does not prove it (finite supports have proper-class many monomials and an augmentation), nor does divisibility alone (only targets of positive characteristic are handled). The set-sized thresholds are exact for the five constructions and the two fixed-group examples treated; no classification of exponent groups or support restrictions is claimed, and generator counts follow no general law stated here. The large-quotient results describe specified subalgebras of  $\mathbf{Oz}/(1 + \omega^a)$  and conditional statements about primes; no existence theorem for nonarithmetic class primes is claimed. The later additions do not change this. The flat and projective dimensions of [Section 13](#) are those of set-sized finite-support cores, of coefficient-lattice modules and of the rings of [Section 12](#); they say nothing about the homological dimensions of the class ring  $\mathbf{Oz}$ , and a flat dimension computed for one subring of  $\mathbf{Oz}$  bounds nothing for another ([Remark 13.16](#)). The finite-support cores are not invisible to set-sized rings, only to finite ones

([Proposition 13.2](#)). The fresh-scale theorems concern polynomials and rational functions with finitely many coefficients and use the whole surreal exponent class.

As an internal ring,  $\mathbf{Oz}$  has all surreal scales available through its fraction field, a nonzero proper-class ideal  $\Pi$ , nonzero class-valued derivations and nonarithmetic prime quotients; as a source of representations on sets it carries no more scalar information than  $\mathbb{Z}$ . The gap is caused not by ignoring limits but by finite identities that become available once the source contains the appropriate Hahn sums (06). Adapting 06’s list of hypotheses that must remain visible to all the constructions, the proofs use four different size restrictions, each for its own reason: individual supports are sets, so that a common positive gap exists; the full class of smaller scales supplies arbitrarily large collision families; in the set-sized models an uncountable bound admits the countable witnesses, and regularity (or cofinality, or the finite-coordinate invariant) bounds unions of fewer than the relevant number of supports; and cardinal arithmetic enters only in computing the size of the ring, which is the exact threshold. Dropping any of them needs a new argument: finite supports lack the witnesses ([Section 9](#)), full supports over a fixed set group need not have gaps ([Remark 11.21](#)), and singular support bounds may preserve field closure but not the simultaneous gap property.

## 18.2 Non-claims of the sources

Every limitation stated by a source survives. They are listed and numbered by source; several recur in the text where the corresponding result is proved.

**03.** (1) Not peer reviewed; no Lean declarations. (2) “Developed here” is relative to the inspected material; no certified priority. (3) Neither Conway’s general refinement problem nor the general factorization problems for omnific integers are settled. (4) No classification of exponent groups or support restrictions; the cardinal theorem is a realization theorem for its family. (5) cf  $\kappa = \aleph_0$  is excluded. (6) Only the lower bound  $\mu$  on the number of algebra generators of  $\mathcal{N}_\kappa$  is proved. (7) No complete integrality test for mixed supports. (8) No descent equality  $\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa$ . (9) No statement on proper-class-valued valuations. (10) Infinite set-sized images of the normalizations are not classified, and the absence of finite images is a general consequence of containing the real algebraic integers. (11) No class tensor products and no class prime-ideal theorem; all Tor is over set-sized rings. (12) The finite checks do not verify the cardinal or class arguments. (13) Restricted Hahn fields, truncation integer parts, complete-integral-closure terminology and the small-unit lemma are standard or elementary, with no priority claim; Kuhlmann is context only. (14) The full-class theorem, common divisors and clearing are credited to the companion manuscript (05). (15) Density produces no convergent set-indexed net: every set-generated intermediate algebra is uniformly discrete, so the dense normalization needs more than any set of generators. (16)  $\frac{1}{2}\mathcal{N}[i]$  is not asserted to be a ring. (17) The flatness result is an application of standard module theory, not a new general flatness theorem. (18) Density does not impose continuity on abstract ring homomorphisms. (19) No claim about unrestricted surcomplex analytic continuation is made.

**04.** (1) Unrefereed; no Lean formalization; the universal theorems are candidate contributions without certified priority. (2) No classification of all class ideals, no class Zorn argument, no claim that proper class ideals extend to maximal ones. (3) The module equivalence covers set-sized modules only, not class modules, group representations of  $\mathrm{GL}_n(\mathbf{Oz})$ , or additive homomorphisms. (4) It does not say that  $P(\mathrm{ct}(x)) = 0$  implies  $P(x) = 0$ . (5) It does not classify infinite points of curves; the earlier project’s curve questions are not declared solved. (6) Nothing is inferred about fraction fields of the bounded-support rings. (7) The fixed-workspace results are lower bounds

only. (8) No named open conjecture is solved. (9) Additive coefficient maps escape the theorem and are not counterexamples. (10) The telescope has a direct precursor [16, Remark 3.4.4]; the constant term, finite quotients and common shifts come from the prior manuscript (02 of [30]). (11) The existence of standard part is classical and in the repository’s infrastructure. (12) The support-threshold theorem differs from fixed-group completion or omitted-type questions. (13) The finite checks do not prove normal forms, infinite identities, class factorizations or novelty. (14) The quotient, module, derivation and localization results are a connected theorem package deduced from the universal theorems, not unrelated discoveries.

**06.** (1) Unrefereed, not Lean-verified; priority not established by the targeted searches. (2)  $\text{pd } \mathbb{Z}$  is shown only to be at least 2. (3) Residue fields of nonarithmetic primes are not classified; the prime-reservoir proposition is a necessary condition, not an existence theorem. (4) The primality of  $\omega^{\sqrt{2}} + \omega + 1$  is imported; no factorization, refinement, GCD or unique-factorization conjecture is used or claimed. (5) No automorphisms, class ideals or factorization patterns are described. (6) No elementarity claim for  $A_\kappa^{06} \subseteq \mathbf{Oz}$ ; no compactness argument for a proper-class theory. (7) The Euler derivation is not asserted to be the Berarducci–Mantova derivation or compatible with exponentiation. (8) Real closedness of bounded-support Hahn fields is not a novelty claim, and regularity is not claimed necessary for field closure. (9) The idea of a universal set-sized quotient is not new to the repository. (10) The module correspondence is not a Morita equivalence of unrestricted module theories. (11) The invisible prime quotient does not say that a particular maximal class ideal fails to exist. (12) The polynomial-test theorem does not assert that a proper-class coordinate ring defines an ordinary scheme. (13) The finite-congruence observation is elementary and not a strong novelty claim. (14) Numerical tests and a successful build are not proofs; the formalization route was not executed.

**07.** (1) Conway’s refinement conjecture is neither settled nor used (the Abramov development [1] is cited for its claim and caveats only); no historical priority; no referee; no Lean. (2) The Cantor algebra is a specified subalgebra; neither the idempotents nor the prime ideals of  $\mathcal{Q}_a$  are classified, and  $\mathcal{Q}_a$  is not claimed to be a function ring. (3) The statement on nonarithmetic primes is conditional; no existence is asserted. (4) Surreal self-similarity is credited to Bagayoko–van der Hoeven. (5)  $\Theta_a$  is not claimed to preserve the ordinal embedding, birthdays, simplicity or the exponential;  $\vartheta$  is not additive and  $\Psi$  not multiplicative. (6) Quotient, completion and module classifications are consequences, not separate major discoveries. (7) The small-image theorem does not say that no set-valued invariant sees positive exponents. (8) The module equivalence is not a homological-dimension statement over class modules. (9) The global theorem is not a theorem about small approximations of  $\mathbf{Oz}$ . (10) The finite checks do not verify the transfinite or ambient-support arguments. (11) The CRT isomorphisms at different stages need not be coherent; evaluation at a point of  $\mathbb{Z}_2$  is not claimed to extend to  $\mathcal{Q}_a$ ; no topology on  $\mathcal{E}_a$  is used. (12) Class prime ideals are not classified, and no Zorn argument for a proper class of ideals is used. (13) No exponential function on surcomplex arguments is assumed.

**08.** (1) Candidate-original, not certified priority, not refereed, no Lean; the broad factorization questions for  $\mathbf{Oz}$  are not addressed. (2)  $\text{Frac}(A_\kappa^{08}) = F_\kappa^{08}$  is not claimed. (3) Residue-field isomorphism types are not classified. (4) No closure under derivations or exponentials, no initial-segment or birthday condition on the embedding. (5) The collision mechanism has published precedents ([16, Proposition 8.2.1]); the constant retraction and finite reductions are not principal novelty claims. (6) Fixed Hahn workspaces lie outside the theorem (an Archimedean group has

no small scales at all). (7) The non-integral-closure corollary is not a factorization theorem. (8) Class-sized modules are not classified; the theorem is not an isomorphism with  $\mathbb{Z}$ , and individual monomials can be represented faithfully in small rings. (9) The Hahn derivations  $D_\eta$  are not an exponential-compatible calculus. (10) No maximal-ideal theorem for proper-class rings; no assertion that proper class ideals lie in maximal ones. (11) The quantitative criterion gives sufficient conditions only. (12) The finite checks prove neither the class theorem, nor summability, nor real closedness. (13) The primality of  $\omega^{\sqrt{2}} + \omega + 1$  is an imported published theorem.

**09.** (1) Novelty provisional after a targeted comparison; no referee; no proof-assistant verification. (2) No named factorization conjecture solved; primality imported. (3) The number of maximal ideals and the isomorphism types of the residue fields of  $A_{\lambda,D}^{09}$  are not determined. (4) No assertion about the unrestricted (uncountable-support) ring over  $\Gamma_\lambda^{09}$ . (5) No class Zorn argument; the Jacobson statements concern set-sized rings. (6)  $\mathbf{Oz}[i]$  is a subring of the surcomplex field, not the field. (7) The checks do not verify infinite supports, cardinal arithmetic or the existence of maximal ideals. (8) The geometric identity is elementary with precedents; profinite, derivation and flatness statements are consequences; the Gaussian 2-torsion computation is standard. (9) No global surcomplex exponential, sine, integration or topology is used; the arithmetic quotient is not an ordered quotient. (10) The rotation quotient is a conceptual predecessor, not imported. (11) No new exponential derivation is constructed and no uniqueness of the Berarducci–Mantova derivation is claimed. (12)  $I/I^2 = 0$  is not a finite-presentation or smoothness claim.

**10.** (1) Unrefereed and not machine-checked; targeted review only; no named factorization conjecture solved. (2) For  $r \geq 1$  the  $p$ -adic completion is not a profinite completion. (3) No classification of all class ideals, automorphisms or factorizations. (4) No transfer principle from  $\mathbb{Z}$  to  $\mathbf{Oz}$ . (5)  $\text{Int}(\mathbb{Z}[i]^r)$  is not replaced by a binomial basis. (6) The derivation  $D_h$  is not the Berarducci–Mantova derivation. (7) The universe-relative reading must carry the size restriction in its statement. (8) The rational classification is not a classification of arbitrary functions or of coordinatewise conjugation. (9) The congruence theorem does not make reduction modulo an infinite omnific integer an ordinary finite congruence, and does not classify congruence-preserving class functions. (10) Classical ingredients (Newton interpolation, binomial rings, the least-common-multiple pattern,  $p$ -adic approximation) are credited, not claimed. (11) The grid test assumes exact surreal input; no effective encoding is claimed. (12) The finite checks do not test class functions, support lemmas or priority.

**11.** (1) The main theorem is relative to one fixed universe, not to fragments viewed in a larger universe. (2) No classification of prime or maximal class ideals. (3) No class module of Kähler differentials. (4) The presentation theorem is not quantifier elimination or a reconstruction of class-valued solutions. (5) Sharpness of the coinitality theorem is stated under  $\kappa^{<\kappa} = \kappa$  and  $|K| \leq \kappa$  as explicit extra assumptions (the unconditional value is a merge result, [Proposition 11.23](#)). (6) Necessary closure conditions for intermediate rings are not given. (7) The ordinal answer is not claimed hard or unknown to specialists. (8) Bounded-support coefficient rings and lower-scale geometric division have precedents in [16, Section 8]. (9) Targeted novelty assessment only; the finite checks prove no infinite-support or class-size statement. (10) The rotation-group theme is acknowledged, not originated. (11) The only answer to the MathOverflow question concerns transcendence degree. (12) No primality or refinement conjecture is newly settled; the ideals, class-valued representations and factorization theory of quotients invisible to set-sized rings are not

determined, and ordinary factorization questions about  $\mathbf{Oz}$  remain outside the conclusions.

**12.** (1) Nothing is claimed about the homological dimensions of the full proper-class ring  $\mathbf{Oz}$ ; all rings are set-sized and all embeddings use finite normal forms. (2) The weak global dimension of  $A_d$  is not computed; only the lower bound  $d + 1$  is proved. (3) Projective dimensions are given only as bounds (the exact values in rank one are merge results). (4) Arbitrary non-simplicial cones, arbitrary ideals and all homological localizations of the cores are not classified. (5) The cores are not integer parts of  $\mathbf{No}$ , not initial in the simplicity order, not preserved by all surreal automorphisms and not Hahn rings; the countable-rank ring still consists of finite sums. (6)  $\hat{A}$  is the profinite completion, not a Hahn completion or a completion in a surreal-valued topology. (7) The  $D + M$  framework (with the pullback literature, for instance [25]), filtered-colimit flatness, coherence criteria and the idempotent-kernel mechanism of formal étaleness are classical and credited; the proposed contributions are the Boolean Tor computation, its dimension consequences, the countable-rank realization and the ordered-cone comparison. (8) No named conjecture is claimed solved; the originality check was targeted, not an exhaustive priority audit. (9) Not refereed and not checked in Lean. (10) The finite audit uses  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$  on a five-point grid and proves nothing over  $\mathbb{Z}$  or for all exponents. (11) The Gaussian version uses only the quadratic extension  $\mathbf{No}[i]$ ; no analytic continuation and no topology on a proper class is used. (12) “Nilpotent” is a test condition on algebras; the cores are domains.

**13.** (1) The theorems are relative to size; nothing is claimed about all maps out of a fixed set-sized Hahn ring, whose identity is a counterexample ( $\mathbb{Z} + X\mathbb{R}[X]$  with evaluation at 1 is 13’s example). (2) No classification of class quotients, class residue objects or class ideals, and no maximal-ideal theorem or Zorn argument for class rings. (3) The ordinal result answers only the quotient addendum; the original transcendence-degree question has Wofsey’s answer; 13’s priority as a first answer is not established. (4) The integer divisibility of the purely infinite part is known (L’Innocente–Mantova); equation-only transfer along a retraction, filtered-colimit flatness and the algebra of idempotent quotient ideals are standard, and the homological and lifting statements are standard consequences, with no étaleness claimed under finite presentation. (5) The fiber theorem (printed in [30]) classifies only nonzero fibers of decomposable polynomials; there is no general classification of fibers, no Thue or Baker finiteness, and no enumeration of the ordinary solutions of its cubic. (6) The cardinal criterion is a necessary condition, not an existence theorem for detecting maps. (7) The statement on Hilbert’s tenth problem is a transfer of the classical theorem, with no complexity claim for Gaussian or number-field problems. (8) No claim about Conway’s refinement conjecture; no refinement, primality or unique-factorization property is used. (9) In the two-universe reading “not generated by a set” means not generated by a family from the smaller universe. (10) Unrefereed; no Lean; the finite checks verify identities only; the literature and repository searches were bounded, and the repository was neither audited nor built.

**14.** (1) Arbitrary finitely generated omnific ideals, with generators at several support levels, are not classified. (2) The rank-one modules with double dual  $B$  are not all listed. (3) No coefficient ring other than a principal ideal domain is treated. (4) The exact weak global dimension of  $A_0$  is not claimed. (5) The ideal-class group under  $\odot$  is not a subgroup of  $\text{Pic}(A)$ , and its identity is  $L(\mathfrak{o})$ , not  $A$ . (6) Full-class tensor products are explicit carriers only, and ordinary flat dimensions are asserted only over  $A_0$ . (7) No novelty is attached to noncoherence; the  $D + M$  context is credited to Dobbs–Papick, and no valuation-ring hypothesis is used. (8) The number field spanned by a multiplier ring need not be totally real. (9) The unimodular-direction criterion concerns coordinate



representatives, not all points of a projective scheme over a ring, and its example  $[1 : \sqrt{2}]$  is not new. (10) The matrix theorem concerns constant matrices at one common monomial scale. (11) Constant-term arithmetic, flat idempotent support ideals and cardinal obstructions are credited to companion manuscripts and are not presented as new. (12) No Conway factorization or refinement claim; no peer review and no new Lean verification; priority is provisional; the finite checks are supplementary and certify no arbitrary support, class universal property, flatness statement or Eisenstein criterion in all degrees.

**15.** (1) The main theorem is for the full class only; the finite-support ring and fixed Hahn workspaces are counterexamples. (2) Its local cardinal criterion is sufficient only. (3) No classification of class prime or maximal ideals; its prime classification carries a set-sized residue hypothesis. (4) The primality of  $\omega^{\sqrt{2}} + \omega + 1$  is imported from L’Innocente–Mantova and is not a new solution of Gonshor’s question. (5) No claim about Conway’s refinement problem; the repository `gaearon/conway-refinement` was noticed but neither built, audited nor relied on. (6) Only set-sized modules are classified; this is not a Morita statement. (7) Universe parameters must stay separate. (8) Geometric factorization and the congruence calculations are classical or elementary, the categorical deductions are not new techniques, and the rotation-group theorem is a precedent, not a dependency. (9) Blocking one extension of the finite-support augmentation is weaker than the theorem. (10) Embeddings of set-sized rings into  $\mathbf{Oz}$  are not excluded, and  $\mathbf{Oz}$  and  $\mathbb{Z}$  are not claimed to have the same first-order theory. (11) Candidate originality rests on a bounded search; no referee; no Lean; the checks cover finite identities only.

**16.** (1) Rational-to-polynomial collapse is not new in principle (D-rings: Gunji–McQuillan, as recalled by Liu; related framework in [8]). (2) Surjectivity rigidity covers one variable and rational functions only; conjugation, multivariate triangular bijections, class functions, definable maps and formal power series are excluded, and not every bijection of  $\mathbf{Oz}$  is claimed affine. (3) The fresh-scale arguments do not transfer to a fixed Hahn field without dominant exponents. (4) The certificates are relative to exact coefficient data, zero tests and a certificate of freshness; they are not a decision procedure for arbitrary surreal inputs. (5) No convergence, analytic continuation, surreal exponential, Berarducci–Mantova derivation, fine-topological completeness or factorization is used. (6) Newton interpolation and binomial bases are classical, and the Gaussian test is not claimed minimal. (7) The fiber theorem classifies the particular fresh fiber, not every fiber. (8) Priority rests on a bounded search; no referee; no Lean; the finite checks are not proofs.

**The merge.** (1) The results marked [merge] are consequences of the sources’ lemmas; they are no more refereed than the sources and are not formalized. (2) No general law is asserted for the generator counts. (3) For full-support Hahn rings over a prescribed set group, exact thresholds are computed only in Examples 10.4 and 10.5; the general questions of 04 and 10 stay open. (4) The support-threshold theorem is stated for  $k \in \{\mathbb{R}, \mathbb{C}\}$  as in 04 and 11, not for every set field. (5) No Gaussian analogue of 10’s congruence-preservation criterion is given. (6) Nothing is claimed about the homological dimensions of the class ring  $\mathbf{Oz}$ ;  $\mathrm{pd}_{A_d} D$  for  $d \geq 2$  and the weak global dimensions of the cores remain undetermined. (7) The finite-support cores of 12 and 14 are outside Theorem 5.1: they have set-sized images detecting the tail. (8) Proposition 13.12(v) needs a countably generated tail; among the five constructions it settles only  $A_\kappa^{08}$  with  $\mathrm{cf} \kappa = \aleph_0$ .



### 18.3 Open questions and their status

The sources pose the following questions, each as a question suggested by the proofs rather than as an established open problem. Several coincide and are merged.

**Question 18.1** (Least detecting targets over a fixed exponent group; 03, 04, 07, 08, 09, 10, 11, 13, 15). For a set-sized ordered group  $\Gamma$ , a family of admissible supports and a specified element (or the whole positive-support ideal), what is the least cardinality of a ring target detecting it, and which intrinsic support conditions characterize it?

*Status.* Open in general. The field bound (Proposition 10.3) is a lower bound that is exact for the five constructions and for the two examples of Section 10.2; there it answers 04’s “optimal small targets” question for its own example, and for 10’s question “when is the bound attained” it shows that the bound need not be attained (in the rational example, and for  $A_{\aleph_1}^{03}$  if CH fails). 08’s criterion and 11’s coinitiality theorem give sufficient conditions (07’s “sharp set-sized analogues” and 09’s “quantitative problem” are the same question). 13’s question (how large a target must be to detect a prescribed series, and whether its divisibility bound is optimal) and 15’s “optimal local cardinal bounds” are this question as well: their bounds are the monomial count of Proposition 10.1(ii), which the field bound dominates; the threshold theorem (Theorem 10.10) answers them for the five constructions, and Example 10.5 shows that the monomial bound need not be attained.

**Question 18.2** (Cardinal arithmetic without the matching hypothesis; 06). For  $\kappa$  uncountable regular with  $\kappa^{<\kappa} > \kappa$ , what is the least size of a target detecting a nonzero element of  $\Pi_\kappa^{06}$ , and is there an integer part of size exactly  $\kappa$  with the same lower bound?

*Status.* Answered. The least size is  $\kappa^{<\kappa}$  in ZFC (Theorem 11.11, a merge result from 06’s collision lemma); 08 constructs, for every infinite  $\kappa$  in ZFC, an integer part of size exactly  $\kappa$  with threshold  $\kappa$  (Theorem 11.16), without claiming its fraction field.

**Question 18.3** (Higher projective dimension; 06). What is  $\text{pd}_A \mathbb{Z}$  for the set-sized integer parts, and how does it depend on the cofinality of the principal-ideal chain?

*Status.* Partly answered.  $\text{fd}_A D = 1$  and  $\text{pd}_A D \geq 2$  are known for all five constructions. If the tail is countably generated, that is if the principal-ideal chain has countable cofinality, then  $\text{pd}_A D = 2$  (Proposition 13.12(v), a merge result from 12’s telescope); among the five constructions this is the case exactly for  $A_\kappa^{08}$  with  $\text{cf } \kappa = \aleph_0$ . For chains of uncountable cofinality (03, 06, 09, 11, and 08 with  $\text{cf } \kappa > \aleph_0$ ) the question stays open. For comparison, the ordered-cone cores have  $\text{pd } D = 2$  and the coordinate-cone cores  $d \leq \text{pd}_{A_d} D \leq d + 1$  (Question 18.12).

**Question 18.4** (Nonarithmetic primes and their residue fields; 06, 07). Which prime or maximal class ideals of  $\mathbf{Oz}$  have nonzero constant-term image and zero intersection with  $\mathbb{Z}$ , and which class fields occur as  $\text{Frac}(\mathbf{Oz}/P)$  for primes  $P \not\subseteq \Pi$ ? Can the embedded field  $\mathcal{F}_a$  distinguish such primes?

*Status.* Open. Necessary conditions: Corollary 14.10, Proposition 14.12, and Remark 14.13.

**Question 18.5** (The ambient binomial quotient; 07). Which normal-form invariants describe the elements and ideals of  $\mathcal{Q}_a$  beyond the Cantor algebra; which idempotents lie outside it; how are cyclotomic decompositions at different scales related?

*Status.* Open.

**Question 18.6** (Residue fields of the set-sized models; 08, 09). Which characteristic-zero fields of the threshold cardinality occur as maximal residue fields of  $A_\kappa^{08}$  and of  $A_{\lambda,D}^{09}$ , and how many maximal ideals are there?

*Status.* Open; existence, cardinality and at least  $\lambda$  nonarithmetic maximal ideals (for 09) are proved.

**Question 18.7** (Extra structure on the workspace; 08). Can prescribed residue-field behaviour be combined with closure under a surreal derivation or an exponential, or with an initial embedding, without destroying the finite-coordinate invariant or the cardinality?

*Status.* Open.

**Question 18.8** (Intermediate rings; 04, 11, 15). Which rings between the finite-support ring and  $D + \Pi_k$  (for instance, given by allowed families of support order types) still have set-sized reflection  $D$ ? Is enough closure under subordinate division also necessary?

*Status.* Open; [Theorem 9.1](#) settles the case of cardinal support bounds. 09's question on support bounds below a regular cardinal is addressed for the groups of 06 and 11 ([Sections 11.2](#) and [11.5](#)), where the support sizes of inverses and quotients are checked.

**Question 18.9** (Beyond ring observations; 04). Which weaker multiplicative compatibility conditions still force factorization through the residue, and which structured families of set-sized local models can retain a specified family of scales?

*Status.* Open.

**Question 18.10** (Invisible class domains and automorphisms; 10, 11, 15). What are the ideals, class-valued representations and factorization theory of proper-class quotient domains invisible to all set-sized rings? How much can an automorphism of  $\mathbf{Oz}$  do inside  $\Pi$  while fixing  $\text{ct}$ ?

*Status.* Open.

**Question 18.11** (Normalization; 03). Give a membership test for  $\mathcal{N}$  on mixed supports; describe the integral closure over a fixed finitely generated exponent subgroup; decide the descent  $\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa$ ; find the algebra generation number of  $\mathcal{N}_\kappa$  between  $\text{cf } \kappa$  and  $\kappa^{\aleph_0}$ ; find a class-sized replacement for the valuative criterion of integrality; describe the infinite set-sized images of  $\mathcal{N}$ .

*Status.* Open. 04's question on Diophantine fibers belongs to the sibling report [\[30\]](#), and so does 13's question on fibers of polynomials with nonlinear factors.

**Question 18.12** (Homological dimensions of the cores; 12, 14, merge). For  $d \geq 2$ , is  $\text{pd}_{A_d} D$  equal to  $d$  or to  $d + 1$ ? What are the weak global dimensions of  $A_d$  (at least  $d + 1$ ) and of  $A_0 = A_1$  (at least 2)? How do cores over non-simplicial cones, their arbitrary ideals and their homological localizations behave?

*Status.* Open. In rank one  $\text{pd } D = 2$ , and over every ordered-cone core  $\text{pd } D = 2$  ([Proposition 13.12](#)).

**Question 18.13** (Beyond one coefficient lattice; 14). Replace the single coefficient lattice by a finite filtration of lattices indexed by support levels and determine which extension data survive multiplication; characterize all finitely generated rank-one modules with double dual  $B$ ; determine how coefficient rings that are not principal ideal domains change the tensor kernel.

*Status.* Open.

**Question 18.14** (Fresh scales; 16). (a) Can the factorial-residue Gaussian certificate be replaced by an optimal or near-optimal grid, and how does a regular basis of  $\text{Int}(\mathbb{Z}[i])$  interact with the omnific splitting? (b) Under what hypotheses are the formal inverse branches of a polynomial map in several variables at independently fresh coordinates classified by finitely many forbidden support blocks (triangular automorphisms exclude a direct analogue of one-variable surjectivity rigidity)? (c) For a fixed set-sized Hahn subfield and its nonnegative-support ring, which coefficient subgroups admit enough dominant extensions inside the field to recover the image gap and the rational certificates?

*Status.* Open.

## 19 Verification record and formalization routes

### 19.1 Finite checks

Each manuscript ships a program that checks finite identities with exact arithmetic. They are kept byte-identical in `code/` with their recorded outputs in `data/`; all were rerun for the two merges on copies (Python 3.14.4, SymPy 1.14.0) and reproduce the recorded outputs, up to line endings and the recorded Python version.

| Source | What is checked                                                                                                                                                                                                 | Checks   | Needs  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| 03     | finite telescopes in a symbolic lattice, the unit series coefficients, lexicographic comparisons                                                                                                                | 2,362    | stdlib |
| 04     | 1,950 finite telescopes in $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ with remainders and positive supports, 600 constant-term and 900 augmentation products, 3 support failures and 3 distinguishing examples | 5,406    | stdlib |
| 06     | geometric remainders and supports, constant terms, boundary cases, Euler derivation, real complex-structure blocks                                                                                              | 17,586   | stdlib |
| 07     | weighted geometric identities, binomial remainders, first idempotents, dyadic refinement maps                                                                                                                   | 621      | SymPy  |
| 08     | telescopes, constant terms, Hahn Leibniz identities, binomial powers, root corrections                                                                                                                          | 1,277    | stdlib |
| 09     | telescopes, truncated comaximality, coefficient maps, finite-support evaluation, Gaussian derivation, gcd identities                                                                                            | 597      | stdlib |
| 10     | divided differences, Newton reconstruction, digit indicators and lifts, telescopes, rational differences                                                                                                        | 171,395  | SymPy  |
| 11     | finite division identities, positive quotient supports, augmentation and conjugation identities, Gaussian derivation                                                                                            | 1,858    | stdlib |
| 12     | homology of the Boolean tensor complex for $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$ on a five-point grid in ranks 1–4, with $d \circ d = 0$ ( <a href="#">Theorem 13.7</a> )                                   | 780      | SymPy  |
| 13     | 51 finite geometric remainders and their positivity, a cubic norm determinant, a Laurent norm identity, a rank-deficient point, a constant-term product                                                         | 6 groups | stdlib |
| 14     | ranks for $\alpha^d = 2$ , homogenized Rees equations, quadratic torsion generators, $P^2 = 2\mathbb{Z}[\sqrt{10}]$ and the mod-5 obstruction, action associativity and tensor balancing                        | 2,189    | SymPy  |
| 15     | finite telescopes and divisor truncations, constant-term products, finite-support evaluation, Gaussian matrix evaluation                                                                                        | 820      | stdlib |
| 16     | Newton reconstruction, lower-set interpolation matrices, Gaussian counterexample, 150 centered defects, 22 formal inverse branches                                                                              | 1,354    | SymPy  |

These checks catch sign, index and coefficient errors in finite special cases. None of them verifies an infinite support, a class-size argument, a cardinal computation, the existence of a maximal ideal, an Ext computation, a primality statement, or novelty; those rest on the written proofs (all sources). The merge results were not machine-checked. For the second merge an independent exact computation, not shipped, rechecked [Theorem 13.7](#) in 1,626 cases (multidegree and subset  $F$ , including every proper  $F$ , over  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$  in ranks up to 3) and the formulas of [Corollary 13.23](#) for  $d \leq 8$ ,  $n \leq 12$ ; like the shipped checks it verifies finite cases only.

## 19.2 Formalization routes

No statement of this report is formalized, and the repository has no omnific module. The sources propose compatible routes, recorded here as proposals only. First formalize the scaled-field collision lemma and the separation lemma, which need no surreals (06, 04, 07). Then the countable telescope in a set-sized Hahn workspace, with its two support inequalities, proved coefficientwise and not by convergence of partial sums (04, 06, 07, 08, 10). The set-sized criteria ([Theorems 10.7](#) and [10.8](#)) avoid proper-class syntax and are good first targets (08). The full-class theorem needs an explicit universe interface in which the target and the collision index

are small and the exponent class supplies smaller scales below every small support; a universe-polymorphic theorem about an arbitrary field does not supply them (04, 08, 10, 11). For the constructions, the finite-coordinate invariant of 08 must be preserved through field generation, real closure and truncation, and the module bound at a cardinal must use cyclic quotients or the direct collision, not an endomorphism ring (08, 06). The homological part comes last, over set-sized rings only (06). For 07's quotient, the support characterization of  $\mathbf{Oz}[\omega^{-a}]$  and the exact binomial kernel should be kept separate from the finite Chinese-remainder algebra. The repository's small-normal-form modules (`SmallNormalForm.lean`, `SmallNormalFormTruncation.lean`, `SmallNormalFormEvaluation.lean`, `SignSequenceValuationBalls.lean`, see [32]) are possible prerequisites (04); they are not a certificate for any result here. The later manuscripts add three proposals. 12 would define its cores as subrings of a monoid algebra cut out by the constant-coefficient predicate, with the directed-union equality, exactness of the finite-support simplex complex and the degreewise classification of [Theorem 13.7](#) as the main obligations, and the surreal embedding last. 14 would start from an abstract domain  $A = D \oplus P$  with  $P$  a  $k$ -linear ideal in which every element is a product of two, a hypothesis the finite-support model satisfies. 16 separates a finite part (simple-root recursion, centering, the leading-term argument, finite differences), a Hahn part (the block embedding) and an actual-surreal part (a common set-sized exponent workspace and a fresh exponent). None was executed.

## References

- [1] D. Abramov, *Conway-refinement*, public Lean development, <https://github.com/gaearon/conway-refinement>, inspected by 07 on 22 September 2026 for its proof claim and stated review caveats only; nothing from it is used.
- [2] V. Bagayoko and J. van der Hoeven, Surreal substructures, *Fundamenta Mathematicae* **266** (2024), 25–96, [doi:10.4064/fm231020-23-2](https://doi.org/10.4064/fm231020-23-2); arXiv:2305.02001.
- [3] V. Barucci, S. Gabelli and M. Roitman, Complete integral closure and strongly divisorial prime ideals, arXiv:math/0302223, 2003. Cited for terminology only.
- [4] A. Berarducci, *Surreal numbers, exponentiation and derivations*, arXiv:2008.06878, 2020 (Sections 3 and 8–10 for cuts, normal forms and the Hahn support calculus).
- [5] A. Berarducci and V. Mantova, Surreal numbers, derivations and transseries, arXiv:1503.00315, 2015.
- [6] P.-J. Cahen and J.-L. Chabert, *Integer-Valued Polynomials*, Mathematical Surveys and Monographs **48**, American Mathematical Society, 1997.
- [7] P. Cegielski, S. Grigorieff and I. Guessarian, Characterizing congruence preserving functions  $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$  via rational polynomials, arXiv:1506.00133, 2015.
- [8] M. M. Chems-Eddin, B. Feryouch, H. Mouanis and A. Tamoussit, On rings of integer-valued rational functions, arXiv:2410.16142v2, 2024. A related framework cited by 16; no result is used.
- [9] J. H. Conway, *On Numbers and Games*, Academic Press, 1976; second edition, A K Peters, 2001.
- [10] D. E. Dobbs and I. J. Papick, When is  $D + M$  coherent?, *Proceedings of the American Mathematical Society* **56** (1976), 51–54. Cited by 14 for context; no criterion is imported.
- [11] J. Elliott, Binomial rings, integer-valued polynomials, and  $\lambda$ -rings, *Journal of Pure and Applied Algebra* **207** (2006), 165–185, [doi:10.1016/j.jpaa.2005.09.003](https://doi.org/10.1016/j.jpaa.2005.09.003).
- [12] J. Elliott, Transcendence degree of the surreals over the subfield generated by the ordinals, MathOverflow question 188430, posted 30 November 2014, <https://mathoverflow.net/questions/188430>. Its final addition asks the quotient question answered in [Theorem 17.1](#); the answer by E. Wofsey concerns transcendence degree.

- [13] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series **110**, Cambridge University Press, 1986, doi:10.1017/CBO9780511629143.
- [14] H. Gunji and D. L. McQuillan, On rings with a certain divisibility property, *Michigan Mathematical Journal* **22** (1976), 289–299.
- [15] F.-V. Kuhlmann, Dense subfields of henselian fields, and integer parts, arXiv:1003.5681, 2010. Context only.
- [16] S. L’Innocente and V. Mantova, A factorisation theory for generalised power series and omnific integers, *Advances in Mathematics* **442** (2024), article 109513, doi:10.1016/j.aim.2024.109513; arXiv:1710.07304v5, 22 January 2024. Cited items: Fact 2.1.1 (closedness of Hahn fields), Remark 3.4.4 (the geometric product), Fact 8.0.1 (Gonshor’s Theorem 8.6) with the example on p. 46, Proposition 8.2.1 (truncation criterion for divisibility), Theorem B (primality of  $\omega^{\sqrt{2}} + \omega + 1$ ).
- [17] B. Liu, Ring structure of integer-valued rational functions, arXiv:2208.09935v3, 26 February 2024 (Proposition 1.12 recalls the D-ring criterion of [14]).
- [18] B. Poonen, Maximally complete fields, *L’Enseignement Mathématique* **39** (1993), 87–106 (Corollary 4).
- [19] The Stacks Project Authors, *The Stacks Project*, Section 10.39, Tag 00H9, Lemma 10.39.3, <https://stacks.math.columbia.edu/tag/00H9>.
- [20] The Stacks Project Authors, *The Stacks Project*, Tag 060H (a formally étale ring map that is not flat), <https://stacks.math.columbia.edu/tag/060H>.
- [21] The Stacks Project Authors, *The Stacks Project*, Tag 05CU (coherent rings), <https://stacks.math.columbia.edu/tag/05CU>.
- [22] The Stacks Project Authors, *The Stacks Project*, Tag 00DM (tensor algebra), <https://stacks.math.columbia.edu/tag/00DM>.
- [23] The Stacks Project Authors, *The Stacks Project*, Tag 052P (blow-up algebras), <https://stacks.math.columbia.edu/tag/052P>.
- [24] The Stacks Project Authors, *The Stacks Project*, Section 10.50, Tag 00I8, Lemmas 10.50.3 and 10.50.11, <https://stacks.math.columbia.edu/tag/00I8>.
- [25] F. Wang, H. Kim and T. Xiong, Finitistic weak dimensions of pullbacks, *Journal of Pure and Applied Algebra* **224** (2020), no. 6, 106274, doi:10.1016/j.jpaa.2019.106274. Cited by 12 for context; no theorem is used.
- [26] *Surreal* repository, foundations report, docs/foundations-and-computation/foundations/ (label found:eq:omnific, the definition  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ , and the fine-topology size obstructions).
- [27] *Surreal* repository, *Euclidean Three-Space over the Surreal Numbers*, docs/surreal/euclidean-three-space/, Part III (labels e3:cut:thm:smallquotient, e3:cut:cor:onlyimage) and 10-rotation-quotients-SOURCE\_AUDIT.md.
- [28] *Surreal* repository, trigonometry report, docs/surcomplex/trigonometry/ (label trigonometry:eq:split).
- [29] *Surreal* repository, *The First- $\kappa$  Coefficients*, docs/surcomplex/first-kappa-coefficients/ (label fkc:thm:main).
- [30] *Surreal* repository, sibling report on the Diophantine geometry of the omnific integers, docs/surreal/omnific-diophantine-geometry/, merged from manuscripts 01, 02 and 05 of the first batch and later additions; written concurrently with this report. Its rationality theorem for primitive constant directions and its exact decomposable fibers are the results cited here.
- [31] *Surreal* repository, docs/NOTATION.md.
- [32] *Surreal* repository, docs/NORMAL\_FORM\_BRIDGE.md.



- [33] *Surreal* repository, gamma and zeta functions report, docs/surcomplex/gamma-and-zeta-functions/.
- [34] *Surreal* repository, *A square-root obstruction to evaluating increasing Hahn series at  $\omega$* , docs/surreal/hahn-evaluation-at-omega/.
- [35] *Surreal* repository, differential equations report, docs/surcomplex/differential-equations/ (label diff:warn:ct).